ROTATION INVARIANCE OF THE THREE-DIMENSIONAL ABELIAN HIGGS MODEL: THE CURVATURE SECTOR

Abstract. King gave a construction of the continuum limit and the infinite-volume limit of the lattice $U(1)$ Higgs model in three dimensions, with a massive non-compact photon, on lattices of a single orientation, whose printed proof asserts several steps, and stated a theorem comparing two lattices whose orientations differ by $\theta_0 = \arctan(3/4)$ on periodic cubes. As printed that statement is false; a corrected form, for curvature sources read as 2-forms in each lattice's own frame and for the two lattices at the same spacing, is proved elsewhere as an implication from explicitly listed renormalisation-group inputs. We prove that King's infinite-volume state on curvature sources, that is, the generating functional of all curvature Schwinger functions, built on the cubes on which King's renormalisation group is run, is $O(3)$-invariant, as an implication from an explicit list of inputs, each proved here or in a companion manuscript and checked independently, or a statement of the literature printed with its proof, stated in mathematical form in one place; the three hypotheses and one proviso of earlier versions are proved. The proof passes through two statements, the corrected torus comparison on walled tori and a closing statement which says that King's state is the lower envelope of the walled tori. Both are proved as implications: the first by the corrected torus theorem from its renormalisation-group inputs, the second from a heavy-shell decoupling estimate at one configuration, which in turn follows from four shell estimates for King's block-spin flow with the shell mass inside the covariance. Every input is stated, with its grade, in one numbered list. Each is proved here, or proved in a companion manuscript whose proof has been checked independently, or is a statement of the literature cited at its printed grade. The theorem holds on all of King's range of the couplings, every positive (e, λ, m, μ_0) at block factor 5, with no further hypothesis: the bounds at long trees for the graphs with block vertices in the families with one inserted external field, the fluctuation estimates for the in-block contours rooted at the block centres and for the contours of the hybrid flow, the positivity condition on the constant that relates the two counterterm conventions on the walled tori, and the large-field bound on the skew tori at the final scale of Bałaban's bounds, which earlier versions carried as hypotheses, are proved; where the positivity condition fails, the invariance follows by right-continuity of King's state in the scalar mass. Among the inputs proved in companion manuscripts are a bound, at each renormalisation step and on the propagator domains of Bałaban's upper bound as well as on the torus, for the first-order term, in the photon fluctuation field, of the determinant of the step, uniform in the shell mass and with no bound on the size of the background, which has two independent proofs; on the skew tori of the corrected torus comparison, the decay, with the depth of its two points, of the difference of Bałaban's propagators on two nested domains; on King's cubes, the bound at each of two cutoffs, by the cut of the long lines, of the graphs whose trees are long, which gives their difference with no rate; and the transfer of the perturbative bound to Bałaban's propagator domains at order fourteen, for the families that the removal of the tail of the background produces. The theorem thus reads: King's curvature state is $O(3)$-invariant on all of King's range, and then invariant under the full Euclidean group; the statements of the literature that it uses are printed with their proofs. The steps that Bałaban and King assert without a printed argument, among them Bałaban's estimates at block factor 5, the renormalisation bound for the operator with a scalar wall included, King's cut of the long lines of his renormalised graphs on his cubes, and King's cube limit, are proved in companion manuscripts, in the forms consumed, and so are the six readings of Bałaban's papers behind them, the bounds for the vector-field propagators and the vector half of one completion included (Section 9). Proved here are the following: an exact involution reduces the torus comparison to one-sided bounds at a single spacing, correlation inequalities compare King's cubes with walled tori, the photon boundary condition is changed only across a matter-free collar where the change is controlled uniformly in the spacing, a two-level sandwich turns the heavy shell into a higher wall, and a continuity lemma for convex functionals carries invariance from a dense subgroup to $O(3)$. We prove that no argument built from the correlation inequalities alone can supply the closing statement, that no bounded region can be aligned with both lattices, and that the heavy-shell estimate holds at every fixed lattice spacing. Every input is graded, and the grades are collected in a table. An appendix lists the corrections to King's papers that the argument needs.

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1. Introduction

1.1. **Background.** The lattice $U(1)$ Higgs model couples a two-component scalar field $\phi = re^{\mathrm{i}\vartheta}$ to a non-compact abelian gauge field A. In three dimensions King gave a construction of its continuum limit [K1, K2], whose printed proof asserts several steps (items (5k) and (13) of Section 1.4), using the ultraviolet stability bounds of Bałaban [Ba1, Ba2, Ba3, Ba4], for the version of the model in which the photon is written in Feynman form and carries a mass $\mu_0 > 0$ [K2, (2.5)]. In [K1] the continuum limit is taken on tori. In [K2] it is taken in the cubes C_J with Neumann boundary conditions for A and Dirichlet boundary conditions for ϕ, realised on the lattice as the cubes C_J^K on which King's renormalisation group is run (Definition 2.15), and the infinite-volume limit $J \to \infty$ is then obtained from the correlation inequalities of Brydges, Fröhlich and Seiler [BFS], which make the cube expectations monotone in J [K2, Thm. 3.2] (with a J-independent Wick constant, in the form King's proof proves, for the boxes of that proof and for the cubes on which his renormalisation group is run; Lemmas 2.14 and 2.16). King states that the infinite-volume generating functional is translation invariant, reflection positive and entire in the coefficients of the sources [K2, Thms. 2.2–2.3]; on curvature sources we prove the parts that we use (Section 3).

Rotation invariance is not automatic, since all lattices in the construction have the same orientation. Theorem 2.4 of [K2] states that on the tori D_J the lattices $\varepsilon_k\mathbb{Z}^3$ and $R(\varepsilon_k\mathbb{Z}^3)$, with R the rotation by $\theta_0 = \arctan(3/4)$ in a coordinate plane, have the same continuum limit; since θ_0/π is irrational these rotations generate a dense subgroup of $SO(3)$. As printed the statement is false in two ways (Appendix A): the source it pairs with the plaquettes is one scalar function on both lattices, which couples the field to two different 2-forms, and the Wick constant it uses misses an order-e^2 logarithm. We therefore call the printed statement *King's torus conjecture*, and we use instead the corrected statement, for 2-form sources sampled in each lattice's own frame and for the two lattices at the same spacing, which is proved in [CT] as an implication from an explicit list of renormalisation-group inputs. Moreover, even granting the torus comparison, two steps remain between it and rotation invariance of King's state: the infinite-volume limit along tori, where correlation inequalities give no monotonicity, and its identification with the limit along cubes, which King writes can be expected from a cluster expansion [K2, (2.24)]. The ultraviolet analysis of [K1, K2] needs boundaries that are lattice hyperplanes [K2, p. 326], and no bounded region has that property for both lattices (Proposition 8.2). Where to place the two-lattice comparison is therefore the whole question. This paper places it on periodic tori carrying a smooth mass wall, and proves $O(3)$ invariance of King's state in the curvature sector as an implication from

an explicit numbered list of inputs (Section 1.4), each with a one-line mathematical statement and a grade. The list is the complete set of inputs that are not printed theorems whose proof has been read at the source; no input is asserted as a fact. Every input is proved here, proved in companion manuscripts and checked independently, or is a statement of the literature at its printed grade; the theorem holds on all of King's range. Among the inputs of the shell estimates is the bound (DOM) of Section 1.5 for the first-order term of the step determinants on Bałaban's propagator domains, which has two independent proofs; three statements inside the proofs of the torus comparison, of King's cube limit and of the shell estimates, (P7)–(P9) of Section 1.5, are proved in companion manuscripts as well, each by more than one argument. Section 9 restates in plain words every input that is not proved here.

1.2. **The torus route.** On each lattice separately King's cubes are compared with tori carrying a *smooth mass wall*. Fix R in the generating set $\mathcal{R}_0 = \{R_{\mu\nu}(\pm\theta_0)\}$, a real 2-form f and a function $g \geq 0$, both smooth with compact support. For a walled torus $\mathfrak{c} = (J', w)$, the periodic model on $D_{J'}$ in which the scalar mass squared is raised by $2w$ with $w \geq 0$ smooth and vanishing near the supports, the chain is

$$Z_{C_L}(\mathrm{i}\,R\cdot f, R\cdot g) \;=\; Z^{R^{-1}\Lambda}_{R^{-1}C_L}(\mathrm{i}f, g) \;\leq\; T^{R^{-1}\Lambda}_{\mathfrak{c}}(f,g) \;\xrightarrow[\varepsilon\to 0]{}\; \mathcal{W}_{\mathfrak{c}}(f,g) \;\xleftarrow[\varepsilon\to 0]{}\; T^{\Lambda}_{\mathfrak{c}}(f,g). \tag{1.1}$$

Here Z_{C_L} is King's cube functional on the axis lattice Λ, the first equality is a passive rotation, the inequality is a correlation inequality on the rotated lattice $R^{-1}\Lambda$, for which $R^{-1}C_L$ is a cube with faces on lattice hyperplanes, and the two continuum limits agree by the torus comparison. What is missing is an upper bound for walled tori on the axis lattice by King's state:

$$\text{for every } \eta > 0 \text{ there is an admissible } \mathfrak{c} \text{ with } \mathcal{W}_{\mathfrak{c}}(f,g) \leq Z_\infty(\mathrm{i}f, g) + \eta. \tag{1.2}$$

This is Hypothesis (TIGHT) of Section 5. It involves one lattice and one orientation, and the reverse inequality holds in the limit (Lemma 5.2), so (TIGHT) says that King's state is the lower envelope of the walled tori. Granting it, letting $L \to \infty$ in (1.1) gives $Z_\infty(\mathrm{i}\,R\cdot f, R\cdot g) \leq Z_\infty(\mathrm{i}f, g)$, and applying this to $R^{-1} \in \mathcal{R}_0$ gives equality. The passage from the dense subgroup to $O(3)$ uses a continuity lemma for convex functionals (Lemma 3.7), so no regularity of the Schwinger functions is needed.

Hypothesis (TIGHT) does not follow from the correlation inequalities: Remark 5.8 gives a family of Gaussian functionals with all the other properties used in the proof whose analogue of King's state is not rotation invariant. We therefore isolate the local estimate that implies (TIGHT). In the walled torus let the sources sit in a box H and let U be a thick shell around H inside the heavy plateau, where the wall is at least 1. Raise the scalar mass squared on the shell further by $2K$, through a second smooth profile, and let $\Psi_k(n, K)$ be the resulting expectation at lattice spacing ε_k, with $\Psi_k(n, \infty)$ the expectation with the matter removed on the shell. The hypothesis, (RES) of Section 7, asks that for one such configuration and one finite K

$$\liminf_{k\to\infty} \big[\Psi_k(n, K) - \Psi_k(n, \infty)\big] \leq \eta,$$

with the width of the shell chosen so that a photon boundary term is below η as well. At every fixed spacing the bracket tends to 0 as $K \to \infty$ (Proposition 8.1); the content of (RES) is that the heavy-shell limit and the continuum limit can be interchanged, at one geometry and with no rate. A two-line sandwich (Lemma 7.7) shows that the heavy shell is bounded below by a higher wall, and once the matter on the shell is dead the photon on the far side can be pinned and then replaced by King's Neumann photon on a larger cube at a cost exponentially small in the width of the shell, uniformly in the spacing (Lemma 7.1); so (RES) implies (TIGHT) (Theorem 7.10). Both $\lim_k \Psi_k(n, K)$ and, under one further existence statement, $\lim_k \Psi_k(n, \infty)$ exist, and (RES) is equivalent to the statement that the monotone limit $\lim_K \lim_k \Psi_k(n, K)$ lies within η of $\lim_k \Psi_k(n, \infty)$ (Theorem 7.15); this is the continuum form (RES^0), in which no lattice spacing appears. In the

curvature sector (RES_0) at one configuration is itself proved as an implication (Theorem 7.12): the shell mass κ is put inside the matter covariance of King's block-spin flow, and four estimates on that flow, two-sided bounds on the stopping-scale integral uniformly in κ, a disc bound for the heavy shell, a κ-Cauchy property of the reduced small-field action, and a threshold clause, give the tail bound at one configuration. Each of the four is proved as an implication in a companion manuscript [SW1, SW3]; the residues inside those proofs are listed in Section 1.4, each with its proof.

The torus comparison is reduced to one-sided bounds at a single lattice spacing (Section 4). The passive map $S = R^{-1}\sigma_\nu$ is an involution that exchanges the two lattices and maps the torus D_J to the torus $R^{-1}D_J$. A forward bound of King's type on each of the two tori therefore gives both directions of the comparison on D_J. The forward bounds and the existence of the axis-lattice limits are the content of [CT], whose own closing supplies them with the large-field clause of King's upper bound in place of King's torus bound [K2, (4.2)], which is vacuous at a non-zero wall; we state its theorem and the grades of its inputs. An alternative placement of the comparison, on a Neumann–Dirichlet box aligned with one lattice and a staircase for the other, is possible in principle; its hypothesis is open and it is not used here.

1.3. **Grades.** Every statement in this paper carries one of the grades [PROVED] (a complete proof here, or a precise citation of a printed proof that we have checked), [PROVED-LIT] (printed with proof in the literature, page and equation given; for the constructive literature [K1, K2, Ba1, Ba2, Ba3, Ba4, BFS, Re1, Re2] the statement and the printed proof have been read at the source and their scope checked against our use; textbook and other well-established results are cited by name, with their exact hypotheses checked against our use), [DERIVED] (the argument is complete at the level of a careful sketch, every step named), [SKETCH] (an argument exists but a step is not written out), [SKETCH-LIT] (asserted in the literature without a printed argument, a printed proof one of whose steps is asserted without justification, or a printed theorem used outside its printed hypotheses), [OPEN], [FALSE]. "[PROVED] as an implication" means proved from named inputs whose own grades are listed; for such an input the residues of its proof are themselves listed as named inputs, so that the list of Section 1.4 is closed under this operation. A statement proved in a companion manuscript is cited at the grade stated there, and we say when that manuscript has been checked independently in full; a statement whose proof passes through a flagged step (Section 9) is graded "[PROVED] as an implication given" that step, never [PROVED] outright. The *printed grade* of a statement of the literature is the grade its print supports: [PROVED-LIT] if its printed proof has no asserted step, [SKETCH-LIT] otherwise. A printed proof that is complete except for named asserted steps carries a split grade, [SKETCH-LIT] through those steps and [PROVED-LIT] otherwise, and each such step is named. King's renormalisation-group theorems consumed here, [K1, Thms. 2.1, 3.1, 3.4, 3.5, Prop. 3.6] and [K2, Thm. 2.1], rest on more than one asserted step; their printed grade is [SKETCH-LIT], and their asserted steps, named in Section 9, are proved in the forms consumed in companion manuscripts, the readings (B1)–(B6) of Bałaban's papers behind them included (Section 9.2). King's correlation inequalities [K2, Prop. 3.1, (3.24)–(3.26)] are printed with proofs, and [K2, Thm. 3.2] holds in the form its proof proves, with one J-independent Wick constant, for the boxes of that proof and for King's renormalisation-group cubes (Lemmas 2.14 and 2.16). An input proved as an implication from a hypothesis would be graded "[PROVED] as an implication given" that hypothesis; no input of Sections 1.4 and 1.5 carries such a hypothesis (the flagged steps of Section 9.6 are carried at the grades of their completions). The grades of all inputs are collected in Section 11.

1.4. **Inputs.** We fix $e > 0$, $\lambda, m, \mu_0 > 0$ in King's range and King's block factor 5 (King's L; below, L denotes only the index of a King cube C_L). The proof of the theorem of this paper runs through two statements about King's walled tori, (K_0) and (TIGHT_0) (Hypotheses 4.1 and 5.5):

(K_0) *The torus comparison, curvature sector*: for every $R \in \mathcal{R}_0$, every walled configuration $\mathfrak{c}$ of the class (6) below and every real smooth periodic 2-form G, $\lim_k \langle e^{F(G)} \rangle_{\mathfrak{c},\Lambda_k}$ and $\lim_k \langle e^{F(G)} \rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ exist and are equal.

(TIGHT_0) *The closing statement, curvature sector*: $\mathcal{W}^{\mathrm{env}}(f,0) \le Z_\infty(\mathrm{i}f,0)$ for every real 2-form $f \in C_c^\infty$; that is, King's state is the lower envelope of the walled tori.

Neither is assumed. (K_0) is [PROVED] as an implication from the inputs (1)–(5) of the following list, on the class (6), by the corrected torus theorem (Theorem 4.9); (TIGHT_0) is [PROVED] as an implication from (K_0), the heavy-shell estimate (RES_0) at one configuration and the dead-shell chain (10c) at that configuration (Theorem 7.10; also written out in [TE]), and (RES_0) is [PROVED] as an implication from the inputs (8)–(11), with their literature inputs (12), by Theorem 7.12, which is item (7). The list (1)–(13) and (15), together with the statements of Section 1.6 (Propositions 1.2, 1.3, 1.4, with their inputs), is the complete set of inputs of the theorem that are not printed theorems whose proof has been read at the source; item (13), King's cube limit, is an input because the theorem's object is defined through it, item (14) is proved here from (13), and item (15), the right-continuity of King's state in the scalar mass, is used at the points of E_W only. The objects are defined in Sections 2–7 and in [CT, H6, RS, SW1, SW3], whose notation is used in (1)–(12). Every item of the list is proved here, proved in a companion manuscript and checked independently, or is a statement of the literature at its printed grade; where a proof uses a statement of the literature at the printed grade [SKETCH-LIT], the item says so. No item depends on a hypothesis of the theorem beyond King's range. Section 9 restates in plain words every input that is not proved here.

(1) **$(\mathbf{BAL})_5$.** Every estimate of [Ba1, Ba2, Ba3, Ba4] that is asserted there without a printed argument (Section 9.2) holds as asserted at blocking factor 5, with per-step constants independent of the initial spacing, including the large-field clause of King's upper bound [K1, (3.4)] on the skew torus: the large-field contribution to the partition function at scale k is at most $\exp(-c_* p(\delta_k)^2 + C|T| + o(K^2))$, with $p(\delta) = b_0(1+\log\delta^{-1})^p$ at the fixed integer p of Convention P (Section A.1) and a constant $c_* \in (0,1]$, $c_* = 1$ at zero scalar source and $c_* < 1$ at a non-zero wall (every consumer uses only $c_* > 0$: $p(\delta_k)^2$ grows at least quadratically in k). Printed grade [SKETCH-LIT]; [PROVED] as an implication in [AS], at zero scalar source, from Bałaban's printed proofs, checked independently, at every background, the replacement of the propagators in the interaction terms of [Ba2, Prop. 2.6] included (Section 9.2; on the skew tori of [CT] through the change of domain (P7) of Section 1.5), in the forms used by (K_0) and by the shell estimates, among them the renormalisation bound for the operator with the scalar wall in the form (P1) of Section 1.5; six readings of Bałaban's papers, (B1)–(B6) of Section 9.2, are not covered by [AS] and are [PROVED] as implications and checked independently [SUP]. The dependence on the blocking factor, and the repair at block factor 5 of four printed steps (one of which is not covered by its printed proof at any block factor), are proved in [CT] (the two repair steps (5l)(v), (vi) [PROVED] as implications and checked independently); the re-runs of the same steps for the model with the shell mass inside the covariance are item (8e). The repair at block factor 5 of Bałaban's cube step [Ba3, p. 433] uses an end-functional bound of those re-runs at zero shell mass, and through it item (1), and with it (K_0), rests on the readings (B3), (B5), (B6) and on the regrouping of the cut families, all [PROVED] as implications. At a non-zero wall the large-field clause is [PROVED] as an implication, checked independently [SUP], from the clause at zero scalar source, an upper bound for the walled partition function obtained by exact shifts of the scalar mass to a mass M_1 with (e, λ, M_1, μ_0) in King's range and correlation inequalities, and a Hölder bound for the large-field part, with King's upper bound at fixed sources in its proved form (Section 9.3); the mass M_1 exists at every point of King's range at which the positivity condition of Section 1.6 holds, and at the remaining

points, a set that is a graph over the couplings (e, λ, μ_0) up to ties, Theorem A is obtained by right-continuity in the scalar mass (Section 1.6). For the rotated lattice, whose flows in [CT] use in-block contours rooted at the block centres, and for the composite contours of the hybrid flow, the estimates of [Ba1, Ba2, Ba3] and of King's upper bound are [PROVED] as implications and checked independently [SUP], by a census of every occurrence in the printed proofs of the operators built from the contours (Section 1.6); on the skew torus the clause holds at every J of the class (6) under the cap on the coarsest scale that the class carries (Remark 5.6).

(2) **(H1*).** For some θ_1 with $0 < \theta_1 < \min(\alpha, 1)$, Bałaban's renormalisation bound [Ba3, Prop. 1] holds with one element weakened by the degree θ_1, in all five sectors, for the renormalised families of [K1, §3.5] with lines taken from the standard and from the hybrid flow. [PROVED] as an implication in [CT] (its consolidated proof has been checked independently, and so has the restricted rule (2b), which it uses), from (5) and:

(2a) the first-layer slice-kernel bound $|\nabla_r \partial_b S_j^y(r, b)| \le C\ell_j^{-3} e^{-\delta|r-b|/\ell_j}$ for the slice kernels of [CT] at scale j with two derivatives, which [CT] obtains from the cubic-Hermite cardinal interpolation of the lattice kernels: [PROVED], checked independently (the extension by the sharp box-zone formula, whose decaying form is false, is not used);

(2b) the restricted contraction rule and the local-term chain of the triangle and overlapping-triangle compounds under it: in the contraction estimate for the two-flow families, the moment-removal contraction at a bare vertex is applied only when the vertex's photon field is external or a photon line outside the component; when it is a line inside, the component is closed by the common-moment bracket instead: [PROVED] as an implication, checked independently [SUP].

The one-flow instance is what King asserts in the proof of [K1, Prop. 3.6], p. 665.

(3) **($\mathbf{B}_\Delta$).** The hybrid block-spin kernels of [CT] obey the decay and regularity bounds of the standard kernels [K1, Thm. 3.3], and the difference elements of the two flows (including contour lines, contour vertices and the constant-background phases) carry a positive power of the scale ratio. [PROVED] in [CT] given the flagged step (F2) of Section 9.6, checked independently, the steps (5l)(viii), (ix) included. Two sub-steps, the scale-uniformity of the analyticity strip in the slice-kernel decay lemma and a uniform Hölder constant (proved with a factor $C_\alpha(1 + \log 5^k)^{3\alpha}$), are [DERIVED] and are used by no consumer.

(4) **(H6W).** For every configuration $\mathfrak{c} = (J, \psi, n)$ of the class (6) and every real smooth periodic 2-form G the axis-lattice limit $\lim_K \langle e^{tF(G)} \rangle_{\mathfrak{c}, X_K}$ exists for all real t, where $X_K := \Lambda_K$ is the axis lattice of spacing ε_K (Section 2), and it is the same number in the free-boundary conventions and in King's torus conventions at the shifted constant source. This is Theorem 2.1 of [K1] with the wall as King's scalar source, read on the normalised ratio; its proof is written out in [H6]: [PROVED] as an implication there (checked independently in full; its vacuum-polarisation step is Bałaban's resummed identity, Section 9.3) from (1), (5), the convention bridge (CT) of [CT], the polynomial lower bound (H5) at the wall, the restricted form of King's renormalisation theorem [RS] (item (5h)) and King's estimate [K1, Prop. 3.6] in its one-flow instance (item (5g)). Of its literature inputs, (1) and King's asserted steps (5k) are [PROVED] as implications in [AS]; two statements of King that earlier versions cited at their printed grade are now [PROVED] as implications and checked independently [SUP]: his readings of the sources, in the form of the small-field half of his upper bound [K1, Thm. 3.1] at every fixed source of bounded size, for King's flow and for the centred flow, and the source-free strict degree property of King's renormalised graphs as rewritten by Bałaban ([Ba3, (3.5)], [K1, (3.78)]), which King asserts at [K1, p. 668], in the componentwise form of [Ba3, (3.5)] (Section 9.3). Its residues, each [PROVED] as an implication and checked independently [SUP], are:

(4a) the Taylor step for the self-energy in the order-$(2,0)$ vacuum family, applied with its counterterm partner although the whole graph has degree 0 [RS, Thm. R-iii(b)];

(4b) Bałaban's ordering-restricted rewriting applied to the field-free members of the classes R-ii and R-iii(a) of [RS] (on the classes with an external leg or a curvature vertex it is [PROVED] as an implication in [RS]);

(4c) the torus one-point Wick bound beyond one loop, at every coupling in King's range, with the uniformity of its constants on a compact interval of masses. The rate $|\Delta_T(s) - \Delta_T(0)| \le Cs$ of the convention bridge, which (4) consumes and no other item does, is [PROVED] as an implication at one and at two loops, checked independently ([CT], (CT)(b); [H6]);

(4d) the change between two cutoffs of the non-polynomial terms of King's effective action [K1, (3.14)] at the frozen wall, at the rate that (4) uses, for which [H6] cites lines of [K1, (3.96), (3.98)] that are [FALSE] as printed; these lines are not used.

Item (4) uses the large-field clause of (1) at the wall, on the axis lattice with King's own blocks; through it (4) is [PROVED] as an implication at every point of King's range at which the positivity condition of Section 1.6 holds; at the remaining points Theorem A is obtained by right-continuity in the scalar mass, and item (4) is not used there.

(5) **Literature inputs of the torus theorem**, each at its own grade:

(5a) decay and regularity of the block-spin operators of the standard flow, [K1, Thm. 3.3], proved in [Ba4, Lemmas 2.1, 2.2, 2.4, Cor. 2.3]: [PROVED-LIT], except three steps that are asserted without an argument, the scale-uniformity of the analyticity neighbourhood in [Ba4, Lemma 2.4] (p. 586) and the identities [Ba1, (2.39)–(2.42)] (p. 612) behind the zero-background case of [Ba4, Lemma 2.2], which are [PROVED] as implications at block factor 5 under (1), and the photon clause of [K1, Thm. 3.3] (p. 656, "of course with $A = 0$"), a specialisation of two lines on the torus that is not written in print, [PROVED] under (5k); the use of [Ba4] at the coefficients $a_k \in [a(1-5^{-2}), a]$ and uniformly in a constant $m^2 \ge 0$, which the printed statement leaves implicit, is proved under (1) as well; the transfer of [Ba4, §§2–4] to the tori of [CT] is (5l)(iv);

(5b) the polynomial lower bound at zero wall [Ba1, (1.14)], whose proof in [Ba1, §§3–4] is a frame with asserted steps: printed grade [SKETCH-LIT]; [PROVED] as an implication at block factor 5 under (1), given (CT);

(5c) the zero-momentum counterterm conditions [K1, (3.32)–(3.33)] through order four (definitions), the Ward identities and the vanishing of the one-loop vacuum-polarisation moments for full propagators [Ba3, (2.18)–(2.28)], and Proposition 2.1 and (2.8) of [Ba3] with the ordering device (2.7) there: [PROVED-LIT];

(5d) the local limit law for the simple random walk [LL, Chap. 2]: a textbook theorem, cited by name;

(5e) Reisz's power-counting and renormalisation theorems [Re1, Thm. 1], [Re2, Renormalization Theorem, p. 92], in the corrected three-dimensional form of [CT]: three printed statements need repair, and the naturalness clause of the printed theorem is false as printed and is replaced by a condition proved for the graphs consumed; the forest identity [Re2, Lemma 5.1], whose proof there is a pointer to Zimmermann's paper, is Zimmermann's forest formula (Section 9.7), with its hypothesis [Re2, (2-24)] checked; and the membership of the subtracted integrands in Reisz's class, asserted at [Re2, (5-1)] and on p. 97 there, is proved in Remark 9.1. [PROVED] as an implication in [CT] from the printed proofs, checked independently;

(5f) the uniformity of the constant of King's upper bound [K1, Thm. 3.1] under a shift of the scalar source by a bounded constant (King's statement that the constant depends on g, h and $|T|$), used by the convention bridge (CT)(b) for (H3), (H4): [PROVED] as an implication under (1) and (5k), from the proof of [K1, Thm. 3.1], which refers

to [Ba1, Ba2, Ba3, Ba4] and to its Appendix; King's statement that the small-field analysis includes the sources [K1, p. 677] is [PROVED] as an implication and checked independently [SUP], in the consumed form: the small-field half of [K1, Thm. 3.1], in King's vertex form [K1, (3.14)–(3.21)], at every fixed source of bounded size, on King's aligned tori and on the rotated torus, for King's flow and for the centred flow; for the hybrid flow the lower half, which is the half consumed, with a cap depending on the number of scales;

(5g) King's estimate [K1, Prop. 3.6], stated on p. 662, whose proof on pp. 663–665 ends in an assertion (p. 665) and uses [K1, Props. 3.8, 3.9, Lemma 4.5]: its one-flow instance is [PROVED] as an implication in [CT, H6], checked independently, from [Ba3, (3.5)], [K1, Props. 3.7–3.9] (item (5k)) and a degree count of the subgraphs, for the classes with block vertices with the split count of (5l)(x); its degree count is [PROVED]: for King's flow the strict degree property named in (4), in the componentwise form of [Ba3, (3.5)], is [PROVED] as an implication and checked independently [SUP]; for the hybrid flow the unweakened form (degree $D \geq 0$ with a bounded count of the zero-degree subgraphs) is [PROVED] as an implication, and the strict form $D' > 0$ is not consumed;

(5h) King's renormalisation theorem [K1, Thm. 3.5(i), (3.38)] is [FALSE] as printed on the h-linear vacuum family of order $(2, 0)$, and is not used; its restricted form [RS], Theorems R-i–R-iii, is [PROVED] as an implication there (checked independently in full), with the degree count of its merged counterterm vertices, with the split count (5l)(x) for the classes with block vertices, and with the residue (4b), both [PROVED] as implications and checked independently;

(5i) the hybrid lower bound (H4) of [CT], the analogue of [K2, Prop. 4.1] for the hybrid flow (whose printed (4.5) has a sign slip and is replaced in [CT]): [PROVED] as an implication in [CT], checked independently, from (1), (3), (CT), the skew instance [SKEW-L], its last input being Bałaban's lower-bound argument at block factor 5 on the torus of the step, with its own steps (5l)(i)–(iii), and the lower half of King's small-field bound for the hybrid flow at the fixed sources it uses (item (5f)); the bridge from King's vertex form of that bound to the Cameron–Martin form of the effective action in which [CT] poses (H3) and (H4) is [PROVED] as an implication, with the insertion bound for the background legs at every scale, which has two independent proofs, each checked independently [SUP];

(5j) the further residues of [CT], printed there at grade [DERIVED] and [PROVED] as implications and checked independently: the standard bound on the free massive lattice propagator and its first three differences; the completions of the places where [Ba1, Ba2, Ba3] sketch (the depth inventory, the cube count and the constants of [Ba2, §3C], whose values at block factor 5 are (5l)(vii)), which are part of (1); and the decay bound for the coarse covariance, whose written derivation by the Paley–Wiener theorem is flagged and which holds by a second derivation without that theorem ((F1) of Section 9.6);

(5k) **King's asserted steps of the torus construction**, each named in Section 9.3: [K1, Thm. 3.1], whose proof refers to [Ba1, Ba2, Ba3, Ba4] and to its Appendix; the propagator bounds [K1, Props. 3.7–3.10, Lemma 4.5, (3.72), (3.88)] and the remainder bounds of [K1, Thm. 3.3], of which [K1, Prop. 3.7] is a printed sketch and the proofs of [K1, Props. 3.8, 3.9] are printed for the last line of (3.71) and the first line of (3.73) only; the photon clause of [K1, Thm. 3.3] (p. 656); and King's readings of the sources, the componentwise replacement of the sources (p. 665) and their inclusion in the small-field analysis (p. 677), with the uniformity of his constants over the source family (pp. 664–665). Printed grade [SKETCH-LIT]; [PROVED] as implications in the forms consumed in [AS], checked independently; King's readings of the sources (the

componentwise replacement, p. 665, the inclusion in the small-field analysis, p. 677, and the uniformity of his constants over the source family, pp. 664–665) are [PROVED] as implications and checked independently [SUP], in the consumed forms of item (5f) and Section 9.3. Of the steps behind [K1, Lemma 4.5], the coercivity bound (4.33) is proved here (Proposition A.4).

(5l) **The own steps of** [CT]**.** The closing implication of [CT], from the inputs (H1)–(H6), (SK), $(\mathrm{BAL})_5$ and (CT) of Section 4.3 to (K_0), is [PROVED] and checked independently. The steps of [CT] on the chain of that implication that are printed there below the grade [PROVED] are the following, together with the Taylor step (4a) and the Hölder bound (F3); each of them is [PROVED] as an implication and checked independently [SUP]:

- (i) the cumulant bound that replaces the cumulant lemma of Benfatto et al. (Section 9.2), a bound on the cumulants of the fluctuation integral of one renormalisation step with constants free of the spacing and of the shell mass, with its consumed forms on cubic tori, with and without the shell mass, and its form on the skew tori of [CT]. It is also an input of (8);
- (ii) in the hybrid lower bound (H4), the absorption of the additional lower-bound term of (i) into the factor $e^{-C\delta_k^{\sigma}|T|}$;
- (iii) Bałaban's renormalisation bound [Ba3, Prop. 1] for the hybrid flow, with its instances on D_J and on RD_J used in (H4), in the form consumed (the hybrid-flow lemma as printed in [CT] has a gap and is not used); its degree input for the hybrid flow is the unweakened form of the degree count, [PROVED] as an implication (item (5g));
- (iv) the transfer of Bałaban's bounds for the lattice Green's functions [Ba4, §§2–4] to the admissible lattice tori of [CT], read at the printed page;
- (v) in the repair at block factor 5 of the steps of [Ba3, pp. 433–445], the independence of the domain after the split;
- (vi) in the repair of the identification step, the two identities that identify Bałaban's all-small-field term with King's effective action, the bound for the determinants that enter them included;
- (vii) the constants of the completion of [Ba2, §3C] at block factor 5, and the uniformity in the scale index of the bounds that [Ba2] writes as $O(1)$ without qualification; they are also inputs of (8). Their vector half is the reading (B4) of [Ba2, p. 591], [PROVED] as an implication and checked independently [SUP] (Section 9.2). The count [Ba2, (3.45)] and the first inequality of [Ba2, (3.46)] are [FALSE] as printed, at every block factor, and are not used: a repaired count of the cubes replaces them;
- (viii) two statements of (3): the double contour lines of [K1, (3.26)], with both ends on contours, as a further type of difference element of the two flows, and the bounds of [K2, Prop. 5.6] for the inverses of the hybrid operators restricted to a union of large blocks, which (H4) uses;
- (ix) four steps of lower weight in (3): the cover of the skew tori by cubes; the polynomial size of the constant in the convergent remainder of the degree count of the appendix of [CT]; the domain difference of the hybrid kernels in the second clause of (3), in the form consumed (a lift-distance clause printed with it is [FALSE] and is not used); and the first clause of (3) at the first two levels of the hybrid flow, where [CT] states it for the later levels only; and with them the regularity across the seam of the periodic cell, in the form consumed;

(x) for the classes of graphs with block vertices, the degree counts behind the restricted renormalisation theorem (5h), the ordering-restricted rewriting (4b) and the one-flow instance of King's estimate (5g): King's credit of one degree for each photon field contracted outside a subgraph [K1, p. 664, (3.68)–(3.70)] is [FALSE] as printed for the counts of subgraphs, and the conclusions are carried by a count that splits the degree, with the step that the split degree governs the multiscale bound. These counts enter (4), and through the restricted theorem also (10d) and (10g)(i).

(4a) enters (K_0) through (4), and (F3) through (iii).

(6) **The configuration class.** A definition, printed because the infimum in (TIGHT_0) is taken over it: $J \in 5^{k_*}\mathbb{N}$ with $J \geq \max(J_0(\varepsilon_0), J_{\mathrm{pos}})$ (k_* the number of large-field scales of the rotated flow, $J_0(\varepsilon_0)$ the non-wrapping threshold of the skew torus at the coarsest scale, J_{pos} the threshold of the positivity condition of Proposition 1.4, above which its margin holds; the consumer takes J above it); $\mathfrak{c} = (J, \psi, n)$ with $\psi \geq 0$ smooth and D_J-periodic, $\psi = 0$ on the 2-neighbourhood of $R^{-1}C_L$, and $n > 0$ (a mass-increasing wall); the wall condition $n\langle\psi\rangle_{X_K} > \Delta_*|T|/2$ with Δ_* the bound of the convention bridge (CT)(b) of [CT]; block alignment: H' and T are unions of blocks of the final scale, whose side we denote ℓ_f; and the grid offset: with $M = 5M'$ and the prime $\mathfrak{p} \neq 5$ dividing M' of Definition 2.1, every corner coordinate of H' is $c\,\ell_f$ with $c \equiv M'/\mathfrak{p} \pmod{M'}$ (the sides $50MJ$ of D_J are multiples of $5M\ell_f$). Theorem 7.10 produces configurations in this class (Remark 5.6).

(7) **The heavy-shell theorem.** For King's walled torus with the shell $U = T \setminus H'$ (the whole exterior of the source box) and the shell mass $\kappa\mathbf{1}_U$ inside the matter covariance, the four estimates (8)–(11) imply (RES_0) at one two-level configuration $(\mathfrak{s}, \chi)$ with plateau height $n \geq n_*$ (n_* the threshold of Section 7.4) and one finite K (Theorem 7.12, proved in Section 7.4). [PROVED] as an implication.

(8) **$(\mathbf{SW1}^\circ)_{\mathrm{int}}$.** For every stopping scale $\eta_0 \leq \eta_1$ there are $a(\eta_0), b(\eta_0) < \infty$, with $a(\eta_0), b(\eta_0) \to 0$ as $\eta_0 \to 0$ at fixed T, such that, uniformly in $\kappa \geq c_0/\eta_0^2$ and in $\varepsilon \leq \varepsilon_0(\eta_0)$, for $z \in [-1, 1]$ and $g' \in \{0, g\}$, the partition function with sources (zf, g') and the capped stopping-scale integral I_κ^{cap} of [SW1] satisfy $e^{-a} I_\kappa^{\mathrm{cap}}(zf, g') \leq Z_\kappa(zf, g') \leq e^{a} I_\kappa^{\mathrm{cap}}(zf, g') + b\, Z_\kappa(0, 0)$, for the κ-model of Definition 2.4 (free-boundary δm_ε^2), that is the model of [SW1] with its counterterm offset by $\Delta_T(\varepsilon)$, $|\Delta_T| \leq \Delta_*$ (the shell-side bridge $(\mathrm{CT})_{\mathrm{sh}}$, [SW1, Lemma M(d)]). [PROVED] as an implication in [SW1] given the flagged steps (F2), (F3), (F5) of Section 9.6 (the manuscript checked independently in full, at block factor 5), from the following residues, each [PROVED] as an implication and checked independently; the determinant clause of (8k)(ii) uses for the first-order term of the step determinants the bound (DOM) of Section 1.5, on the torus and, on the histories of the upper bound with proper propagator domains, on those domains; on the same histories the removal of the tail of the background uses the sub-domain form of the perturbative bound at order fourteen, (P9) of Section 1.5, through (8i) and the difference form of (P2):

(8a) the frozen-scale sub-domain bound in scale-free form, [SW1, Lemma subdom(a)(iii)], with a constant independent of c_0 for every $c_0 > 0$ (the form with constant 16 and a rate in the penetration depth is false at small c_0 and is not used): [PROVED] as an implication, checked independently;

(8b) the statements written in [SW1] as short extensions of proved arguments: the face clause at one block [SW1, Lemma subdom(b)]; the tangentially localised amplitude on every sub-domain [SW1, Lemma DVlocN]; the sub-domain form of the perturbative bound with its boundary-difference clause, in Bałaban's restricted form [SW1, Prop. TDsub, Lemma dG]; the legs in the half-amplitude norm and the derivative clause for small-field data [SW1, Prop. legs(a),(d)]; the norm form and the mismatch

factor of the perturbative bound [SW1, Prop. TD(b),(c)]; King's Appendix class for the shell model with the background-gradient and block-mismatch clauses [SW1, Lemmas appclass, G, P], with the indicated remainder of the expansion in the background field; the e^2 and two-loop parts of the mismatch vertex and its third clause [SW1, Lemma M]; the light-scale constant on far blocks; the payment of the seventh family past the stopping scale under block alignment; the dominated-convergence step at the endpoint $\kappa = \infty$; the image barrier with Neumann faces; and the use of the external field of the Appendix class in the proof of the finite-domain perturbative bound (F5): [DERIVED] in [SW1], and [PROVED] as implications and checked independently, or not used (the image barrier, which the tangentially localised amplitude supersedes); the endpoint step at $\kappa = \infty$, which Theorem 7.12 does not use, enters item (15) with the counterterm offset graded in the couplings and is [PROVED] as an implication and checked independently there; the payment of the far blocks [SW1, D7(c)] is (8k)(vi);

(8c) the near-face argument of [K2, §5.2], printed as a one-page sketch, transplanted to the shell model with the contour run from the first light layer: [PROVED] as an implication, checked independently, with Bałaban's [Ba2, Lemma 2.7, (2.113)], whose proof is omitted in print, [PROVED] at block factor 5 under (1);

(8d) the second part of the proof of the perturbative bound, Bałaban's enumeration [Ba3, §§2–3] used as a frame: printed grade [SKETCH-LIT]; [PROVED] as an implication in [AS], checked independently; its load-bearing first clause is [PROVED] and checked independently, with two residues, the constants of the pairing lemma and the shell side of the photon two-leg lemma of [SW1], [DERIVED] in [SW1] and [PROVED] as implications in [SW3];

(8e) the re-runs, for the model with the shell mass inside the covariance, of the four steps of [Ba2, Ba3] that are repaired at block factor 5 (item (1)): [PROVED] as an implication, checked independently, with the renormalisation bound for the walled operator at zero shell mass in the form (P1) of Section 1.5, and with the bound for the mixed classes with two scalar legs and one photon leg, which rests on the sub-domain form (8k)(i) and on the local coefficient (B3), [PROVED] on the torus (Section 9.2) (the order of these steps is stated at the end of Section 1.5);

(8f) the counterterm offset: if the counterterm is $\delta m^2 + c(\varepsilon)$ with $\sup|c| \le \Delta_c$, graded by coupling order, everything holds with constants depending also on Δ_c [SW1, Lemma M(d)]: [PROVED] as an implication, checked independently; it is used with $c = \Delta_T$, $\Delta_c = \Delta_*$ (no sign of Δ_T and no equality of states across the two conventions is used).

(8g) **(K-pt)**, the pointwise difference bounds of the line kernels: for the lines G of step j, uniformly in κ on both sides of ∂U, $|D_A G(x,y)| \le C|x-y|_\eta^{-2} e^{-c|x-y|}$ and, for the terms of the current two-point function with both differences on one line, $|D_A^x D_A^y G(x,y)| \le C|x-y|_\eta^{-3} e^{-c|x-y|}$. With the undifferentiated bound $|G(x,y)| \le C|x-y|_\eta^{-1} e^{-c|x-y|}$, which is proved in [SW1] with an argument of [SW3], they give the bound $|\Pi_j^\kappa(x,y)| \le C|x-y|^{-4}$ on the remainder of the photon Taylor step [SW1, Lemma ward(b)], and through it the norm form of the perturbative bound [SW1, Prop. TD(b)]. The bounded-source estimates of [SW1] do not imply these pointwise bounds. They are [PROVED] as an implication by two independent arguments on the torus, at every scale, for every $\kappa \in [0,\infty]$ and on both sides of ∂U; on the Neumann sub-domains used in [SW1] they hold in the restricted form used there, with both points at distance at least one block from the complement of the sub-domain, by the second of these arguments (near a re-entrant edge the first-difference bound fails, and it is not used there); the proofs use the interior, first-difference and amplitude estimates of [SW1], a

bound for the mixed differences at the far end of a line, Kato's inequality, the random-walk estimates of [LL] with the gradient bound (8k)(iii), and [K1, (2.17), (3.5)]; both arguments are checked independently, and [SW1] cites them;

(8h) The regularity of the reflected gauge field [SW1, Lemma refl(a)] is not an input. The statement made there, that for a Neumann face F of a sub-domain the gauge field reflected across F (tangential components even, the normal component odd, and 0 on the bonds crossing F) is regular at its scale in the sense of [K1, (3.5)], which bounds every first difference and not only the curvature, is [FALSE]: across F the normal difference is of size $|A_m|/\eta$, and a constant normal component is a counterexample. What [SW1] uses is that the reflected field is regular after a gauge transformation, which a smoothed gauge on the face layer supplies, and the coercivity of the doubled operator for the mirrored field, which holds whether that field is regular or not: both are [PROVED], checked independently, and [SW1] states the lemma in this form. The interior estimates up to flat faces and convex corners [SW1, Lemma refl(b)] are used only at zero gauge field, in the proof of (8i), where the reflected field vanishes and they reduce to the interior estimates of [SW1] for the operator without gauge field;

(8i) **(BAL)**$^{\kappa}_{5}$, the clause of (1) for the model with the shell mass inside the covariance (this is (8e), named in [SW1] as an input of its main theorem): Bałaban's steps of the lower and the upper bound, and the cube step of the finite-domain perturbative bound, re-run for that model on the four objects that [SW1] replaces. That the four objects do not depend on κ is [PROVED]; the transfer is [PROVED] as an implication and checked independently, all its clauses included: the identification of Bałaban's all-small-field term with King's action, with bounds for the derivatives in the gauge field of the terms of that identification in the shell model, and its block-vertex part, which is moved into a determinant on the block lattice. Its steps of the lower and the upper bound expand the determinant of each renormalisation step through the determinant clause of (8k)(ii), with the bound (DOM) of Section 1.5 for its first-order term, on the torus and on the propagator domains of the upper bound, and on the histories of the upper bound with proper propagator domains its removal of the tail of the background uses, for the families of order fourteen that it produces, the sub-domain form (P9) of Section 1.5; its cube step and its identification step do not;

(8j) **(LOC-R)**, the light-side regularity estimates [LOC] on sub-domains $\Omega \neq T$, at points whose distance from the complement of Ω is at least a fixed multiple of the block size and near which every cube of Bałaban's decomposition that meets the boundary of U is a flat-face cube (the flat-face condition (L_x)), as consumed in [SW1] on sub-domains, among them by the derivative clause for small-field data [SW1, Prop. legs(d)]. It is named in [SW1] as an input and is supplied by [LOC, Prop. global(b),(c)], which is stated for every union of large blocks with Neumann boundary conditions, with no condition on the faces of the sub-domain, at such points: [PROVED] as an implication, checked independently, with the reading (8k)(iv) of the wall inside it. Near the edges and corners of H', where (L_x) fails, (LOC-R) is not used: the forms that the consumers use are supplied by a localisation lemma [LT], which compares, at a point of depth at least 4ρ in any Neumann union of blocks, Bałaban's local inverse with the torus inverse for the data restricted to the cube of side 8ρ around the point, with an error $Ce^{-\delta\rho}$ times the size of the data in the supremum and in the first difference, uniformly in κ, the spacing, the scale and the domain: [PROVED] as an implication, checked independently. It covers every consumer, the payment of the far blocks near the edges of H' included, (8k)(vi);

(8k) **The further steps of** [SW1], besides (8e) = (8i), each [PROVED] as an implication and checked independently [SUP], the determinant clause of (ii) with the bound (DOM) of Section 1.5:

(i) the sub-domain form of the finite-domain perturbative bound [SW1, Prop. TD-sub]: the transfer of its torus form to the propagator domains, at vertices of depth at least Bałaban's radius, and its norm form on those domains, together with the tangentially localised amplitude and the bounds of the amplitude theorem at depth at least Bałaban's radius, uniformly in κ and in the spacing, on the domains every face of which lies at distance at least two blocks of the scale from every face of ∂U parallel to it. For the large-block integer M of Definition 2.1 and the corners of H' of (6) every propagator, splitting and localisation domain of every scale is of this kind; for other choices, for instance M a power of 5, domains with a face on a plane of ∂U occur, generically at the scales at which $M\ell_k$ divides ℓ_f, and on them the bound is not proved. Its difference steps for several cut lines regroup the cut families of Bałaban's renormalised classes by cut datum, into products of the full classes of each label of the components, which are unions of orbits of the completion moves; at a non-zero background a regrouping along single orbits does not suffice. This regrouping is [PROVED] as an implication and checked independently, and the closure of the per-label full classes under the completion moves, with the bound for every closed family, on which it rests, is [PROVED] as an implication and checked independently (Section 9.2). (8k)(i) uses (ii) only through the norm form and the even sector of the mismatch clause at zero background, which contain no determinant; it uses the readings (B5) and (B6) of Bałaban's construction, [PROVED] as implications (Section 9.2). A localisation statement of the first written proof is [FALSE] as worded and is replaced by a weaker form that suffices. The transfer is stated for the families of order at most 13; at order 14, for the families with one more vertex that the removal of the tail of the background produces on the histories of the upper bound with proper propagator domains, it is (P9) of Section 1.5;

(ii) in the norm form of the perturbative bound [SW1, Prop. TD(b)], the bound on the members cut by the resummation in the vanishing of the zeroth moment of the photon two-leg kernel along a line, which also enters its block-data form, and the mismatch clause [SW1, Prop. TD(c)] in both of its sectors, at King's backgrounds (a form for every regular background is not claimed); the block-data form [SW1, Prop. TD(a)] is obtained from the norm form and the mismatch clause, so that the cut members that carry the mismatch vertex on window lines are used by no consumer. The block-data form holds on King's small-field set [K1, (3.2)] as printed, which gives no bound $|A_k| \le Cp$ uniform in the scale: the background enters the families only as the field inside the lines, the transporters and the block averages, through its first differences and their Hölder quotients, which the derivative clause of [K1, (3.2)] bounds; and the photon legs that occur are Bałaban's fields, the legs of his cube step after the constant part of the background is removed by a gauge transformation, and small differences, for which the light-side first moment of the photon two-leg kernel [SW1, Lemma ward(c)] holds with the printed constants, up to factors of order one that depend on μ_0. The frame in the odd sector is (P3), the vacuum families with the mismatch vertex at King's backgrounds are bounded by (P2), and the device for divergent subgraphs with two scalar legs uses (P1) (Section 1.5); the mixed classes with two scalar legs and one photon leg use

Bałaban's local coefficient (B3), [PROVED] on the torus for the window coefficient (Section 9.2). The determinant of the matter integral of a renormalisation step, which is kept exactly in the effective action, enters the bounds only per step, in Bałaban's form [Ba1, (3.46)–(3.47)], [Ba2, (2.99)–(2.107)]: at each step the determinant at the background plus the fluctuation is the determinant at the background, carried to the next scale, times a factor that is expanded in the photon fluctuation field, and the coefficients of that expansion enter as kept vertices, with bounds per block (the *determinant clause*). The determinant clause is [PROVED] as an implication, on the torus and on the propagator domains of the upper bound. The first-order term, which on a propagator domain is the first-order coefficient of Bałaban's kept polynomial, composed over the earlier steps [Ba2, (2.104)–(2.107)], is bounded block by block, uniformly in the shell mass and with no bound on the size of the background, by (DOM) of Section 1.5, on the torus and on those domains alike; its zero-shell-mass part is also Bałaban's per-step bound [Ba1, Prop. 3.2, (3.58)], [Ba2, pp. 578–579] under (1). The second-order term, whose two legs are fluctuation fields, is bounded as the photon two-leg classes of the norm form, the third-order term on the frame (P3), and the terms of order at least four have positive degree; on the propagator domains the terms of order at least two are bounded in the same way, with (8k)(i) in place of the torus forms. The absolute ratio of the determinants at the background and at zero is used only at the final scale, at base point 0, where the background is bounded and the odd orders vanish by charge conjugation [Ba1, (3.66)–(3.69)] ([PROVED] as an implication); King's interpolation of the determinant [K1, (3.94)–(3.98)] is not used (Remark 1.1);

(iii) the gradient bound for the simple random walk behind the pair estimates of [SW1]: its written proof uses a bound at each fixed number of steps that fails by parity; the statement holds, by an argument that pairs consecutive steps;

(iv) in [LOC], the reading of the wall $n\psi$ as constant on each cube of Bałaban's decomposition, with the profile varying inside the cubes;

(v) the cumulant bound (5l)(i), the Hölder bound (F3) and the constants (5l)(vii), which (8) uses as well;

(vi) the payment of the far blocks at frozen scales on the propagator domains [SW1, D7(c)]: its radius R_0 is taken from a bound free of the large-block integer, since the rate cited for it in [SW1] has a defect of constant order (that rate is not used); and at the points near the edges and corners of H' where (L_x) fails and whose distance from U exceeds R_0, the smallness $|a_i[G_i^\kappa - G_i^0]Q_i^*\phi_i(x)| \le \frac{1}{4}|\phi_i(x)|$ of the κ-difference of the background is supplied by the localisation lemma of (8j).

Two further points of [SW1] are settled and are not inputs. The order-$(2,0)$ h-linear vacuum family, on which [K1, (3.38)] fails (item (5h)), carries the source g', so it does not occur among the families consumed in [SW1, Cor. TDcons(v)] at $g = 0$, where the chain of Theorem A runs. King's identities [K1, (A.3)–(A.6)], which are photon identities, are cited in [SW1] for matter backgrounds ((F5)(f)) but carry no load there: the matter content at those places is Bałaban's [Ba2, Lemma 2.4], whose derivative clauses are printed as following similarly and are completed in [CT], with a comparison for the wall profile ([PROVED] as implications, checked independently); one display of [SW1] at that place is false as written for a non-constant wall and has an exact replacement.

(9) **(SW2$^{\min}$).** The disc bound for the heavy shell: $\sup_{|z|\le 2} |Z_\kappa(zf,g)/Z_\kappa(0,0)| \le M < \infty$ uniformly in $\kappa \in [0,\infty]$ and in ε. At $g = 0$ its matter moment factor (the supremum over κ

of the expectation of the Wick-source weight) equals 1 and the bound is the Gaussian domination of Lemma 2.13(a), with no input: [PROVED] (Step 0 of the proof of Theorem 7.12).

(10) **(SW3°)$_{\mathrm{int}}$.** On the capped stopping-scale set, for every $\eta_0 \le \eta_2$, every κ, κ' with $\max(c_0, 1)/\eta_0^2 \le \kappa \le \kappa' < \infty$, every $z \in [-1, 1]$ and $g' \in \{0, g\}$, the mixed differences of the reduced small-field action satisfy $\sup_{b,b'} |\Delta_z(b) - \Delta_0(b')| \le 2R(\kappa, \kappa') \le 2r(\kappa; \eta_0)$ with $r(K; \eta_0) \to 0$ as $K \to \infty$, uniformly in $\varepsilon \le \varepsilon_0(\eta_0)$, for the same model ($(\mathrm{CT})_{\mathrm{sh}}$, [SW3, Lemma mismatch]). [PROVED] as an implication in [SW3] given the flagged step (F6) of Section 9.6 (the manuscript checked independently in full), from (10a), (10b), (10d)–(10g), and from the statements of [SW1] that it cites, among them the perturbative bound and the photon Taylor step, which carry the inputs (8g), (8i)–(8k) at the places where they rest on them; each of its residues below is [PROVED] as an implication and checked independently:

(10a) the tail of the block-free determinant factor on the torus [SW3, Thm. LD$_T$]: for $\kappa \ge 2m_V^2$, uniformly in $\varepsilon \le \varepsilon_0 \le w_U$ and in the background A with $|\partial_\mu A_\nu| \le C_A$ and any torus holonomy, $0 \le \Phi^{\mathrm{bf}}(\kappa) - \Phi^{\mathrm{bf}}(\infty) \le \tau(\kappa)$ with

$$\tau(\kappa) := Ce^2C_A^2|U|\kappa^{-1/2} + C(T, H')\, e^2C_A^2\, m_V^{-1}\kappa^{-1/2}\big(1+\log(\kappa/m_V^2)\big)^c + C(T, H', m_V)\Big[e^{-c\, w_U\sqrt{\kappa}} + \big(1+\tfrac{\varepsilon_0^2\kappa}{8d}\big)^{-w_U/(2\varepsilon_0)}\Big],$$

$c \le 5$, $d = 3$, m_V the mass on the source box, w_U the width of the shell; the bracket collects the loops winding through U and, since $w_U \ge \varepsilon_0$ under the block alignment of (6), is at most $C\kappa^{-1/2}$, so τ is bounded by an ε-free nonincreasing function of κ that tends to 0 as a power of κ at fixed (T, H', m_V, η_0), which is all that (10) uses: [PROVED] as an implication given the flagged step (F6), checked independently; inside it the bookkeeping of the crossing and winding loops, whose bracket is the form uniform in ε in the frozen regime (the first term alone is not), is [PROVED] as an implication as well. The form without the winding bracket and with a right side proportional to C_A^2 is [FALSE] on the torus (a constant background has $C_A = 0$ while the winding loops contribute a positive tail) and is not used [SW3, Remark LDfalse];

(10b) the heavy-kernel bounds for the inverse G^κ of the matter operator with the shell mass $w = \kappa\ell^2$ in block units [SW3, Lemma Yp(a)–(g), Prop. p39(a)–(d)]: the kernel and mass bounds, the far pairs and the heavy far field in both regimes, the bounds on the inserted line, and the differentiated light estimate with its lattice-regime display, together with the near pairs of the full covariance with a heavy endpoint in the form with $\mathrm{dist}(x, y)^{-2}$ and in L^1 over the heavy end: [PROVED] as implications from the operator bounds of [SW1], the heavy far field and the lattice-regime display checked independently. Not used, and not claimed: a sup bound for the full covariance next to ∂U in the lattice regime and a form of the near pairs with $\mathrm{dist}(x, y)^{-3}$ (the former is false), and the Yukawa form of the heavy-pair clause, which is false for first-layer pairs [SW3, Remark Yrev];

(10c) the steps of the dead-shell chain of Theorem 7.6 for the shell $U = T \setminus H'$ with one convex face: [PROVED] as an implication, checked independently. This item is not an input of [SW3]: it is consumed by Theorem 7.10, through Theorem 7.6 at the configuration of Theorem 7.12;

(10d) the κ-derivative bound for the deep-U linear part of an insertion on a line used as a leg, for deep clusters closed in matter [SW3, Lemma Σ(iv′)(d)]. For deep clusters with a scalar connection the deep payment holds ([PROVED] as an implication, checked independently); for a cluster all of whose connections are photonic it fails, because the photon factors do not see the shell mass, and the heavy-matter photon subfamily containing the insertion must be bounded as a whole, with the insertion unsplit. The bound is: for the members of King's renormalised graphs with one insertion of the

κ-derivative of the matter operator whose matter component through the insertion is closed (no vertex of it carries a scalar leg), a class that contains the deep clusters closed in matter, the sum over the members, less its field-free vacuum part, is at most $C\,|U|\,\hat\kappa^{-3/2}(1+\log\hat\kappa)^{\bar m}$ per unit of $\hat\kappa=\kappa\eta_0^2$, in the units of the stopping scale, uniformly in the spacing and in both regimes, so that it is integrable in $\hat\kappa$. It is [PROVED] as an implication, checked independently, from the inputs that [SW3] applies to every graph with one insertion, among them (8g), (8i), (10g)(i), (10g)(vi), the split count (5l)(x) and the enumeration (10f); the proof applies one gauge identity to each class of members with the same full lines, and expands the classes into King's members only after that, with the background rebased for each member. A route through a photon Ward identity for each fixed ordering, or fixed scale assignment, of King's graphs is not available: that identity is [FALSE] (it holds for full lines and in Bałaban's resummed organisation, with a convergent remainder). The step (10g)(viii) is settled;

(10e) the inputs of [SW3] at grade [DERIVED]: the restricted form of [K1, (3.38)] for the shell model, with the degree count of [RS] re-run for the shell vertex set and the shell model's propagator bounds; the Taylor-leg placement for lines used as legs at vertices in H' or near ∂U; the source-decaying clause of the inserted-leg lemma on U; the positions in U of the one-loop z^2-term in the lattice-frozen regime; and the membership of the capped set in the class of the lower bound [SW1, Lemma appclass(d)(i)]. All five are [PROVED] as implications and checked independently, the restricted form of [K1, (3.38)] and the positions of the one-loop z^2-term being (10g)(i), (ii);

(10f) Bałaban's all-orders enumeration [Ba3, §§2–3] to order 13 with one inserted vertex, as in the proofs of [K1, Thms. 3.4, 3.5]: printed grade [SKETCH-LIT]; [PROVED] as an implication in [AS], checked independently, on tori and on King's cubes, with King's vertex list including the seagull vertex of order two (Section A.5);

(10g) **The further steps of** [SW3], besides the winding bookkeeping inside (10a), each [PROVED] as an implication and checked independently [SUP]; those that rest on (8k)(ii) use its forms for families of graphs, which contain no determinant:

- (i) the restricted form of [K1, (3.38)] for the shell model (10e), with King's bound read clause by clause at the ends of the lines, the members produced by interpolating the single-scale inserted covariance and the grouping of near pairs included; the winding of the shifted constant on the torus is handled by a rebasing at a fixed radius;
- (ii) the positions in U of the one-loop z^2-term (10e), in the lattice-frozen and in the continuum regime, including the part of the one-loop z^2-term and of the tadpole in g' carried by King's block fluctuation line; one input of the block part, in both regimes, is the bound for the vector-field propagators [Ba3, p. 426], the reading (B1) of Section 9.2, [PROVED] as an implication on the torus and checked independently [SUP];
- (iii) the Hölder remainder of the Taylor step on the cubes of U for lines used as legs [SW3, Prop. taylor(ii)], when the matter component through the insertion is not closed;
- (iv) the photon-leg classes with an insertion [SW3, Lemma photon(b),(c)]: the mixed class with two scalar legs and one photon leg, and the κ-residual of the photon first moment with one more insertion; these classes and the Hölder remainder (iii) use Bałaban's local coefficient (B3), [PROVED] on the torus for the window coefficient (Section 9.2);
- (v) the scale sum of the Hölder-leg remainder terms [SW3, Lemma Σ(iii)] in the lattice regime;

(vi) for the divergent subgraphs without insertion whose external photon fields are lines, in the model with the shell mass inside the covariance, the bound that (10d) uses, in the form of a first-moment bound, uniform in κ, for the two-photon kernel built from full lines; the bound on the remainder of Bałaban's regrouped Ward identity with the gain in degree claimed for it, whose form at $\kappa = 0$ is Bałaban's [Ba3, p. 441], is [FALSE] as so posed and is not used;

(vii) the bound for the members of a source vertex with no open photon end, which are linear in the source and so are not part of the field-free vacuum part, at every placement, the light local vertex of the Taylor split at all orders and in the lattice regime included, when the support of g' has distance at least η_0 from U;

(viii) the field-dependent members of the closed class of (10d) that the light local vertex of (vii) may produce: they are bounded by the count of (10d).

(11) **(SWT).** The threshold clause between adjacent representations of the stopping-scale integral; for the capped object there is one test set on $[\max(c_0, 1)/\eta_0^2, \infty)$ (the threshold κ_0 of Section 7.4) and the clause is empty: [PROVED].

(12) **Literature inputs of the shell estimates**, each at its own grade:

(12a) the correlation inequalities [K2, Prop. 3.1, (3.24)–(3.26)], the monotonicity in the cube [K2, Thm. 3.2] (with a J-independent Wick constant, in the form its proof proves, for the boxes of that proof and for King's cubes, Lemmas 2.14 and 2.16), [BFS, Thms. 6.1, 6.2(1)–(2)] and [BFS, §3.3, (3.7), (3.16)–(3.17)]: [PROVED-LIT] (the extension of the last to a site-dependent mass, used in [SW1], is flagged in (F5)); the reversed inequalities [BFS, Thm. 6.2(3)–(4)], whose printed proof is a sketch, and [K2, (3.28)], which is stated there without a proof and is not among the inequalities printed in [BFS], are proved here (Lemma 2.12(d));

(12b) the infinite-volume limit [K2, (3.40)–(3.44)] is not an input: at $h = 0$ what is used is proved here from (13) and Lemmas 2.14 and 2.16 (Theorem 3.8(i)–(iv), Proposition 3.11), where [K2, (3.44)] is the monotone limit of quantities bounded by 1 and [K2, (3.42)–(3.43)] are not used; at $\int h_+ > 0$ the bound [K2, (3.43)], asserted in print by "a standard argument (see [GJ1])" (not printed), fails in King's Wick ordering at order e^2 in perturbation theory (Remark 3.1), and [K2, (3.44)] rests on it; neither is used;

(12c) the estimates of Bałaban's lower and upper bounds that [SW1] re-runs, [Ba1, (3.26)–(3.69)], [Ba2, (2.2)–(3.54)], [Ba3, Prop. 1, (1.33)] and [Ba4, (1.8), (1.15)–(1.16), (1.20), (2.5)–(2.33)]: each range contains steps asserted without a printed argument (among them [Ba3, Prop. 1] itself, [Ba1, Props. 3.1–3.2, (3.66)–(3.69)], [Ba4, (1.8)] on general domains, whose proof is not printed and is completed in [CT, SW1], and the transfer of [Ba4, Cor. 2.3] to the mixed operators with its consequences [Ba4, (1.15)–(1.16), (1.20)], the domain-difference clause (1.20) not being argued): printed grade [SKETCH-LIT]; their asserted steps are [PROVED] at block factor 5 under (1), the readings (B1)–(B6) of Section 9.2 [PROVED] as implications and checked independently [SUP]; within [Ba4, (2.5)–(2.33)] the printed grade is split, [PROVED-LIT] except the two asserted steps named in (5a), on which the proof of [Ba4, Lemma 2.2] rests and which are proved likewise (the lower bound π^2 for the gap in [Ba4, (2.27)] is false as printed, and the correct value $4N^2 \sin^2(\pi/2N)$ suffices); for the transfer of [Ba4, §§2–4] to the tori of [CT] see (5l)(iv);

(12d) King's near-face analysis [K2, (5.41)–(5.49)]: printed grade [SKETCH-LIT]; its transplant to the shell model is (8c), and its use on King's cubes, at the first scale included, is part of (13), both [PROVED] as implications;

(12e) King's Appendix [K1, (A.1)–(A.8)]: printed grade [SKETCH-LIT]. Its argument asserts the expansion to order 13 with the claim that (3.54) still suffices (p. 676) and the inclusion of sources (p. 677), and (A.5)–(A.6) use the photon clause of [K1, Thm. 3.3]; the expansion and the photon clause are [PROVED] as implications in [AS] (item (5k)), the other steps of (A.2)–(A.8) are read, two of them (the regularity of the background in (A.8), and "the other bounds in (A.1) follow similarly") being completed in [SW1, Lemma appclass] and [PROVED] as implications (item (8b)), and the inclusion of sources is [PROVED] as an implication in the form consumed (item (5f), Section 9.3); for the shell model the class is [SW1, Lemma appclass] (item (8b)). The photon legs of the per-step constants of the field-independent part of the perturbative bound of the shell model [SW1, Prop. TD(d)] are handled by (KA*): the photon slice lines and King's photon legs are Bałaban's external fields in the norm [Ba3, (1.32)], [PROVED] as an implication from the bounds for the vector-field propagators on the periodic torus (the reading (B1) of Section 9.2, itself [PROVED] as an implication and checked independently [SUP]), componentwise and uniformly in the spacing, the scale and the scale mass, in that norm, checked independently [SUP]; no moment bound of [K1, (A.2)–(A.6)] is read;

(12f) the all-orders enumeration with one inserted vertex: item (10f).

(13) **King's cube limit for 2-form sources.** [K2, Thm. 2.1] at $h = 0$ on King's cubes: for every King cube C_J^K (Definition 2.15) and every finite family of real smooth 2-forms supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$, sampled at plaquette centres in the lattice's own frame, the limit $\lim_k \langle e^{\sum_j z_j F_{\Lambda_k}(f_j)} \rangle_{k,J}^K$ exists in $(0, \infty)$ for real z, with constants depending on the sources through $\|f_{j,\mu\nu}\|_\infty$ and $\|\nabla f_{j,\mu\nu}\|_\infty$ only (Remark 2.9). Printed grade [SKETCH-LIT]: its proof [K2, §3.1] transfers the torus proof of [K1] to Neumann–Dirichlet cubes and states that "all the results of [K1] go through" (p. 332), the boundary analysis being printed for one counterterm graph only and, for the photon, one ordering of the renormalised graphs at a time [K2, pp. 331–332], a flagged step (Section 9.3); the transfer uses the image-sum identity [K2, (3.9)], which together with [K2, (3.3), (3.8)] is false as printed for the weight printed there and holds with the Dirichlet weight f_D, which suffices for the transfer and with which King's flow on C_J^K is run. [PROVED] as an implication in [CA], checked independently: the limit follows from the ultraviolet construction on C_J^K with the flow of weight f_D, the convergence of the scalar boundary remainders, the line bounds and the source-free strict degree property on that flow [K2, App. A], King's transfer [K1, (3.90)–(3.98)] of the quadratic form and of the determinants to the cube, and the cube clause of the split degree count with its difference clause, whose part for the long trees is (P8) of Section 1.5, each [PROVED] as an implication and checked independently from (1), (5k) and (10f), with the bound at long trees for the families with one inserted external field whose graphs contain block vertices, which the ultraviolet construction on C_J^K uses and which is [PROVED] as an implication and checked independently [SUP] (Section 1.6). The bounds for the classes with curvature vertices on the cube, for the tails of the localisations near the faces of the cube and for the source-free classes of the restricted theorem [RS, Thm. R-i] on the cube are proved in the same way for the localisations, at a scale of the flow with blocks of size δ, whose tree has length at most $p(\delta)$ in units of these blocks, p being King's small-field function, and away from the faces up to a length of order $\delta^{-1/2}/(e\, p(\delta))$, and at greater lengths by the cut of the long lines (P5) of Section 1.5. The regrouping of the cut families along the orbits of the completion moves of Bałaban's classes, which the difference steps for several cut lines use, is proved as well, as an implication, with the closure of his per-label classes under the moves and checked independently. The ultraviolet construction on the cube uses the sub-domain form (8k)(i) of the perturbative

bound at zero shell mass, and with it the readings (B5) and (B6); the boundary analysis uses Bałaban's bounds (B2) for the massive lattice Green's function and its differences; all three readings are [PROVED] as implications and checked independently (Section 9.2). The photon boundary step uses the zeroth moment of the full photon kernel of the cube and no identity for a fixed ordering. Behind it stand [K1, Thm. 3.1, Prop. 3.6] (items (1), (5k), (5g)), the strict degree property named in (4), for one class with curvature and scalar vertices, and King's componentwise treatment of the sources [K1, p. 665], both [PROVED] as implications (items (4), (5g), (5k)). The further statements [K2, Thms. 2.2, 2.3(iii)] and the parts of [K2, (3.40)–(3.44)] at $h = 0$ that we use are not inputs: they are proved here from (13) and Lemmas 2.14 and 2.16 (Theorem 3.8(i)–(iv), Proposition 3.11); [K2, (3.42)–(3.43)] are not used. On the boxes $C_J \cap \Lambda_k$ of Definition 2.5(a) the fixed-J limit is neither known nor used; Lemma 2.16 gives the same infinite-volume state from them (Corollary 2.17).

(14) **Translations.** Theorem 3.8(v) at $h = 0$, [PROVED] here from (13), the cutting lemma and Lemma 2.16 (Proposition 3.11); King's printed argument [K2, Thm. 2.3(i)], which appeals to uniqueness of the limit and is written for centred cubes only, is not used. Used for the $E(3)$ clause only.

(15) **Right-continuity in the scalar mass.** At every point (e, λ, m, μ_0) of the set E_W of Proposition 1.4, for every finite family of real 2-forms $f_j \in C_c^\infty$, the functional $Z_\infty(\sum_j z_j f_j, 0)$ of item (13) at the scalar mass M converges to its value at m, locally uniformly in $z \in \mathbb{C}^N$, as M decreases to m, and the nearby masses M lie off E_W. [PROVED] as an implication and checked independently [SUP], from: the monotonicity of the walled and the cube states in the bare mass coefficient, [BFS, Thm. 6.1] and [K2, (3.25)], and in the cube, [K2, Thm. 3.2] (item (12a)), printed with their proofs; the existence of the limit of the differences $\alpha_k(M) - \alpha_k(m)$ of the bare mass coefficients of Proposition 1.4, equal to $\alpha_\infty(M) - \alpha_\infty(m)$ for a function α_∞ real-analytic in M, which puts the nearby masses off E_W; a comparison of King's cube with the dead-shell torus, through the photon boundary switch across a matter-free collar (as in Lemma 7.1) and the correlation inequalities, and of the dead-shell torus at a fixed stopping scale with the model at the nearby mass, through the endpoint step at $\kappa = \infty$ of [SW1] (item (8b)) with the counterterm offset graded in the couplings and the comparison of the counterterm offsets at a fixed stopping scale on the dead-shell torus; and item (13) at the nearby masses. Used for Theorem A at the points of E_W only, for the $E(3)$ clause as for the rotations; it does not enter (K_0), (TIGHT_0) or (13).

Not carried: the Wick-sector hypotheses $(\mathrm{W})_a$, $(\mathrm{K})_a$, $(\mathrm{TIGHT})_a$, $(\mathrm{RES})_a$ of Sections 3–7, which enter only for scalar sources and are outside the theorem of this paper; the general-source extension through ($\mathrm{W}^\sharp$) (Section 10); the alternative sufficient forms (SH), (HEL), (RES^0) of Section 7, which are not on the chain; and King's expected identity [K2, (2.24)], which is never used. The general-source extension uses in addition the statements (P4)–(P6) and (P8) of Section 1.5 and the rate (SRC) of Section 10, which enters through the difference form of (P6); it has no hypothesis of its own (Section 10).

1.5. **The step determinants on the propagator domains, and nine further inputs.** This subsection states, in the forms in which the chain uses them, ten inputs that are proved in companion manuscripts and checked independently: first (DOM), a bound for the first-order term of the determinant of a renormalisation step on the propagator domains of Bałaban's upper bound and on the torus, uniform in the shell mass, an input of the shell estimates (8); then six inputs of the shell estimates, of King's cube limit and of the general-source extension, (P1)–(P6); then three statements inside the proofs of item (1) on the skew tori, of King's cube limit (13) and of the upper bound of (8), (P7)–(P9); and last the order in which the residues of (8) are proved.

The first-order term of the step determinants. In Bałaban's upper bound at scale i the factors $Z^{(j)}$, $j<i$, of the earlier steps are kept on the domains of the kept terms, composed there and expanded in the photon fluctuation field [Ba2, (2.104)–(2.107)]; the term of first order is the object of the following bound.

(DOM) *The first-order term of the step determinants on the propagator domains* [SUP]. In the model with the shell mass $\kappa\mathbf{1}_U$ inside the matter covariance, let i be a scale of the flow, at most the final scale, and use the units of the blocks of scale i, of side ℓ_i; let $e_i^2 := e^2\ell_i$ be the coupling at that scale and p King's small-field function. Let Ω be a union of blocks of scale i with Neumann conditions on $\partial\Omega$: one of the two domains of the kept terms of Bałaban's upper bound at scale i, $B^i(\Lambda_0^{(i)})$ [Ba2, (2.104)–(2.106)] and $B^i(\Lambda_2^{(i)})$ [Ba2, (2.107)], or the torus T, whose sides are at least M/ℓ_i blocks [K1, (2.21)]. On Ω the factors of the earlier steps compose [Ba2, (2.105)]: in the shell model their product is $c\,(\det M_\Omega^\kappa(A))^{-1/2}$, where $M_\Omega^\kappa(A)$ is the operator of the composed step on Ω at the photon field A (the covariant Laplacian with Neumann conditions on $\partial\Omega$, the scalar potential with the wall and the shell mass, and the block term of scale i) and c depends neither on A nor on κ. Put $F_\Omega^\kappa := \ln\det M_\Omega^\kappa$; the term of first order in the fluctuation field a of the kept polynomial at the background B is $-\frac12 DF_\Omega^\kappa(B)[a]$, with $(B,a)=(\tilde B,\theta_{i+1}A'^{(i)})$ on $B^i(\Lambda_0^{(i)})$ and $(B,a)=(B^{(i+1),\eta},g_iA'^{(i)})$ on $B^i(\Lambda_2^{(i)})$ in the notation of [Ba2, p. 579]. Let B be a background on Ω whose first differences are at most $Cp(\ell_i)$ on Ω [Ba2, (2.98)], with no bound on $|B|$ itself, and let $|a|\le Cp(\ell_i)$ [Ba2, (2.86)], with the first differences of a at most $Cp(\ell_i)$ [Ba3, p. 412]. Bałaban's final block size being taken small enough (so that the first differences of B obey the regularity bound of [K1, (3.5)]), there are $C, C_\sigma, \delta>0$, independent of $\kappa\in[0,\infty]$, of the spacing, of i, of Ω and of B (they may depend on the parameters of the model, among them m, the height n of the wall and μ_0, and C_σ on σ), such that:

(a) $DF_\Omega^\kappa(B)[a] = \sum_{x\in\Omega} d_x^\kappa$, with $d_x^\kappa := -\int_0^\infty Dg_x^{\kappa,s}(B)[a]\,ds$, where $g_x^{\kappa,s}(B) := (\varepsilon/\ell_i)^3\,\mathrm{tr}(M_\Omega^\kappa(B)+s)^{-1}(x,x)$ is invariant under every gauge transformation of B on the sites of Ω; and for every block Q of scale i in Ω, the blocks at the boundary of Ω, next to its holes and at its re-entrant edges included, every $r\ge1$ and every $\sigma>0$,

$$\sum_{x\in Q}|d_x^\kappa| \le C\,e_i^2\Big[\sup_{Q_r}|\partial B|\sup_{Q_r}|\partial a| + e^{-\delta r/2}\sup_{\Omega}|\partial B|\sup_{\Omega}|\partial a|\Big] + C_\sigma\,e_i^2\,p(\ell_i)^2\,\ell_i^\sigma,$$

Q_r being the set of points of Ω joined to Q by a path in Ω of length at most r; in particular the left side is at most $Ce_i^2p(\ell_i)^2$. The same holds for the shell-mass residual, with $d_x^\kappa - d_x^0$ in place of d_x^κ. No Hölder quotient of B or of a enters.

(b) On the two domains of the kept terms write $DF_\Omega^\kappa(B)[b]=\sum_{y\in\Omega}b(y)\cdot j_B^\kappa(y)$. Every vertex of the kept terms lies in a block Q of scale i at distance greater than $r(\ell_i)-1$ from the complement of $B^i(\Lambda_0^{(i)})$, respectively at least $5r(\ell_i)-5$ from the complement of $B^i(\Lambda_2^{(i)})$ [Ba2, pp. 578–579], where $r(\cdot)$ is Bałaban's localisation function [Ba2, (2.7)]. At every such block Q, if in addition the Hölder quotients of the first differences of B are at most $Cp(\ell_i)$ at the points at distance less than $r(\ell_i)$ from Q, as they are for Bałaban's background at that depth ((F3)), then for every $\sigma>0$

$$\sum_{y\in Q}\big|j_B^\kappa(y)-j_B^0(y)\big| \le C\,e_i^2\,p(\ell_i)+C_\sigma\,\ell_i^\sigma,$$

and the zero-shell-mass current j_B^0 obeys Bałaban's per-step bound [Ba1, Prop. 3.2, (3.58)], [Ba2, pp. 578–579], under (1).

(c) On the torus T, besides (a), which holds there at every background as above, the bound of (b) holds at every block of T at King's backgrounds, whose size is bounded by [K1, (3.2), (A.1)].

The term of first order is paired with the fluctuation field, so (a) bounds the first-order vertex of the determinant, and its shell-mass residual, by $Ce_i^2 p(\ell_i)^2$ per block, at every block of the domain; this is the form in which the determinant clause of (8k)(ii) uses it, the photon legs being measured there in a norm that dominates their first differences. (b) gives the same bound through the L^1 norm of the current, at the blocks that carry the vertices. (DOM) bounds the first-order term on each of the two domains; the replacements of the domains in the kept terms, of the domains of the factors by $B^i(\Lambda_0^{(i)})$ [Ba2, p. 578] and of $B^i(\Lambda_0^{(i)})$ by $B^i(\Lambda_2^{(i)})$ [Ba2, (2.107)], are terms with one difference of two propagators at vertices deep inside the domains, bounded by the boundary-difference clause of the sub-domain form of the perturbative bound ((8b), (8k)(i)), and, for the first-order term, also by the comparison with the torus of the argument for (b) below.

Part (a) and part (b) are [PROVED] as implications in [SUP] by two independent arguments, each checked independently; each part alone gives the bound that the determinant clause of (8k)(ii) uses, which therefore has two independent proofs, and a third argument, also checked independently, which compares the domain with the torus at the blocks of the vertices with no use of (a), gives (b) again. None of them uses a photon Ward identity for a fixed ordering of King's graphs. The arguments for (a) and for (b) both start from the covariance of M_Ω^κ under every gauge transformation on the sites of Ω, whatever the shape of Ω: the Neumann form sees only the bonds inside Ω, and the blocks of Ω are whole blocks. The argument for (a) writes the first-order term as an integral over a uniform mass shift $s \geq 0$ of the derivatives of the diagonal of the resolvents of $M_\Omega^\kappa + s$; at each site it subtracts from the background a constant field that is a large gauge transformation, so that the value of the background at the site becomes small, removes the rest near the site by a gauge transformation that is affine there, and bounds the columns of the resolvents at every point of Ω, its boundary included, by an energy estimate from Bałaban's coercivity bound on Ω [Ba4, (1.8)] under (1) and an on-diagonal bound for the reflecting random walk on Ω; the shell mass enters only as a non-negative potential. The argument for (b) transforms the background, on a cube of side of order $r(\ell_i)$ around Q, by a gauge transformation into a field that vanishes at a point of Q and is bounded on the cube through its first differences alone, cuts the result off smoothly to a field on the torus that is regular everywhere, compares the coefficient on Ω with the coefficient on the torus at that field up to an error smaller than every power of ℓ_i, and bounds the torus coefficient through the first moments of the two-photon kernel [SW1, Lemma ward(c)], with the kernel bounds (8g), (8k)(iii), the Hölder bound (F3) and the frame (8d), (P3); on the torus at King's backgrounds this last step is the whole argument. In the torus bound that the arguments for (b), the third one included, and for (c) use, the far region, at distance more than a quarter of the side $N \geq M/\ell_i$ of the torus from Q, is summed slice by slice over the scales $j \leq i$ of the lines: the slice of scale j decays at a rate at least $c\,5^{i-j}$ in the units of the blocks of scale i, so that the sum over the slices is at most $CN^3 e^{-cN/4}$ times the size of the background, with no factor of the number of scales, and this part is at most $C_\sigma \ell_i^\sigma$, uniformly in the spacing, as in (b) and (c) (a bound of each slice at the rate c alone would leave a factor of the number of scales, which is not bounded at a fixed block size). The argument for (a) has no such sum.

(DOM) is used at one place, the determinant clause of (8k)(ii): on the torus by the lower bound of (8) and by the upper bound on its other histories, and on the domains of the kept terms by the upper bound of (8) on its histories with proper propagator domains, at the steps of Bałaban's bounds re-run in (8i). The corrected torus comparison (K_0), with the repaired cube step of (1), and King's cube limit (13) are statements at zero shell mass, where $d_x^\kappa - d_x^0 = 0$ and only Bałaban's per-step bound enters. The other residues of the shell estimates, (8k)(i) with its form (P9) at order fourteen, the bound for the mixed classes, the cube step and the identification step of (8e) = (8i),

(P2), (P3), (10d), (10e) and (10g), use the forms of the perturbative bound for families of graphs, which contain no determinant; the residues of (8) among them come before the determinant clause in the order of proof stated at the end of this subsection.

Remark 1.1 (King's interpolation of the determinant). King controls the ratio $\ln[Z_k(A^{(k)})/Z_k(0)]$ of the determinants of the matter integral of a renormalisation step at the background $A^{(k)}$ and at zero by interpolating between them in $R_k := 1 + [(\mu_0\ell_k)^{-1}]$ steps [K1, (3.90), (3.94)–(3.97)]. This serves his convergence theorem [K1, Thm. 3.4]: [K1, (3.98)] bounds only the difference of these ratios at two cutoffs. In the model with the shell mass the first-order terms summed over the interpolation points equal, up to second-order corrections, the absolute ratio at the stopping scale, which no step of the shell estimates bounds at a scale before the final one: the chain keeps the determinant exactly in the effective action, expands it per step in the fluctuation field (the determinant clause of (8k)(ii)), and expands the absolute ratio only at the final scale, at base point 0. In particular no bound is needed, at King's interpolated backgrounds, for the part of the shell-mass residual of the two-photon kernel that is odd under the exchange of its two points; a bound of that part through the supremum of the background would lose the factor R_k, since $|A^{(k)}|$ can be as large as a constant times $p(\ell_{k-1})(\mu_0\ell_k)^{-1}$ on King's small-field set.

Nine further inputs proved in companion manuscripts. The following statements are [PROVED] as implications and checked independently, in the forms stated here, which are the forms in which the chain uses them; the difference form (F-diff) of (P6), for its members with block vertices, uses the rate (SRC) of Section 10, proved in companion manuscripts, which no step of Theorem A uses, and (P4) at a non-zero scalar source uses the bound at long trees for the families with one inserted field whose graphs contain block vertices, [PROVED] as an implication (Section 1.6).

(P1) *The renormalisation bound for the operator with the scalar wall* [AS]. Consider the model at zero shell mass whose scalar field carries a potential $0 \le v \le \bar{v}$, among them the wall $m^2 + n\psi$, and whose photon carries none, with the model's own counterterms (Bałaban's rule applied to the graphs with the walled lines) and bounded finite vertices.

(a) Bałaban's renormalisation bound [Ba3, Prop. 1, (1.33)] at block factor 5 holds for every renormalised family of Bałaban of order at most 13 with at least one external field, at every regular background, the vacuum families being cancelled by the model's own vacuum counterterm up to bounded image terms, with constants polynomial in $\bar{v}$ and independent of the spacing, of the scale, of the domains, of the background and of the torus.

(b) At zero background, on an aligned torus of side at least one and at every step, the bound (1.33) holds also for the classes of graphs closed under Bałaban's completion moves at each node (passing to the counterterm partner of a divergent subgraph, permuting the external scalar legs of the divergent subgraphs with two scalar legs and one vector leg [Ba3, (2.18)–(2.22)], and forming the classes with two vector legs [Ba3, p. 431]), with constants polynomial in $\bar{v}$ and independent of the spacing, of the scale, of the labels and of the torus. This includes the bounds for the zeroth-order coefficients of the parts with two scalar legs and one vector leg whose leg vertex is pinned at the root; for such a part with a bridge the bound fails for the single member under the wall and holds for the member taken together with its counterterm partner.

Part (a) is used by the block-data form of the perturbative bound (8k)(ii), at the background of the block data, by the transfer (8e) = (8i) at the two places where it evaluates walled objects (Bałaban's [Ba2, Lemma 2.4] at the walled background and the cube step with walled lines), and, in its stated form, by the device of (8k)(ii) for divergent subgraphs with two scalar legs. Part (b) is used only through the bounds, at every step, for the finite parts of the counterterm vertices of the walled model, which enter (a) as sizes of its finite vertices and the wall term at zero shell mass of the mismatch bound [SW1,

Lemma mismatch(b)]. Through these uses (P1) reaches (8k)(i), (10g)(i) and King's cube limit (13).

(P2) *The vacuum families with the mismatch vertex at King's backgrounds* [SUP]. In the model with the shell mass $\kappa \mathbf{1}_U$ inside the matter covariance, let F be a renormalised vacuum family of [SW1, Prop. TD(d)] of order at most 13 that contains the mismatch vertex M of [SW1, Lemma mismatch], for instance one in which a tadpole carrying M is joined by a photon line to a tadpole without it, and let Q be a cube of scale k, of side $\ell_k = 5^k\varepsilon$ (the δ_k of (P5) and (P6)); $\beta_0 > 0$ and $\delta_0 > 0$ are the exponent of the coefficient bounds and the decay rate of the perturbative bound [SW1, Prop. TD]. *King's class* at scale k is the set of backgrounds whose first differences and their Hölder quotients are at most $Cp(\ell_{k-1})$ and whose supremum is at most $Cp(\ell_{k-1})(\mu_0\ell_{k-1})^{-1}$, in the units of the blocks of scale k; it contains King's backgrounds $uA^{(k)}$, $u \in [0,1]$, Bałaban's cut background and the segment between them. On the torus, at King's backgrounds $B = uA^{(k)}$, $|F(B;Q) - F(0;Q)| \le C\ell_k^{\beta_0}$, and for two backgrounds B, $B+b$ of King's class, $|F(B+b;Q) - F(B;Q)| \le C\ell_k^{3/2} N_k(b)$, where $N_k(b)$ is the sum of the suprema of the first differences of b, of their Hölder quotients and of $e^{-\delta_0 N/16}|b|$, N the side of the torus in blocks of scale k. On a propagator domain Ω of Bałaban's upper bound, with the lines of F taken on Ω at the background of the step and with Q at the depth at which the sub-domain form (8k)(i) holds, the first bound holds in the same form, $F(0;Q)$ being the value of the torus family at zero background: the cube step of (8e) = (8i), applied on Ω to the whole vacuum class, writes $F(B;Q)$ on Ω as the value of the torus family at zero background plus a sum of classes at zero background with at least one external field, which the norm form of the perturbative bound controls, and the values at zero background on Ω and on the torus are compared through (8k)(i). The constants do not depend on $\kappa \in [0,\infty]$, on the spacing, on the scale, on the domain or on the torus. The value $F(0;Q)$ at zero background is carried as it is, and no bound on it uniform in the spacing is used. These families are used in [SW1]: in the absolute form on the propagator domains of the upper bound, by [SW1, Cor. TDcons(iii)] on the collars at every scale and by [SW1, Cor. TDcons(v)] at the final scale (the final step of the lower bound, on the torus, expands at zero background and meets them only through classes at zero background), and in the difference form by two steps of the upper bound, the replacement of the cut-off photon background by King's background in the identification step and the removal of the tail of the background at each step; on the histories with proper propagator domains the second of these uses (8k)(i) at order 14, for the families with one more vertex that it produces: this is (P9). The torus bound is not used on the propagator domains. The proof uses (P3) only for the difference form, and uses the cube step of the re-runs (8e) = (8i), carrying the readings on which these rest. Bałaban's sentence on the divergent vacuum graphs [Ba3, p. 445] is not used (Section 9.2).

(P3) *Bałaban's enumeration block by block at higher orders* [SUP]. In the odd sector of the mismatch clause [SW1, Prop. TD(c)], after the closed odd matter blocks of a graph are rewritten by charge conjugation on full lines, the resulting families have order at most $14+n_o$, n_o the number of rewritten blocks, and block backgrounds in the class of fields whose first differences and their Hölder quotients are bounded as in [K1, (3.5)]. For these families Bałaban's bound for generalised expressions [Ba3, Prop. 2.2] and his enumeration [Ba3, §§2–3], read block by block, hold with constants that depend on the order, uniformly in $\kappa \in [0,\infty)$, in the spacing and in the interpolation parameters. The proof is from the frame (8d) at its grade, with the cube step of (8e) = (8i), (P1) and the bound for the mixed classes, which uses the local coefficient (B3), and it carries the readings (B3), (B5), (B6) on which these rest, each [PROVED] as an implication.

(P4) *The identification step at a non-zero scalar source* [SUP]. On King's cube, at a non-zero scalar source s, the identification of Bałaban's all-small-field term, taken at the cut-off photon background A_θ, with King's effective action $S^{(i)}$, up to an error $C\ell_i^{s'}|C_J|\ell_i^{-3}$, $|C_J|\ell_i^{-3}$ being the number of blocks of scale i in C_J, holds at every scale $i \le k$ of the flow, not only at the stopping scale k, with the source vertex of [K1, (3.21)] included, whose scalar background depends on the photon field. It is used for the small-field half of the cube bound at non-negative scalar sources (Section 10). Its proof uses, at the source s, the bound at long trees for the families with one inserted field whose graphs contain block vertices (Section 1.6), [PROVED] as an implication, so that (P4) is [PROVED] as an implication.

(P5) *King's cut of the long lines on King's cubes* [CA]. On King's cube C_J^K with the flow of weight f_D, the free-boundary δm_ε^2 and the free Wick constant, let $\varepsilon = 5^{-K}$, $\delta_k := 5^{k-K}$ and $0 \le k \le K - J_*$, where J_* is a fixed integer, the largest of the finitely many thresholds for the smallness of δ_k that the proof uses (each use of (P5) and (P6) takes its range of scales inside this one), and let $p(\delta)$ be King's small-field function. Let τ be a tuple of localisation points whose tree has length $d_\tau > p(\delta_k)$ in units of the blocks of scale k, and let H be a member of one of the following classes: the classes of the restricted theorems [RS, Thms. R-i, R-ii, R-iii(a)] on the cube, those that depend on a scalar source included; the classes with curvature vertices that King's cube limit uses, those with curvature and scalar vertices included; and the source-free field-dependent classes. Then, at both cutoffs of a diagonal pair, the sum $E(H_{\mathrm{ren}};\tau)$ of the renormalised class of H at τ obeys King's bound [K1, (3.38)] in its restricted form,

$$|E(H_{\mathrm{ren}};\tau)| \le C\,(1+K-k)^{p_0}\,\delta_k^{\rho(H)}\,p(\delta_k)^{n_1}\,e^{-\delta d_\tau},$$

with $\rho(H)$, n_1, p_0, δ and C as in the torus form [RS, Thm. R-ii], independent of k and K. On the torus this is King's treatment of every line longer than a fixed multiple of the block size as an external line [K1, p. 666], made exact: the tuple splits into clusters of bounded size, each cluster is reduced on its own box of size $O(p(\delta_k))$, and the lines between clusters are bounded by their decay [AS]. On the cube one box containing the tree suffices only for trees of length at most $p(\delta_k)$, and away from the faces up to a length of order $\delta_k^{-1/2}/(e\,p(\delta_k))$, below which the expansion of the lines in the background as a Neumann series, the reading of [K1, (3.52)] used on the torus, converges on it; beyond that length it fails in general, and King's finite form of [K1, (3.52)] leaves remainder lines at the full background, which the one-box bounds do not cover. The proof on the cube cuts the tuple into clusters as on the torus, reduces the clusters away from the faces as there, on the flow with the weight f_D, and the clusters at the faces by the boundary bound of the cube with the lines between clusters as external legs; it uses Bałaban's bounds (B2), [PROVED] as implications (Section 9.2). (P5) is used in the proof of (13) in [CA], for the classes with curvature vertices on the cube, the tails of the localisations near the faces and the source-free classes of the restricted theorem, and in (P6). Its use for the localisations with long trees in the difference clause of the split degree count of (13), at the faces of the cube and away from them, whose clusters at the faces carry the boundary bound in the forms that this clause uses, is (P8). The torus statements, and with them (K_0), do not use it.

(P6) *The restricted renormalisation bound on King's cubes for general sources* [SUP]. On King's cube C_J^K with the flow of weight f_D, the free-boundary δm_ε^2 and the free Wick constant, let $\varepsilon = 5^{-K}$, $\delta_k := 5^{k-K}$ and $0 \le k \le K - J_*$ with the integer J_* of (P5), and let $p(\delta)$ be King's small-field function. Take a real scalar source h, smooth and supported at distance > 1 from ∂C_J, with $\|h\|_\infty + \|\nabla h\|_\infty$ bounded, and any curvature source, on

which the members below do not depend; let N^h consist of the field-dependent parts of the renormalised graphs of [K1, §3.5] without curvature vertices that contain a vertex of the scalar source, and of the field-independent such graphs, except the h-linear graphs of order in $P_0 = \{(0,0),(0,1),(2,0)\}$. For a localisation of a member H, with value $E(H)$, let R be the length of its tree in units of the blocks of scale k, and call the localisation interior if its small-field domain does not meet the boundary of the cube. Then, with $\rho(H)$, n_1, p_0, δ, γ', β', σ' as in the torus form [RS, Thm. R-ii] and constants C, $\delta' > 0$ independent of k and K:

(F-one) at each cutoff of a diagonal pair, the sum of $|E(H)|e^{\delta' R}$ over the localisations with a source vertex in a fixed block of scale k is at most $C(1+K-k)^{p_0}\delta_k^{\rho(H)}p(\delta_k)^{n_1}$, and every interior localisation obeys $|E(H)| \le C(1+K-k)^{p_0}\delta_k^{\rho(H)}p(\delta_k)^{n_1}e^{-\delta R}$, King's bound [K1, (3.38)] in its restricted form; if H has external fields, $\rho(H) - s(H)/4 \ge 3/2$, with $s(H)$ the number of its external scalar legs;

(F-diff) every localisation with $R \le p(\delta_k)$ is interior and obeys King's difference bound [K1, (3.39)] between the cutoffs (K,k) and $(K+1,k+1)$ of a diagonal pair, and the one-step difference summed over the positions obeys, for every member and for the sum of the members, $|E_{K+1}^{(k+1)}(H) - E_K^{(k)}(H)| \le C[5^{-\gamma' k}\delta_k^{-\beta'} + \delta_k^{\sigma'}]|C_J|$ on King's small-field set.

The members of the restricted list of [RS, Thm. R-ii] that do not depend on the scalar source are bounded in the proof of (13) in [CA]. (P6) is the cube form of the torus statement [RS, Thm. R-ii]. It is used in the forms (F-one) and (F-diff), for the small-field half of the cube bound at non-negative scalar sources and for the inputs of the general-source cube lower bounds, at the given mass and at a second, heavier mass (Section 10), and by no step of Theorem A. Its proof uses (P5) for the long trees and for every localisation that carries the scalar source near the faces, and (P8) for the long trees of its members with block vertices, in the part of (P8) for $d_\tau > p(\delta_k)$ that covers the classes with a vertex of a scalar source. (F-one) uses no rate and is [PROVED] as an implication; in (F-diff) the members with block vertices take the difference clause of the split degree count at the localisations with $R \le p(\delta_k)$, which at a non-zero scalar source is the rate (SRC) of Section 10, [PROVED] as an implication in companion manuscripts [SUP], so that (F-diff) is [PROVED] as an implication.

(P7) *The change of domain of Bałaban's propagators on the skew tori* [AS]. On each skew torus of [CT], at every scale k of the flow and in the units of the blocks of scale k, let $\Omega_2 \subset \Omega$ be two unions of large blocks, A a background regular on Ω in the sense of [K1, (3.5)] (King's backgrounds A_θ are regular in this sense on the torus and on the cube, with a fixed Hölder exponent $\beta \in (0,1)$ and constants depending on β and e, [PROVED] as an implication and checked independently [SUP]), $G_k(\Omega, A)$ and $G_k(\Omega_2, A)$ Bałaban's propagators of scale k on the two domains, and $d(x)$ the distance of $x \in \Omega_2$ from the complement of Ω_2. There are $C, \delta > 0$, independent of k, of the spacing, of the domains and of A, such that for all $x, x' \in \Omega_2$ with $d(x), d(x') \ge D - 1$, where $D \ge R_0 + 2$ is a fixed integer and R_0 the constant of Bałaban's localisation function [Ba2, (2.7)],

$$\big|\partial_x^\sigma \partial_{x'}^{\sigma'} \big[G_k(\Omega, A) - G_k(\Omega_2, A)\big](x, x')\big| \le C\, e^{-\delta(d(x)+d(x')+|x-x'|)}, \qquad |\sigma|, |\sigma'| \le 1,$$

and the Hölder quotients of exponent α_0 of these first differences obey the same bound. On the cubic tori, and on the domains of the shell estimates, this bound is [PROVED] as an implication in [AS], from Bałaban's bounds for the lattice Green's functions and the block operators [Ba4, §2] at every scale and the scope statements of his construction. On the skew tori it is [PROVED] as an implication by two independent arguments, each checked independently. The first transfers that proof line by line: the proof uses no symmetry of the cubic lattice and no aligned axis or periodicity of the torus, its axial gauge and its contours

lying inside single blocks, which are cubes of the lattice on the skew tori as on the cubic ones; and its inputs, Bałaban's bounds for the Green's functions and the block operators and the scope statements of his construction, hold on the skew tori (item (5l)(iv)). The second obtains the bound on a skew torus from the bound on the aligned cubic torus that covers it, with a group of deck transformations of order 125, with the same rate and the constant multiplied by 125. (P7) is the factor that, with the renormalisation bound [Ba3, (1.33)] for the full classes of the cut components, gives Proposition 2.6 of [Ba2] at a non-zero background in the form stated in Section 9.2; the change of the background on the smaller domain is exact there. It enters only the torus comparison: through Proposition 2.6 of [Ba2] in item (1) on the skew tori, in the skew instance [SKEW-U] of King's upper bound, and through it (K_0).

(P8) *The long trees in the difference clause of the split degree count on King's cubes* [CA]. In the proof of King's cube limit (13), on C_J^K with the flow of weight f_D, $\varepsilon = 5^{-K}$, $\delta_k := 5^{k-K}$ and $0 \le k \le K - J_*$ as in (P5), the cube clause of the split degree count (5l)(x) for the classes with block vertices has a difference clause: the change of the sum of such a class over its localisations between the cutoffs (K, k) and $(K+1, k+1)$ of a diagonal pair is at most $C[5^{-\gamma' k}\delta_k^{-\beta'} + \delta_k^{\sigma'}]|C_J|$ on King's small-field set, with $\gamma', \beta', \sigma' > 0$ as in (P6). At the localisations τ whose tree has length $d_\tau \le p(\delta_k)$, in units of the blocks of scale k, the clause holds with its rate on one box of size of order $p(\delta_k)$. At those with $d_\tau > p(\delta_k)$, in the zone of the faces of the cube and away from it, each of the two cutoffs obeys the bound of (P5),

$$|E(H_{\mathrm{ren}}; \tau)| \le C\,(1 + K - k)^{p_0}\,\delta_k^{\rho(H)}\,p(\delta_k)^{n_1}\,e^{-\delta d_\tau},$$

with constants independent of k and K, the clusters of the tree at the faces being bounded by the boundary bound of the cube for the scalar boundary remainders in the forms that the difference clause uses; with the bound $e^{-\delta d} \le C\delta_k^{3+\sigma} e^{-\delta d/2}$ for $d > p(\delta_k)$ [K1, (3.41)] and the sum over the trees of the remaining factor $e^{-\delta d/2}$, these localisations contribute at most $C_N \delta_k^N |C_J|$ at each cutoff, for every N, so that the difference clause holds there with no rate in k. The clause is [PROVED] as an implication and checked independently; its part for $d_\tau > p(\delta_k)$ has three proofs, each checked independently, one of which covers also the classes that carry a vertex of a scalar source. At zero scalar source (P8) enters King's cube limit (13), and through it the definition of King's state (Definition 3.9), (TIGHT_0) and Theorem A; it does not enter (K_0) or the shell estimates (8)–(11). The general-source extension uses its part for $d_\tau > p(\delta_k)$ also at a non-zero scalar source; its part for $d_\tau \le p(\delta_k)$ at a non-zero scalar source is the rate (SRC) of Section 10, proved in companion manuscripts. The families with one inserted external field whose graphs contain block vertices are not classes of (P8); their long trees are the bound of Section 1.6, [PROVED] as an implication and checked independently.

(P9) *The sub-domain form of the perturbative bound at order fourteen* [SUP]. At each step i of the upper bound of (8), on its histories with proper propagator domains, Bałaban's background is freed of its tail [Ba2, (2.89)–(2.90)], written $\tilde{B}^{(i)}$ as in [SW1] (this tail is not the background $\tilde{B}$ of (DOM)), and the change of the perturbative part is written as $P^{(i)}[\mathrm{bg} + \tilde{B}^{(i)}] - P^{(i)}[\mathrm{bg}] = \int_0^1 \frac{d}{ds} P^{(i)}[\mathrm{bg} + s\tilde{B}^{(i)}]\,ds$, with bg the configuration of the step without the tail: a sum of families with one more vertex, whose additional external photon field is the tail, which lies at distance greater than $r(\ell_i)/2$, in units of the blocks of scale i, from every place where the operators of the step on the propagator domain differ from those on the torus. These families have order at most 14, the vacuum families of (P2) with the inserted vertex among them. The transfer (8k)(i) of the torus form of the perturbative bound to the propagator domains, in its norm form and in its mismatch clause [SW1, Prop. TD(c)], holds for them, with and without the mismatch vertex, with

constants independent of $\kappa \in [0,\infty)$, of the spacing, of the scale and of the domain. This is [PROVED] as an implication by two arguments, each checked independently. The first proves the transfer at every fixed order, with the depth condition of (8k)(i) taken at that order, that is, with Bałaban's radius and the smallness of his final block size chosen for order fourteen, and with constants that depend on the order only through the size of the families, by the order-uniform form of Bałaban's enumeration that (P3) uses; it covers all the families above, those with a vertex of a source included. The second shows that for these families the depth and the constants of order 13 suffice: it runs the cube step of (8e) = (8i) on the propagator domain at the background $\mathrm{bg} + s\tilde{B}^{(i)}$, after which the perturbative bound is used only with the lines at zero background, where the odd sector of the mismatch clause is empty and its even sector is bounded at zero background at the size of these families; the families with a vertex of a source are covered by the first argument. (P9) enters only the upper bound of (8), through (8i) and the difference form of (P2), and through (8) it reaches (RES_0), (TIGHT_0) and Theorem A; (K_0) and (13) do not use it.

None of the statements of this subsection uses a photon Ward identity for a fixed ordering or a fixed scale assignment of King's graphs.

The order of proof. The residues of (8) that cite one another are proved in the following order, so that no statement is used before it is proved: first the norm form of the perturbative bound (8k)(ii) and the even sector of its mismatch clause at zero background, with the frame (8d), (P1) and the cube step of the re-runs (8e) = (8i); then the sub-domain form (8k)(i), which uses these, with its form (P9) at order fourteen, and the bound for the mixed classes, which uses (8k)(i); then the identification step of (8e) = (8i), which uses the mixed classes; then (P3), the odd sector of the mismatch clause at King's backgrounds, which uses (P3), and (P2); then the block-data form and the determinant clause of (8k)(ii), the latter with the bound (DOM), whose argument for (a) uses no residue of (8); and last the steps of Bałaban's lower and upper bounds in (8). The sub-domain form (8k)(i) and the bound for the mixed classes use the perturbative bound only through its norm form and the even sector of its mismatch clause at zero background, (P2) uses the mismatch clause only for families with an external field, and (P3) is the frame of the odd sector only.

1.6. **Four statements formerly carried as hypotheses, and the step formerly open.** Earlier versions of this paper stated Theorem A given three hypotheses and one proviso. All four are now [PROVED] as implications and checked independently [SUP], or replaced by a proved statement; they are recorded here in the forms in which the chain uses them, because the plain-words restatements of Section 9 and the table of Section 11 refer to them. The step of the chain that the preceding version carried as open, the odd-block clause of the degree census of the shell model, is proved as well and is recorded at the end of this subsection.

Proposition 1.2 (block vertices at long trees in the families with one inserted field)**.** *On King's cube C_J^K with the flow of weight f_D, let $\varepsilon = 5^{-K}$, $\delta_k := 5^{k-K}$, $0 \le k \le K - J_*$ and $p(\delta)$ be as in (P5), and let the scalar source be zero or a source h as in (P6). Consider the families of the perturbative part at a background B on C_J^K that is regular in the sense of* [K1, (3.5)]*, among them the backgrounds $A_\theta + \sigma X_0$, $\sigma \in [0,1]$, with A_θ Bałaban's cut-off photon background and X_0 its difference from King's background, that carry one inserted vertex whose additional leg is an external field X (the field X_0, in that case), and whose graphs, with every block vertex split into its two summation points as in the split count (5l)(x), contain a block vertex. Then at every localisation τ of such a family whose tree has length $d_\tau > p(\delta_k)$ in units of the blocks of scale k, in the zone of the faces of the cube and away from it, the sum $E(\tau)$ of the renormalised family at τ obeys at each of the two cutoffs of a diagonal pair the bound of (P5) multiplied by $\|X\|_{1,\alpha_0}$, the supremum of X, of its first differences and of their Hölder quotients of exponent α_0, with constants independent of*

k, K, B and X; so that, by [K1, (3.41)]*, these localisations contribute at most* $C_N \delta_k^N \|X\|_{1,\alpha_0} |C_J|$ *for every* N*.*

[PROVED] as an implication and checked independently [SUP]: the bound is (P8) for these families, with the three further rows that the inserted field requires, one cluster for each product of block vertices, whose block points stay inert and whose cut lines are paired at the ends of the products, the cut photon lines with an end on a contour, and the ordering bound with the split count, each carried at the per-tuple bound of (P5) times $\|X\|_{1,\alpha_0}$ and summed by [K1, (3.41)]. For the same families without block vertices the bound is [PROVED] as an implication and checked independently [SUP], and for King's classes with block vertices, which carry no inserted field, it is the part of (P8) for long trees. At zero scalar source the proposition enters the ultraviolet construction on C_J^K of King's cube limit (13), through the norm form of the perturbative bound for the derivative of the perturbative part in the photon background at Bałaban's cut-off background, and hence the definition of King's state (Definition 3.9), (TIGHT_0) and Theorem A; at a non-zero scalar source it enters (P4), and through it the general-source extension. It does not enter (K_0) or the shell estimates (8)–(11).

Proposition 1.3 (the fluctuation estimates for centre-rooted and hybrid in-block contours)**.** *In Bałaban's construction the fields are transported inside each block along a contour* $\Gamma_{r,x}$ *from a point* x *of the block to its reference point* r [Ba1, (2.1)–(2.3)]*, and King's flow uses axial contours rooted at a corner of the block* [K1, (2.12)]*. The flows of* [CT] *use in-block contours of two other families: the axial contours rooted at the centre of each block, of length at most* $\frac{3}{2}$ *times the side of the block, the coarse variable being attached to the centre; and the composite contours of the hybrid flow, of length at most* 2.1 *times the side of the block of each level. The statement: every estimate of the fluctuation integrals of* [Ba1, Ba2, Ba3]*, among them* [Ba3, Prop. 1]*, and King's upper bound* [K1, Thm. 3.1]*, in the forms that* [CT] *consumes, holds for these two families in place of King's, with constants that depend on the family only through the bound* $c_\Gamma \in \{\frac{3}{2}, 2.1\}$ *on the length of its contours in units of the block side.*

[PROVED] as an implication and checked independently [SUP]. The printed proofs use the contour family through a short list of its properties, which the three families share (each contour lies in its block; the block average built from the contours is gauge covariant, satisfies $QQ^* = 1$ and, at a constant field, does not depend on the contours; and the line bounds see the family only through the length of the contours). The proof is a complete census of the occurrences, in the printed proofs, of the contours and of the operators built from them, the block averages, their adjoints and the propagators and block operators of the flow, each occurrence being checked against the list; for the composite contours of the hybrid flow the census finds one estimate used beyond the list, Bałaban's bound for the fluctuation field ([Ba2, (2.65)–(2.66)]) in the regime of [Ba3, p. 421], which is proved for that family directly, on two independent routes; for the centre-rooted family it has three independent proofs. The bounds for the Green's functions and the block operators [K1, Thm. 3.3], [Ba4] are printed for arbitrary oriented contours in the blocks [Ba4, p. 573] and are not part of the proposition. It enters (K_0): through King's upper bound (H3) for the rotated lattice on both tori, the large-field clause of (1) at the wall for that lattice, the hybrid lower bound (H4) and the comparison of the effective actions of the two flows in [CT]. It does not enter (13), the walled limit (4), whose flow uses King's contours, or the shell estimates (8)–(11).

Proposition 1.4 (the positivity condition on the constant of the convention bridge)**.** *For a scalar mass* $M > 0$ *let* $\alpha_k(M) := M^2 + \delta m_{T,k}^2(M)$*, where* $\delta m_{T,k}^2(M)$ *is King's mass counterterm* [K1, (3.32)–(3.33)] *at the mass* M*, computed with the propagators of the torus* T *at spacing* ε_k*, and let* $\Delta_T(s)$*,* $s \le s_0$*, be the constants of the convention bridge (CT)(b) on* T*. For each* $R \in \mathcal{R}_0$ *and each of the three pairs of a lattice and a torus used, the axis lattice on* D_J *and the lattice rotated by* R *on* D_J *and on* RD_J*, there are* $p_H > 1$*, a mass* $M_1 > 0$ *with* (e, λ, M_1, μ_0) *in King's range and a*

threshold, depending on the couplings, the block integer M, p_H, M_1 and R (equal to 1 if $\Delta_T(s) \geq 0$ for all $s \leq s_0$ and every J), such that for every J above it either $\Delta_T(s) \geq 0$ for all $s \leq s_0$, or

$$\liminf_{k\to\infty} \big(\alpha_k(m) - \alpha_k(M_1)\big) > p_H \sup_{s\leq s_0} \max\{0, -\Delta_T(s)\};$$

J_{pos} denotes the maximum of these thresholds over the finite set $\mathcal{R}_0$ and the three pairs. This holds at every point of King's range, $e > 0$, outside the set E_W of the points at which the scalar mass m minimises, at fixed (e, λ, μ_0), the function of the mass that the bridge constants define; E_W is the set of the masses minimising that function: off a smaller set of couplings it is the graph of one function of (e, λ, μ_0), so that it meets every line of fixed (e, λ, μ_0) in one scalar mass, up to ties.

[PROVED] as an implication and checked independently [SUP], by three independent arguments, one of which finds M_1 without the lower bounds at the auxiliary mass. The constant c_* of the large-field clause of (1) at a non-zero wall is then $1 - 1/p_H$ ($c_* = \frac{1}{2}$ if $\Delta_T \geq 0$). The condition is needed only where the constant $-\frac{1}{2}\Delta_T$ that the bridge adds to the wall is positive, that is, where the bare mass is lowered: an upper bound there is obtained from a mass M_1 whose bare mass squared lies below that of m. At order λ the constant Δ_T is positive and the condition is void; the counterexample at $e = 0$ lies outside King's range. At the points of E_W the chain of (K_0) is not run: there Theorem A follows from its statement at the nearby scalar masses by right-continuity of King's state in the scalar mass (item (15); the nearby masses approaching from above and lying off E_W), which is [PROVED] as an implication and checked independently [SUP] from the stability of the ultraviolet construction under a change of the scalar mass at a fixed stopping scale (its last input, the comparison of the counterterm offsets at a fixed stopping scale on the dead-shell torus, is [PROVED]). The proposition enters the large-field clause of (1) at a non-zero wall on the three pairs, and through it King's upper bound (H3) and the walled limit (4); hence (K_0), and through (4) and Theorem 7.10 also (TIGHT_0). It does not enter (13) or the shell estimates (8)–(11).

The final-scale cap. Bałaban's lower and upper bounds [Ba1, (1.14)], [Ba2, (1.3)] for the partition functions at the fixed masses end at a final spacing $t_K \in (\varepsilon_0/5, \varepsilon_0]$, and the large-field clause of (1) at the wall on the skew torus D_J takes one step at that spacing. Earlier versions carried, as a proviso of the theorem, that the skew tori do not wrap at that scale. The proviso is replaced by a cap that does not depend on J and is closed downward: the coarsest scale ε_0 of the flow is at most a threshold $\varepsilon_{\mathrm{sys}}(M, R, \rho, \delta_0)$ fixed by the large-block integer M, the rotation R, Bałaban's radius ρ and the finest block size δ_0; under it the non-wrapping threshold at the final scale equals 1 for every J in the residue class of (6), so that every configuration of the class (6) meets it ([PROVED] and checked independently [SUP]). The cap is a proviso of the corrected torus theorem of [CT] on its parameters, carried by the class (6) (Remark 5.6); its consumers are the fixed-mass bounds on the tori of the comparison, King's upper bound at the wall on D_J (item (1), (H3)) and the skew-torus instances (SK). It is not a hypothesis of Theorem A.

The step formerly open. The κ-Cauchy estimate of [SW3] (item (10)) uses a degree census of the renormalised graphs of the model with the shell mass inside the covariance, at zero source, component by component. Its clause for the closed matter blocks with one photon end, the blocks of degree -2 or -1 with one photon line attached, states that after the rewriting these blocks appear as classes with two photon legs, one background leg and one line slot, of degree exactly -1 or 0, imported as black boxes into the enumeration, and that the renormalised members have no proper prefix sub-component of degree zero outside the black boxes beyond those that one local vertex at the window scale absorbs. This clause was the one step of the chain open in the preceding version of this paper. It is proved: the census clause itself, and the tail of the coarse counterterm at orders two to four that its one-loop form leaves, piece by piece, each piece by at least three independent arguments and the whole confirmed by a further independent review; it is [PROVED] as an implication and checked independently [SUP]. It enters item (10) only, hence (RES_0) and (TIGHT_0); it does not enter (K_0) or King's cube limit (13).

1.7. **Results.** Let $Z_\infty(\cdot,0)$ denote King's state on curvature sources (Definition 3.9, Proposition 3.11): the entire function $z \mapsto Z_\infty(\sum_j z_j f_j, 0)$ of the coefficients of real 2-forms $f_j \in C_c^\infty$, whose derivatives at $z=0$ are the curvature Schwinger functions $\langle F(f_1)\cdots F(f_N)\rangle$. For $R \in O(3)$ and $b \in \mathbb{R}^3$, sources are transported as 2-forms, $R\cdot f$ and $\tau_b f$ (Definition 2.8).

Theorem A ($O(3)$ invariance of the curvature sector)**.** *Let $e > 0$ and $\lambda, m, \mu_0 > 0$ be any point of King's range, at block factor 5. Then* (K_0) *and* (TIGHT_0) *hold outside the set E_W of Proposition 1.4, and King's state on curvature sources (Definition 3.9) is $O(3)$-invariant at every point of King's range, at the points of E_W by right-continuity in the scalar mass: for every $R \in O(3)$, all real 2-forms $f_1, \dots, f_N \in C_c^\infty$ and all $z \in \mathbb{C}^N$,*

$$Z_\infty\Big(\sum_j z_j\, R\cdot f_j,\, 0\Big) = Z_\infty\Big(\sum_j z_j f_j,\, 0\Big),$$

so all curvature Schwinger functions are $O(3)$-covariant. With the translations (14), proved from (13), the same holds for every Euclidean motion $x \mapsto Rx + b$, with $\tau_b R\cdot f_j$ in place of $R\cdot f_j$, at the points of E_W again by right-continuity in the scalar mass (item (15)): the curvature sector is $E(3)$-invariant. For the curvature two-point function the weaker closing hypothesis (TIGHT_2) *suffices (Corollary 6.5).*

Theorem A rests on inputs that are proved, here or in companion manuscripts checked independently, and on statements of the literature at the printed grade [PROVED-LIT]; it has no other hypothesis, and in particular none on the couplings beyond King's range. The three hypotheses and the proviso that earlier versions carried are proved, or replaced by proved statements (Section 1.6): the bound at long trees for the families with one inserted field enters King's cube limit (13), which defines the object; the fluctuation estimates for the two further contour families and the positivity condition enter (K_0), through item (1) and the inputs (H3), (H4) of Theorem 4.9, the positivity condition also through (4) and hence (TIGHT_0); the final-scale cap is a proviso of the torus theorem on its parameters, met by the class (6). At the points of E_W, where the positivity condition fails, the chain of (K_0) is not run and the invariance follows from the nearby scalar masses by right-continuity of King's state in the scalar mass (item (15), Proposition 1.4). The implication (1)–(5) $\Rightarrow$ (K_0), on the class (6), is Theorem 4.9, the implication (8)–(11) $\Rightarrow$ (RES_0), with the literature inputs (12), is Theorem 7.12 (item (7)), (RES_0), (K_0) and the dead-shell chain (10c) give (TIGHT_0) by Theorem 7.10, and (K_0) and (TIGHT_0) give the invariance by Theorem 6.4; item (13) defines the object, King's state on curvature sources on King's cubes (Definition 3.9, Proposition 3.11). Every item of the list (1)–(15) is proved here, or proved as an implication in a companion manuscript and checked independently, or is a statement of the literature cited at its printed grade. Among the inputs proved in companion manuscripts, the change of domain (P7) of Bałaban's propagators on the skew tori enters (K_0), through item (1) on the skew tori; the long trees (P8) in the difference clause of the split degree count enter King's cube limit (13), which defines the object; and the sub-domain form (P9) of the perturbative bound at order fourteen enters the upper bound of (8), and through (8) (RES_0) and (TIGHT_0). The bound (DOM) for the first-order term of the step determinants on Bałaban's propagator domains and on the torus, uniform in the shell mass, has two independent proofs (Section 1.5); it enters (8) through the determinant clause of the perturbative bound (8k)(ii), and its shell-mass residual does not enter (K_0) or King's cube limit (13), which are statements at zero shell mass. King's cut of the long lines on his cubes, which King's cube limit (13) uses, is proved, (P5). No statement of the literature is consumed in a form asserted in print without a printed argument or read off a printed construction. Two readings of Bałaban's papers that earlier versions cited at that grade, (B1) and (B4) of Section 9.2, the bounds for the vector-field propagators [Ba3, p. 426] on the torus and the vector half of the completion of [Ba2, §3C] on p. 591 there, are [PROVED] as implications and checked independently [SUP]; (K_0) and (8) use (B4), and (10) uses (B1). King's

two statements, the source-free strict degree property of his renormalised graphs as rewritten by Bałaban [K1, p. 668] and his treatment of the sources [K1, pp. 664–665, 677], the photon legs of King's Appendix (Section 9.3) and the readings (B2), (B3), (B5), (B6) of Bałaban's papers, with the closure of his renormalised classes under the completion moves, which earlier versions cited at their printed grade, are likewise [PROVED] as implications and checked independently [SUP], in the forms consumed. Section 9 restates all the inputs in plain words, with the flagged steps of the companion manuscripts and their completions.

Theorem B (the heavy-shell hypothesis)**.** (i) (RES) *implies* (TIGHT)*, and* (RES_0) *implies* (TIGHT_0)*, under* $(\mathrm{K})_a$*, respectively* (K_0) *(Theorem 7.10).*

(ii) *For every two-level configuration,* $\Psi^0(n,K) := \lim_k \Psi_k(n,K)$ *exists under* $(\mathrm{K})_a$ *for* $K < \infty$ *and is non-increasing in* K*, and* (RES) *at that configuration is equivalent to* $\lim_{K\to\infty} \Psi^0(n,K) \le \limsup_k \Psi_k(n,\infty) + \eta$*. If the dead-shell torus value* $\Psi^0(n,\infty) = \lim_k \Psi_k(n,\infty)$ *exists, this is the continuum statement* $\lim_K \Psi^0(n,K) \le \Psi^0(n,\infty) + \eta$*, and the reverse inequality always holds (Theorem 7.15).*

(iii) *At every fixed lattice spacing,* $\Psi_k(n,K) - \Psi_k(n,\infty) \to 0$ *as* $K \to \infty$ *(Proposition 8.1); so any failure of* (RES) *is carried by shell matter whose mass squared tends to infinity with the inverse spacing.*

(iv) (TIGHT) *does not follow from the correlation inequalities, the monotonicity in the volume, the torus comparison and passive covariance (Remark 5.8); and no bounded region of* $\mathbb{R}^3$ *has all its faces on lattice hyperplanes of both* Λ_k *and* $R^{-1}\Lambda_k$ *(Proposition 8.2), so a comparison of a cube with a torus must go through a smooth wall.*

(v) *In the curvature sector,* (RES_0) *at one two-level configuration with the shell* $U = T \setminus H'$ *holds as an implication from the four shell estimates (8)–(11) of Section 1.4 (Theorem 7.12).*

Theorem C (the torus comparison at one spacing)**.** *Let* $R \in \mathcal{R}_0$*. Two one-sided bounds of King's type* (F)*, one on* D_J *and one on* $R^{-1}D_J$*, with vacuum constants* v, v' *and source-linear constants* c, c'*, imply* $e^{v'+c'\int h}(1+o(1)) \le \mathcal{Z}_{R^{-1}\Lambda_k,D_J}(s)/\mathcal{Z}_{\Lambda_k,D_J}(s) \le e^{-v-c\int h}(1+o(1))$ *for every smooth periodic source* $s = (G,h)$*; the constants are forced to satisfy* $v + v' \le o(1)$*, and* $c + c' \to 0$ *once* $v + v' \to 0$*. Under* (F) *and* (V) *all walled curvature ratios and all curvature moments of the plain tori agree asymptotically on the two lattices, with no condition on the source-linear constants; with* (C) *in addition, all ratios agree (Theorem 4.5). The bounds* (F_0) *and* (V) *for walled curvature ratios are supplied by the closing of* [CT] *from the hypotheses (1)–(5), on the class (6), with the large-field clause of (1) in place of King's torus bound* [K2, (4.2)]*, which is vacuous at a non-zero wall (Section 4.3).*

Theorem A is Theorem 6.4 with Theorems 4.9, 7.12 and 7.10; Theorem B collects Theorems 7.10, 7.15, 7.12, Propositions 8.1, 8.2 and Remark 5.8; Theorem C is Theorem 4.5 with its corollaries. The proofs are complete given the inputs of Section 1.4, and those inputs are proved as stated there, given the statements of the literature at their printed grade [PROVED-LIT], on which Theorem A rests. Their status is discussed in Remark 4.10, Section 8 and Remark 7.13, and tabulated in Section 11. In one sentence: (K_0) is a one-sided comparison of effective actions at equal spacing of the kind sketched in [K2, §4] for King's torus conjecture, carried out in [CT] as an implication from (1)–(5), on the class (6), with Bałaban's asserted estimates (1), its closing implication and its own steps (5l) proved and checked independently, the reading (B4) included, the estimates (1) on the skew tori with the change of domain (P7), and its flagged steps completed (Section 9), outside the set E_W of Proposition 1.4; (RES_0) is the uniform-in-the-spacing decoupling of a heavy weakly coupled shell from a light core across a heavy plateau, which holds at every fixed spacing, which every soft argument fails to supply, and which is proved as an implication from four estimates on King's block-spin flow with the shell mass inside the covariance, (8)–(11), whose own residues

(8a)–(8k), (10a)–(10g) are proved as implications and checked independently (given the flagged steps named in Section 9.6, with their completions), the determinant clause of (8k)(ii) with the bound (DOM), and the upper bound of (8) with the sub-domain form (P9).

Definition 3.9 states the object, and the scope of the paper, in one place. The relation of King's state to infinite-volume limits along periodic boxes, King's expected identity [K2, (2.24)], is never used.

1.8. **Organisation.** Section 2 defines the model, the states and the sources, collects the correlation inequalities, and defines King's renormalisation-group cubes, which it compares with the boxes. Section 3 introduces Wick schemes and constructs King's state; at $g = 0$ only Proposition 3.11 is used, and the scheme hypothesis $(\mathrm{W})_a$ enters for scalar sources alone, outside the theorem of this paper. Section 4 states the torus hypothesis, reduces it to one-sided bounds at one spacing, and recalls the corrected torus theorem with its inputs (1)–(5), on the class (6). Section 5 introduces walled tori and the closing hypothesis (TIGHT), and Section 6 proves the conditional invariance theorem. Section 7 contains the heavy-shell hypothesis (RES), the two-level sandwich, the proof that (RES) implies (TIGHT), the heavy-shell theorem which proves (RES_0) from the shell estimates (8)–(11), and, for completeness, the continuum form of (RES) and two sufficient forms that are not on the chain. Section 8 collects what is proved about the torus and heavy-shell hypotheses: the fixed-spacing statement and the forcedness of the wall. Section 11 is the table of grades. Appendix A collects the corrections to [K1, K2].

2. The model

2.1. **Geometry.**

Definition 2.1. Let $\theta_0 \in (0, \pi/2)$ with $\cos\theta_0 = \frac{4}{5}$, $\sin\theta_0 = \frac{3}{5}$. For $1 \le \mu < \nu \le 3$ let $R_{\mu\nu}(\theta)$ be the rotation with $Re_\mu = \cos\theta\, e_\mu + \sin\theta\, e_\nu$, $Re_\nu = -\sin\theta\, e_\mu + \cos\theta\, e_\nu$, $Re_\rho = e_\rho$ for $\rho \ne \mu, \nu$. Put $\mathcal{R}_0 := \{R_{\mu\nu}(\pm\theta_0) : 1 \le \mu < \nu \le 3\}$ and let G_0 be the group generated by $\mathcal{R}_0$. Let $\varepsilon_k := 5^{-k}$ and $\Lambda_k := \varepsilon_k\mathbb{Z}^3$, $k \ge 0$. Let M be the large-block integer of [K1, §2.3], fixed throughout; we take $M = 5M'$ with M' divisible by a prime $\mathfrak{p} \ne 5$ and $M' \ge 2\mathfrak{p}$, for instance $M = 20 \cdot 5^b$ with b large. This is admissible: every condition that the constructions of [K1, K2, Ba1, Ba2, Ba3, Ba4] place on M is a lower bound, apart from divisibility conditions on the sides of King's cubes and tori, such as that the side $2MJ$ of C_J be a multiple of every block side [K2, (3.4)], which hold for every M by the definitions of C_J and D_J below; it does not change King's state (Definition 3.9), and it is used, with the grid offset of the box H' in the class (6) of Section 1.4, by the sub-domain bound (8k)(i). The *King cubes* are $C_J := \{x \in \mathbb{R}^3 : -MJ \le x_\mu < MJ,\ \mu = 1, 2, 3\}$, $J \in \mathbb{N}$ [K2, (2.7)], and the *King tori* are $D_J := \mathbb{R}^3/(50MJ\,\mathbb{Z})^3$ with fundamental cell $\Gamma_J := [-25MJ, 25MJ)^3$ [K2, (4.1)]. We write $Q_b := [-b, b]^3$, $N_r(X) := \{y : \mathrm{dist}_\infty(y, X) \le r\}$, and $\bar{B}_\rho$ for the closed Euclidean ball of radius ρ about 0.

Lemma 2.2. (a) *For $R \in \mathcal{R}_0$ the matrix $5R$ has integer entries, $R^{-1} \in \mathcal{R}_0$, and $\sigma_\nu R_{\mu\nu}(\theta)\sigma_\nu = R_{\mu\nu}(-\theta)$, where σ_ν is the reflection $x_\nu \mapsto -x_\nu$.*

(b) *For $R \in \mathcal{R}_0$, $k \ge 0$ and $J \ge 1$ the lattices $R\Lambda_k$ and $R^{-1}\Lambda_k$ are invariant under translations by $50MJ\,\mathbb{Z}^3$, so they descend to D_J.*

(c) *For $R \in \mathcal{R}_0$, $|Rx|_\infty \le \frac{7}{5}|x|_\infty$, and $Q_{5b/7} \subset R^{-1}Q_b \subset Q_{7b/5}$ for $b > 0$.*

(d) *θ_0/π is irrational, and G_0 is dense in $SO(3)$.*

Proof. (a) The entries of R are $0, 1, \pm\frac{3}{5}, \pm\frac{4}{5}$, and $R_{\mu\nu}(\theta)^{-1} = R_{\mu\nu}(-\theta)$; conjugation by σ_ν reverses the orientation of the (x_μ, x_ν)-plane and fixes the third axis. (b) For $p \in 50MJ\,\mathbb{Z}^3$ we have $R^{\pm1}p = \frac{1}{5}(5R^{\pm1})p \in 10MJ\,\mathbb{Z}^3 \subset \Lambda_k$. Hence $R^{-1}\Lambda_k + p = R^{-1}(\Lambda_k + Rp) = R^{-1}\Lambda_k$, and likewise for $R\Lambda_k$. (c) The sums of the absolute values of the entries in each row of R are at most $\frac{7}{5}$. If $Rx \in Q_b$ then $|x|_\infty = |R^{-1}(Rx)|_\infty \le \frac{7}{5}b$, since $R^{-1} \in \mathcal{R}_0$; if $|x|_\infty \le \frac{5}{7}b$ then $|Rx|_\infty \le b$. (d) If $\theta_0 \in \pi\mathbb{Q}$,

then $e^{\mathrm{i}\theta_0} = (4+3\mathrm{i})/5$ is a root of unity, hence an algebraic integer. Its minimal polynomial over $\mathbb{Q}$ is $x^2 - \frac{8}{5}x + 1$, which does not have integer coefficients; this is a contradiction (compare [K2, pp. 337–338]). Hence $\{n\theta_0\}_{n\in\mathbb{Z}}$ is dense modulo 2π, and the closure of $\{R_{\mu\nu}(\theta_0)^n\}_{n\in\mathbb{Z}}$ is the group of all rotations in the (x_μ, x_ν)-plane. Every element of $SO(3)$ is a product $R_{12}(\alpha)R_{23}(\beta)R_{12}(\gamma)$ (Euler angles), so $\overline{G_0} = SO(3)$. □

2.2. **Lattices, fields and action.**

Definition 2.3 (lattices and fields)**.** A *lattice of spacing* ε_k is $\Lambda = O\Lambda_k$ with $O \in O(3)$, equipped with the frame $f_\alpha := Oe_\alpha$, $\alpha = 1,2,3$, and with the nearest-neighbour bonds $\langle x, x+\varepsilon_k f_\alpha\rangle$. A field configuration on a set X of sites consists of $\phi(x) = r_x e^{\mathrm{i}\vartheta_x} \in \mathbb{C}$ and $A_\alpha(x) \in \mathbb{R}$, $x \in X$; the bond $\langle x, x+\varepsilon f_\alpha\rangle$ carries $A_{\langle x,x+\varepsilon f_\alpha\rangle} := A_\alpha(x)$ and the reversed bond carries $-A_\alpha(x)$ (Feynman form of the vector field, [K2, (2.5)]). A plaquette is $p = (x;\alpha,\beta)$ with $\alpha < \beta$; its centre is $c(p) := x + \frac{\varepsilon}{2}(f_\alpha + f_\beta)$ and its field strength is

$$F(p) := \varepsilon^{-1}\big(A_\alpha(x) + A_\beta(x+\varepsilon f_\alpha) - A_\alpha(x+\varepsilon f_\beta) - A_\beta(x)\big) \qquad \text{[K2, (2.11)]}.$$

Definition 2.4 (King's action)**.** Write $\varepsilon = \varepsilon_k$. The formal action on a lattice Λ of spacing ε is [K2, (2.5)]

$$\begin{aligned} S(A,\phi) = {} & \frac{1}{2}\sum_{\alpha,\beta}\sum_{x} \varepsilon\,\big(A_\alpha(x+\varepsilon f_\beta) - A_\alpha(x)\big)^2 + \frac{\mu_0^2}{2}\sum_{\alpha}\sum_{x}\varepsilon^3 A_\alpha(x)^2 \\ & + \frac{1}{2}\sum_{\beta}\sum_{x}\varepsilon\,\big|e^{\mathrm{i}e\varepsilon A_\beta(x)}\phi(x+\varepsilon f_\beta) - \phi(x)\big|^2 + \sum_{x}\varepsilon^3\Big[\tfrac{1}{2}\big(m^2+\delta m_\varepsilon^2\big)|\phi(x)|^2 + \lambda|\phi(x)|^4\Big], \end{aligned}$$

where $|e^{\mathrm{i}e\varepsilon A}\phi(y) - \phi(x)|^2 = r_x^2 + r_y^2 - 2r_x r_y\cos(\vartheta_x - e\varepsilon A - \vartheta_y)$. The mass counterterm δm_ε^2 is King's [K1, (3.32)–(3.34)], computed with propagators on the infinite lattice $\varepsilon\mathbb{Z}^3$ (free boundary conditions, as in [K2, p. 330]); it is a polynomial in e^2 and λ whose coefficients depend only on ε, m and μ_0. King's action also carries the normalisation constant E_0 and the vacuum-energy counterterm E_1 [K1, (2.4), (3.35)], [K2, p. 325]; both are field-independent, they cancel in every normalised expectation, and we omit them. The *free Wick constant* is $\mathsf{C}_\varepsilon := (-\Delta_{\varepsilon\mathbb{Z}^3} + m^2)^{-1}(0,0)$.

Definition 2.5 (states)**.** Let Λ be a lattice of spacing ε, and $d\nu := \prod_x \prod_\alpha dA_\alpha(x)\, d^2\phi(x)$ over the sites in question.

(a) *Neumann–Dirichlet boxes.* For a finite set X of sites of Λ, $\langle\cdot\rangle^{\mathrm{ND}}_{\Lambda,X}$ is the normalised measure $Z^{-1}e^{-S^{\mathrm{ND}}_{\Lambda,X}}d\nu$ on the fields on X, where $S^{\mathrm{ND}}_{\Lambda,X}$ is obtained from S by setting $\phi = 0$ at all sites outside X (Dirichlet condition for the scalar field), and by keeping the photon gradient terms only for pairs $x, x+\varepsilon f_\beta$ that both lie in X and the photon mass term only for $x \in X$ (Neumann condition for the vector field). For $\Lambda = \Lambda_k$ and $X = C_J \cap \Lambda_k$ this is King's state $\langle\cdot\rangle^{\varepsilon_k}_{N,D;C_J}$ of [K2, (2.9)–(2.10)] in the convention of King's proof of [K2, Thm. 3.2] (pp. 333–334); we write $\langle\cdot\rangle^{\mathrm{ND}}_{k,J}$. The cube on which King's renormalisation group is run differs from it by the matter on one face layer (Definition 2.15).

(b) *Tori.* If Λ descends to D_J, $\langle\cdot\rangle_{\Lambda,D_J}$ is the normalised measure with the action S summed over the sites and bonds of the finite graph $\Lambda/(50MJ\,\mathbb{Z})^3$ (periodic conditions for both fields).

(c) *Walled tori.* A *walled configuration* is $\mathfrak{c} = (J', w)$ with $J' \in \mathbb{N}$ and $w \in C^\infty(D_{J'};[0,\infty))$, the *wall*. If Λ descends to $D_{J'}$, $\langle\cdot\rangle_{\mathfrak{c},\Lambda}$ is the torus state of (b) with the density multiplied by $\prod_x \exp(-\varepsilon^3 w(x)|\phi(x)|^2)$. The wall raises the scalar mass squared by $2w$ and acts in the single-site measures. We write $(J',\psi,n) := (J', n\psi)$ for $n \geq 0$.

(d) *Killed sites.* For any of the states above and a set U of sites, the state *with the matter killed on* U is obtained by replacing the single-site measure $d^2\phi(x)$ times its weight, for $x \in U$,

by the point mass at $\phi(x) = 0$. All hopping terms with an endpoint in U then reduce to their diagonal parts at the other endpoint; the photon variables are unchanged.

For a lattice Λ descending to a torus T we also use the *unnormalised* generating functional $\mathcal{Z}_{\Lambda,T}(G,h) := \int e^{-S(A,\phi)+F_\Lambda(G)+:\phi^2:_\Lambda(h)}\,d\nu$ (Definition 2.8), with King's constants E_0, E_1 omitted. Two lattices of the same spacing descending to T have the same number of sites in a fundamental cell, so any field-independent constant per site multiplies both by the same factor.

The definition of δm_ε^2 with free boundary conditions is the one King uses in cubes [K2, p. 330]; it makes all states on all lattices of spacing ε carry the same local counterterm.

Remark 2.6 (the counterterm). King's δm_ε^2 is a single number: it is fixed by King's two-point graphs of order at most 4 in (e,λ) on the infinite lattice $\varepsilon\mathbb{Z}^3$ at zero momentum [K1, (3.32)–(3.33)], with the propagators of the infinite lattice [K2, p. 330], and it is the same at every site of every box and every torus. We use only this; it is the reading under which King's cut of a cube out of a larger cube is "exactly" the smaller Neumann–Dirichlet state [K2, p. 334]. The theorems of [K1] are proved for the torus conventions of [K1, (3.33), (2.8), (3.21), (3.35)]; the bridge between the two conventions is the input (CT) of [CT] (Section 4.3) on the King side and $(\mathrm{CT})_{\mathrm{sh}}$ ([SW1, Lemma M(d)]: the offset Δ_T is a constant added to the counterterm-mismatch vertex; no cross-convention equality of states is used) on the shell side (items (8f) and (10) of Section 1.4).

Lemma 2.7 (killing)**.** *Let $\langle\cdot\rangle$ be a state of Definition 2.5 (a)–(c), U a set of sites, $\chi : U \to (0,\infty)$, and $\langle\cdot\rangle_{U,s}$ the state with the single-site measures at $x \in U$ multiplied by $e^{-s\varepsilon^3\chi(x)r_x^2}$, $s \geq 0$. Then for every continuous function Θ of the fields with $|\Theta| \leq e^{C\sum_x r_x^2}$, $\langle\Theta\rangle_{U,s} \to \langle\Theta\rangle_U$ as $s \to \infty$, where $\langle\cdot\rangle_U$ is the state with the matter killed on U.*

Proof. At fixed lattice spacing all integrals are finite-dimensional. Substituting $\phi(x) = (s\varepsilon^3\chi(x))^{-1/2}\zeta(x)$ for $x \in U$, the normalised single-site measures at $x \in U$ converge weakly to the point mass at 0, the hopping terms $\varepsilon r_x r_y \cos(\cdot)$ with $x \in U$ are bounded by $\frac{\varepsilon}{2}(s\varepsilon^3\chi(x))^{-1}|\zeta_x|^2 + \frac{\varepsilon}{2}r_y^2$, and the quartic terms at the other sites dominate; dominated convergence applies to numerator and denominator. □

2.3. **Sources and observables.**

Definition 2.8 (sources)**.** A *curvature source* is a real 2-form $f \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$, and a *scalar source* is a real $h \in C_c^\infty(\mathbb{R}^3)$; on a torus, smooth periodic sources are used. On a lattice Λ of spacing ε with frame (f_α) put

$$F_\Lambda(f) := \sum_p \varepsilon^3\, f_{c(p)}(f_\alpha, f_\beta)\, F(p),$$

$$\mathsf{S}_\Lambda(h) := \sum_x \varepsilon^3 h(x), \qquad :\phi^2:_\Lambda(h) := \sum_x \varepsilon^3 h(x)\big(|\phi(x)|^2 - 2\mathsf{C}_\varepsilon\big),$$

the sums running over the plaquettes $p = (x;\alpha,\beta)$, respectively the sites, of the region (of one fundamental cell on a torus). For finite families and $(z,w) \in \mathbb{C}^{2N}$ put

$$\Theta\Big(\sum_j z_j f_j, \sum_j w_j h_j\Big) := \exp\Big(\sum_j z_j F_\Lambda(f_j) + \sum_j w_j :\phi^2:_\Lambda(h_j)\Big).$$

For $O \in O(3)$ and $b \in \mathbb{R}^3$ let $(O\cdot h)(x) := h(O^{-1}x)$, $(O\cdot f)_x(u,v) := f_{O^{-1}x}(O^{-1}u, O^{-1}v)$, $(\tau_b h)(x) := h(x-b)$ and $(\tau_b f)_x := f_{x-b}$. The *cone* is

$$\mathcal{K} := \{(f,g) :\ f \text{ a real 2-form},\ g \geq 0,\ \text{both in } C_c^\infty\},$$

with the observable $\Theta(\mathrm{i}f, g)$. $\mathcal{K}$ is invariant under $O(3)$, under translations and under $(f,g) \mapsto (tf, tg)$, $t \geq 0$, and under sums.

All states of Definition 2.5 are invariant under $(A,\vartheta)\mapsto(-A,-\vartheta)$, which maps $F_\Lambda(f)$ to $-F_\Lambda(f)$ and fixes r. Hence $\langle\Theta(\mathrm{i}f,g)\rangle=\langle\cos F_\Lambda(f)\,e^{:\phi^2:_\Lambda(g)}\rangle$ is real.

Remark 2.9 (the frame convention). In [K1, K2] the curvature source is written $F(g)=\sum_p\varepsilon^3 g(p)F(p)$ with one function g on all plaquettes [K2, (2.12)]. On one lattice a 2-form sampled as above is a function on plaquettes, and King's proofs treat the source terms componentwise ("the sources $g_{\mu\nu}(x'),h(x')$ ... can be replaced by $g_{\mu\nu}(x),h(x)$", [K1, p. 665]; the sentence is asserted there without an argument). We use King's single-lattice statements at $h=0$ for 2-form sources in this sense: Theorem 2.1 of [K2] holds, on King's cubes C_J^K (Definition 2.15), for real smooth 2-form sources sampled at plaquette centres in the lattice's own frame, periodic or compactly supported, with constants that depend on the source only through $\|G_{\mu\nu}\|_\infty$ and $\|\nabla G_{\mu\nu}\|_\infty$, and the limit does not depend on the sampling rule (item (13) of Section 1.4: printed grade [SKETCH-LIT], [PROVED] as an implication in [CA], with the bound at long trees of Proposition 1.2); the statements of [K2, Thms. 2.2, 2.3, (3.40)–(3.44)] that we use, all at $h=0$, are proved here from it and from Lemmas 2.14 and 2.16 (Section 3), and [K2, (3.42)–(3.43)] are not used, and the torus statement [K1, Thm. 2.1] at a wall is item (4). The two-lattice statement is different: Proposition A.2 shows that it fails for a single scalar function on all plaquettes, and Hypothesis 4.1 is formulated for 2-forms sampled in each lattice's own frame.

Lemma 2.10 (passive covariance)**.** *Let $O\in O(3)$, let Λ be a lattice of spacing ε with frame (f_α), and let X be a box, a torus or a walled torus for Λ. Equip $O\Lambda$ with the frame (Of_α) and OX with the transported boundary conditions and wall $O\cdot w$. Then*

$$\big\langle\Theta(z\,O\cdot f,w\,O\cdot h)\big\rangle_{O\Lambda,OX}=\big\langle\Theta(zf,wh)\big\rangle_{\Lambda,X},\qquad \mathsf{S}_{O\Lambda}(O\cdot h)=\mathsf{S}_\Lambda(h),$$

and likewise for finite families. Moreover, on every torus the periodic model on Λ does not change if the frame (f_α) is replaced by $(\pm f_{\pi(\alpha)})$, for a permutation π and any choice of signs.

Proof. The map $\phi'(Ox)=\phi(x)$, $A'_\alpha(Ox)=A_\alpha(x)$, with components taken in the frames (f_α) and (Of_α), is a bijection of the configuration spaces preserving $d\nu$. The action is written in the frame, and δm_ε^2 and C_ε depend on ε only. Plaquettes, their centres and field strengths are transported, and $(O\cdot f)_{Oc}(Of_\alpha,Of_\beta)=f_c(f_\alpha,f_\beta)$. For the second statement, a permutation of the frame relabels the components; reversing f_α replaces $A_\alpha(x)$ by $-A_\alpha(x-\varepsilon f_\alpha)$, a bijection of the torus variables that preserves the action, and changes the sign of both $F(p)$ and $f_{c(p)}(f_\alpha,f_\beta)$ while preserving the set of plaquette centres. □

2.4. **Correlation inequalities.**

Definition 2.11 (BFS measures)**.** A *BFS measure* on a finite set V of sites is a probability measure of the form

$$Z^{-1}\exp\Big(\sum_{\{x,y\}\in B}c_{xy}\,r_xr_y\cos(\vartheta_x-\vartheta_y-\ell_{xy}(A))\Big)\prod_{x\in V}d\rho_x(r_x)\,d\vartheta_x\;\;d\gamma(A),$$

where B is a set of pairs of sites, $c_{xy}\ge0$, ρ_x are finite measures on $[0,\infty)$ with all exponential moments of r^2 finite (point masses at 0, that is killed sites, included), γ is a non-degenerate centred Gaussian measure on $\mathbb{R}^m$, and the ℓ_{xy} are linear functionals. For a real linear functional F of A and $g\ge0$ on V put $\Theta_{F,g}:=e^{\mathrm{i}F}e^{\sum_x g_xr_x^2}$.

Every state of Definition 2.5 is a BFS measure: $c_{xy}=\varepsilon$ for the bonds present, the single-site measures contain $e^{-\varepsilon^3[\frac12(m^2+\delta m_\varepsilon^2)r^2+\lambda r^4]}$, the diagonal parts $\frac{\varepsilon}{2}r_x^2$ of all bonds at x (including, for boxes, those leading out of X), and the wall; the photon form is positive definite because $\mu_0>0$. The same holds for every measure obtained from such a state by multiplying some c_{xy} by $t\in[0,1]$, multiplying photon gradient terms by $t\in[0,1]$, adding $s|\ell(A)|^2$ $(s\ge0)$ to the action,

or multiplying single-site measures by e^{-sr^2} ($s \geq 0$), or killing sites. Curvature sources $F_\Lambda(f)$ are real linear functionals of A, and $e^{:\phi^2:(g)}$ is a positive constant times $e^{\sum_x \varepsilon^3 g(x) r_x^2}$.

Lemma 2.12 (Brydges–Fröhlich–Seiler, King)**.** *Let $\langle\cdot\rangle$ be a BFS measure, F a real linear functional of A, $g \geq 0$, and $\Theta = \Theta_{F,g}$. Write $\langle X;Y\rangle := \langle XY\rangle - \langle X\rangle\langle Y\rangle$. Then $\langle\Theta\rangle$ is real, and*

(a) *$\langle\Theta; |\ell(A)|^2\rangle \leq 0$ for every complex linear functional ℓ; hence adding $s|\ell(A)|^2$ to the action, $s \geq 0$, does not decrease $\langle\Theta\rangle$, and deleting a photon gradient term does not increase it;*

(b) *$\langle\Theta; r_x r_y \cos(\vartheta_x - \vartheta_y - \ell_{xy}(A))\rangle \geq 0$; hence decreasing c_{xy} does not increase $\langle\Theta\rangle$;*

(c) *$\langle\Theta; r_x^2\rangle \geq 0$; hence multiplying ρ_x by e^{-sr^2}, $s \geq 0$, does not increase $\langle\Theta\rangle$, and killing a site does not increase it;*

(d) *for $\Theta^- := e^F e^{-\sum_x g_x r_x^2}$ the three inequalities hold with the opposite signs;*

(e) *if the pairs in B and the Gaussian γ do not couple the fields at the sites of $X \subset V$ (and the photon variables attached to them) to the others, the measure is a product, and for Θ depending only on fields in X, $\langle\Theta\rangle$ equals the expectation in the X-factor.*

Proof. (a)–(c) are [BFS, Thms. 6.1, 6.2(1)–(2)] (stated on an arbitrary finite lattice, with an arbitrary Gaussian measure for A and arbitrary single-site measures, [BFS, pp. 275–276]); compare [K2, Prop. 3.1, (3.24)–(3.26)]. (c) is Theorem 6.1 of [BFS] with $P = r_x^2$. For (d), [BFS, Thm. 6.2(3)–(4)] states the photon and the hopping inequality with a sketched proof, and the r^2-inequality, [K2, (3.28)], is stated in [K2] without a proof and is not among the inequalities printed in [BFS]; we prove all three. Let μ be the unnormalised measure and ω' an independent copy of $\omega = (r, \vartheta, A)$; then $(\int d\mu)^2 \langle\Theta^-; Y\rangle = \iint \Theta^-(\omega)(Y(\omega) - Y(\omega'))\, d\mu(\omega)\, d\mu(\omega')$. Write $r = u - v$, $r' = u + v$ (so $u_x \geq |v_x|$), $\vartheta = \vartheta_+ + \vartheta_-$, $\vartheta' = \vartheta_+ - \vartheta_-$, $A = a_+ + a_-$, $A' = a_+ - a_-$. Then a_+, a_- are independent with one centred Gaussian law, ϑ_+, ϑ_- are independent and uniform, and the radial law $\prod_x d\rho_x(u_x - v_x)\, d\rho_x(u_x + v_x)$ is invariant under $v_x \mapsto -v_x$ at each site. With $\chi_{xy} := \vartheta_{+,x} - \vartheta_{+,y} - \ell_{xy}(a_+)$ and $\psi_{xy} := \vartheta_{-,x} - \vartheta_{-,y} - \ell_{xy}(a_-)$,

$$r_x r_y \cos(\chi + \psi) + r'_x r'_y \cos(\chi - \psi) = 2(u_x u_y + v_x v_y)\cos\chi\cos\psi + 2(u_x v_y + v_x u_y)\sin\chi\sin\psi,$$

$$r_x r_y \cos(\chi + \psi) - r'_x r'_y \cos(\chi - \psi) = -2(u_x v_y + v_x u_y)\cos\chi\cos\psi - 2(u_x u_y + v_x v_y)\sin\chi\sin\psi,$$

$r_x^2 - r_x'^2 = -4u_x v_x$, $\ell(A)^2 - \ell(A')^2 = 4\ell(a_+)\ell(a_-)$ for real ℓ, and

$$\Theta^-(\omega) = e^{F(a_+)} e^{F(a_-)} e^{-\sum_x g_x(u_x^2 + v_x^2)} e^{2\sum_x g_x u_x v_x}.$$

Expanding the combined hopping weight (the first line) and $e^{2\sum g_x u_x v_x}$ in power series, every term of the integrand of $\iint \Theta^-(\omega)(Y(\omega) - Y(\omega'))$ is a sign σ times a monomial in (u, v) with a non-negative coefficient, times the even radial factor $e^{-\sum g_x(u_x^2 + v_x^2)}$, times $V(\vartheta_+, a_+)V(\vartheta_-, a_-)$ with one and the same real function V (a product of e^F, of factors $\cos\chi_b$, $\sin\chi_b$, and of ℓ for the photon slot). The angular and photon integral of such a term is the square of the integral of V, and the radial integral vanishes if the degree in v_x is odd at some x and is non-negative otherwise, since $u_x \geq |v_x|$; all series converge absolutely at fixed lattice spacing. The sign is $\sigma = -1$ for $Y = r_x^2$ and for the hopping term (the second line), and $\sigma = +1$ for $Y = \ell(A)^2$ (for complex ℓ apply this to its real and imaginary parts). This is (d). Each monotonicity statement in a parameter follows by differentiating in the parameter, the interpolating measures being BFS measures. The statements on killing need no limit. Let $x \in V$, and let $\langle\cdot\rangle_x$ be the measure with ρ_x replaced by the point mass at 0, in which every term $c_{xy} r_x r_y \cos(\cdot)$ vanishes. Decreasing c_{xy} to 0 for every pair $\{x, y\} \in B$ does not increase $\langle\Theta\rangle$, by (b), and does not decrease $\langle\Theta^-\rangle$, by (d). In the resulting measure the variables (r_x, ϑ_x) are coupled to nothing, so by (e) it is the product of $\rho_x(dr_x)\, d\vartheta_x$ and of the law of the other variables, which is that of $\langle\cdot\rangle_x$; the expectations of Θ and Θ^- in it are therefore $m_x^+\langle\Theta\rangle_x$ and $m_x^-\langle\Theta^-\rangle_x$, with $m_x^\pm := \int e^{\pm g_x r^2} d\rho_x(r) \big/ \int d\rho_x$, so that $m_x^- \leq 1 \leq m_x^+$. Since $\Theta^- > 0$, $\langle\Theta^-\rangle_x \geq m_x^-\langle\Theta^-\rangle_x \geq \langle\Theta^-\rangle$. And $\langle\Theta\rangle_x \geq 0$: multiplying the single-site measures by $e^{g_y r_y^2}$ gives a BFS measure, in which the expectation of $e^{\mathrm{i}F}$ is non-negative by Lemma 2.13(a), whose proof

does not use the present lemma; hence $\langle\Theta\rangle_x \le m_x^+\langle\Theta\rangle_x \le \langle\Theta\rangle$. Several sites are killed one at a time. No condition on ρ_x beyond Definition 2.11 is used, and Lemma 2.7 is not needed here. (e) is clear. □

Lemma 2.13 (current representation and Gaussian domination)**.** *Let $\langle\cdot\rangle$ be a BFS measure whose Gaussian has covariance C. Integrating out the matter at fixed A, the matter integral is $\Psi(A) = \sum_q w(q)\cos\langle\kappa(q), A\rangle$, where q runs over integer-valued divergence-free currents on B, $w(q) \ge 0$, $\sum_q w(q) < \infty$, and $\kappa(q) \in \mathbb{R}^m$ is defined by $\langle\kappa(q), A\rangle := \sum_{xy} q_{xy}\ell_{xy}(A)$. Let $\mathbb{E}^\sharp$ be the probability measure on currents with weights proportional to $w(q)e^{-\langle\kappa(q),C\kappa(q)\rangle/2}$. Then for $v \in \mathbb{R}^m$:*

(a) $\langle e^{\mathrm{i}\langle v,A\rangle}\rangle = e^{-\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp\cosh\langle Cv,\kappa\rangle$ *and* $\langle e^{s\langle v,A\rangle}\rangle = e^{s^2\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp\cos(s\langle Cv,\kappa\rangle) \le e^{s^2\langle v,Cv\rangle/2}$ *for $s \in \mathbb{R}$;*

(b) $\langle\langle v,A\rangle^2\rangle = \langle v,Cv\rangle - \mathbb{E}^\sharp\langle Cv,\kappa\rangle^2 \in [0,\langle v,Cv\rangle]$.

Proof. Expanding $e^{c\cos t} = \sum_{q\in\mathbb{Z}} I_q(c)e^{\mathrm{i}qt}$ with $I_q(c) \ge 0$ for $c \ge 0$ and integrating over the angles gives Ψ; the weights are even in q because $I_{-q} = I_q$. For a Gaussian γ with covariance C, $\int e^{\mathrm{i}\langle v,A\rangle}\cos\langle\kappa,A\rangle\,d\gamma = e^{-\langle v,Cv\rangle/2-\langle\kappa,C\kappa\rangle/2}\cosh\langle v,C\kappa\rangle$ and, by the Cameron–Martin formula, $\int e^{s\langle v,A\rangle}\cos\langle\kappa,A\rangle\,d\gamma = e^{s^2\langle v,Cv\rangle/2-\langle\kappa,C\kappa\rangle/2}\cos(s\langle v,C\kappa\rangle)$; this gives (a). Replacing v by tv in the first formula of (a) and comparing the terms of order t^2 (all sums converge absolutely at fixed lattice spacing) gives (b). □

The following consequence is the form in which the correlation inequalities are used.

Lemma 2.14 (cutting)**.** *Let $\langle\cdot\rangle_Y$ be a box, torus or walled-torus state on a lattice Λ, possibly with killed sites, and let X be a set of sites of Λ such that X and its nearest neighbours lie in one fundamental cell. Let $\langle\cdot\rangle_X^{\mathrm{ND}}$ be the Neumann–Dirichlet state on X whose single-site measures (wall, killed sites and diagonal parts of the bonds leaving X included) are those of $\langle\cdot\rangle_Y$. Let f, g be sources with $g \ge 0$ such that every plaquette and site at which they do not vanish lies in X. Then, at every lattice spacing,*

$$\langle\Theta(\mathrm{i}f,g)\rangle_X^{\mathrm{ND}} \le \langle\Theta(\mathrm{i}f,g)\rangle_Y, \qquad \langle\Theta(f,-g)\rangle_X^{\mathrm{ND}} \ge \langle\Theta(f,-g)\rangle_Y.$$

If the wall of Y vanishes on X and no site of X is killed, $\langle\cdot\rangle_X^{\mathrm{ND}} = \langle\cdot\rangle_{\Lambda,X}^{\mathrm{ND}}$ is the state of Definition 2.5(a).

Proof. Delete from Y every photon gradient term coupling a site of X to a site outside X, and the cosine parts of all hopping terms between X and its complement, keeping their diagonal parts $\frac{\varepsilon}{2}(r_x^2 + r_y^2)$. By Lemma 2.12(a),(b),(d) this does not increase the first expectation and does not decrease the second. The resulting measure is a product (Lemma 2.12(e)); its X-factor is $\langle\cdot\rangle_X^{\mathrm{ND}}$, since the diagonal parts of the cut bonds are exactly the Dirichlet terms and the remaining photon terms are the Neumann ones. The free Wick constants and δm_ε^2 are the same in both states. □

With $Y = C_{J+1}\cap\Lambda_k$ and $X = C_J\cap\Lambda_k$ this is King's monotonicity in the volume [K2, Thm. 3.2, (3.31)–(3.33)] in the form his proof proves, with one Wick constant for all J. King states it with the Dirichlet Wick constant of [K2, (2.13)], which increases with J; then the c-number factor $\exp[\mp 2\sum_x \varepsilon^3 h(x)(C^\infty - C^J)(x,x)]$ (C^J the Dirichlet covariance of C_J at coinciding points) moves against the monotone factor and the monotonicity does not follow. That form is not used, in either convention for the cube. King's proof [K2, pp. 333–334] is written for the boxes $C_J\cap\Lambda_k$ of Definition 2.5(a), and it begins by replacing the Dirichlet subtraction by the free one: "the advantage of doing this is that $:\phi^2:(h)$ is then independent of J" (p. 333). For the cubes on which King's renormalisation group is run the same monotonicity, with the same J-independent Wick constant, follows from Lemma 2.16 below, with one kill step that King does not print (Corollary 2.17).

2.5. **King's renormalisation-group cubes.**

Definition 2.15 (King's cubes)**.** For $k \geq 0$ and $J \in \mathbb{N}$ put $X_J := C_J \cap \Lambda_k$, and let $T_J := \{x \in X_J : x_\mu = MJ - \varepsilon_k \text{ for some } \mu\}$ be the union of its three top face layers. *King's cube* C_J^K at spacing ε_k is the Neumann–Dirichlet box X_J of Definition 2.5(a) with the matter killed on T_J (Definition 2.5(d)). We write $\langle\cdot\rangle_{k,J}^K$ for its state and $Z_{k,J}^{a,K}$ for its functionals in a scheme a. Its photon is the Neumann photon of X_J, its scalar field is alive on $X_J \setminus T_J = \{-MJ, \dots, MJ - 2\varepsilon_k\}^3$, and its scalar Dirichlet planes are $x_\mu = -MJ - \varepsilon_k$ and $x_\mu = MJ - \varepsilon_k$, at distance $2MJ$.

The translation by $(MJ + \varepsilon_k)(1,1,1)$ carries X_J onto the box $\{0 < x_\mu \leq 2MJ\}$ of [K2, (3.3)] with $L_\mu = 2MJ$, the Neumann images of the photon onto those of [K2, (3.7)], about the planes half a spacing outside the box, and the scalar Dirichlet condition onto that of [K2, (3.9)], on the planes $x_\mu = 0$ and $x_\mu = L_\mu$, the second of which is a layer of the box; we read (3.9) as setting the scalar field to zero on these planes. The side $2MJ$ is a multiple of every block side, as [K2, (3.4)] requires, and King's block-spin flow [K2, §3.1] is defined on C_J^K with the Dirichlet weight f_D for the block averages of the scalar field, the weight for which the image representation [K2, (3.9)] holds (it does not hold for the weight printed in [K2, (3.8)]), and with the weight 1 and blocks tiling C_J for the photon. For the box X_J itself the scalar Dirichlet planes are at distance $2MJ + \varepsilon_k$, which is a multiple of no block side, and no flow with Dirichlet images is defined on it. The box is the cube of King's proof of the monotonicity in the volume [K2, pp. 333–334], and King's cube C_J^K is the cube of his renormalisation-group construction [K2, §3.1, pp. 328–329]; the two differ by the matter on T_J only.

Lemma 2.16 (interlacing of the two cube families)**.** *Let $k \geq 0$, $J \geq 1$, and let f be a real 2-form and $g \geq 0$, both in C_c^∞, such that every plaquette of Λ_k at which f does not vanish has its four corners in X_J and every site at which g does not vanish lies in $X_J \setminus T_J$; this holds if* $\operatorname{supp} f \cup \operatorname{supp} g$ *has ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$. Then*

$$\langle\Theta(\mathrm{i}f, g)\rangle_{k,J}^K \leq \langle\Theta(\mathrm{i}f, g)\rangle_{k,J}^{\mathrm{ND}} \leq \langle\Theta(\mathrm{i}f, g)\rangle_{k,J+1}^K,$$
$$\langle\Theta(f, -g)\rangle_{k,J}^K \geq \langle\Theta(f, -g)\rangle_{k,J}^{\mathrm{ND}} \geq \langle\Theta(f, -g)\rangle_{k,J+1}^K,$$

and the same holds for the functionals Z^a in every scheme a.

Proof. The plaquettes of the sources have their corners in X_J, whose photon variables are present in all three states, and the sites of g are alive in all three; C_ε and δm_ε^2 do not depend on the site. So the observables are the same functions of the shared variables, times the same positive constants. The state $\langle\cdot\rangle_{k,J}^K$ is obtained from $\langle\cdot\rangle_{k,J}^{\mathrm{ND}}$ by killing the sites of T_J; by the killing clauses of Lemma 2.12(c), (d) this does not increase $\langle\Theta(\mathrm{i}f, g)\rangle$ and does not decrease $\langle\Theta(f, -g)\rangle$, which is the first inequality of each chain. For the second, the sites of T_{J+1} have a coordinate $M(J+1) - \varepsilon_k > MJ - \varepsilon_k$, so no site of X_J is killed in $\langle\cdot\rangle_{k,J+1}^K$, and X_J with its nearest neighbours lies in X_{J+1}. Lemma 2.14, with Y the state of C_{J+1}^K and $X = X_J$, gives the inequalities, its last sentence identifying the cut-out state with $\langle\cdot\rangle_{k,J}^{\mathrm{ND}}$: a bond from a site of X_J to a killed site of T_{J+1} carries its diagonal part at its alive end, which is the Dirichlet term of X_J. For Z^a use Lemma 3.3(a). A detailed account, with the conventions compared at King's printed pages, is [G]. □

Corollary 2.17 (King's cubes in the volume)**.** *Let f be a real 2-form and $g \geq 0$, both in C_c^∞ and supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$.*

(a) *For all $k \geq 0$ and $J \geq J_0$, $\langle\Theta(\mathrm{i}f, g)\rangle_{k,J}^K \leq \langle\Theta(\mathrm{i}f, g)\rangle_{k,J+1}^K$ and $\langle\Theta(f, -g)\rangle_{k,J}^K \geq \langle\Theta(f, -g)\rangle_{k,J+1}^K$, and likewise for $Z^{a,K}$: King's monotonicity in the volume* [K2, Thm. 3.2]*, in the form his proof proves, with one J-independent Wick constant, holds on King's cubes.*

(b) *At $g = 0$ and every fixed k, the families $J \mapsto \langle e^{\mathrm{i}F(f)}\rangle^K_{k,J}$ and $J \mapsto \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J}$ are nondecreasing on $J \geq J_0$, bounded by 1, and have the same limit as $J \to \infty$.*

(c) *If $Z^K_J(\mathrm{i}f, 0) := \lim_k \langle e^{\mathrm{i}F(f)}\rangle^K_{k,J}$ exists for every $J \geq J_0$, then $\lim_J \liminf_k \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J} = \lim_J \limsup_k \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J} = \lim_J Z^K_J(\mathrm{i}f, 0)$; if instead $\lim_k \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J}$ exists for every $J \geq J_0$, the same holds with the two families exchanged.*

Proof. (a) The hypothesis of Lemma 2.16 holds for every $J \geq J_0$; chain its two inequalities. (b) Both families are nondecreasing, by (a) and by Lemma 2.14 with $Y = C_{J+1} \cap \Lambda_k$, $X = C_J \cap \Lambda_k$; they are real and bounded by 1; and two nondecreasing sequences with $a_J \leq b_J \leq a_{J+1}$ (Lemma 2.16) have the same limit. (c) Lemma 2.16 at every k gives $Z^K_J \leq \liminf_k \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J} \leq \limsup_k \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J} \leq Z^K_{J+1}$, and $J \mapsto Z^K_J(\mathrm{i}f, 0)$ is nondecreasing and bounded by (a); for the exchanged statement use $\langle\cdot\rangle^{\mathrm{ND}}_{k,J-1} \leq \langle\cdot\rangle^K_{k,J} \leq \langle\cdot\rangle^{\mathrm{ND}}_{k,J}$, $J \geq J_0 + 1$. □

Remark 2.18 (what the interlacing gives). Lemma 2.16 relates the two families at every spacing and every J, so that they have the same infinite-volume state (Corollary 2.17), but it gives no information on the fixed-J continuum limit of one family from that of the other: the two bounds $\langle\cdot\rangle^K_{k,J}$ and $\langle\cdot\rangle^K_{k,J+1}$ differ macroscopically. The existence statement (13) is therefore posed on the cubes on which King's renormalisation group is run, and every other comparison of cubes in this paper is transferred to them through Lemma 2.16, at the cost of one index at each end: the comparisons of a cube with walled tori (Theorems 6.4 and 7.6), the monotonicity and the translations (Proposition 3.11), and the reflection (Lemma 6.2). For the reflection $x_3 \mapsto -x_3$ the comparison of C^K_J with $C^K_{J\mp1}$ is also obtained directly, by deletions and kills, when $M \geq 2\varepsilon_k$, which holds at every k for the M of Definition 2.1, since $M \geq 20$ [G]; the proof of Lemma 6.2 uses the comparison with $C^K_{J\mp2}$, which needs no condition. The same account [G] carries out the transfer for two comparisons that this paper does not use, and finds the printed cube sizes sufficient: the comparison of a cube with a reflection-positive box, whose upper cube needs no enlargement because every site of the box lies in the alive set of the larger cube, and the cell bound for negative scalar sources, whose cut-off sources vanish within distance 2 of every cell wall and so never reach a killed site.

3. Wick schemes and King's state

3.1. **The Wick square.**

Remark 3.1 (a logarithm in the Wick square). With the free Wick constant alone, the expectation of the Wick square of Definition 2.8 diverges at order e^2: in perturbation theory $\langle{:}\phi^2{:}_{\varepsilon\mathbb{Z}^3}(h)\rangle = \frac{e^2}{4\pi^2}\ln(1/\varepsilon)\int h + O(1)$ at order e^2 (Proposition A.3). Consequently, with King's printed Wick ordering the generating functionals with $\int h \neq 0$ cannot have finite limits beyond the curvature sector, at least in perturbation theory. We therefore allow an additional vacuum counterterm, linear in h, that depends only on the lattice spacing. The curvature sector $h = 0$ is not affected.

3.2. **Schemes.**

Definition 3.2 (schemes)**.** A *scheme* is a real sequence $a = (a_k)_{k\geq 0}$ such that a_k depends only on ε_k and on (e, λ, m, μ_0). For a state $\langle\cdot\rangle_{\Lambda,X}$ of Definition 2.5 on a lattice of spacing ε_k put

$$Z^a_{\Lambda,X}\Big(\sum_j z_j f_j, \sum_j w_j h_j\Big) := e^{-\sum_j w_j a_k \mathsf{S}_\Lambda(h_j)}\Big\langle \Theta\Big(\sum_j z_j f_j, \sum_j w_j h_j\Big)\Big\rangle_{\Lambda,X},$$

that is, the expectation of $\exp(\sum_j z_j F(f_j) + \sum_j w_j \sum_x \varepsilon^3 h_j(x)[|\phi(x)|^2 - 2\mathsf{C}_\varepsilon - a_k])$. We write Θ^a for the corresponding observable. For walled tori $\mathfrak{c} = (J', w)$ and (f, g) with $w = 0$ on the support

of g put

$$T^a_{\mathfrak{c},\Lambda}(f,g) := \langle \Theta^a(\mathrm{i}f,g)\rangle_{\mathfrak{c},\Lambda} = \frac{Z^a_{\Lambda,D_{J'}}(\mathrm{i}f, g-w)}{Z^a_{\Lambda,D_{J'}}(0,-w)}.$$

Examples. (i) The printed scheme $a \equiv 0$. (ii) The scheme $\Theta^\sharp$: $a_k^\sharp := e^2 X_{\varepsilon_k}$, where X_ε is the coefficient of e^2 in the perturbative expansion of $\langle|\phi(0)|^2\rangle$ for the model on $\varepsilon\mathbb{Z}^3$ at $\lambda = 0$, with the order-e^2 part of δm^2_ε inserted; we write $\Theta^\sharp := \Theta^{a^\sharp}$. (iii) The *self-normalised scheme*: fix J_0 and $0 \le h_0 \in C_c^\infty$, $h_0 \not\equiv 0$, supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, and put $a_k^{sn} := \mathsf{S}_{\Lambda_k}(h_0)^{-1} \ln Z^0_{\Lambda_k, C_{J_0}\cap\Lambda_k}(0,h_0)$ when $\mathsf{S}_{\Lambda_k}(h_0) > 0$, and $a_k^{sn} := 0$ otherwise.

The second expression for $T^a_{\mathfrak{c}}$ agrees with the first because

$$e^{-:\phi^2:(w)} = e^{2\mathsf{C}_\varepsilon \mathsf{S}(w)} \prod_x e^{-\varepsilon^3 w(x)|\phi(x)|^2},$$

and the multipliers $e^{a_k \mathsf{S}(w)}$ of numerator and denominator cancel. At order e^2 the scheme $\Theta^\sharp$ removes the logarithm of Remark 3.1 identically on $\varepsilon\mathbb{Z}^3$: by translation invariance the order-$e^2\lambda^0$ part of $\langle :\phi^2:(h)\rangle$ there equals $e^2 X_\varepsilon \mathsf{S}(h)$.

Lemma 3.3. *Let a be a scheme.*

(a) *For real w the multiplier is a positive constant depending only on k, w and $\mathsf{S}_\Lambda(h)$. Hence the inequalities of Lemmas 2.12 and 2.14 between two states on the same lattice with the same sources hold for Z^a in the same direction.*
(b) $Z^a_{O\Lambda,OX}(z\,O\cdot f, w\,O\cdot h) = Z^a_{\Lambda,X}(zf,wh)$ *for $O \in O(3)$, and likewise for families.*
(c) $T^a_{\mathfrak{c},\Lambda}(f,g) = e^{-a_k \mathsf{S}_\Lambda(g)} \langle \Theta(\mathrm{i}f,g)\rangle_{\mathfrak{c},\Lambda}$*; at $g = 0$ the walled value does not depend on the scheme.*
(d) $|Z^a(\zeta)| \le Z^a(\mathrm{Re}\,\zeta)$*, and $\ln Z^a$ is convex in the real coefficients.*
(e) $Z^a(f,h)\,Z^a(-f,-h) \ge 1$ *for real (f,h).*
(f) *For real f and real $k_1, k_2 \ge 0$: $Z^a(f, k_1 - k_2) \le Z^a(2f, -2k_2)^{1/2}\, Z^a(0, 2k_1)^{1/2}$.*

Proof. (a) is immediate (compare the remark in [BFS, p. 281] that the inequalities are not affected by Wick ordering). (b) follows from Lemma 2.10. (c) was shown above. (d) $|\Theta(\zeta)| = \Theta(\mathrm{Re}\,\zeta)$, and the multiplier has modulus $e^{-\sum \mathrm{Re}\, w_j a_k \mathsf{S}(h_j)}$; convexity is Hölder's inequality. (e) $\langle e^X\rangle\langle e^{-X}\rangle \ge 1$ by Cauchy–Schwarz, and the multipliers cancel. (f) By Cauchy–Schwarz, $\langle e^{F+:\phi^2:(k_1)-:\phi^2:(k_2)}\rangle \le \langle e^{2F-2:\phi^2:(k_2)}\rangle^{1/2}\langle e^{2:\phi^2:(k_1)}\rangle^{1/2}$, and $e^{-a\mathsf{S}(k_1-k_2)} = (e^{2a\mathsf{S}(k_2)})^{1/2}(e^{-2a\mathsf{S}(k_1)})^{1/2}$. □

3.3. **Admissible schemes.**

Hypothesis 3.4 ((W)$_a$: admissibility of the scheme a)**.** (W1) For every King cube C_J, every finite family of curvature sources f_j and scalar sources h_j supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, and every $(z,w) \in \mathbb{R}^{2N}$, the limit $\lim_{k\to\infty} Z^a_{\Lambda_k, C_J\cap\Lambda_k}(\sum z_j f_j, \sum w_j h_j)$ exists in $(0,\infty)$.
(W2) For every $0 \le q \in C_c^\infty(\mathbb{R}^3)$, $\sup_{J \ge J_q} \lim_k Z^a_{\Lambda_k, C_J\cap\Lambda_k}(0,q) < \infty$, where J_q is the least J such that $\mathrm{supp}\, q$ has distance > 1 from $\mathbb{R}^3 \setminus C_J$.

(W1) is the analogue in the scheme a of [K2, Thm. 2.1], and (W2) of the bound [K2, (3.43)], which King asserts for the Neumann–Dirichlet cubes from (3.42), reflection positivity and ultraviolet stability by "a standard argument (see [GJ1])" (not printed) and which, in King's Wick ordering, fails for $\int h_+ > 0$ at order e^2 in perturbation theory (Remark 3.1, Section A.3). At $h = 0$, (W1) on King's cubes C_J^K is King's theorem (Remark 2.9, item (13)); on the boxes $C_J \cap \Lambda_k$ it is not known, and Proposition 3.11 runs on King's cubes instead; (W2) is not needed (Proposition 3.11).

Lemma 3.5. (a) *If a is admissible and $a'_k - a_k \to \kappa \in \mathbb{R}$, then a' is admissible, and* $\lim_k Z^{a'}_{\Lambda_k,C_J}(\zeta) = e^{-\kappa\sum_j w_j \int h_j} \lim_k Z^a_{\Lambda_k,C_J}(\zeta)$.
(b) *If a and a' are admissible, then $a'_k - a_k$ converges.*
(c) *If some scheme is admissible, then a^{sn} is admissible.*

Proof. (a) $Z^{a'} = e^{-\sum_j w_j (a'_k - a_k)\mathsf{S}_{\Lambda_k}(h_j)} Z^a$ and $\mathsf{S}_{\Lambda_k}(h_j) \to \int h_j$. (b) For $0 \le h \in C^\infty_c$ with $\int h > 0$, supported well inside C_J, the ratio $Z^{a'}_{\Lambda_k,C_J}(0,h)/Z^a_{\Lambda_k,C_J}(0,h) = e^{-(a'_k-a_k)\mathsf{S}_{\Lambda_k}(h)}$ converges to a number in $(0,\infty)$, and $\mathsf{S}_{\Lambda_k}(h) \to \int h > 0$. (c) If a is admissible, $a^{sn}_k - a_k = \mathsf{S}_{\Lambda_k}(h_0)^{-1} \ln Z^a_{\Lambda_k,C_{J_0}}(0,h_0)$ converges; apply (a). □

Remark 3.6 (status of (W)). By Lemma 3.5, (W) holds for some scheme if and only if it holds for a^{sn}, and admissible schemes form a single class modulo convergent sequences; $\Theta^\sharp$ is admissible if and only if some scheme is and $a^\sharp - a^{sn}$ converges. If the logarithm of Remark 3.1 is present non-perturbatively, the printed scheme $a \equiv 0$ is not admissible. (W) is expected from King's renormalisation group with the counterterm $a_k\mathsf{S}(h)$ carried as an additional h-linear vacuum vertex; it is not proved here or in the literature: [OPEN]. It is not a hypothesis of Theorem A: at $g = 0$ every quantity is scheme-free (Lemma 3.3(c)), and King's printed state on curvature sources is Proposition 3.11.

3.4. **The continuity lemma.**

Lemma 3.7 (continuity of convex functionals)**.** *Let E be a Fréchet space and $\Phi : E \to \mathbb{R}$ a convex Borel function. Then Φ is continuous.*

Proof. $E = \bigcup_{j\in\mathbb{N}}\{\Phi \le j\}$, so by the Baire category theorem some $S := \{\Phi \le j\}$ is non-meagre. Being Borel, S has the Baire property: there are a non-empty open set O and a meagre set $\mathcal{M}$ with $O \setminus \mathcal{M} \subset S$. Pick $x_0 \in O$ and a balanced open neighbourhood W of 0 with $x_0 + W + W \subset O$. Let $w \in W$ and $y := x_0 + w$. Then $y + W \subset O$, and the point reflection $\iota(x) := 2y - x$ maps $y + W$ onto itself and preserves meagre sets. The sets $S \cap (y + W)$ and $\iota(S \cap (y + W))$ are comeagre in the non-empty open set $y + W$, so they intersect: there is $u \in S$ with $2y - u \in S$. By convexity $\Phi(y) \le \frac{1}{2}(\Phi(u) + \Phi(2y - u)) \le j$. Hence $\Phi \le j$ on the neighbourhood $x_0 + W$ of x_0. A convex function on a topological vector space that is bounded above on a neighbourhood of one point is continuous. □

3.5. **King's state in an admissible scheme.** For $\rho > 0$ let $E_\rho := C^\infty_c(\bar B_\rho; \Lambda^2) \times C^\infty_c(\bar B_\rho)$ be the space of pairs of smooth real sources supported in the closed ball $\bar B_\rho$, with the C^∞ topology; it is a Fréchet space and it is invariant under $O(3)$.

Theorem 3.8 (King's state in the scheme a)**.** *Let a be admissible, and write $Z^a_J(\zeta) := \lim_k Z^a_{\Lambda_k,C_J\cap\Lambda_k}(\zeta)$.*

(i) *For C_J and families as in (W1), $Z^a_{\Lambda_k,C_J} \to Z^a_J$ locally uniformly on $\mathbb{C}^{2N}$, and Z^a_J is entire.*
(ii) *For $(f,g) \in \mathcal{K}$ supported at distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, $J \mapsto Z^a_J(\mathrm{i}f, g)$ is nondecreasing and $J \mapsto Z^a_J(f,-g)$ is nonincreasing for $J \ge J_0$.*
(iii) *For every finite family of curvature and scalar sources the limit $Z^a_\infty(\zeta) := \lim_{J\to\infty} Z^a_J(\zeta)$ exists locally uniformly on $\mathbb{C}^{2N}$ and is entire; $Z^a_\infty(f,h) > 0$ for real sources; and $Z^a_J \le Z^a_\infty$ on $\mathcal{K}$.*
(iv) *For every $\rho > 0$, $\Phi^a(s) := \ln Z^a_\infty(s)$ is finite, convex and continuous on E_ρ.*
(v) *$Z^a_\infty(\tau_b \cdot \zeta) = Z^a_\infty(\zeta)$ for all $b \in \mathbb{R}^3$, where τ_b acts on all sources of the family.*
(vi) *If a' is another admissible scheme, then $a' - a \to \kappa$ and $Z^{a'}_\infty(\zeta) = e^{-\kappa\sum_j w_j \int h_j} Z^a_\infty(\zeta)$.*

Proof. (i) At fixed k the functionals are entire (a finite-dimensional integral with Gaussian decay in A and decay $e^{-\lambda\varepsilon^3 r^4}$ in r). Fix a compact $K \subset \mathbb{C}^{2N}$ and a real box $Q \supset \operatorname{Re} K$. By Lemma 3.3(d),

$\sup_K |Z^a_{\Lambda_k,C_J}| \le \sup_Q Z^a_{\Lambda_k,C_J}$, and by convexity the latter is the maximum over the vertices of Q, at which the sequences converge by (W1). So the family is locally bounded; by Montel's theorem and since $\mathbb{R}^{2N}$ is a set of uniqueness for entire functions, it converges locally uniformly, and the limit is entire (compare [K2, (3.40)–(3.44)]: Lemma 3.3(d) plays the role of (3.40); King's (3.42)–(3.43) are not used, (W2) replacing (3.43)).

(ii) Lemma 2.14 with Lemma 3.3(a) at fixed k, then $k \to \infty$.

(iii) Split each $h_j = h_j^+ - h_j^-$ with $h_j^- := \|h_j\|_\infty \chi_j$ and $h_j^+ := h_j + h_j^-$, where $\chi_j \in C_c^\infty$ takes values in $[0,1]$, equals 1 on $\operatorname{supp} h_j$ and is supported in $N_{1/2}(\operatorname{supp} h_j)$. Let J_0 be such that all supports have distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, and for $J \ge J_0$ and $(z, w^+, w^-) \in \mathbb{C}^{3N}$ let $F_J(z,w^+,w^-) := Z^a_J(\sum z_j f_j, \sum (w_j^+ h_j^+ + w_j^- h_j^-))$, entire by (i). *Local bound.* For real $\zeta = (z,w^+,w^-)$ let $k_1 := \sum_j [(w_j^+)_+ h_j^+ + (w_j^-)_+ h_j^-]$ and $k_2 := \sum_j [(w_j^+)_- h_j^+ + (w_j^-)_- h_j^-]$, where $x_\pm = \max(\pm x, 0)$; both are smooth and non-negative, and the scalar source is $k_1 - k_2$. By Lemma 3.3(f), $F_J(\zeta) \le Z^a_J(2\sum z_j f_j, -2k_2)^{1/2} Z^a_J(0, 2k_1)^{1/2}$. The first factor is nonincreasing in J by (ii), hence at most its value at J_0; the second is bounded uniformly in J by (W2). By convexity and Lemma 3.3(d) the family $\{F_J\}_{J \ge J_0}$ is locally bounded on $\mathbb{C}^{3N}$. *Convergence on a set of uniqueness.* On $\Omega_+ := (\mathrm{i}\mathbb{R})^N \times (0,\infty)^{2N}$ the sources lie in $\mathcal{K}$, so $J \mapsto F_J$ is nondecreasing there by (ii) and bounded, hence convergent. Ω_+ is a non-empty open subset of the totally real subspace $(\mathrm{i}\mathbb{R})^N \times \mathbb{R}^{2N}$ of maximal dimension, hence a set of uniqueness. By Montel's theorem $F_J \to F_\infty$ locally uniformly on $\mathbb{C}^{3N}$, with F_∞ entire. Setting $w^+ = w$ and $w^- = -w$ gives the claim for the original family; the limit is $\lim_J Z^a_J(\sum z_j f_j, \sum w_j h_j)$ and does not depend on the splitting. Positivity follows from Lemma 3.3(e) in the limit, and the last claim from (ii).

(iv) Finiteness and positivity by (iii); convexity as a pointwise limit of convex functions. For J so large that $N_1(\bar{B}_\rho) \subset C_J$, and each k, the function $s \mapsto \ln Z^a_{\Lambda_k,C_J}(s)$ on E_ρ depends continuously on finitely many sampled values of s, hence is continuous; so $\Phi^a = \lim_J \lim_k$ of continuous functions is Borel. Apply Lemma 3.7.

(v) Let $b \in 5^{-k_0}\mathbb{Z}^3$ with $|b|_\infty \le M$. For $k \ge k_0$, τ_b preserves Λ_k and S_{Λ_k}, so $Z^a_{\Lambda_k,C_J}(\tau_b \cdot s) = Z^a_{\Lambda_k,(C_J - b)}(s)$. As sets of sites, $C_{J-1} \cap \Lambda_k \subset (C_J - b) \cap \Lambda_k \subset C_{J+1} \cap \Lambda_k$, so Lemma 2.14 (with $X = C_{J-1} \cap \Lambda_k$ in the state on $(C_J - b) \cap \Lambda_k$, then $X = (C_J - b) \cap \Lambda_k$ in the state on $C_{J+1} \cap \Lambda_k$) gives, for $s \in \mathcal{K}$ and $J \ge J_0 + 1$, where all sources of s are supported at distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$ (so that, the corners of a plaquette lying within ℓ^∞-distance $\varepsilon/2 \le \frac{1}{2}$ of its centre, every plaquette and site at which s does not vanish lies in C_{J-1}), $Z^a_{\Lambda_k,C_{J-1}}(s) \le Z^a_{\Lambda_k,C_J}(\tau_b \cdot s) \le Z^a_{\Lambda_k,C_{J+1}}(s)$. Let $k \to \infty$ and then $J \to \infty$. The identity extends to all coefficients by the argument of (iii), to all $b \in \bigcup_k 5^{-k}\mathbb{Z}^3$ by iteration, and to all $b \in \mathbb{R}^3$ at real coefficients by (iv), since $b \mapsto \tau_b \cdot s \in E_{\rho'}$ is continuous; then to complex coefficients by the identity theorem. (King states translation invariance as [K2, Thm. 2.3(i)]; his argument is not used.)

(vi) Lemma 3.5(a),(b) and (iii). □

Definition 3.9 (the object of Theorem A: King's printed state)**.** Fix $e > 0$, $\lambda, m, \mu_0 > 0$ in King's range, the block factor 5 and the large-block integer M, of the form $M = 5M'$ with M' divisible by a prime $\mathfrak{p} \ne 5$ and $M' \ge 2\mathfrak{p}$ (Definition 2.1), and let $\Lambda_k = \varepsilon_k \mathbb{Z}^3$, $\varepsilon_k = 5^{-k}$. For $J \in \mathbb{N}$ let $\langle\cdot\rangle^K_{k,J}$ be the state of King's cube C^K_J at spacing ε_k (Definition 2.15): the Neumann photon on $C_J \cap \Lambda_k$, and the scalar field with Dirichlet conditions on the planes $x_\mu = -MJ - \varepsilon_k$ and $x_\mu = MJ - \varepsilon_k$, that is, the cube of [K2, (3.3), (3.7), (3.9), p. 329] with $L_\mu = 2MJ$, where (3.9) is read, after the translation that follows Definition 2.15, as setting the scalar field to zero on the planes $x_\mu = 0$ and $x_\mu = L_\mu$ (a reading of the printed page, marked as such in [G]), for the action of Definition 2.4, whose mass counterterm $\delta m^2_{\varepsilon_k}$ is the free-boundary one, computed on the infinite lattice $\varepsilon_k\mathbb{Z}^3$ and the same at every site (Remark 2.6). For finite families of real 2-forms $f_j \in C_c^\infty$, sampled at plaquette centres in the lattice's own frame (Definition 2.8), and $z \in \mathbb{C}^N$, *King's state on curvature*

sources is

$$Z_\infty\Big(\sum_j z_j f_j, 0\Big) := \lim_{J\to\infty}\lim_{k\to\infty}\Big\langle \exp\Big(\sum_j z_j F_{\Lambda_k}(f_j)\Big)\Big\rangle^K_{k,J}.$$

The inner limit is King's cube limit [K2, Thm. 2.1] at $h=0$ along the spacings 5^{-k}, at fixed μ_0 and fixed J, taken for the J at which the sources have ℓ^∞-distance >1 from $\mathbb{R}^3\setminus C_J$; King's proof runs his block-spin flow on C^K_J, whose block averages of the scalar field carry the Dirichlet weight f_D, and the limit is item (13) of Section 1.4, proved as an implication in [CA], with the bound at long trees of Proposition 1.2. The outer limit runs along the family $(C^K_J)_J$. Both limits exist, locally uniformly in z, given item (13) (Proposition 3.11), and the same entire function is obtained from the boxes $C_J\cap\Lambda_k$ of Definition 2.5(a) in place of King's cubes, through the iterated limits of Corollary 2.17(c) on imaginary coefficients. The state does not depend on the admissible choice of M: the states $\langle\cdot\rangle^K_{k,J}$ depend on M only through MJ, so if M_1 divides M_2 the cubes of M_2 are the cubes $C^K_{JM_2/M_1}$ of M_1, and the limits along the two families agree, on imaginary coefficients by Corollary 2.17(a) and then everywhere by Proposition 3.11; two admissible values divide their least common multiple, which is again admissible. This is King's printed state [K2, (2.15)] at $h=0$, and it is the object of Theorem A. We do not address the regularity of the Schwinger functions (they are distributions), the limit $\mu_0\to 0$, or the uniqueness of the vacuum. For scalar sources, outside Theorem A, the object is the functional Z^a_∞ of Theorem 3.8 in an admissible scheme a, whose restriction to $h=0$ is Z_∞ (Corollary 2.17(c)).

Remark 3.10 (scope). The scheme changes Z^a_∞ only by the factor in Theorem 3.8(vi), which affects the one-point function of the Wick square; joint cumulants of order ≥ 2 do not depend on the scheme. The relation of King's state to infinite-volume limits along periodic boxes, King's expected identity [K2, (2.24)], is never used.

Proposition 3.11 (the curvature sector)**.** *At $h=0$ the conclusions of Theorem 3.8, with $\Phi(f) := \ln Z_\infty(f,0)$ on $C^\infty_c(\bar B_\rho;\Lambda^2)$, hold without (W) for King's printed state $Z_\infty(f,0) := \lim_J\lim_k\langle e^{F_{\Lambda_k}(f)}\rangle^K_{k,J}$ of Definition 3.9, with King's cubes C^K_J in place of the boxes $C_J\cap\Lambda_k$; in (v) the cut sandwich is taken with $C^K_{J\mp 2}$.*

Proof. No Wick constant enters. (W1) at $h=0$ on King's cubes is [K2, Thm. 2.1] (in the sense of Remark 2.9; item (13)). In the proof of Theorem 3.8, (i) and (iv) are unchanged; in (ii) Corollary 2.17(a) replaces Lemma 2.14; and in (iii) no splitting is needed: $|Z_J(\zeta)|\leq Z_J(\operatorname{Re}\zeta)\leq Z_{J_0}(\operatorname{Re}\zeta)$, since real curvature sources belong to the nonincreasing class of (ii), and on $(\mathrm{i}\mathbb{R})^N$ the functions are nondecreasing in J and bounded by 1. In (v), let $b\in 5^{-k_0}\mathbb{Z}^3$ with $|b|_\infty\leq M$ and $k\geq k_0$, write $\langle\cdot\rangle^{\mathrm{ND},b}_{k,J}$ and $\langle\cdot\rangle^{K,b}_{k,J}$ for the states of the box and of King's cube translated by $-b$, which are again states on Λ_k, and let $s\in\mathcal{K}$ at $g=0$ be supported at distance >1 from $\mathbb{R}^3\setminus C_{J_0}$. Lemma 2.16, applied to the cubes and to their translates, and the cut sandwich of the proof of Theorem 3.8(v), applied at J and at $J-1$, give for $J\geq J_0+3$

$$\langle\Theta(s)\rangle^K_{k,J-2}\leq\langle\Theta(s)\rangle^{\mathrm{ND}}_{k,J-2}\leq\langle\Theta(s)\rangle^{\mathrm{ND},b}_{k,J-1}\leq\langle\Theta(s)\rangle^{K,b}_{k,J}\leq\langle\Theta(s)\rangle^{\mathrm{ND},b}_{k,J}\leq\langle\Theta(s)\rangle^{\mathrm{ND}}_{k,J+1}\leq\langle\Theta(s)\rangle^K_{k,J+2},$$

and $\langle\Theta(s)\rangle^{K,b}_{k,J}=\langle\Theta(\tau_b\cdot s)\rangle^K_{k,J}$. The rest of the proof of (v) is unchanged. □

4. The torus comparison

4.1. **The hypothesis.**

Hypothesis 4.1 ((K)$_a$ and (K$_0$): the torus comparison)**.** **(K)**$_a$: for every $R\in\mathcal{R}_0$, every King torus D_J and every finite family of real smooth D_J-periodic 2-forms G_j and functions h_j, and

every $(z,w)\in\mathbb{R}^{2N}$, the limits

$$\lim_{k\to\infty} Z^a_{\Lambda_k,D_J}\Big(\sum_j z_jG_j,\sum_j w_jh_j\Big) \qquad \text{and} \qquad \lim_{k\to\infty} Z^a_{R^{-1}\Lambda_k,D_J}\Big(\sum_j z_jG_j,\sum_j w_jh_j\Big)$$

exist in $(0,\infty)$ and are equal. **($\mathbf{K_0}$)**: for every $R\in\mathcal{R}_0$, every walled configuration $\mathfrak{c}$ (of the class (6) of Section 1.4 when (K_0) enters Theorem A; Remark 5.6) and every real smooth periodic 2-form G, $\lim_k\langle e^{F(G)}\rangle_{\mathfrak{c},\Lambda_k}$ and $\lim_k\langle e^{F(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ exist and are equal.

Here $R^{-1}\Lambda_k$ carries the frame $(R^{-1}e_\alpha)$ and descends to D_J by Lemma 2.2(b); both lattices have spacing ε_k and carry the same a_k, $\delta m^2_{\varepsilon_k}$ and $\mathsf{C}_{\varepsilon_k}$. Since $\mathcal{R}_0=\mathcal{R}_0^{-1}$, (K) for all $R\in\mathcal{R}_0$ is the same as the statement for the lattices $R\Lambda_k$ used in [K2]. Read with a single scalar function on all plaquettes, as printed in [K2, (2.12), Thm. 2.4], the comparison is false already at $e=0$ (Proposition A.2); the 2-form reading is consistent with King's proof, which treats the source components separately (Remark 2.9).

Lemma 4.2. (a) *Under $(K)_a$ the convergence holds locally uniformly on $\mathbb{C}^{2N}$, the two limits are the same entire function, and for every walled configuration $\mathfrak{c}$ and (f,g) supported in one cell with $w=0$ on* $\operatorname{supp} g$, $\lim_k T^a_{\mathfrak{c},\Lambda_k}(f,g)$ *and* $\lim_k T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f,g)$ *exist and are equal. The same holds for (K_0) at complex z.*

(b) *$(K)_a$ implies (K_0), and $(K)_a$ implies $(K)_{a'}$ if a'_k-a_k converges.*

(c) *If the statement of $(K)_a$, respectively (K_0), holds for $R=R_{12}(\theta_0)$, it holds for every $R\in\mathcal{R}_0$.*

(d) *Under $(K)_a$, for every King torus D_J and every $0\le q\in C^\infty_c$ supported in one cell, $C_{D_J}(q):=\sup_k Z^a_{\Lambda_k,D_J}(0,q)<\infty$.*

Proof. (a) As in Theorem 3.8(i): local boundedness from Lemma 3.3(d) and convexity, Montel's theorem, and $\mathbb{R}^{2N}$ as a set of uniqueness. The walled ratios are quotients of such limits (with the family $(f;g,w)$ and coefficients $(\mathrm{i};1,-1)$, respectively $(0;0,-1)$), and the denominators have positive limits. For (K_0) use $|\langle e^{zF}\rangle|\le\langle e^{\operatorname{Re} z\,F}\rangle$. (b) $\langle e^{F(G)}\rangle_{\mathfrak{c},\Lambda}=Z^a_{\Lambda,D}(G,-w)/Z^a_{\Lambda,D}(0,-w)$. For the second claim, the factors $e^{-\sum w_j(a'_k-a_k)\mathsf{S}(h_j)}$ have the same limit on both lattices, since $\mathsf{S}_{\Lambda_k}(h)$ and $\mathsf{S}_{R^{-1}\Lambda_k}(h)$ both tend to $\int_{\Gamma_J}h$. (c) Let $\sigma_2=\operatorname{diag}(1,-1,1)$. It preserves Λ_k and D_J, and changes the frame only by a sign, which does not change the periodic model (Lemma 2.10); moreover $\sigma_2R_{12}(\theta_0)^{-1}\sigma_2=R_{12}(\theta_0)$ (Lemma 2.2(a)). By Lemma 2.10 with $O=\sigma_2$ the statement for $(R_{12}(\theta_0),\text{sources } s)$ is equivalent to that for $(R_{12}(-\theta_0),\sigma_2\cdot s)$; as s ranges over all sources, so does $\sigma_2\cdot s$. For the other planes, $R_{\mu\nu}(\pm\theta_0)=\Pi R_{12}(\pm\theta_0)\Pi^{-1}$ for a permutation matrix Π mapping e_1,e_2 to e_μ,e_ν; apply Lemma 2.10 with $O=\Pi$, which preserves Λ_k and D_J and permutes frames. (d) A convergent sequence with a finite limit is bounded. □

4.2. **Reduction to one-sided bounds at one lattice spacing.** King's proof of his Theorem 2.4 compares the lattice of spacing ε with the rotated lattice of spacing 5ε, and obtains the reverse inequality by "switching around the argument" [K2, p. 337]. We compare the two lattices at the *same* spacing, and obtain the reverse inequality from an exact symmetry.

Lemma 4.3 (the involution)**.** *Let $R=R_{\mu\nu}(\pm\theta_0)\in\mathcal{R}_0$, let $\sigma:=\sigma_\nu$ be the reflection $x_\nu\mapsto-x_\nu$, and put $S:=R^{-1}\sigma$. Then $S^2=1$, $S\Lambda_k=R^{-1}\Lambda_k$ and $S(R^{-1}\Lambda_k)=\Lambda_k$, and S maps $D_J=\mathbb{R}^3/P_J$, $P_J=(50MJ\mathbb{Z})^3$, onto the torus $R^{-1}D_J:=\mathbb{R}^3/R^{-1}P_J$. Both lattices descend to both tori, and for all real smooth D_J-periodic sources $s=(G,h)$,*

$$\mathcal{Z}_{R^{-1}\Lambda_k,D_J}(s)=\mathcal{Z}_{\Lambda_k,R^{-1}D_J}(S\cdot s),\qquad \mathcal{Z}_{\Lambda_k,D_J}(s)=\mathcal{Z}_{R^{-1}\Lambda_k,R^{-1}D_J}(S\cdot s),$$
$$\mathsf{S}_{R^{-1}\Lambda_k,D_J}(h)=\mathsf{S}_{\Lambda_k,R^{-1}D_J}(S\cdot h).$$

Proof. By Lemma 2.2(a), $\sigma R\sigma = R^{-1}$, so $S^2 = R^{-1}(\sigma R^{-1}\sigma) = R^{-1}R = 1$; $\sigma\Lambda_k = \Lambda_k$ gives $S\Lambda_k = R^{-1}\Lambda_k$, and then $S(R^{-1}\Lambda_k) = S^2\Lambda_k = \Lambda_k$. Also $\sigma P_J = P_J$, so $SP_J = R^{-1}P_J$. The lattice $R^{-1}\Lambda_k$ is invariant under $R^{-1}P_J$ trivially, and Λ_k is invariant under $R^{-1}P_J$ because $R^{-1}p \in 10MJ\mathbb{Z}^3$ for $p \in P_J$ (Lemma 2.2(b)). The map S carries the frame $(R^{-1}e_\alpha)$ to $(SR^{-1}e_\alpha) = (\sigma e_\alpha)$ and the frame (e_α) to $(R^{-1}\sigma e_\alpha)$; these differ from the frames of Λ_k and $R^{-1}\Lambda_k$ only by signs, which do not change the periodic models (Lemma 2.10). Lemma 2.10 with $O = S$, applied to the unnormalised integrals (the bijection of configurations preserves $d\nu$), gives the identities. □

Hypothesis 4.4 ((F), (V) and (C): one-sided bounds at one spacing)**.** Let $R \in \mathcal{R}_0$ and let $J \in \mathbb{N}$. Write $T_1 := D_J$, $T_2 := R^{-1}D_J$. **(F)**: for $i = 1, 2$ there are real numbers $v_k(T_i)$ and $c_k(T_i)$, $k \geq k_0(J)$, independent of the fields and the sources, with $|c_k(T_i)| \leq C\varepsilon_k^{-q}$ for some $q < \infty$ and $v_k(T_i)$ bounded, such that for every real smooth T_i-periodic source $s = (G, h)$,

$$\mathcal{Z}_{R^{-1}\Lambda_k,T_i}(s) \leq \big(1 + \delta_k(T_i, s)\big) \exp\big(- v_k(T_i) - c_k(T_i)\, \mathsf{S}_{\Lambda_k,T_i}(h)\big)\, \mathcal{Z}_{\Lambda_k,T_i}(s), \qquad \lim_{k\to\infty} \delta_k(T_i, s) = 0.$$

(V): $v_k(T_1) + v_k(T_2) \to 0$. **(C)**: $c_k(T_1) - c_k(T_2) \to 0$. **($\mathbf{F}_0$)** is (F) restricted to sources with $h = 0$ and to the sources $(G, \pm h_0)$, where $h_0 = -w$ is a wall; it involves no Wick scheme.

On each torus, (F) bounds the functional of the rotated lattice from above by that of the axis lattice. The rotated lattice is the one whose flow is King's standard block-spin flow in its own frame, and the axis lattice is the one run through King's hybrid flow; this is the direction of [K2, (4.7)–(4.9)]. Since smooth periodic functions have rapidly convergent Riemann sums, $|\mathsf{S}_{\Lambda,T}(h) - \int_\Gamma h| \leq C_N(h)\varepsilon_k^N$ for every N and every lattice Λ of spacing ε_k descending to T, Γ a fundamental cell; with $N > q$ the error $c_k(\mathsf{S}_{\Lambda,T}(h) - \int h)$ tends to 0.

Theorem 4.5 (closing at one spacing)**.** *Let $R \in \mathcal{R}_0$, $J \in \mathbb{N}$, and assume* (F)*. Write $v = v_k(T_1)$, $v' = v_k(T_2)$, $c = c_k(T_1)$, $c' = c_k(T_2)$, and $\rho_k(s) := \mathcal{Z}_{R^{-1}\Lambda_k,D_J}(s)/\mathcal{Z}_{\Lambda_k,D_J}(s)$.*

(a) *For every real smooth D_J-periodic source $s = (G, h)$, $e^{v'+c'\int h}(1 + o(1)) \leq \rho_k(s) \leq e^{-v-c\int h}(1 + o(1))$ as $k \to \infty$.*

(b) *$v + v' \leq o(1)$. If* (V) *holds, then $c + c' \to 0$.*

(c) *If* (V) *holds, then $\rho_k(s) = e^{-v-c\int h}(1 + o(1))$ for every s. Consequently, for sources s_0, s_1 with $\int h_0 = \int h_1$,*

$$\frac{\mathcal{Z}_{R^{-1}\Lambda_k,D_J}(s_1)}{\mathcal{Z}_{R^{-1}\Lambda_k,D_J}(s_0)} = \frac{\mathcal{Z}_{\Lambda_k,D_J}(s_1)}{\mathcal{Z}_{\Lambda_k,D_J}(s_0)}\,(1 + o(1)).$$

(d) *If* (V) *and* (C) *hold, then $c, c' \to 0$, and the identity in (c) holds for all s_0, s_1.*

No convergence of either family and no Wick scheme is assumed.

Proof. (a) The upper bound is (F) on T_1, with $c\,\mathsf{S}_{\Lambda_k,D_J}(h) = c\int h + o(1)$. Apply (F) on T_2 to the source $S \cdot s$ and use Lemma 4.3: $\mathcal{Z}_{R^{-1}\Lambda_k,T_2}(S \cdot s) = \mathcal{Z}_{\Lambda_k,D_J}(s)$, $\mathcal{Z}_{\Lambda_k,T_2}(S \cdot s) = \mathcal{Z}_{R^{-1}\Lambda_k,D_J}(s)$ and $\mathsf{S}_{\Lambda_k,T_2}(S \cdot h) = \mathsf{S}_{R^{-1}\Lambda_k,D_J}(h) = \int h + O(\varepsilon_k^N)$. This is the lower bound. (b) The interval in (a) is non-empty, so $(v + v') + (c + c')\int h \leq o(1)$ for every s. Apply this to (G, h) and to $(G, -h)$ and add: $v + v' \leq o(1)$. The same two inequalities give $|c + c'|\,|\int h| \leq o(1) - (v + v')$; under (V) the right side tends to 0, and choosing h with $\int h \neq 0$ gives $c + c' \to 0$. (c) Under (V) and (b), the ratio of the two ends of the interval in (a) tends to 1. Dividing the statement for s_1 by the one for s_0, the prefactors cancel when $\int h_1 = \int h_0$. (d) Now $c + c' \to 0$ and $c - c' \to 0$. □

The proof uses (F) only for the sources in question and their images under $h \mapsto -h$. In particular, under (F_0) and (V) the conclusions (a)–(c) hold for the sources $(G, \pm h_0)$ (where $\int h_0 \neq 0$ unless $w = 0$) and, with no condition on c, c', for the sources with $h = 0$.

Corollary 4.6 (walled curvature ratios)**.** *Assume* (F_0) *and* (V) *for R and J'. Then for every walled configuration $\mathfrak{c} = (J', w)$ and every real smooth periodic 2-form G, $\langle e^{zF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ –*

$\langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda_k} \to 0$ *locally uniformly in* $z \in \mathbb{C}$. *If moreover* $\lim_k \langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda_k}$ *exists for real* t, *then* (K_0) *holds for* $\mathfrak{c}$ *and* G.

Proof. With $h_0 = -w$, $\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda} = \mathcal{Z}_\Lambda(tG, h_0)/\mathcal{Z}_\Lambda(0, h_0)$ (the Wick constants cancel), a balanced ratio. By Theorem 4.5(c), $u_k(t) := \langle e^{tF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}/\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda_k} \to 1$ for real t. By Lemma 2.13(a) and the symmetry $(A,\vartheta) \mapsto (-A,-\vartheta)$, $1 \le \langle e^{tF(G)}\rangle \le e^{t^2\sigma_\Lambda/2}$ for real t, where $\sigma_\Lambda = \langle J, CJ\rangle$ is the variance of $F(G)$ under the free periodic photon on the lattice; $\sigma_\Lambda \le \mu_0^{-2}\|\delta_\varepsilon G\|^2$ is bounded uniformly in k, δ_ε the lattice codifferential. Hence the difference of the two functions tends to 0 on $\mathbb{R}$, and it is bounded on every strip $|\operatorname{Re} z| \le r$ uniformly in k, because $|\langle e^{zF}\rangle| \le \langle e^{\operatorname{Re} z\, F}\rangle$. Vitali's theorem gives locally uniform convergence to 0 on $\mathbb{C}$. The last statement follows by the same argument. □

Corollary 4.7 (the full comparison)**.** *Let* a *be a scheme with* $|a_k| \le C(1+k)$. *Assume* (F), (V) *and* (C) *for* $R = R_{12}(\theta_0)$ *and every* J, *and that for every King torus and every finite family of real smooth periodic sources the limit* $\lim_k Z^a_{\Lambda_k,D_J}$ *exists in* $(0,\infty)$ *for real coefficients. Then* $(\mathrm{K})_a$ *holds.*

Proof. $Z^a_{\Lambda,D_J}(s) = e^{-a_k \mathsf{S}_\Lambda(h)}\mathcal{Z}_\Lambda(s)/\mathcal{Z}_\Lambda(0)$, and the multipliers of the two lattices differ by $e^{-a_k(\mathsf{S}_{R^{-1}\Lambda_k}(h) - \mathsf{S}_{\Lambda_k}(h))} = 1 + O(k\varepsilon_k^N)$. Theorem 4.5(d) with $s_0 = 0$ gives $Z^a_{R^{-1}\Lambda_k,D_J}(s)/Z^a_{\Lambda_k,D_J}(s) \to 1$ for every real source, which with the assumed convergence gives $(\mathrm{K})_a$ for $R_{12}(\theta_0)$; Lemma 4.2(c) gives the other generators. □

Remark 4.8 (what this subsection supplies). Lemma 4.3, Theorem 4.5 and its corollaries are a reduction of the torus comparison to the abstract one-sided bounds (F), (V), (C): they are [PROVED], and they supply none of the bounds. In the curvature sector the bounds (F_0) and (V) that Corollary 4.6 consumes come from the closing of [CT] itself (Section 4.3), in which King's torus bound [K2, (4.2)] is not used, because at a non-zero wall it is vacuous; its place is taken by the large-field clause of King's upper bound at blocking factor 5, item (1) of Section 1.4, and at a mass-increasing wall the polynomial lower bound (H5) is the wall instance proved in [CT]. Corollary 4.7 concerns scalar sources and is not used by Theorem A.

4.3. **The corrected torus theorem.** The bounds (F_0), (V) and the existence of the axis-lattice limits for walled curvature ratios are the content of the corrected torus theorem of [CT], which we now recall in the form used. Write $X_K := \Lambda_K$ and $Y_K := R\Lambda_K$, $R = R_{12}(\theta_0)$, so that (K_0) for R^{-1} is the statement for the pair (X_K, Y_K). Configurations are taken in the class of item (6) of Section 1.4: $J \in 5^{k_*}\mathbb{N}$ with $J \ge \max(J_0(\varepsilon_0), J_{\mathrm{pos}})$, $\mathfrak{c} = (J,\psi,n)$ with $n > 0$, ψ vanishing on the 2-neighbourhood of $R^{-1}C_L$, the wall condition $n\langle\psi\rangle_{X_K} > \Delta_*|T|/2$, block alignment and the grid offset of the corners of H'.

Theorem 4.9 (the corrected torus theorem, curvature sector; [CT])**.** *Let* $e^2, \lambda, m^2, \mu_0^2 > 0$ *be in King's range, the block factor* 5, *and* $\mathfrak{c} = (J,\psi,n)$ *a walled configuration of the class (6) with wall source* $h_0 = -n\psi$. *Assume the hypotheses (1)–(3) and (5) of Section 1.4; they supply the hypotheses* (H1)–(H5) *of* [CT] *for the sources* (tG, h_0), $t \in \mathbb{R}$, *on* D_J, *for their* S^**-images on* RD_J, *and for their images under the coordinate reflections and permutations on* D_J *together with the* S^**-images of those on* RD_J. *Then for every real smooth* D_J*-periodic* 2*-form* G:

(i) $\langle e^{zF(G)}\rangle_{\mathfrak{c},Y_K} - \langle e^{zF(G)}\rangle_{\mathfrak{c},X_K} \to 0$ *as* $K \to \infty$, *locally uniformly in* $z \in \mathbb{C}$; *in particular* $\langle F(G)^n\rangle_{\mathfrak{c},Y_K} - \langle F(G)^n\rangle_{\mathfrak{c},X_K} \to 0$ *for every* n;

(ii) *if in addition (4) holds (existence of* $\lim_K \langle e^{tF(G)}\rangle_{\mathfrak{c},X_K}$ *for real* t*), both families converge, locally uniformly on* $\mathbb{C}$, *to the same entire function* $W_{\mathfrak{c}}(zG)$, *and all moments converge to the same limits;*

(iii) *the same holds with* R *replaced by any* $R' \in \mathcal{R}_0$.

At $e = 0$ *the statement holds with no hypothesis.*

The chain of [CT] is: exact passive-rotation reductions and the involution of Lemma 4.3; an equal-spacing comparison of the effective actions produced by the two block-spin flows (the standard flow of Y_K and King's hybrid flow of X_K, whose first step averages over rotated blocks) on King's small-field domain, in total form, that is with every field-independent constant included; a one-sided forward bound of the form (F) on each torus, obtained by inserting the comparison into King's upper bound [K1, Thm. 3.1] for Y_K and the hybrid lower bound [K2, Prop. 4.1] for X_K, with the additive large-field term, in the form of the large-field clause of item (1), absorbed on the left by the polynomial lower bound (H5), which at a mass-increasing wall is the wall instance proved in [CT]; and the closing by Theorem 4.5, in which the vacuum constants are absent because the comparison is in total form, so that (V) is automatic. King's torus bound [K2, (4.2)] is not used anywhere in this chain. The inputs of [CT], in the names used there, and the items of Section 1.4 that supply them, are:

(H1) Bałaban's renormalisation bound [Ba3, Prop. 1] with one element weakened by a small degree, for the renormalised families of [K1, §3.5] with lines taken from the two flows: in the form consumed, the statement (H1*) of item (2) for some $\theta_1 > 0$, [PROVED] as an implication in [CT] from (2a) ([PROVED]) and (2b) ([PROVED] as an implication); the one-flow instance is what King asserts in the proof of [K1, Prop. 3.6], p. 665.

(H2) Decay and regularity of the block-spin operators, [K1, Thm. 3.3]: for the standard flow [Ba4], item (5a) ([PROVED-LIT], its three asserted steps [PROVED] as implications under (1) and (5k)); for the hybrid flow, [PROVED] from the hybrid-kernel bounds of item (3).

(H3) King's upper bound [K1, Thm. 3.1, (3.4)] for Y_K, with the additive large-field term in the form of the large-field clause of item (1): [PROVED] as an implication on RD_J from items (1) and (5k), given (CT) (the proof of [K1, Thm. 3.1] refers to [Ba1, Ba2, Ba3, Ba4]); on D_J, which is skew in Y's frame, the instance [SKEW-U], [PROVED] as an implication relative to the same bounds on aligned tori at blocking factor 5 (item (1)), the replacement of the propagators of [Ba2, Prop. 2.6] on the skew torus with the change of domain (P7) of Section 1.5; for the contours of the flow of Y_K, rooted at the block centres, with Proposition 1.3, and at the wall with Proposition 1.4, for $J \geq J_{\mathrm{pos}}$, on D_J under the final-scale cap of the class (6), through the large-field clause of (1) (Section 1.6).

(H4) The hybrid lower bound, the analogue of [K2, Prop. 4.1, (4.5)]: [PROVED] as an implication from $(\mathrm{BAL})_5$, (CT), (B_Δ) and, on D_J, [SKEW-L] (item (5i)), its own steps (5l)(i)–(iii) included, with Proposition 1.3 for the composite contours of the hybrid flow.

(H5) A polynomial lower bound $\mathcal{Z}_{Y_K}(T;s) \geq e^{-C_1|T|}\varepsilon_K^{q(s)}$: at $h = 0$ Bałaban's lower bound [Ba1, (1.14)] (item (5b), printed as a frame with asserted steps and [PROVED] as an implication under (1)) on RD_J, and on D_J [PROVED] as an implication relative to it given $(\mathrm{BAL})_5$ and (CT); at the mass-increasing wall source $h_0 = -n\psi$, the wall instance proved in [CT] as an implication from the source-free bounds at two other fixed masses and exact identities, for $J \geq J_{\mathrm{pos}}$ outside E_W (Proposition 1.4): [PROVED] as an implication.

(H6) Existence of $\lim_K \langle e^{tF(G)} \rangle_{\mathfrak{c},X_K}$: item (4), King's Theorem 2.1 of [K1] at a wall, read on the normalised ratio, in which the h-linear c-numbers cancel: [PROVED] as an implication in [H6], with the residues (4a)–(4d) [PROVED] as implications, outside the set E_W of Proposition 1.4 and for $J \geq J_{\mathrm{pos}}$, through the large-field clause of (1) at the wall; King's two statements that it uses are [PROVED] as implications (item (4), Section 9.3).

(SK) The skew-torus instances [SKEW-U], [SKEW-L] of King's and Bałaban's bounds, for the rotated lattice on D_J, which is not aligned with its frame: [PROVED] as implications relative to the aligned bounds at blocking factor 5 (item (1)), given the skew-torus form of the cumulant bound (5l)(i), and, for [SKEW-U], the change of domain (P7) of Section 1.5.

$(\mathrm{BAL})_5$ Item (1): printed grade [SKETCH-LIT]; [PROVED] as an implication at block factor 5 [AS], checked independently.

(CT) The bridge between the torus counterterm conventions of [K1] and the free-boundary conventions of [K2, p. 330] (Remark 2.6): (a) exact algebra [PROVED]; (b) a bounded factor for the extensive bounds, in the form consumed (a single bound Δ_* on $|\Delta_T(s)|$ for $s \le s_0$, which is also the constant of the wall condition in (6)), [PROVED] as an implication, with the uniformity (5f) of King's constant under a constant source shift ([PROVED] as an implication, King's statement on the inclusion of sources in its consumed form included); its rate clause $|\Delta_T(s) - \Delta_T(0)| \le Cs$ (item (4c)) enters only through (4); (c) the existence transfer is not an input, (H6) being read on the normalised ratio. On the shell side the same bound Δ_* is the only input of $(\mathrm{CT})_{\mathrm{sh}}$ (item (8f)).

Theorem 4.9 is [PROVED] as an implication from the items (1)–(5). Its closing implication, from (H1)–(H6), (SK), $(\mathrm{BAL})_5$ and (CT) to (i)–(iii), is [PROVED] and checked independently, and so are its own steps (5l), with (4a) and (F3), its other residues, the items (2b), (4b), (4c), (5i), (5j), which [CT, H6] grade [DERIVED], and the completions of its flagged steps (F1), (F2); Bałaban's and King's asserted steps (1) and (5k), with those of (5a), (5b), (5f), (5g) under them, are [PROVED] as implications at block factor 5 [AS], except two statements of King, which are cited at their printed grade [SKETCH-LIT] in their asserted form and [PROVED] in the forms consumed (items (4), (5k); Section 9.3); the reading (B4) of [Ba2, p. 591] behind the constants (5l)(vii) is [PROVED] as an implication and checked independently [SUP] (Section 9.2); on the skew tori the replacement of the propagators of [Ba2, Prop. 2.6] in (1) is proved through the change of domain (P7) of Section 1.5; the repaired cube step in (1) uses an end-functional bound of the re-runs (8e) = (8i) at zero shell mass, which rests on the readings (B3), (B5), (B6) and on the regrouping of the cut families, all [PROVED] as implications (Section 9.2); and the forms of (1) for the contour families of its flows and at the walls are proved with Propositions 1.3 and 1.4 of Section 1.6, outside the set E_W there, under the final-scale cap of the class (6), Proposition 1.3 also through (H4) and the comparison of the effective actions. None of its inputs has been found false in the form consumed. The sector in which the sources are read through a single scalar function on all plaquettes is false (Proposition A.2) and is not what the theorem states. The scalar (Wick-source) sector of [CT], which needs the convergence of the Wick-source functionals in the scheme $\Theta^\sharp$ on the torus, is outside the theorem of this paper.

Remark 4.10 (status of (K)). Hypothesis 4.1 is [K2, Thm. 2.4] with the corrections of Appendix A, for arbitrary smooth periodic sources (walls included) and with free-boundary counterterms. By Corollaries 4.6–4.7 its two-lattice content is (F) and (V), plus (C) for scalar sources, and its one-lattice content is the existence of the torus limits on the axis lattice. By Theorem 4.9, (K_0) holds for every $R \in \mathcal{R}_0$ and every configuration of the class (6), given the items (1)–(5) of Section 1.4, which are proved as implications, (1) on the skew tori with the change of domain (P7), with Propositions 1.3 and 1.4 of Section 1.6, outside the set E_W there; they reach the readings (B3), (B5), (B6), [PROVED] as implications, only through the end-functional bound of the re-runs (8e) = (8i) that the repaired cube step of (1) uses; they use no statement of the literature at the printed grade [SKETCH-LIT]: the reading (B4) of Section 9.2 is [PROVED] as an implication and checked independently [SUP]. (K_0) is not reduced to any second-order statement, and no part of Section 4 is used to supply it. The rotation enters the whole argument of this paper only here.

Remark 4.11 (what the torus comparison alone does not give). Given (K_0), the walled values of a configuration $\mathfrak{c}$ on D_J agree on the two lattices, and by passive covariance the walled value of $R \cdot \mathfrak{c}$ on the rotated torus $R\,D_J$ for the axis lattice equals the walled value of $\mathfrak{c}$ on D_J for the rotated lattice. Rotation invariance of a limit along tori would therefore follow from (K_0) if the limit did not depend on the shape and orientation of the tori, which is the content of King's expected identity [K2, (2.24)] between torus and cube limits. That identity is a uniqueness statement for

the infinite-volume limit, and we do not use it. The walls of Section 5 replace it by a one-sided statement about one lattice, (TIGHT), whose local form (RES) is the hypothesis of Theorem 7.10.

5. Walled tori and the closing hypothesis

Definition 5.1 (admissible walls and walled values)**.** For a bounded set $X \subset \mathbb{R}^3$ let $\mathcal{A}(X)$ be the set of walled configurations $\mathfrak{c} = (J', w)$ with $N_1(X) \subset (-25MJ', 25MJ')^3$ and $w = 0$ on $N_1(X)$; the cell $(-25MJ', 25MJ')^3$ is a fundamental domain of $D_{J'}$ for both lattices, $R^{-1}\Lambda_k$ descending to $D_{J'}$ by Lemma 2.2(b), and sources supported in X are periodised from it. If $X \subset Y$ then $\mathcal{A}(Y) \subset \mathcal{A}(X)$. Assume $(\mathrm{K})_a$. For $(f, g) \in \mathcal{K}$ supported in X and $\mathfrak{c} \in \mathcal{A}(X)$, the sources restricted to the cell and periodised are smooth, and by Lemma 4.2(a) the *walled value*

$$\mathcal{W}^a_{\mathfrak{c}}(f, g) := \lim_{k\to\infty} T^a_{\mathfrak{c},\Lambda_k}(f, g) = \lim_{k\to\infty} T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f, g) \qquad (R \in \mathcal{R}_0)$$

exists. For $b > 0$ and $(f, g) \in \mathcal{K}$ supported in Q_b let $I^a_b(f, g) := \inf_{\mathfrak{c}\in\mathcal{A}(Q_b)} \mathcal{W}^a_{\mathfrak{c}}(f, g)$, which is nondecreasing in b, and

$$\mathcal{W}^{a,\mathrm{env}}(f, g) := \lim_{b\to\infty} I^a_b(f, g).$$

At $g = 0$ these definitions need only (K_0), and they do not depend on the scheme; we then write $\mathcal{W}_{\mathfrak{c}}(f, 0)$ and $\mathcal{W}^{\mathrm{env}}(f, 0)$.

Lemma 5.2 (the free direction)**.** *Assume $(W)_a$ and $(K)_a$.*

(a) *If $(f, g) \in \mathcal{K}$ is supported at distance > 1 from $\mathbb{R}^3 \setminus C_K$ and $\mathfrak{c} \in \mathcal{A}(\bar{C}_K)$, then $\mathcal{W}^a_{\mathfrak{c}}(f, g) \geq Z^a_K(\mathrm{i}f, g)$.*

(b) *$\mathcal{W}^{a,\mathrm{env}} \geq Z^a_\infty$ on $\mathcal{K}$; in particular $\mathcal{W}^{a,\mathrm{env}} \in (0, \infty]$.*

Proof. (a) At each k, Lemma 2.14 with $X = C_K \cap \Lambda_k$ and Lemma 3.3(a) give $T^a_{\mathfrak{c},\Lambda_k}(f, g) \geq Z^a_{\Lambda_k,C_K}(\mathrm{i}f, g)$; let $k \to \infty$. (b) If $\bar{C}_K \subset Q_b$ then $\mathcal{A}(Q_b) \subset \mathcal{A}(\bar{C}_K)$, so $I^a_b \geq Z^a_K$; let $b \to \infty$ and then $K \to \infty$ (Theorem 3.8(iii)). □

Remark 5.3. Lemma 5.2(a) compares a walled value with a cube of fixed size. It does not assert $\mathcal{W}^a_{\mathfrak{c}} \geq Z^a_\infty$ for an individual configuration, and this need not hold: for instance, at $e = \lambda = 0$, $f = 0$ and $g \geq 0$ small, as $n \to \infty$ the walled value of (J', ψ, n) decreases (Lemma 2.12(c)) towards the value of the Dirichlet state on the set where $\psi = 0$, which is smaller than the infinite-volume value, as one checks with the Gaussian formula for $\langle e^{:\phi^2:(g)}\rangle$.

Lemma 5.4 (independence of the orientation)**.** *Assume $(K)_a$. For $R \in \mathcal{R}_0$ and $(f, g) \in \mathcal{K}$, $\lim_{L\to\infty} \inf_{\mathfrak{c}\in\mathcal{A}(R^{-1}Q_{ML})} \mathcal{W}^a_{\mathfrak{c}}(f, g) = \mathcal{W}^{a,\mathrm{env}}(f, g)$, the infimum being defined as soon as $\operatorname{supp} f \cup \operatorname{supp} g \subset Q_{5ML/7}$.*

Proof. By Lemma 2.2(c), $Q_{5ML/7} \subset R^{-1}Q_{ML} \subset Q_{7ML/5}$, hence $\mathcal{A}(Q_{7ML/5}) \subset \mathcal{A}(R^{-1}Q_{ML}) \subset \mathcal{A}(Q_{5ML/7})$, and $I^a_{5ML/7} \leq \inf_{\mathcal{A}(R^{-1}Q_{ML})} \mathcal{W}^a_{\mathfrak{c}} \leq I^a_{7ML/5}$. Both bounds tend to $\mathcal{W}^{a,\mathrm{env}}$. □

Hypothesis 5.5 ($(\mathrm{TIGHT})_a$, (TIGHT_0), (TIGHT_2): the closing hypothesis)**.** **$(\mathbf{TIGHT})_a$**: $\mathcal{W}^{a,\mathrm{env}} \leq Z^a_\infty$ on $\mathcal{K}$. Equivalently: for every $b > 0$, every $(f, g) \in \mathcal{K}$ supported in Q_b and every $\eta > 0$ there is $\mathfrak{c} \in \mathcal{A}(Q_b)$ with $\mathcal{W}^a_{\mathfrak{c}}(f, g) \leq Z^a_\infty(\mathrm{i}f, g) + \eta$. **$(\mathbf{TIGHT}_0)$**: (TIGHT) for $g = 0$, that is, $\mathcal{W}^{\mathrm{env}}(f, 0) \leq Z_\infty(\mathrm{i}f, 0)$ for every real 2-form f; it involves only (K_0) and King's printed state. Where (K_0) is supplied on a sub-class of the walled configurations only, the class (6) of Section 1.4 (Remark 5.6), (TIGHT_0) is read with the infimum over that sub-class, whose conditions other than the vanishing of ψ near $R^{-1}C_L$ do not depend on R (J_{pos} being the maximum over the finite set $\mathcal{R}_0$ and the three torus pairs, Proposition 1.4): for every $b > 0$, every real 2-form f supported in Q_b and every $\eta > 0$ there is a configuration $\mathfrak{c}$ of the class (6) in $\mathcal{A}(Q_b)$ with $\mathcal{W}_{\mathfrak{c}}(f, 0) \leq Z_\infty(\mathrm{i}f, 0) + \eta$; for $b \geq \frac{7}{5}ML + 2$ such a configuration has $\psi = 0$ on the 2-neighbourhood of $R^{-1}C_L$ for every $R \in \mathcal{R}_0$ (Lemma 2.2(c)), so it lies in the class (6) for R and L. **$(\mathbf{TIGHT}_2)$**:

for every real 2-form $f \in C_c^\infty$, $\lim_{b\to\infty} \sup_{\mathfrak{c}\in\mathcal{A}(Q_b)} \sigma_{\mathfrak{c}}^2(f) \geq S_2(f)$, where $\sigma_{\mathfrak{c}}^2(f) := \lim_k \langle F_{\Lambda_k}(f)^2\rangle_{\mathfrak{c},\Lambda_k}$ and $S_2(f) := \lim_J \lim_k \langle F_{\Lambda_k}(f)^2\rangle_{k,J}^K$, on King's cubes (Definition 2.15). Where (K_0) is supplied on the class (6) only, the same reading applies to (TIGHT_2): the supremum is over the configurations of the class (6) in $\mathcal{A}(Q_b)$.

The equivalence in $(\mathrm{TIGHT})_a$ holds because I_b^a is nondecreasing in b. By Lemma 5.2, $(\mathrm{TIGHT})_a$ is equivalent to $\mathcal{W}^{a,\mathrm{env}} = Z_\infty^a$ on $\mathcal{K}$: *King's state is the limit of the lower envelopes of walled tori, the continuum limit being taken first.* The limits defining $\sigma_{\mathfrak{c}}^2$ and S_2 exist under (K_0), respectively by Proposition 3.11, since second derivatives of locally uniformly convergent entire functions converge.

Remark 5.6 (the configuration class). (K_0) is supplied by Theorem 4.9 for configurations of the class (6) of Section 1.4, which is smaller than $\mathcal{A}(X)$: it asks $J \in 5^{k_*}\mathbb{N}$, $J \geq \max(J_0(\varepsilon_0), J_{\mathrm{pos}})$, $n > 0$, the wall condition $n\langle\psi\rangle_{X_K} > \Delta_*|T|/2$, block alignment and the grid offset of the corners of H'; the threshold J_{pos} of Proposition 1.4 belongs to the class because the margin of the positivity condition is proved for $J \geq J_{\mathrm{pos}}$ only. This costs nothing: in the proof of Theorem 6.1 below the infimum over $\mathcal{A}(R^{-1}Q_{ML})$ is used only through an upper bound by the walled value of one configuration supplied by (TIGHT_0), so the infimum may be taken over any sub-class that contains that configuration, and (TIGHT_0) is then read with the same sub-class. The configuration that Theorem 7.10 produces lies in it: its wall is $(n+K)\psi$ with $n + K \geq n_* > 0$, mass-increasing; the wall condition is linear in the height and holds for K large at fixed T, which Remark 7.11(iii) allows; J (the J' of Theorem 7.12), the block alignment of H', T and the grid offset of the corners of H' are choices of the torus and of the box that Theorem 7.12 makes ($J' \geq \max(J_0, J_{\mathrm{pos}})$ among them), the last moving H' by less than $M'\ell_f$; and Lemma 5.4 uses only shrinking families, so it holds on the intersection. The large-field clause of (1) at the wall on the skew torus takes one step at the final spacing of Bałaban's fixed-mass bounds; the class carries the cap of Section 1.6 on the coarsest scale, $\varepsilon_0 \leq \varepsilon_{\mathrm{sys}}(M, R, \rho, \delta_0)$, which does not depend on J and is closed downward, and under it the non-wrapping threshold at that spacing equals 1 for every J of the residue class, so that no further restriction of J is needed.

Lemma 5.7. (a) *If a, a' are admissible and $(K)_a$ holds, then $(TIGHT)_a$ and $(TIGHT)_{a'}$ are equivalent.*

(b) *$(TIGHT)_a$ implies ($TIGHT_0$).*

(c) *Under (K_0), ($TIGHT_0$) implies ($TIGHT_2$), and $\lim_b \sup_{\mathcal{A}(Q_b)} \sigma_{\mathfrak{c}}^2(f) \leq S_2(f)$ always; so ($TIGHT_2$) is an equality.*

Proof. (a) If $a' - a \to \kappa$ (Lemma 3.5(b)), then $T_{\mathfrak{c}}^{a'} = e^{-(a'_k - a_k)\mathsf{S}(g)} T_{\mathfrak{c}}^a$, so $\mathcal{W}_{\mathfrak{c}}^{a'} = e^{-\kappa\int g}\mathcal{W}_{\mathfrak{c}}^a$ with a factor independent of $\mathfrak{c}$; the same factor relates $Z_\infty^{a'}$ and Z_∞^a (Theorem 3.8(vi)). (b) At $g = 0$ all quantities are scheme-free (Lemma 3.3(c)). (c) Fix f, b and $\eta' > 0$. All functions of t below are real and even. Since $\cos x \geq 1 - \frac{x^2}{2}$ and $\cos x \leq 1 - \frac{x^2}{2} + \frac{x^4}{24}$ at every lattice level, and derivatives converge in the limits,

$$\mathcal{W}_{\mathfrak{c}}(tf, 0) \geq 1 - \tfrac{t^2}{2}\sigma_{\mathfrak{c}}^2(f), \qquad Z_\infty(\mathrm{i}tf, 0) \leq 1 - \tfrac{t^2}{2}S_2(f) + \tfrac{t^4}{24}M_4(f),$$

with $M_4(f) := \partial_t^4 Z_\infty(tf, 0)|_{t=0} < \infty$. By ($\mathrm{TIGHT}_0$) applied to tf, for every $t > 0$ and $\eta > 0$ there is $\mathfrak{c} \in \mathcal{A}(Q_b)$ with $\mathcal{W}_{\mathfrak{c}}(tf, 0) \leq Z_\infty(\mathrm{i}tf, 0) + \eta$, hence $\sigma_{\mathfrak{c}}^2(f) \geq S_2(f) - \frac{t^2}{12}M_4(f) - \frac{2\eta}{t^2}$. Choosing t with $t^2 M_4(f)/12 < \eta'/2$ and then $\eta = t^2\eta'/4$ gives $\sup_{\mathcal{A}(Q_b)} \sigma_{\mathfrak{c}}^2(f) \geq S_2(f) - \eta'$. For the second claim, let $\mathfrak{c} \in \mathcal{A}(\bar{C}_L)$ with f supported well inside C_L. At every k, Lemma 2.16 and Lemma 2.14, as in the proof of Lemma 5.2(a), give $T_{\mathfrak{c},\Lambda_k}(tf, 0) \geq \langle e^{\mathrm{i}tF(f)}\rangle_{k,L}^K$; so $\mathcal{W}_{\mathfrak{c}}(tf, 0) \geq Z_L^K(\mathrm{i}tf, 0) := \lim_k \langle e^{\mathrm{i}tF(f)}\rangle_{k,L}^K$ for all real t, with equality at $t = 0$, and both sides are even and smooth, so $\sigma_{\mathfrak{c}}^2(f) \leq \langle F(f)^2\rangle_{C_L^K} := \lim_k \langle F(f)^2\rangle_{k,L}^K$. Hence $\sup_{\mathcal{A}(Q_b)} \sigma_{\mathfrak{c}}^2(f) \leq \langle F(f)^2\rangle_{C_L^K}$ whenever $\bar{C}_L \subset Q_b$. By Proposition 3.11 (Theorem 3.8(ii) at $g = 0$ on King's cubes), comparing second derivatives at $t = 0$, the cube variances are nonincreasing in L, and they converge to $S_2(f)$. □

Remark 5.8 (the role of (TIGHT)). Apart from (TIGHT), the proof of Theorem 6.1 below uses only: monotonicity of the cube functionals in J (on $\mathcal{K}$ and on the decreasing class), the comparison "cube $\leq$ walled torus" on $\mathcal{K}$ and "cube $\geq$ walled torus" on the decreasing class on each lattice, the equality of walled values on the two lattices, passive covariance, and the existence of the limits. All of these hold for the following family of Gaussian functionals, in which the analogue of King's state is not rotation invariant. Let $A = \mathrm{diag}(1,2,3)$, $\mu > 0$, and let C_I, C_R, C_U be the translation-invariant operators on $L^2(\mathbb{R}^3)$ with Fourier symbols $\hat{c}_I(k) = (\langle k, Ak\rangle + \mu^2)^{-1}$, $\hat{c}_R(k) = \hat{c}_I(Rk)$ and $\hat{c}_U = \min(\hat{c}_I, \hat{c}_R)$. Let the "cube of size J" on the lattice $X \in \{I, R\}$ have covariance $C_X + J^{-1}$, let every "walled torus" have covariance C_U on both lattices, and let a covariance C define the functionals $e^{-\frac{1}{2}\langle f, Cf\rangle}$ on the cone and $e^{\frac{1}{2}\langle f, Cf\rangle}$ on the decreasing class. Every relation listed reduces to $C_U \leq C_X + J^{-1}$ or to the monotonicity of $C_X + J^{-1}$ in J, and passive covariance holds since $\langle f, C_R f\rangle = \langle R\cdot f, C_I\, R\cdot f\rangle$. But $e^{-\frac{1}{2}\langle R\cdot f, C_I R\cdot f\rangle} \neq e^{-\frac{1}{2}\langle f, C_I f\rangle}$ in general, and (TIGHT) fails since $C_U \neq C_I$. Hence (TIGHT) is not a consequence of the other properties, and some input asserting "walled torus $\leq$ cube + small" is needed. (The same family shows that any placement of the two-lattice comparison on a finite box must also carry an input beyond the correlation inequalities: there the input is the two-lattice comparison itself, which fails for $C_R \neq C_I$.)

6. Rotation invariance

Theorem 6.1 (invariance under the generators)**.** *Let a be admissible, and assume $(K)_a$ and $(TIGHT)_a$. Then for every $R \in \mathcal{R}_0$ and every $(f,g) \in \mathcal{K}$,*

$$Z^a_\infty(\mathrm{i}\, R\cdot f, R\cdot g) = Z^a_\infty(\mathrm{i} f, g),$$

and consequently the same holds for every $R \in G_0$.

Proof. Fix $R \in \mathcal{R}_0$ and $(f,g) \in \mathcal{K}$ supported in Q_ρ.

Step 0 (geometry). Take L with $ML \geq \frac{7}{5}\rho + 2$. By Lemma 2.2(c), $R\cdot(f,g)$ is supported in $Q_{7\rho/5}$, at distance > 1 from $\mathbb{R}^3 \setminus C_L$, and (f,g) is supported in $Q_{5ML/7} \subset R^{-1}Q_{ML}$. Let $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$. Then $R^{-1}\Lambda_k$ descends to $D_{J'}$ (Lemma 2.2(b)), and $B'_k := R^{-1}(C_L \cap \Lambda_k)$ is a set of sites of $R^{-1}\Lambda_k$ which, with its nearest neighbours, lies in the open cell, and on which $w = 0$. Since $R\cdot(f,g)$ has Euclidean distance > 1 from $\mathbb{R}^3 \setminus C_L$, every plaquette and site of $R^{-1}\Lambda_k$ at which (f,g) does not vanish lies in B'_k.

Step 1 (cutting on the rotated lattice). By Lemma 2.14 and Lemma 3.3(a), at every k,

$$Z^a_{R^{-1}\Lambda_k, B'_k}(\mathrm{i} f, g) \leq T^a_{\mathfrak{c}, R^{-1}\Lambda_k}(f,g).$$

Step 2 (passive rotation). By Lemma 3.3(b) with $O = R$, $Z^a_{R^{-1}\Lambda_k, B'_k}(\mathrm{i} f, g) = Z^a_{\Lambda_k, C_L\cap\Lambda_k}(\mathrm{i}\, R\cdot f, R\cdot g)$.

Step 3 (continuum limits). As $k \to \infty$ the left side converges by (W1) (Theorem 3.8(i)) and the right side by $(K)_a$ (Definition 5.1):

$$Z^a_L(\mathrm{i}\, R\cdot f, R\cdot g) \leq \mathcal{W}^a_{\mathfrak{c}}(f,g) \qquad \text{for every } \mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML}).$$

Step 4 (infimum, volume limit, closing hypothesis). Take the infimum over $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$ and let $L \to \infty$. By Theorem 3.8(iii), Lemma 5.4 and $(TIGHT)_a$,

$$Z^a_\infty(\mathrm{i}\, R\cdot f, R\cdot g) = \lim_{L\to\infty} Z^a_L(\mathrm{i}\, R\cdot f, R\cdot g) \leq \mathcal{W}^{a,\mathrm{env}}(f,g) \leq Z^a_\infty(\mathrm{i} f, g).$$

Step 5 (the inverse). $R^{-1} \in \mathcal{R}_0$ and $R\cdot(f,g) \in \mathcal{K}$, so Steps 0–4 for $(R^{-1}, R\cdot(f,g))$ give $Z^a_\infty(\mathrm{i} f, g) \leq Z^a_\infty(\mathrm{i}\, R\cdot f, R\cdot g)$. Since $\mathcal{K}$ is G_0-invariant, the claim extends to G_0. □

The order of limits is: at fixed R, (f,g), L and $\mathfrak{c}$, first $k \to \infty$; then the infimum over $\mathfrak{c}$; then $L \to \infty$. The continuum limit is always taken first, at fixed volume and fixed wall, as in King's

construction [K2, p. 336], and (K) is used only as an equality of continuum limits, not at fixed lattice spacing.

Lemma 6.2 (reflection)**.** *Let a be admissible and $\sigma_3 = \mathrm{diag}(1,1,-1)$. Then $Z^a_\infty(\mathrm{i}\,\sigma_3 \cdot f, \sigma_3 \cdot g) = Z^a_\infty(\mathrm{i}f, g)$ for $(f,g) \in \mathcal{K}$. At $g = 0$ the same holds, with no scheme and without (W), for King's printed state: $Z_\infty(\mathrm{i}\,\sigma_3 \cdot f, 0) = Z_\infty(\mathrm{i}f, 0)$ for every real 2-form $f \in C^\infty_c$.*

Proof. σ_3 preserves Λ_k and S_{Λ_k}, and maps $C_J \cap \Lambda_k$ onto $X_+ := (C_J + \varepsilon_k e_3) \cap \Lambda_k$. By Lemma 2.10, $Z^a_{\Lambda_k, C_J}(\mathrm{i}\,\sigma_3 \cdot f, \sigma_3 \cdot g)$ equals the corresponding functional of the Neumann–Dirichlet state on X_+ defined with the frame $(e_1, e_2, -e_3)$. Written in the frame (e_α) (with $A_3'(x) = -A_3(x - \varepsilon e_3)$), this is the BFS measure in which ϕ, A_1, A_2 live on X_+ and A_3 on $X_+ - \varepsilon e_3 = C_J \cap \Lambda_k$, with Dirichlet conditions for ϕ on X_+ and photon gradient terms within these sets. It is obtained from the Neumann–Dirichlet state on $C_{J+1} \cap \Lambda_k$ by deleting photon gradient terms and cosine parts of hopping terms, since $X_+ \cup (X_+ - \varepsilon e_3) \subset C_{J+1}$; and the state on $C_{J-1} \cap \Lambda_k$ is obtained from it in the same way, since $C_{J-1} \subset C_J \cap (C_J + \varepsilon e_3)$. As in Lemma 2.14, for J large, $Z^a_{\Lambda_k, C_{J-1}}(\mathrm{i}f, g) \le Z^a_{\Lambda_k, C_J}(\mathrm{i}\,\sigma_3 \cdot f, \sigma_3 \cdot g) \le Z^a_{\Lambda_k, C_{J+1}}(\mathrm{i}f, g)$. Let $k \to \infty$ and then $J \to \infty$.

At $g = 0$, on King's cubes: the sandwich just proved holds at every k for the boxes, at J and at $J-1$, and Lemma 2.16, applied at its two ends and twice to the source $\sigma_3 \cdot f$ in the middle, gives for J large and every $k \ge 0$

$$\langle e^{\mathrm{i}F(f)}\rangle^K_{k,J-2} \le \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J-2} \le \langle e^{\mathrm{i}F(\sigma_3 \cdot f)}\rangle^{\mathrm{ND}}_{k,J-1} \le \langle e^{\mathrm{i}F(\sigma_3 \cdot f)}\rangle^K_{k,J} \le \langle e^{\mathrm{i}F(\sigma_3 \cdot f)}\rangle^{\mathrm{ND}}_{k,J} \le \langle e^{\mathrm{i}F(f)}\rangle^{\mathrm{ND}}_{k,J+1} \le \langle e^{\mathrm{i}F(f)}\rangle^K_{k,J+2};$$

let $k \to \infty$ by item (13) and then $J \to \infty$ (Proposition 3.11). The comparison of C^K_J with $C^K_{J\mp1}$ is also obtained by deletions and kills when $M \ge 2\varepsilon_k$, which holds at every k for the M of Definition 2.1 [G]; the two-step form used here needs no such condition. □

Theorem 6.3 (closure: from the generators to $O(3)$)**.** *Let a be admissible, and suppose $Z^a_\infty(\mathrm{i}\,R \cdot f, R \cdot g) = Z^a_\infty(\mathrm{i}f, g)$ for all $(f,g) \in \mathcal{K}$ and $R \in \mathcal{R}_0$. Then for every $R \in O(3)$, every N, all real 2-forms f_j and real h_j in C^∞_c, and all $(z,w) \in \mathbb{C}^{2N}$,*

$$Z^a_\infty\Big(\sum_j z_j\, R \cdot f_j, \sum_j w_j\, R \cdot h_j\Big) = Z^a_\infty\Big(\sum_j z_j f_j, \sum_j w_j h_j\Big).$$

The same holds at $g = 0$ for King's printed state $Z_\infty(\cdot, 0)$ with no scheme, under the hypothesis for curvature sources only.

Proof. Complex coefficients. The hypothesis passes to G_0, since $\mathcal{K}$ is invariant under $O(3)$ and $\mathcal{R}_0 = \mathcal{R}_0^{-1}$. For $R \in G_0$ and a finite family, split $h_j = h_j^+ - h_j^-$ as in the proof of Theorem 3.8(iii). The difference of the two sides, as a function of $(z, w^+, w^-) \in \mathbb{C}^{3N}$, is entire and vanishes on $\Omega_+ = (\mathrm{i}\mathbb{R})^N \times (0,\infty)^{2N}$, where all sources lie in $\mathcal{K}$; hence it vanishes identically.

Closure in $SO(3)$. At real coefficients, $\Phi^a(R \cdot s) = \Phi^a(s)$ for $s \in E_\rho$ and $R \in G_0$. The map $R \mapsto R \cdot s \in E_\rho$ is continuous on $O(3)$, and Φ^a is continuous on E_ρ by Theorem 3.8(iv). Hence $G := \{R \in O(3) : \Phi^a(R \cdot s) = \Phi^a(s) \text{ for all } \rho \text{ and } s \in E_\rho\}$ is a closed subgroup of $O(3)$ containing G_0, so $G \supset SO(3)$ by Lemma 2.2(d).

Reflection. Lemma 6.2 and the first step (with σ_3 in place of R) give $\sigma_3 \in G$, so $G = O(3) = SO(3) \cup \sigma_3 SO(3)$.

Conclusion. For $R \in O(3)$ both sides of the claim are entire in $(z,w) \in \mathbb{C}^{2N}$ and agree on $\mathbb{R}^{2N}$. At $g = 0$ the same runs with Proposition 3.11 in place of Theorem 3.8, $(\mathrm{i}\mathbb{R})^N$ as the set of uniqueness and Lemma 6.2 at $g = 0$. □

Theorem 6.4 (the curvature sector)**.** *Assume (K_0) on the class (6) of Section 1.4 and (TIGHT$_0$) read on that class, the sub-class of Remark 5.6 on which (K_0) is proved and whose conditions do not*

depend on R (Hypothesis 5.5). Then King's printed state on curvature sources is $O(3)$-invariant: for every $R \in O(3)$, all real 2-forms $f_j \in C_c^\infty$ and all $z \in \mathbb{C}^N$, $Z_\infty(\sum_j z_j\, R{\cdot}f_j, 0) = Z_\infty(\sum_j z_j f_j, 0)$, and all curvature Schwinger functions are $O(3)$-covariant. Moreover (item (14) of Section 1.4, proved from (13)), for every Euclidean motion $x \mapsto Rx+b$, $Z_\infty(\sum_j z_j\, \tau_b R{\cdot}f_j, 0) = Z_\infty(\sum_j z_j f_j, 0)$.

Proof. At $g = 0$ the walled values are those of (K_0) (Lemma 3.3(c)), and Proposition 3.11 replaces Theorem 3.8. Steps 0–5 of the proof of Theorem 6.1 then run with (K_0) and (TIGHT_0), the infimum being taken over the class of Remark 5.6 (in Step 0 the configuration that (TIGHT_0) supplies for a source supported in Q_b with $b \geq \frac{7}{5}ML + 2$ has $\psi = 0$ on the 2-neighbourhood of $R^{-1}C_L$, Lemma 2.2(c), so it is admissible there), and with King's cubes: in Step 1 the functional of C_L^K for the source $R \cdot f$ is bounded at every k by that of the box $C_L \cap \Lambda_k$ (Lemma 2.16), to which Steps 1 and 2 apply as written, so that $\langle e^{\mathrm{i}F(R\cdot f)}\rangle_{k,L}^K \leq T_{\mathfrak{c},R^{-1}\Lambda_k}(f,0)$; in Step 3 the limit of the left side is item (13); and in Step 4 the limit $L \to \infty$ is that of Proposition 3.11. This gives invariance under $\mathcal{R}_0$ on $(\mathrm{i}\mathbb{R})^N$; Theorem 6.3 applies at $g = 0$, with $(\mathrm{i}\mathbb{R})^N$ as the set of uniqueness and Lemma 6.2 at $g = 0$ for the reflection, and gives $O(3)$ and the entire extension. Translations are Theorem 3.8(v) at $h = 0$ (Proposition 3.11), proved there from (13), Lemma 2.16 and the cut sandwich; King's printed argument [K2, Thm. 2.3(i)] is not used. □

Corollary 6.5 (the curvature two-point function)**.** *Assume (K_0) and (TIGHT_2). Then $S_2(f_1, f_2) := \lim_J \lim_k \langle F_{\Lambda_k}(f_1) F_{\Lambda_k}(f_2)\rangle_{k,J}^K$ satisfies $S_2(R \cdot f_1, R \cdot f_2) = S_2(f_1, f_2)$ for all $R \in O(3)$ and real 2-forms $f_1, f_2 \in C_c^\infty$.*

Proof. Let $R \in \mathcal{R}_0$. Steps 0–3 of the proof of Theorem 6.1 at $g = 0$, with source tf and with King's cubes as in the proof of Theorem 6.4, give $Z_L^K(\mathrm{i}t\, R \cdot f, 0) \leq \mathcal{W}_\mathfrak{c}(tf, 0)$ for all real t and $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$. Both sides are even and smooth in t and equal 1 at $t = 0$, so $\langle F(R\cdot f)^2\rangle_{C_L^K} \geq \sigma_\mathfrak{c}^2(f)$. Since $\mathcal{A}(Q_{7ML/5}) \subset \mathcal{A}(R^{-1}Q_{ML})$ and $b \mapsto \sup_{\mathcal{A}(Q_b)} \sigma_\mathfrak{c}^2(f)$ is nonincreasing,

$$\langle F(R \cdot f)^2\rangle_{C_L^K} \geq \sup_{\mathcal{A}(Q_{7ML/5})} \sigma_\mathfrak{c}^2(f) \geq \lim_{b\to\infty} \sup_{\mathcal{A}(Q_b)} \sigma_\mathfrak{c}^2(f) \geq S_2(f)$$

by (TIGHT_2). The cube variances converge to S_2 (proof of Lemma 5.7(c)), so $S_2(R \cdot f) \geq S_2(f)$, and the inverse generator gives equality; hence S_2 is G_0-invariant. By Lemma 2.13, $\langle F(f)^2\rangle_{k,J}^K \leq \langle v, Cv\rangle_\varepsilon$ with $v = d_\varepsilon^* f$, where C is the photon covariance of the cube, the Neumann photon of $C_J \cap \Lambda_k$; the photon form is bounded below by μ_0^2, so $0 \leq S_2(f) \leq \mu_0^{-2}\|d^*f\|_{L^2}^2$. Since S_2 is a positive semidefinite quadratic form, $|S_2(f) - S_2(f')| \leq S_2(f-f')^{1/2} S_2(f+f')^{1/2}$, and $R \mapsto S_2(R\cdot f)$ is continuous on $O(3)$. Density of G_0 gives $SO(3)$, Lemma 6.2 at $g = 0$ (expanded to second order) gives σ_3, and polarisation gives the bilinear form. □

7. The heavy-shell hypothesis

Throughout this section $\Lambda = \Lambda_k$ is the axis lattice, $\varepsilon = \varepsilon_k \leq 1/\mu_0$, and $m_\varepsilon := \frac{2}{\varepsilon}\operatorname{arcsinh}(\varepsilon\mu_0/2)$; since $t \mapsto \operatorname{arcsinh}(t)/t$ is decreasing, $m_\varepsilon \geq 0.96\,\mu_0$. Let $G_{\mathbb{Z}^3}$ be the Green's function of simple random walk on $\mathbb{Z}^3$ and

$$\mathsf{a} := \sup_{z \in \mathbb{Z}^3 \setminus \{0\}} |z|_\infty\, G_{\mathbb{Z}^3}(0, z),$$

which is finite by the Green's function asymptotics of simple random walk in three dimensions [LL, Chap. 4]. For a set $X \subset \mathbb{R}^3$ put $\mathrm{vol}_\varepsilon(X) := \varepsilon^3 \#(X \cap \Lambda)$.

7.1. **Photon boundary conditions across a matter-free collar.** The photon in Feynman form consists of three independent scalar Gaussian fields A_α, $\alpha = 1, 2, 3$, and on a finite graph $\mathcal{G}$ of sites of Λ each has the precision $K_\mathcal{G} = \varepsilon L_\mathcal{G} + \mu_0^2 \varepsilon^3$, with $L_\mathcal{G}$ the graph Laplacian of $\mathcal{G}$ with unit conductances and $\langle u, v\rangle = \sum_x u(x)v(x)$; King's Neumann photon on a cube is the case of the induced subgraph [K2, (2.5), p. 333], and the periodic photon the case of the torus graph.

Lemma 7.1 (photon boundary switch)**.** *Let $H = Q_{b'}$ be a closed cube and C_L a King cube with $\operatorname{dist}_\infty(H, \mathbb{R}^3 \setminus C_L) = D + 3$, where $D \geq \max(1, 3/\mu_0)$ and $\varepsilon \leq \min(1/\mu_0, 1/4)$. Let $\langle\cdot\rangle_\natural$ and $\langle\cdot\rangle_N$ be BFS measures with the same matter part: the sites of $H \cap \Lambda$, BFS single-site measures (arbitrary; wall and killed sites included), and hopping terms only between neighbouring sites of H, coupled to the photon variables of the bonds of H as in King's action. Their photon parts are:*

- (♮) *the variables $A_\alpha(x)$ with $x, x + \varepsilon e_\alpha \in C_L$, with the law of the periodic photon of a torus whose cell contains $N_1(C_L)$, conditioned on $A_\beta(y) = 0$ for all other (β, y);*
- (N) *King's Neumann photon on $C_L \cap \Lambda$.*

Let $\Theta = e^{\mathrm{i}F(f)} e^{\sum_x \varepsilon^3 g(x) r_x^2 - c}$ with f supported on plaquettes of H, $g \geq 0$ supported in H, and $c \in \mathbb{R}$. Then

$$0 \leq \langle\Theta\rangle_\natural - \langle\Theta\rangle_N \leq \langle|\Theta|^2\rangle_N^{1/2} \big(e^\tau - 1\big)^{1/2}, \qquad \tau \leq C_{\mathrm{GB}}\, \mathsf{a}\, \mathrm{vol}_\varepsilon(N_2(H))\, D^{-1} e^{-\frac{3}{2} m_\varepsilon D},$$

with an absolute constant C_{GB}.

Proof. Step 1 (marginals and order). Fix a component α. Let $X_\alpha := \{x \in C_L \cap \Lambda : x + \varepsilon e_\alpha \in C_L\}$; conditioning a Gaussian on the vanishing of some coordinates leaves the principal submatrix of the precision on the others, so in (♮) the variables $A_\alpha(x)$, $x \in X_\alpha$, have precision $K_\natural = \varepsilon(6 - \mathrm{Adj}_{X_\alpha}) + \mu_0^2 \varepsilon^3$, where Adj_{X_α} is the adjacency matrix of X_α. In (N) the variables are $A_\alpha(x)$, $x \in C_L \cap \Lambda$, with precision $K_N = \varepsilon L_{C_L} + \mu_0^2 \varepsilon^3$. The principal submatrix of K_N on X_α is $\varepsilon(\deg_{C_L} - \mathrm{Adj}_{X_\alpha}) + \mu_0^2 \varepsilon^3 \leq K_\natural$, and the Schur complement of K_N on X_α is at most its principal submatrix; hence $C_\natural := K_\natural^{-1} \leq C_N|_{X_\alpha}$, $C_N := K_N^{-1}$. Let $\hat{X} := \{(\alpha, x) : x, x + \varepsilon e_\alpha \in H\}$. The integrands depend on A only through $A_{\hat{X}}$, whose laws are $N(0, C_\natural|_{\hat{X}})$ and $N(0, C_N|_{\hat{X}})$, with $C_\natural|_{\hat{X}} \leq C_N|_{\hat{X}}$. Put $M := (C_\natural|_{\hat{X}})^{-1} - (C_N|_{\hat{X}})^{-1} \geq 0$. The ♮-measure is obtained from the N-measure, after integrating out the photon variables off $\hat{X}$, by adding $\frac{1}{2}\langle A, M\rangle A$ to the action; by Lemma 2.12(a) along $s \mapsto \frac{s}{2}\langle A, M\rangle A$, $\langle\Theta\rangle_\natural \geq \langle\Theta\rangle_N$.

Step 2 (Cauchy–Schwarz). With $Q := \langle A, M\rangle A$ and $\varsigma := e^{-Q/2}$, $\langle\Theta\rangle_\natural = \langle\Theta\varsigma\rangle_N / \langle\varsigma\rangle_N$, so

$$\langle\Theta\rangle_\natural - \langle\Theta\rangle_N = \frac{\langle(\Theta - \langle\Theta\rangle_N)(\varsigma - \langle\varsigma\rangle_N)\rangle_N}{\langle\varsigma\rangle_N} \leq \langle|\Theta|^2\rangle_N^{1/2} \Big(\frac{\langle e^{-Q}\rangle_N}{\langle e^{-Q/2}\rangle_N^2} - 1\Big)^{1/2} \leq \langle|\Theta|^2\rangle_N^{1/2} \big(e^{\langle Q\rangle_N} - 1\big)^{1/2},$$

by $\langle e^{-Q}\rangle_N \leq 1$ and Jensen's inequality. Writing $M = \sum_j \lambda_j v_j v_j^T$, Lemma 2.13(b) in the N-measure gives $\langle Q\rangle_N \leq \sum_j \lambda_j \langle v_j, C_N v_j\rangle = \mathrm{tr}(M C_N|_{\hat{X}}) =: \tau$.

Step 3 (the trace). Let $\Delta := C_N|_{X_\alpha} - C_\natural \geq 0$, for each α. Then $M C_N|_{\hat{X}} = (C_\natural|_{\hat{X}})^{-1} \Delta_{\hat{X}}$, so $\tau = \mathbb{E}\langle Y, (C_\natural|_{\hat{X}})^{-1} Y\rangle$ with $Y \sim N(0, \Delta_{\hat{X}})$. Since the inverse of a principal block of the covariance is the minimum of the precision form over extensions, $\langle a, (C_\natural|_{\hat{X}})^{-1} a\rangle = \min\{\langle u, K_\natural u\rangle : u|_{\hat{X}} = a\}$. Let $\tilde{Y} \sim N(0, \Delta)$ on X_α and $\eta(x) := (1 - \operatorname{dist}_\infty(x, H))_+$; then $\eta\tilde{Y}$ is an extension of $\tilde{Y}|_{\hat{X}}$, so $\tau \leq \sum_\alpha \mathbb{E}\langle\eta\tilde{Y}, K_\natural \eta\tilde{Y}\rangle$. Let int be the set of $x \in X_\alpha$ whose six neighbours lie in X_α; it contains $N_2(H) \cap \Lambda$. At $x \in \mathrm{int}$ the rows of $K_\natural$, K_N and of the infinite-lattice precision K_∞ coincide, so $(K_\infty \Delta)(x, w) = (K_N C_N)(x, w) - (K_\natural C_\natural)(x, w) = 0$ for all $w \in X_\alpha$, hence $\mathrm{Var}((K_\infty \tilde{Y})(x)) = 0$ and $K_\infty \tilde{Y} = 0$ on int almost surely. For u with $K_\infty u = 0$ on $\mathrm{int} \supset \operatorname{supp}\eta$, summation by parts and the identity $(\eta_x u_x - \eta_y u_y)^2 - (u_x - u_y)(\eta_x^2 u_x - \eta_y^2 u_y) = (\eta_x - \eta_y)^2 u_x u_y$ give $\langle\eta u, K_\natural \eta u\rangle = \langle\eta u, K_\infty \eta u\rangle = \varepsilon \sum_{\langle xy\rangle} (\eta_x - \eta_y)^2 u_x u_y$. Since $|\eta_x - \eta_y| \leq \varepsilon$ and $|\Delta(x, y)| \leq \frac{1}{2}(\Delta(x, x) + \Delta(y, y))$,

$$\tau \leq \sum_\alpha \varepsilon^3 \sum_{\langle xy\rangle \subset N_2(H)} |\Delta(x, y)| \leq 9\, \mathrm{vol}_\varepsilon(N_2(H)) \sup_{\alpha,\ x \in N_2(H)} \Delta(x, x).$$

Step 4 (decay of Δ). For $K = \mathcal{D} - \varepsilon \mathrm{Adj}_{\mathcal{G}}$ with $\mathcal{D}$ diagonal, $K^{-1}(x, y) = \sum_\omega \prod_{(u \to v) \in \omega} (\varepsilon / \mathcal{D}(u)) \cdot \mathcal{D}(y)^{-1}$, the sum over walks ω from x to y in $\mathcal{G}$. For $K_\natural$ and K_N the weights of steps from sites of int coincide. Decomposing walks from $x \in N_2(H)$ at their first visit to $\Sigma := X_\alpha \setminus \mathrm{int}$,

$K^{-1}(x,y) = G^{\mathrm{int}}(x,y) + \sum_{z\in\Sigma} \mathsf{H}(x,z)K^{-1}(z,y)$ with G^{int} and H the same for both precisions. Hence

$$\Delta(x,x) = \sum_{z\in\Sigma} \mathsf{H}(x,z)\big(K_N^{-1}(z,x) - K_{\natural}^{-1}(z,x)\big) \le \Big(\sum_{z\in\Sigma} \mathsf{H}(x,z)\Big) \sup_{z\in\Sigma} K_N^{-1}(z,x).$$

The first factor is the probability that simple random walk from x, killed with probability $\mu_0^2\varepsilon^2/(6+\mu_0^2\varepsilon^2)$ per step, reaches Σ, which lies at ℓ^∞-distance at least D from x. With $\cosh\lambda = 1+\mu_0^2\varepsilon^2/2$, that is $\lambda = \varepsilon m_\varepsilon$, $e^{\pm\lambda S_n^\beta}\mathbf{1}_{\{\text{alive at } n\}}$ is a martingale for each coordinate S^β; optional stopping and a union over the six half-spaces give $\sum_z \mathsf{H}(x,z) \le 6e^{-m_\varepsilon D}$. The same argument, with the decomposition at the first exit from the ℓ^∞-ball of radius $|y|_\infty/2$ and the bound $K^{-1}_{\varepsilon\mathbb{Z}^3,\mu_0} \le K^{-1}_{\varepsilon\mathbb{Z}^3,0} = (6\varepsilon)^{-1}G_{\mathbb{Z}^3}(\cdot/\varepsilon)$ entrywise, gives the free bound

$$G_\varepsilon(y) := K^{-1}_{\varepsilon\mathbb{Z}^3}(0,y) \le 4\mathsf{a}\,|y|_\infty^{-1} e^{-m_\varepsilon|y|_\infty/2} \qquad (|y|_\infty \ge 4\varepsilon).$$

King's Neumann precision on the box $C_L\cap\Lambda$ has the Green's function $K_N^{-1}(z,x) = \sum_{x'} G_\varepsilon(z-x')$, the sum over the images x' of x under the reflections of $\varepsilon\mathbb{Z}^3$ in the planes half a lattice spacing outside the faces of the box (method of images; the mass term is uniform). Every image other than x lies outside C_L at ℓ^∞-distance at least D from every site of C_L, and so does x from $z\in\Sigma$. If $s\ge 2D$ is the side of C_L, at most $8(j+3)^3$ images satisfy $|z-x'|_\infty < D+(j+1)s$; since $m_\varepsilon D \ge 2$,

$$K_N^{-1}(z,x) \le \sum_{j\ge 0} 8(j+3)^3\,\frac{4\mathsf{a}\,e^{-m_\varepsilon(D+js)/2}}{D+js} \le C_1\,\mathsf{a}\,D^{-1}e^{-m_\varepsilon D/2}$$

with an absolute constant C_1. Combining, $\Delta(x,x) \le 6C_1\mathsf{a}D^{-1}e^{-\frac{3}{2}m_\varepsilon D}$, and Step 3 gives the claim with $C_{\mathrm{GB}} = 54C_1$. □

The Gaussian measures in Lemma 7.1 are compared only on the bonds of H, where no live matter crosses the region in which the two photon boundary conditions differ; this is what makes the bound uniform in ε. A comparison in which the photon is pinned or cut on a surface that live matter crosses is not of this type: a one-loop computation shows that the order-e^2 difference of such states diverges logarithmically as $\varepsilon\to 0$, because the free-boundary mass counterterm compensates the photon self-energy of the matter only where the photon has its bulk covariance.

7.2. **Shell configurations, two-level walls, and the shell-kill hypothesis.**

Definition 7.2 (shell configurations and shell profiles)**.** Let $b>0$. A *shell configuration* for Q_b is $\mathfrak{s} = (\mathfrak{c}, b', L)$, where $\mathfrak{c} = (J',\psi,n)$ with $\psi\in C^\infty(D_{J'};[0,\infty))$ and $n\ge 0$ is a walled configuration in $\mathcal{A}(Q_b)$ (so $\psi = 0$ on $N_1(Q_b)$), $H := Q_{b'}$ with $b'\ge b+3$, and C_L is a King cube (the shell's cube, with its own index L; not the source cube C_L of the class (6) and of Theorems 6.1 and 6.4) with $\mathrm{dist}_\infty(H,\mathbb{R}^3\setminus C_L) = D(\mathfrak{s})+3$, $D(\mathfrak{s}) \ge \max(1,3/\mu_0)$, and $N_1(\bar C_L)\subset(-25MJ', 25MJ')^3$. Its *shell* is $U := C_L\setminus H$, and we require the *plateau condition* $\psi\ge 1$ on $N_1(U)$ (compatible with $\psi = 0$ on $N_1(Q_b)$, since $\mathrm{dist}_\infty(U,Q_b)\ge 3$). Put

$$\tau(\mathfrak{s}) := C_{\mathrm{GB}}\,\mathsf{a}\,\mathrm{vol}(N_3(H))\,D(\mathfrak{s})^{-1}e^{-1.44\,\mu_0 D(\mathfrak{s})}.$$

A *shell profile* for $\mathfrak{s}$ is $\chi\in C^\infty(D_{J'};[0,1])$ with $\chi = 1$ on U and $\mathrm{supp}\,\chi\subset N_1(U)$; then $\chi\le\psi$ automatically, and $\chi = 0$ on $N_1(Q_b)$. Its *dead set* is $D_\chi := \{\chi>0\}\subset N_1(U)$; every site of C_L within ε of $\mathbb{R}^3\setminus C_L$, and every site of U, lies in D_χ. We call $(\mathfrak{s},\chi)$ a *two-level configuration* for Q_b.

Definition 7.3 (two-level states and the tail)**.** Let $(\mathfrak{s},\chi)$ be a two-level configuration for Q_b, $(f,g)\in\mathcal{K}$ supported in Q_b, and k with $\varepsilon_k\le\min(1/\mu_0,1/4)$. For $K\in[0,\infty)$ let

$$\Psi_k(n,K) := T^a_{(J',\,n\psi+K\chi),\Lambda_k}(f,g)$$

be the walled value of the configuration with the two-level wall $n\psi + K\chi$, which lies in $\mathcal{A}(Q_b)$; and let $\Psi_k(n,\infty) := \langle \Theta^a(\mathrm{i}f,g)\rangle$ in the walled torus state of $\mathfrak{c} = (J', n\psi)$ with the matter killed on $D_\chi \cap \Lambda_k$ (Definition 2.5(d)). Put $\mathrm{tail}_k(K) := \Psi_k(n,K) - \Psi_k(n,\infty)$.

Lemma 7.4 (elementary properties)**.** (a) *$K \mapsto \Psi_k(n,K)$ is nonincreasing on $[0,\infty)$, $\Psi_k(n,K) \geq \Psi_k(n,\infty)$, and $\Psi_k(n,K) \to \Psi_k(n,\infty)$ as $K \to \infty$. Hence $\mathrm{tail}_k(K) \geq 0$ is nonincreasing in K and tends to 0 as $K \to \infty$, at every fixed k.*
(b) *(two-level sandwich) For $n, K \geq 0$: $\Psi_k(n+K,0) \leq \Psi_k(n,K) \leq \Psi_k(n,0)$.*

Proof. (a) Increasing K multiplies the single-site measures on D_χ by $e^{-(K'-K)\varepsilon^3\chi(x)r_x^2}$; Lemma 2.12(c) with Lemma 3.3(a). Killing D_χ is the limit (Lemma 2.7 with the rates $\chi > 0$ on D_χ), and it does not increase the expectation. (b) The wall of $\Psi_k(n+K,0)$ exceeds that of $\Psi_k(n,K)$ by $K(\psi-\chi) \geq 0$, since $\chi \leq \psi$; the wall of $\Psi_k(n,K)$ exceeds that of $\Psi_k(n,0)$ by $K\chi \geq 0$. Apply Lemma 2.12(c). □

Part (b) is the two-level sandwich (Lemma 7.7 below in the form used); it is the only place where $\chi \leq \psi$ is needed. In words: raising the matter mass on the shell alone, by any amount, gives a state above the one in which the whole plateau is raised by the same amount, and below the one with the shell untouched.

Hypothesis 7.5 ($(\mathrm{SH})_a$ and (SH_0): shell-kill decoupling)**.** **$(\mathbf{SH})_a$**: for every $b > 0$, every $(f,g) \in \mathcal{K}$ supported in Q_b and every $\eta > 0$ there is a two-level configuration $(\mathfrak{s},\chi)$ for Q_b with $\tau(\mathfrak{s}) \leq \eta$ and

$$\liminf_{k\to\infty} \mathrm{tail}_k(0) = \liminf_{k\to\infty} \big[\Psi_k(n,0) - \Psi_k(n,\infty)\big] \leq \eta.$$

$(\mathbf{SH}_0)$ is the same statement for $g = 0$; it does not depend on the scheme.

Since $\tau(\mathfrak{s})$ depends only on H and the width $D(\mathfrak{s})$, the condition $\tau(\mathfrak{s}) \leq \eta$ only asks the shell to be wide enough, logarithmically in b' and $1/\eta$. Hypothesis (SH) involves one lattice, a single walled torus and a single dead set, a fixed wall height, and the lim inf over the lattice spacings; it involves no limit $n \to \infty$ and no comparison between lattices.

Theorem 7.6 (shell-kill decoupling implies the closing hypothesis)**.** *Assume $(\mathrm{W})_a$ and $(\mathrm{K})_a$. Then $(\mathrm{SH})_a$ implies $(\mathrm{TIGHT})_a$. Under (K_0), (SH_0) implies (TIGHT_0). More precisely, for every two-level configuration $(\mathfrak{s},\chi)$ for Q_b, every $(f,g) \in \mathcal{K}$ supported in Q_b and every k with ε_k small,*

$$\Psi_k(n,\infty) \leq Z^a_{\Lambda_k,C_L}(\mathrm{i}f,g) + Z^a_{\Lambda_k,C_L}(0,2g)^{1/2}\big(e^{\tau(\mathfrak{s})}-1\big)^{1/2}, \tag{7.1}$$

and hence $\limsup_k \Psi_k(n,\infty) \leq Z^a_\infty(\mathrm{i}f,g) + Z^a_\infty(0,2g)^{1/2}(e^{\tau(\mathfrak{s})}-1)^{1/2}$.

Proof. Fix k with $\varepsilon = \varepsilon_k$ small, and write $\Theta = \Theta^a(\mathrm{i}f,g)$. Let E be the set of sites of the torus outside C_L.

Step 1 (pinning). In the state with the matter killed on $D_\chi \cap \Lambda_k$, add $s|A_\beta(y)|^2$ for every photon variable on a bond with an endpoint in E, and let $s \to \infty$. By Lemma 2.12(a) the expectation of Θ does not decrease, and in the limit the photon is the conditioned law ($\natural$) of Lemma 7.1. In this state the hopping terms within E do not involve A, and there are no hopping terms between E and live sites: every site of C_L adjacent to E lies in D_χ and is dead. So the matter on E is a factor independent of A and cancels (Lemma 2.12(e)). What remains is the $\natural$-measure of Lemma 7.1, with the matter on H (dead on $D_\chi \cap H$), the wall, and Dirichlet conditions at ∂H, since $U \subset D_\chi$ is dead. Hence $\Psi_k(n,\infty) \leq \langle\Theta\rangle_\natural$.

Step 2 (photon switch). By Lemma 7.1, $\langle\Theta\rangle_\natural \leq \langle\Theta\rangle_N + \langle|\Theta|^2\rangle_N^{1/2}(e^{\tau(\mathfrak{s})}-1)^{1/2}$, and $|\Theta|^2 = \Theta^a(0,2g)$; here $m_\varepsilon \geq 0.96\mu_0$ and $\mathrm{vol}_\varepsilon(N_2(H)) \leq \mathrm{vol}(N_3(H))$ for small ε give $\tau \leq \tau(\mathfrak{s})$.

Step 3 (to the box C_L). The N-measure is the limit $s \to \infty$ of the walled Neumann–Dirichlet state on $C_L \cap \Lambda_k$ with the single-site measures on D_χ multiplied by $e^{-s\varepsilon^3\chi(x)r^2}$ (Lemma 2.7). By Lemma 2.12(c) the expectations of $\Theta^a(\mathrm{i}f,g)$ and $\Theta^a(0,2g)$ do not decrease as s decreases to 0, nor

when the wall is then removed. The result is the box state $\langle\cdot\rangle^{\mathrm{ND}}_{k,L}$ of Definition 2.5(a), with the free Wick constants since $\operatorname{supp} g$ has distance > 1 from ∂C_L. With Lemma 3.3(a) this is (7.1).

Step 4 (limits). The cube functionals converge by (W1), and $Z^a_L \le Z^a_\infty$ on $\mathcal{K}$ (Theorem 3.8(iii)), with $(0, 2g) \in \mathcal{K}$. This gives the lim sup bound. Now let $(\mathfrak{s}, \chi)$ be as in $(\mathrm{SH})_a$. Then $T^a_{\mathfrak{c},\Lambda_k}(f, g) = \Psi_k(n, 0) = \mathrm{tail}_k(0) + \Psi_k(n, \infty)$, the left side converges to $\mathcal{W}^a_{\mathfrak{c}}(f, g)$ by $(\mathrm{K})_a$, and taking $\liminf_k$,

$$\mathcal{W}^a_{\mathfrak{c}}(f, g) \le \eta + Z^a_\infty(\mathrm{i}f, g) + Z^a_\infty(0, 2g)^{1/2}\big(e^\eta - 1\big)^{1/2}.$$

Since $\mathfrak{c} \in \mathcal{A}(Q_b)$ and η is arbitrary, $I^a_b(f, g) \le Z^a_\infty(\mathrm{i}f, g)$ for every b, which is $(\mathrm{TIGHT})_a$. At $g = 0$ the same argument runs with Proposition 3.11 in place of Theorem 3.8 and $\langle|\Theta|^2\rangle = 1$: the box functional $Z_{\Lambda_k, C_L}(\mathrm{i}f, 0)$ of (7.1) is at most $\langle e^{\mathrm{i}F(f)}\rangle^K_{k,L+1}$ by Lemma 2.16, whose limit as $k \to \infty$ exists by item (13) and is at most $Z_\infty(\mathrm{i}f, 0)$. □

7.3. **The main hypothesis.**

Lemma 7.7 (two-level sandwich)**.** *For every two-level configuration, every $(f, g) \in \mathcal{K}$ supported in Q_b, every k and $n, K \ge 0$,*

$$\Psi_k(n + K, 0) \;\le\; \Psi_k(n, K) \;=\; \Psi_k(n, \infty) + \mathrm{tail}_k(K) \;\le\; \Psi_k(n, 0), \qquad \mathrm{tail}_k(K) \ge 0.$$

Proof. Lemma 7.4. □

Hypothesis 7.8 ($(\mathrm{RES})_a$ and (RES_0): heavy-shell decoupling at one configuration)**. (RES)$_a$**: for every $b > 0$, every $(f, g) \in \mathcal{K}$ supported in Q_b and every $\eta > 0$ there are a two-level configuration $(\mathfrak{s}, \chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ and a shell mass $K \in [0, \infty)$ such that

$$\liminf_{k\to\infty} \mathrm{tail}_k(K) = \liminf_{k\to\infty} \big[\Psi_k(n, K) - \Psi_k(n, \infty)\big] \le \eta.$$

(RES$_0$) is the same statement for $g = 0$; it does not depend on the scheme.

In words: in a walled torus, once the scalar mass squared on a thick shell of the heavy plateau has been raised by $2K$, removing the matter on the shell altogether changes the expectation of the observable by at most η in the continuum limit, for one geometry and one K, with no rate in K and no uniformity in anything. By Lemma 7.4(a) the bracket tends to 0 as $K \to \infty$ at every fixed k; the hypothesis says that the heavy-shell limit and the continuum limit may be interchanged at one configuration. The quantifiers are the weakest that Theorem 7.10 accepts: one configuration, one K (with $K = 0$ allowed, which is (SH)), the lim inf along the spacings, and no uniformity in n, L, D or the torus.

Proposition 7.9 ((SH) implies (RES))**.** *For every two-level configuration and every $K \ge 0$, $\mathrm{tail}_k(K) \le \mathrm{tail}_k(0)$. Hence $(\mathrm{SH})_a$ implies $(\mathrm{RES})_a$ with the same configurations and every K, and likewise at $g = 0$.*

Proof. Lemma 7.4(a). □

Theorem 7.10 ((RES) implies (TIGHT))**.** *Assume $(\mathrm{W})_a$ and $(\mathrm{K})_a$. Then $(\mathrm{RES})_a$ implies $(\mathrm{TIGHT})_a$. Under (K_0), (RES_0) implies (TIGHT_0). More precisely, if $(\mathfrak{s}, \chi)$ and K are as in $(\mathrm{RES})_a$ for (f, g) and η, then the single-level configuration $\mathfrak{c}' := (J', \psi, n + K) \in \mathcal{A}(Q_b)$ satisfies*

$$\mathcal{W}^a_{\mathfrak{c}'}(f, g) \le Z^a_\infty(\mathrm{i}f, g) + Z^a_\infty(0, 2g)^{1/2}\big(e^\eta - 1\big)^{1/2} + \eta.$$

Proof. By Lemma 7.7, $T^a_{\mathfrak{c}',\Lambda_k}(f, g) = \Psi_k(n + K, 0) \le \Psi_k(n, \infty) + \mathrm{tail}_k(K)$ at every k. The left side converges to $\mathcal{W}^a_{\mathfrak{c}'}(f, g)$ by $(\mathrm{K})_a$, since $\mathfrak{c}'$ is a walled configuration (Lemma 4.2(a)); hence

$$\begin{aligned}\mathcal{W}^a_{\mathfrak{c}'}(f, g) = \liminf_k \Psi_k(n + K, 0) &\le \limsup_k \Psi_k(n, \infty) + \liminf_k \mathrm{tail}_k(K) \\ &\le Z^a_\infty(\mathrm{i}f, g) + Z^a_\infty(0, 2g)^{1/2}\big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2} + \eta\end{aligned}$$

by Theorem 7.6 and $(\mathrm{RES})_a$, using $\liminf(x_k+y_k)\le\limsup x_k+\liminf y_k$. With $\tau(\mathfrak{s})\le\eta$ this is the displayed bound; as η is arbitrary and $\mathfrak{c}'\in\mathcal{A}(Q_b)$, $I^a_b(f,g)\le Z^a_\infty(\mathrm{i}f,g)$ for every b, which is $(\mathrm{TIGHT})_a$. At $g=0$ use Proposition 3.11 and $Z(0,0)=1$. □

Remark 7.11 (on the proof). (i) The two-level state is only an intermediate inequality; the configuration that enters (TIGHT) is the single-level wall $(n+K)\psi$, and the only limit input is the existence of its walled value, which is part of $(\mathrm{K})_a$, respectively (K_0) (the hypothesis (H6) of Theorem 4.9: King's Theorem 2.1 of [K1] with the wall as King's scalar source, read on the normalised ratio). No existence statement is needed for the two-level state or for the dead-shell state. (ii) The lattice statements of this subsection hold verbatim for the sharp profile $\chi=\mathbf{1}_U$ (then $D_\chi=U$ and $\chi\le\psi$ needs only $\psi\ge1$ on U); smoothness of χ is used only in Theorem 7.15 below, where the two-level walled value must exist. (iii) Since $\mathrm{tail}_k(K)$ is nonincreasing in K, (RES) for one K implies it for every larger K; so K may be taken as large as one likes, which is used to place $n+K$ in any prescribed window of wall heights.

7.4. **The heavy-shell estimate as an implication.** In the curvature sector (RES_0) is not left as a hypothesis. The shell is now taken to be $U:=T\setminus H'$, the whole exterior of the source box $H':=H$ in the torus T of the configuration, so that $\partial U=\partial H'$ is a single convex surface with no re-entrant edge (on a cubic shell $C_L\setminus H$ the derivative clause of King's bound on the block propagators, [K1, Thm. 3.3], fails at the re-entrant edges); the plateau condition of Definition 7.2 is read for this shell, $\psi\ge1$ on $N_1(U)$ with $U=T\setminus H'$, that is, $\psi\ge1$ except at the points of H' at distance more than one from $\partial H'$, which is compatible with $\psi=0$ on $N_1(Q_b)$ since $H'=Q_{b'}$ with $b'\ge b+3$, and the shell profile χ and the dead set D_χ are those of Definition 7.2 for this U; the steps of the dead-shell chain of Theorem 7.6 in this geometry are item (10c) of Section 1.4. For $\kappa\ge0$ let $\Psi_k(n,\kappa)$ be the two-level expectation of Definition 7.3 with the shell mass $\kappa\mathbf{1}_U$ added to the wall (so that the scalar mass squared is raised by 2κ on U, Definition 2.5(c); in the normalisation of [SW1, SW3], whose profile V enters $\frac12\langle\phi,V\phi\rangle$, the wall $n\psi+\kappa\mathbf{1}_U$ is their $2n\psi+2\kappa\mathbf{1}_U$), so that $\Psi_k(n,K)$ and $\Psi_k(n,\infty)$ in (RES) are the cases $\kappa=K$ and $\kappa=\infty$, and let $Z_\kappa(zf,g')$ be the corresponding partition function with sources. King's block-spin flow [K1, §3] is run for the model in which $\kappa\mathbf{1}_U$ is part of the matter covariance at every scale; $\eta_0\in5^{-\mathbb{N}}$ is the stopping scale, $\kappa_0:=\max(c_0,1)/\eta_0^2$ the threshold above which the shell is heavy in block units at the stopping scale and both (8) (stated for $\kappa\ge c_0/\eta_0^2$) and (10) hold, I^{cap}_κ the stopping-scale integral over the capped test set (a set of block fields that does not depend on κ), and $\Delta_z(b)$ the difference of the reduced small-field actions at shell masses κ' and κ at the block field b and source coefficient z; $n_*:=1$, so that for $n\ge n_*$ the plateau height $2n$ of [SW1, SW3] satisfies the condition $2n\ge1$ under which the estimates (8), (10) and (11) are stated there (both assume a plateau height ≥1); the estimates (8)–(11) are read through this dictionary, which changes the threshold κ_0 by a factor 2 and nothing else, since they hold uniformly in $\kappa\ge\kappa_0$ at every fixed plateau height. The dictionary carries the profile shape as well as n and κ: the estimates (8)–(11) are stated in [SW1, SW3] for King's wall profile only, $\psi\colon T\to[0,1]$ with $\psi=0$ on the source region and $\psi=1$ off its 1-neighbourhood, so that $\psi\equiv1$ on U and on its η_0-neighbourhood, at a fixed plateau height $n\ge1$ in their normalisation, and the two-level configurations to which Theorem 7.12 is applied are taken with this profile (one configuration suffices for (RES_0), Theorem 7.10); the notation is that of [SW1, SW3].

Theorem 7.12 (the heavy-shell estimate from the shell bounds)**.** *Let $g=0$ and let $(\mathfrak{s},\chi)$ be a two-level configuration with the shell $U=T\setminus H'$, the torus $T=D_{J'}$ with $J'\in5^{k_*}\mathbb{N}$ and $J'\ge\max(J_0,J_{\mathrm{pos}})$, $J_0=J_0(\varepsilon_0)$ the non-wrapping threshold of the class (6), King's wall profile ($\psi\in[0,1]$, $\psi=0$ on $N_1(Q_b)$, $\psi=1$ off $N_2(Q_b)$), plateau height $n\ge n_*$, H', T unions of blocks of the final scale and the corners of H' in the residue class of (6). Assume $(\mathrm{SW1}^\circ)_{\mathrm{int}}$, $(\mathrm{SW2}^{\mathrm{min}})$, $(\mathrm{SW3}^\circ)_{\mathrm{int}}$ and (SWT) in the forms (8)–(11) of Section 1.4. Then for every $\eta>0$ there are a*

stopping scale η_0 and a shell mass $K<\infty$ with $\limsup_k \mathrm{tail}_k(K)\le\eta$. The quantifiers of ($\mathrm{RES}_0$) *are met in this order: given b, a source f supported in Q_b and $\eta>0$, the configuration $(\mathfrak{s},\chi)$ is chosen first, with $D(\mathfrak{s})$ so large that $\tau(\mathfrak{s})\le\eta$, and then η_0 and K as above; so* (RES_0) *holds at $(\mathfrak{s},\chi)$ with this K. Consequently, under* (K_0), (TIGHT_0) *holds (Theorem 7.10), with the single-level configuration $(J',\psi,n+K)$ in the class (6), K being taken large enough for its wall condition (Remark 7.11(iii)) and the vanishing of ψ near $R^{-1}C_L$, C_L the source cube of the class (6), being the choice of b in Hypothesis 5.5.*

Proof. Fix $(\mathfrak{s},\chi)$, the source f and $g=0$. For $\kappa\in[0,\infty]$ and $z\in\mathbb{C}$ put $\Psi_\kappa(z):=Z_\kappa(zf,0)/Z_\kappa(0,0)$, where Z_∞ is the partition function of the state with the matter killed on D_χ; the shell mass $\kappa\mathbf{1}_U$ is the second level $\kappa\chi$ of the wall, so that by Definition 7.3, at $g=0$, $\Psi_K(\mathrm{i})=\Psi_k(n,K)$, $\Psi_\infty(\mathrm{i})=\Psi_k(n,\infty)$ and $\mathrm{tail}_k(K)=\Psi_K(\mathrm{i})-\Psi_\infty(\mathrm{i})$. For $\kappa\ge\kappa_0$ and real z put $J_\kappa(z):=I^{\mathrm{cap}}_\kappa(zf,0)/I^{\mathrm{cap}}_\kappa(0,0)$. Every constant below is uniform in $\varepsilon\le\varepsilon_0(\eta_0)$.

Step 0: the disc bound. At fixed spacing $z\mapsto\Psi_\kappa(z)$ is entire, a finite-dimensional integral of $e^{zF(f)}$ against a measure with Gaussian photon tails. For $|z|\le2$, $|e^{zF(f)}|\le e^{2F(f)}+e^{-2F(f)}$, and Lemma 2.13(a) with $s=\pm2$ and v the vector with $\langle v,A\rangle=F(f)$ gives $\langle e^{\pm2F(f)}\rangle_\kappa\le e^{2\langle v,Cv\rangle}$ in every state with the photon covariance C, the shell mass and the killing acting on the matter only. Hence $|\Psi_\kappa(z)|\le M:=2e^{2\langle v,Cv\rangle}$ for all $\kappa\in[0,\infty]$, and $\langle v,Cv\rangle$, the variance of $F(f)$ under the free massive photon, is bounded uniformly in ε for smooth f. This is item (9): at $g=0$ the matter moment factor of the disc bound is 1. For real z, $\Psi_\kappa(z)\to\Psi_\infty(z)$ as $\kappa\to\infty$ at fixed ε, by the substitution and dominated convergence of the proof of Lemma 2.7, the factor $e^{zF(f)}$ being integrable against the photon Gaussian by the same bound.

Step 1: the sandwich. Let $\eta_0\le\min(\eta_1,\eta_2)$ be so small that $b(\eta_0)\le\frac12$, and let $\kappa\ge\kappa_0$, $z\in[-1,1]$. By (8) at $(0,0)$, $e^{-a}(1-b)Z_\kappa(0,0)\le I^{\mathrm{cap}}_\kappa(0,0)\le e^aZ_\kappa(0,0)$; by (8) at $(zf,0)$ and these two bounds, $e^{-2a}(1-b)J_\kappa(z)\le\Psi_\kappa(z)\le e^{2a}J_\kappa(z)+b$. Hence $J_\kappa(z)\le2e^{2a}M=:\bar J$ and $|\Psi_\kappa(z)-J_\kappa(z)|\le\bar J(e^{2a}-e^{-2a}+b)+b=:m_1(\eta_0)$, and $m_1(\eta_0)\to0$ as $\eta_0\to0$ because $a,b\to0$.

Step 2: the κ-Cauchy property. Let $\kappa_0\le\kappa\le\kappa'<\infty$ and $z\in[-1,1]$. Write $I^{\mathrm{cap}}_\kappa(zf,0)=\int_{\tilde\chi}e^{-\tilde S_\kappa(b;z)}\,db$ with $\tilde S_\kappa(\cdot;z)$ the reduced small-field action at shell mass κ and $\tilde\chi$ the capped test set, which is the same for all $\kappa\ge\kappa_0$; this is where (11) enters: no threshold term arises between κ and κ'. Then

$$\frac{J_\kappa(z)}{J_{\kappa'}(z)}=\frac{\iint_{\tilde\chi\times\tilde\chi}e^{-\tilde S_\kappa(b;z)-\tilde S_{\kappa'}(b';0)}\,db\,db'}{\iint_{\tilde\chi\times\tilde\chi}e^{-\tilde S_{\kappa'}(b;z)-\tilde S_\kappa(b';0)}\,db\,db'},$$

and the two integrands differ pointwise by the factor $e^{\Delta_z(b)-\Delta_0(b')}$, which lies in $[e^{-2r},e^{2r}]$ with $r:=r(\kappa;\eta_0)$ by (10). So $J_\kappa(z)/J_{\kappa'}(z)\in[e^{-2r},e^{2r}]$ and $|J_\kappa(z)-J_{\kappa'}(z)|\le\bar J(e^{2r}-1)$. Put $\bar r(K;\eta_0):=\sup_{\kappa\ge K}r(\kappa;\eta_0)$, which tends to 0 as $K\to\infty$ at fixed η_0. With Step 1, for all $\kappa'\ge\kappa\ge K\ge\kappa_0$,

$$|\Psi_\kappa(z)-\Psi_{\kappa'}(z)|\le m(K,\eta_0):=2m_1(\eta_0)+\bar J\big(e^{2\bar r(K;\eta_0)}-1\big),\qquad z\in[-1,1],$$

uniformly in $\varepsilon\le\varepsilon_0(\eta_0)$.

Step 3. Letting $\kappa'\to\infty$ at fixed ε (Step 0): $|\Psi_K(z)-\Psi_\infty(z)|\le m(K,\eta_0)$ for $z\in[-1,1]$.

Step 4: two constants. $D:=\Psi_K-\Psi_\infty$ is entire, $|D|\le2M$ on $\{|z|\le2\}$ and $|D|\le m(K,\eta_0)$ on $[-1,1]$. Since $\log|D|$ is subharmonic on $\Omega:=\{|z|<2\}\setminus[-1,1]$ and is dominated there by the harmonic function with boundary values $\log(2M)$ on $|z|=2$ and $\log m(K,\eta_0)$ on the slit, $|D(\mathrm{i})|\le(2M)^{1-\omega}m(K,\eta_0)^\omega$, where $\omega\in(0,1)$ is the harmonic measure of the slit $[-1,1]$ at the point i in Ω, an absolute constant.

Step 5: the real point. The map $(A,\phi)\mapsto(-A,\bar\phi)$ preserves the action, the wall, the shell mass, the gauge fixing and the measure, and reverses the sign of $F(f)$; so $\Psi_\kappa(-z)=\Psi_\kappa(z)$, and since $\Psi_\kappa(\bar z)=\overline{\Psi_\kappa(z)}$, the value $\Psi_\kappa(\mathrm{i})$ is real. Thus $D(\mathrm{i})=\mathrm{tail}_k(K)$ and

$$0\le\mathrm{tail}_k(K)\le(2M)^{1-\omega}\,m(K,\eta_0)^\omega\qquad\text{for every }\varepsilon_k\le\varepsilon_0(\eta_0).$$

Choice of the parameters. Given $\eta > 0$: take $\delta > 0$ with $(2M)^{1-\omega}\delta^{\omega} \le \eta$; then $\eta_0 \le \min(\eta_1, \eta_2)$ with $b(\eta_0) \le \frac{1}{2}$ and $2m_1(\eta_0) \le \delta/2$, which is possible because $a, b \to 0$; then $K \ge \kappa_0$ with $\bar{J}(e^{2\bar{r}(K;\eta_0)} - 1) \le \delta/2$, which is possible because $r(K;\eta_0) \to 0$ as $K \to \infty$ at fixed η_0; and only then $\varepsilon \le \varepsilon_0(\eta_0)$. Then $m(K, \eta_0) \le \delta$ and $\mathrm{tail}_k(K) \le \eta$ for all $\varepsilon_k \le \varepsilon_0(\eta_0)$, so $\limsup_k \mathrm{tail}_k(K) \le \eta$: this is ($\mathrm{RES}_0$) at $(\mathfrak{s}, \chi)$ with this K, in the quantifier order of (RES): one configuration, one K, no rate, the limit along the spacings last. Since tail_k is nonincreasing in K (Lemma 7.4(a)), K may be enlarged at will (Remark 7.11(iii)), which places $(J', \psi, n + K)$ in the class (6) (Remark 5.6); Theorem 7.10 then gives (TIGHT_0) under (K_0). □

Remark 7.13 (status of the shell bounds). (8) and (10) are [PROVED] as implications in [SW1] and [SW3], each from the residues inside its proof, the items (8a)–(8k) and (10a), (10b), (10d)–(10g), and given the flagged steps (F2), (F3), (F5), respectively (F6); both manuscripts have been checked independently in full; (9) and (11) are [PROVED]; the literature inputs (12) are listed at their own grades. Several statements in earlier forms of the shell estimates are false and are not used: the sub-domain clause with the constant 16 and a rate in the penetration depth, which fails at small c_0, while (8) consumes the scale-free form (8a); the tail bound without the bracket for the loops winding through U and with a right side proportional to C_A^2, which is false on the torus (a constant background has $C_A = 0$ while the winding loops contribute a positive tail); the Yukawa form of the heavy-pair clause, which fails for first-layer pairs; and a sup bound for the full covariance next to ∂U in the lattice regime. Theorem 7.12 uses (10a) only through $\tau(\kappa) \to 0$, so a change of the exponent of τ or a logarithmic factor leaves the theorem unchanged. The bound (10d) for deep clusters closed in matter and the pointwise difference bounds (8g) of the line kernels are [PROVED] as implications. The residues (8k) and (10g), with (8e) = (8i), are [PROVED] as implications and checked independently [SUP], in the order stated at the end of Section 1.5, the determinant clause of (8k)(ii) with the bound (DOM) of that section, which is [PROVED] as an implication; on the histories with proper propagator domains the removal of the tail of the background in (8e) = (8i) uses the sub-domain form (P9) at order fourteen, which is [PROVED] as an implication and enters the upper bound of (8) only; (8e) = (8i) rests on the norm form of (8k)(ii) and on (8k)(i) through the bound for the mixed classes, and directly on the walled renormalisation bound (P1), which is proved. They enter (8) directly and (10) through the statements of [SW1] that it cites. A photon Ward identity for each fixed ordering of King's graphs would not supply (10d), since that identity is false; the proof of (10d) uses the identities that hold, those for full lines, and Bałaban's resummed organisation keeps a convergent remainder, in whose place (10d) uses the bound (10g)(vi). The residues (8b)–(8f), (8k), (10e)–(10g) are internal to the proofs of (8) and (10), and (10c) to the use of Theorem 7.6 inside Theorem 7.10 at the configuration of Theorem 7.12: none of them changes the statement of (RES_0) that Theorem 7.10 consumes.

7.5. **The continuum form of the hypothesis.** This subsection and the next are not on the chain of Theorem A: they record an equivalent form and two sufficient forms of (RES) which are not used, and the existence statement (E_∞) and the Hellinger form (HEL) introduced here are not hypotheses of the theorem. Both sides of the bracket in (RES) have continuum limits, the first by $(\mathrm{K})_a$ and the second under one further existence statement. In those terms the hypothesis says that two continuum objects coincide up to η.

Proposition 7.14 (the dead-shell state between two cube states)**.** *Let $(\mathfrak{s}, \chi)$ be a two-level configuration for Q_b and $(f, g) \in \mathcal{K}$ supported in Q_b. Let $B_k := \langle \Theta^a(\mathrm{i}f, g)\rangle$ in the Neumann–Dirichlet state on $C_L \cap \Lambda_k$ (Neumann photon on C_L, Dirichlet matter outside C_L) whose single-site measures are those of the dead-shell walled torus: the wall $n\psi$, and the matter killed on $D_\chi \cap C_L$. Then at every k with ε_k small*

$$B_k \;\le\; \Psi_k(n, \infty) \;\le\; B_k + Z^a_{\Lambda_k, C_L}(0, 2g)^{1/2}\big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2}, \qquad B_k \le Z^a_{\Lambda_k, C_L}(\mathrm{i}f, g).$$

Proof. The lower bound is Lemma 2.14 applied to the dead-shell torus state and $X = C_L \cap \Lambda_k$: the only couplings between X and its complement are photon gradient terms, since every site of C_L adjacent to the complement is dead, and deleting them lowers the expectation and factorises the measure, whose X-factor is the state defining B_k. The upper bound is Steps 1–2 of the proof of Theorem 7.6, since the N-measure there is exactly the state defining B_k and $\langle \Theta^a(0,2g)\rangle_N \le Z^a_{\Lambda_k,C_L}(0,2g)$ by Step 3. The last inequality is Step 3. □

Theorem 7.15 (the continuum form)**.** *Assume* $(\mathrm{K})_a$ *(at* $g = 0$*:* (K_0)*); for the sandwich in (d) at* $g \neq 0$ *assume also* $(\mathrm{W})_a$ *(at* $g = 0$ *Proposition 3.11 and* $Z(0,0) = 1$ *replace it). Let* $(\mathfrak{s}, \chi)$ *be a two-level configuration for* Q_b *and* $(f, g) \in \mathcal{K}$ *supported in* Q_b*.*

(a) *For every* $K \in [0,\infty)$ *the limit* $\Psi^0(n,K) := \lim_k \Psi_k(n,K)$ *exists, equals the walled value* $\mathcal{W}^a_{(J',n\psi+K\chi)}(f,g)$*, and is nonincreasing in* K*; so* $\Psi^0(n,\infty^-) := \lim_{K\to\infty} \Psi^0(n,K)$ *exists.*

(b) $\liminf_k \mathrm{tail}_k(K) = \Psi^0(n,K) - \limsup_k \Psi_k(n,\infty)$ *for every* K*, and* $\limsup_k \Psi_k(n,\infty) \le \Psi^0(n,\infty^-)$*.*

(c) $(\mathrm{RES})_a$ *holds at* $(\mathfrak{s},\chi)$ *with tolerance* η *for some* K *if and only if*

$$\Psi^0(n,\infty^-) = \lim_{K\to\infty} \lim_{k\to\infty} \Psi_k(n,K) \;\le\; \limsup_{k\to\infty} \Psi_k(n,\infty) + \eta.$$

(d) *If the* dead-shell torus value $\Psi^0(n,\infty) := \lim_k \Psi_k(n,\infty)$ *exists, then (c) reads* $\lim_{K\to\infty} \Psi^0(n,K) \le \Psi^0(n,\infty) + \eta$*, the reverse inequality* $\lim_K \Psi^0(n,K) \ge \Psi^0(n,\infty)$ *always holds, and*

$$B^0 \le \Psi^0(n,\infty) \le B^0 + Z^a_\infty(0,2g)^{1/2}\big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2}$$

for every limit point B^0 *of* (B_k)*; in particular* $\Psi^0(n,\infty) \le Z^a_\infty(\mathrm{i}f,g) + Z^a_\infty(0,2g)^{1/2}(e^{\tau(\mathfrak{s})} - 1)^{1/2}$*.*

Proof. (a) $(J', n\psi + K\chi)$ is a walled configuration in $\mathcal{A}(Q_b)$, so Lemma 4.2(a) gives the limit; monotonicity in K passes to the limit (Lemma 7.4(a)). (b) $\liminf_k(x_k - y_k) = \lim_k x_k - \limsup_k y_k$ when x_k converges; and $\Psi_k(n,\infty) \le \Psi_k(n,K)$ for every k and K. (c) By (b), (RES) for some K says $\Psi^0(n,K) \le \limsup_k \Psi_k(n,\infty) + \eta$ for some K, which by monotonicity in K is the same as $\inf_K \Psi^0(n,K) \le \limsup_k \Psi_k(n,\infty) + \eta$. (d) The first two statements are (b), (c) with the limit in place of the lim sup; the sandwich is Proposition 7.14 in the limit along a subsequence on which $B_k \to B^0$, with $Z^a_{\Lambda_k,C_L} \to Z^a_L \le Z^a_\infty$ on $\mathcal{K}$ (Theorem 3.8(i),(iii), under $(\mathrm{W})_a$; at $g = 0$, where $Z_{\Lambda_k,C_L}(0,0) = 1$, $\limsup_k Z_{\Lambda_k,C_L}(\mathrm{i}f,0) \le Z_\infty(\mathrm{i}f,0)$ by Lemma 2.16 and Proposition 3.11). □

Hypothesis 7.16 ((RES^0): the continuum form)**.** For every $b > 0$, $(f,g) \in \mathcal{K}$ supported in Q_b and $\eta > 0$ there is a two-level configuration $(\mathfrak{s},\chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ for which the dead-shell torus value $\Psi^0(n,\infty)$ exists and

$$\lim_{K\to\infty} \Psi^0(n,K) \;\le\; \Psi^0(n,\infty) + \eta.$$

By Theorem 7.15, (RES^0) implies $(\mathrm{RES})_a$, and $(\mathrm{RES})_a$ implies (RES^0) whenever the dead-shell torus values of the configurations in question exist; the two are the same statement about the interchange of the limits $K \to \infty$ and $k \to \infty$, and in (RES^0) no lattice spacing appears. The existence of $\Psi^0(n,K)$ is covered by $(\mathrm{K})_a$ because the two-level wall $n\psi + K\chi$ is a smooth source; with the sharp profile $\mathbf{1}_U$ it would not be. The existence of $\Psi^0(n,\infty)$, the continuum limit of a torus state with an interior region of Dirichlet matter, is a theorem of King's type that King does not print: the Dirichlet region is bounded by lattice hyperplanes only when ∂H is, and the methods of [K2] for Dirichlet matter on a cube apply to a shell between two aligned cubes, but the statement is not in [K1, K2]. We grade it [SKETCH-LIT] ((E_∞) in Section 11); the lattice form (RES) needs neither this nor any limit beyond $(\mathrm{K})_a$.

7.6. **Sufficient forms: a Hellinger distance, and a general identity.** Nothing in this subsection is used by Theorem A. Let $(\mathfrak{s},\chi)$ be a two-level configuration and T the torus of $\mathfrak{c}$. For a site x of $\Lambda_k/(50MJ'\mathbb{Z})^3$ let $d(x)$ be the ℓ^∞-distance on the torus from x to the nearest site outside H; it takes values in $\varepsilon\mathbb{N}$ and $|d(x)-d(y)|\le|x-y|_\infty$. For $t\in\varepsilon\mathbb{N}$ put $Y_t:=\{d=t\}$, $I_t:=\{d>t\}$, $O_t:=\{d<t\}$. Write ω_x for the fields attached to the site x, ω_X for the family over X, and $\mathcal{F}_X$ for the σ-algebra it generates. Note that $D_\chi\subset N_1(U)\subset\{d\le 1\}$ and that the sources are supported in $Q_b\subset\{d\ge 3\}$.

Lemma 7.17 (Markov layer)**.** *Let $t\in\varepsilon\mathbb{N}$ with $1+2\varepsilon\le t\le 3-2\varepsilon$. Let $\mathfrak{F}_t$ be the set of probability measures on the fields of the torus with a density proportional to $e^{-\mathcal{S}_I-\mathcal{G}}$, where $\mathcal{S}_I$ is the sum of the terms of the walled torus action of $\mathfrak{c}$ that involve a site of I_t, and $\mathcal{G}$ is a function of $(\omega_{Y_t},\omega_{O_t})$; killed sites in O_t are allowed. Then for every $P\in\mathfrak{F}_t$, $\mathcal{F}_{I_t}$ and $\mathcal{F}_{O_t}$ are conditionally independent given $\mathcal{F}_{Y_t}$, and the conditional law of ω_{I_t} given ω_{Y_t} is the same for all $P\in\mathfrak{F}_t$. In particular $\hat\Theta:=\mathbb{E}_P[\Theta\mid\mathcal{F}_{Y_t}]$, for $\Theta=\Theta^a(\mathrm{i}f,g)$, is the same function of ω_{Y_t} for every $P\in\mathfrak{F}_t$. The two-level states $\Psi_k(n,K)$, $K\in[0,\infty]$, all belong to $\mathfrak{F}_t$.*

Proof. Every term of King's action (Definition 2.4) involves the fields at the sites of a set of ℓ^∞-diameter at most ε: a single site, or a pair $x,x+\varepsilon e_\beta$. Since d takes values in $\varepsilon\mathbb{N}$ and is 1-Lipschitz, such a set cannot meet both I_t and O_t. Hence $\mathcal{S}_I$ is a function of $(\omega_{I_t},\omega_{Y_t})$ and the density factorises as $\rho_1(\omega_{I_t},\omega_{Y_t})\rho_2(\omega_{Y_t},\omega_{O_t})$, which gives the conditional independence and the common conditional law. The observable Θ depends on the fields at the sites of the plaquettes and sites of $\mathrm{supp}(f,g)\subset Q_b$, which lie in I_t because $t\le 3-2\varepsilon$. The two-level terms $K\chi$ and the kill live on $D_\chi\subset O_t$ because $t\ge 1+2\varepsilon$. □

For $K\in[0,\infty]$ let p_K be the law of ω_{Y_t} under the state $\Psi_k(n,K)$ (a density on a finite-dimensional space), and let $\mathsf{H}(p,q):=(\frac12\int(\sqrt{p}-\sqrt{q})^2)^{1/2}$ be the Hellinger distance.

Hypothesis 7.18 ((HEL): the Hellinger form)**.** For every $b>0$, $(f,g)\in\mathcal{K}$ supported in Q_b and $\eta>0$ there are a two-level configuration $(\mathfrak{s},\chi)$ for Q_b with $\tau(\mathfrak{s})\le\eta$, a layer t as in Lemma 7.17, and $K<\infty$ such that

$$\liminf_{k\to\infty}\ 2\,\mathsf{H}(p_K,p_\infty)\,\big(\mathbb{E}_K\hat\Theta^2+\mathbb{E}_\infty\hat\Theta^2\big)^{1/2}\le\eta.$$

Proposition 7.19 ((HEL) implies (RES))**.** *For every two-level configuration, layer t and $K<\infty$, at every k,*

$$\mathrm{tail}_k(K)\le 2\,\mathsf{H}(p_K,p_\infty)\big(\mathbb{E}_K\hat\Theta^2+\mathbb{E}_\infty\hat\Theta^2\big)^{1/2}\le 2\sqrt2\,\mathsf{H}(p_K,p_\infty)\,C_{D_{J'}}(2g)^{1/2},$$

where $C_{D_{J'}}(2g)=\sup_k Z^a_{\Lambda_k,D_{J'}}(0,2g)$, finite under $(\mathrm{K})_a$ by Lemma 4.2(d), and equal to 1 at $g=0$. Hence (HEL) *implies* $(\mathrm{RES})_a$*, and under $(\mathrm{K})_a$ it suffices that $\liminf_k\mathsf{H}(p_K,p_\infty)\le\eta/(2\sqrt2\,C_{D_{J'}}(2g)^{1/2})$.*

Proof. By Lemma 7.17, $\mathrm{tail}_k(K)=\mathbb{E}_K\Theta-\mathbb{E}_\infty\Theta=\int\hat\Theta\,(p_K-p_\infty)$ with one function $\hat\Theta$. Then

$$\begin{aligned}\Big|\int\hat\Theta(p_K-p_\infty)\Big|&\le\Big(\int\hat\Theta^2(\sqrt{p_K}+\sqrt{p_\infty})^2\Big)^{1/2}\Big(\int(\sqrt{p_K}-\sqrt{p_\infty})^2\Big)^{1/2}\\&\le\big(2(\mathbb{E}_K\hat\Theta^2+\mathbb{E}_\infty\hat\Theta^2)\big)^{1/2}\sqrt2\,\mathsf{H}(p_K,p_\infty),\end{aligned}$$

using $(\sqrt{p}+\sqrt{q})^2\le 2(p+q)$. By conditional Jensen, $\mathbb{E}_X\hat\Theta^2\le\mathbb{E}_X\Theta^2$, and $\Theta^2\le\Theta^a(0,2g)$ pointwise ($\cos^2\le 1$). By Lemma 2.12(c) and Lemma 3.3(a), removing the kill and the walls does not decrease $\langle\Theta^a(0,2g)\rangle$, so $\mathbb{E}_X\Theta^a(0,2g)\le Z^a_{\Lambda_k,D_{J'}}(0,2g)\le C_{D_{J'}}(2g)$. □

At every fixed k, $p_K\to p_\infty$ pointwise as $K\to\infty$ (Lemma 2.7), so $\mathsf{H}(p_K,p_\infty)\to 0$ by Scheffé's theorem; (HEL) is again an interchange of the limits $K\to\infty$ and $k\to\infty$. Conversely (RES) does

not imply (HEL): one observable against a distance. The following identity contains both (HEL) and the form in which a renormalisation-group proof would supply the hypothesis.

Proposition 7.20 (a general sufficient form)**.** *Let $(\mathfrak{s},\chi)$ be a two-level configuration, $K<\infty$, $\mathcal{B}$ any sub-σ-algebra of the fields, and $\mathcal{S}\in\mathcal{B}$. For $X\in\{K,\infty\}$ let ν_X be the restriction of the state $\Psi_k(n,X)$ to $\mathcal{B}$, $h_X:=\mathbb{E}_X[\Theta\mid\mathcal{B}]$ and $m_X:=\mathbb{E}_X[\Theta^a(0,2g)\mid\mathcal{B}]^{1/2}$, so that $|h_X|\le m_X$. Then*

$$\mathrm{tail}_k(K)=\int_{\mathcal{S}}(h_K-h_\infty)\,d\nu_\infty+\int_{\mathcal{S}}h_K\,d(\nu_K-\nu_\infty)+\int_{\mathcal{S}^c}h_K\,d\nu_K-\int_{\mathcal{S}^c}h_\infty\,d\nu_\infty,$$

and

$$\mathrm{tail}_k(K)\le\int_{\mathcal{S}}|h_K-h_\infty|\,d\nu_\infty+\int_{\mathcal{S}}|h_K|\,d|\nu_K-\nu_\infty|+C_{D_{J'}}(2g)^{1/2}\big(\nu_K(\mathcal{S}^c)^{1/2}+\nu_\infty(\mathcal{S}^c)^{1/2}\big).$$

With $\mathcal{B}=\mathcal{F}_{Y_t}$ and $\mathcal{S}$ the whole space this is Proposition 7.19; with $\mathcal{B}$ the σ-algebra of the block fields at a fixed physical scale and $\mathcal{S}$ a small-field set, it is the form in which a renormalisation-group comparison of the two states (a sup-norm comparison of their block-field densities and of their conditional expectations on $\mathcal{S}$, together with a large-field bound $\nu_X(\mathcal{S}^c)\le\delta$) implies (RES).

Proof. $\mathrm{tail}_k(K)=\int h_K\,d\nu_K-\int h_\infty\,d\nu_\infty$; split both integrals over $\mathcal{S}$ and $\mathcal{S}^c$ and add and subtract $\int_{\mathcal{S}}h_K\,d\nu_\infty$. For the bound, Cauchy–Schwarz gives $|\int_{\mathcal{S}^c}h_X\,d\nu_X|\le\nu_X(\mathcal{S}^c)^{1/2}(\int m_X^2\,d\nu_X)^{1/2}$ and $\int m_X^2\,d\nu_X=\mathbb{E}_X\Theta^a(0,2g)\le C_{D_{J'}}(2g)$ as in Proposition 7.19. In the Hellinger case $h_K=h_\infty$ by Lemma 7.17, and $\int|h_K|\,d|\nu_K-\nu_\infty|\le(\int h_K^2(\sqrt{p_K}+\sqrt{p_\infty})^2)^{1/2}(\int(\sqrt{p_K}-\sqrt{p_\infty})^2)^{1/2}$ is the bound of Proposition 7.19. □

8. What is known about the torus and heavy-shell hypotheses

8.1. The fixed-spacing statement and the shape of a counterexample.

Proposition 8.1 (fixed spacing)**.** *For every two-level configuration, every $(f,g)\in\mathcal{K}$ supported in Q_b and every k, $\mathrm{tail}_k(K)\to0$ as $K\to\infty$. Consequently, if* $(\mathrm{RES})_a$ *fails for some (f,g), η, then for every two-level configuration $(\mathfrak{s},\chi)$ for Q_b with $\tau(\mathfrak{s})\le\eta$ and every K, $\mathrm{tail}_k(K)>\eta$ for all large k, while $\mathrm{tail}_k(K)\le\eta$ for $K\ge K_*(k)<\infty$; hence $K_*(k)\to\infty$ as $k\to\infty$.*

Proof. The first statement is Lemma 7.4(a). The second is its contrapositive form: the failure of (RES) at $(\mathfrak{s},\chi)$ means $\liminf_k\mathrm{tail}_k(K)>\eta$ for every K, so $\mathrm{tail}_k(K)>\eta$ for all large k; and $K_*(k):=\inf\{K:\mathrm{tail}_k(K)\le\eta\}$ is finite by the first statement. If $K_*(k)\le K_0$ along a subsequence, then $\mathrm{tail}_k(K_0)\le\eta$ along it, contradicting the previous sentence. □

So any failure of the hypothesis is carried by shell matter whose mass squared tends to infinity with k: a lattice-scale effect that disappears at every fixed cutoff. Quantitatively, an elementary single-site computation (the conditional law of $|\phi(x)|^2$ given the neighbouring fields, with the local moment bound $\mathbb{E}r_x^{2j}\le C_j\varepsilon^{-2j}$ at every site of a BFS state of King's class) gives, for $|\Theta|\le1$, $\mathrm{tail}_k(K)\le Q_\varepsilon|U|/(\varepsilon^6K)$ for $K\ge K_*(\varepsilon)$ with $K_*(\varepsilon)=O(\varepsilon^{-2})$ and $Q_\varepsilon=O(\varepsilon^{-c})$; the threshold ε^{-2} is the inverse square of the spacing, that is the mass scale at which the shell matter is frozen at the lattice scale, and only the prefactor is lossy. We do not reproduce the computation ([DERIVED]). In the free field (Gaussian matter) the tail is explicit: it follows a continuum law of order $(K+n)^{-1/2}$ for $K\lesssim\varepsilon^{-2}$ and falls faster beyond, so that the continuum curve is an ε-uniform envelope approached from below ([DERIVED]; not used).

8.2. Why the wall, and why no soft argument.

Proposition 8.2 (no common lattice hyperplanes)**.** *Let $R=R_{\mu\nu}(\pm\theta_0)\in\mathcal{R}_0$ and let ρ be the third index.*

(a) *There is no lattice $P\subset\mathbb{R}^3$ of rank 3 with $RP=P$.*

(b) *An affine plane is a lattice hyperplane of both Λ_k and $R^{-1}\Lambda_k$ (that is, of the form $\{x_\alpha = c\}$ and of the form $\{(R^{-1}e_\beta)\cdot x = c'\}$) only if it is of the form $\{x_\rho = c\}$.*

(c) *Consequently no bounded region of $\mathbb{R}^3$ has all its faces on lattice hyperplanes of both lattices; the only unbounded regions with that property are the slabs $\{c_1 \le x_\rho \le c_2\}$. The bounded geometries to which both lattices descend are the tori $\mathbb{R}^3/P$ with $P \subset \Lambda_k \cap R^{-1}\Lambda_k$, for instance $P = 50MJ\,\mathbb{Z}^3$; by (a), no such P is R-invariant, so the rotated torus $R^{-1}D_J$ is never the torus D_J.*

Proof. (a) If $RP = P$, then in a $\mathbb{Z}$-basis of P the map R is an integer matrix, whose characteristic polynomial has integer coefficients; but the characteristic polynomial of R is $(x-1)(x^2 - \frac{8}{5}x + 1)$. (b) The normals are e_α and $R^{-1}e_\beta$; $e_\alpha = \pm R^{-1}e_\beta$ means $Re_\alpha = \pm e_\beta$, and for $\alpha \in \{\mu,\nu\}$ the vector Re_α has the components $\pm\frac{3}{5}, \pm\frac{4}{5}$ in the (μ,ν)-plane. (c) A bounded region with faces on hyperplanes needs face normals spanning $\mathbb{R}^3$; by (b) only e_ρ is available. Both lattices descend to $\mathbb{R}^3/P$ exactly when $P \subset \Lambda_k \cap R^{-1}\Lambda_k$, which contains $50MJ\mathbb{Z}^3$ by Lemma 2.2(b). □

The ultraviolet analysis of [K1, K2] needs boundaries that are lattice hyperplanes of the lattice being analysed ([K2, p. 326]: "These hold only when the boundary is a lattice hyperplane orthogonal to a lattice coordinate direction. Clearly the two lattices in Fig. 1 never share such a boundary. Our solution of this problem is to use periodic boundary conditions."). Proposition 8.2 makes this precise: the objects on which the two lattices can be compared by King's method are tori with smooth sources, and smooth walls are the only admissible way to localise a torus; a Dirichlet region cannot be aligned with both lattices. The correlation inequalities give exactly "cube ≤ walled torus" on the cone and never the reverse (Lemma 2.14; Remark 5.8), so the comparison "walled torus ≤ King's state $+\eta$" must be supplied by an estimate on King's measure; in the curvature sector that estimate is Theorem 7.12.

Remark 8.3 (the content of (RES)). (i) At every fixed spacing (RES) holds (Proposition 8.1); its content is uniformity in the spacing. (ii) It is a decoupling statement for a heavy, weakly coupled region: on the shell the dimensionless couplings are of order $\lambda/\sqrt{K}$ and $e^2/\sqrt{K}$, and the plateau mass n is free. The difficulty is not the shell but the comparison of two states that differ on the shell, seen through the interacting plateau and the light core, uniformly in the spacing. (iii) No correlation inequality bounds the tail from above: Lemma 2.12 gives signs and monotonicity, and the one comparison through a relative density that is uniform in ε is the Gaussian photon switch across a matter-free collar (Lemma 7.1); for the matter, the couplings across the boundary of the shell contribute to the logarithm of the relative density a term whose variance is of order $|\partial U|\varepsilon^{-2}$. (iv) In the curvature sector the estimate is proved as an implication (Theorem 7.12) by a renormalisation-group expansion of the walled model of [K1, K2] with the shell mass kept inside the matter covariance at every scale, run down to a stopping scale at which the plateau is massive in block units, the dependence on the shell mass being controlled through the κ-Cauchy property of the reduced small-field action on a test set that does not depend on κ; the four estimates (8)–(11) are the pieces of that expansion. No such expansion for the symmetric massive phase of the three-dimensional model is in the printed literature; Bałaban, Imbrie and Jaffe [BIJ] develop a multiscale expansion of the high-frequency modes of the abelian Higgs model in the Higgs regime and postpone the large-field and decoupling estimates. The proof carries a window $n \ge n_*$ for the plateau height, which costs nothing for the theorem since n is free. (v) At $e = 0$ the photon is independent of the matter, and (RES_0) holds trivially ($\mathrm{tail}_k \equiv 0$ at $g = 0$), while (RES) for $g > 0$ is a decoupling statement for the two-component φ^4_3 lattice model with a heavy shell. Nothing proved at $e = 0$ has been carried to $e > 0$ in this problem, since King's mass counterterm at order e^2 is negative and of order e^2/ε, so that every comparison of the $e > 0$ model with an $e = 0$ model at the bare level is a comparison with a model deep in its broken phase.

9. The inputs of the theorem, in plain words

Theorem A is an implication. This section states, in plain words and grouped by kind, every input of it that is not proved here: the steps completed in companion manuscripts, among them the inputs stated in Section 1.5, and the statements of the literature with their printed grade; the numbers are those of Section 1.4. A step of a printed proof that is asserted without a justification is called *flagged*; every flagged step on the chain is named below, and it is carried at the grade [SKETCH-LIT] unless one of the companion manuscripts completes it.

9.1. **The steps completed in companion manuscripts.** Three statements inside the proofs of the torus comparison, of King's cube limit and of the shell estimates are proved in companion manuscripts, each by more than one argument and each argument checked independently, (P7)–(P9) of Section 1.5. In plain words: (P7) says that on the skew tori of the corrected torus theorem, as on the cubic tori, the difference of Bałaban's propagators on two nested domains decays with the depth of its two points in the smaller domain; this is what makes his replacement of the propagators of the interaction terms at a non-zero background small. Its proof on the cubic tori goes over to the skew tori, since it uses no symmetry of the cubic lattice and its inputs hold there, and a second proof obtains it from an aligned cubic torus that covers the skew torus; only the torus comparison uses it. (P8) says that in King's cube limit the graphs whose trees are long, at the faces of the cube and away from them, are bounded at each of two cutoffs by the cut of the long lines, so that their change between the two cutoffs needs no rate; it is used by the cube limit, and so by the definition of King's state, and by the general-source extension. (P9) says that the transfer of the perturbative bound to Bałaban's propagator domains holds for the families of order fourteen that the removal of the tail of the background produces, as it does up to order thirteen; only the upper bound of the shell estimates uses it.

(DOM) of Section 1.5 is [PROVED] as an implication, by two independent arguments, each checked independently. In plain words: the shell estimates expand the determinant of each renormalisation step in the photon fluctuation field, on the torus and, in the upper bound, on the domains on which Bałaban defines his propagators; (DOM) says that the first-order term of that expansion, taken block by block, is bounded by the first differences of the background and of the fluctuation field, uniformly in the shell mass and with no bound on the size of the background, up to a term smaller than every power of the block size, at every block of the domain, its boundary included. The full determinant is invariant under every gauge transformation on the sites of the domain. The first argument uses this invariance at each point, together with a large gauge transformation of the background, and controls the resolvent at every point by an energy estimate; the second compares the coefficient at the blocks that carry the vertices, which lie deep inside the domain, with a coefficient on the torus at a regular field built from the background near the block. The torus comparison and King's cube limit are statements at zero shell mass, where the shell-mass residual of this term vanishes. The renormalisation bound for the operator with the scalar wall, the vacuum families with the mismatch vertex at King's backgrounds, Bałaban's enumeration read block by block at higher orders, the identification step at a non-zero scalar source, King's cut of the long lines on his cubes and the restricted renormalisation bound on his cubes for general sources are proved, in the forms (P1)–(P6) of Section 1.5, the identification step at a non-zero scalar source with the bound at long trees of Proposition 1.2.

The four statements of Section 1.6 that earlier versions carried as hypotheses are proved. In plain words: Proposition 1.2 says that on King's cube the graphs with block vertices, in the families that carry one inserted external field, are bounded at long trees, at each of two cutoffs, by the cut of the long lines, as the same graphs without the inserted field are by (P8); it is used by the ultraviolet construction of King's cube limit, and so by the definition of King's state. Proposition 1.3 says that Bałaban's and King's estimates of the fluctuation integrals, printed for King's in-block contours rooted at a corner of each block, hold for the contours rooted at the block

centres and for the composite contours of the hybrid flow, which the corrected torus theorem uses. Proposition 1.4 says that where the change of counterterm convention lowers the bare scalar mass on a walled torus, there is a mass in King's range whose bare mass squared lies below that of King's mass by more than this lowering, uniformly in the spacing, at every point of King's range except those at which the scalar mass minimises the bridge constants' function of the mass; it is used by the large-field bound at the wall, and at the exceptional points the invariance follows by right-continuity in the scalar mass (item (15)), proved in a companion manuscript from the monotonicity of the state in the bare mass coefficient and in the cube, the real-analyticity in the mass of the limiting difference of the bare mass coefficients, and a comparison of King's cube with the dead-shell torus at a fixed stopping scale, the endpoint step at $\kappa = \infty$ of [SW1] included. The final-scale cap says that the coarsest scale of the flow lies below a threshold fixed by the geometry, under which this large-field bound holds on every skew torus of the configurations used. The first enters King's cube limit; the other three enter the torus comparison, and the positivity condition also the walled limit (4). The step formerly open, the odd-block clause of the degree census of the shell model, is proved and enters the κ-Cauchy estimate (10) only.

(10d) is [PROVED] as an implication. In the κ-Cauchy estimate of [SW3] the κ-derivative of King's effective action is King's sum of graphs with one insertion of the shell mass. Consider an insertion on a line used as a leg, deep inside the shell, and the cluster of short lines around it. If the cluster is joined to the rest of the graph through a scalar leg or a scalar line, the heavy matter pays for it (the deep payment, proved as an implication and checked independently). If all its connections are photon lines or photon legs, the photon factors do not see the shell mass and the deep payment fails. The bound then treats the photon subfamily containing the insertion as a whole, with the insertion unsplit: the members whose matter component through the insertion is closed are grouped into classes with the same full lines, one gauge identity is applied to each class, and only then is each class expanded into King's members, each with its own background; its base layer is the first-moment bound (10g)(vi). The route through a photon Ward identity for each fixed ordering, or each fixed scale assignment, of King's graphs is closed: that identity is false, the violation being of order one in exact computations. The Ward identities that hold are those for full propagators and for Bałaban's resummed organisation, which sums over the scales first and keeps a convergent remainder. King states the per-ordering cancellation [K1, p. 666] by citation only.

(8g) is [PROVED] as an implication. The perturbative bound of [SW1] expands the effective action around the background and Taylor-expands the photon legs; the remainder of that step is controlled by the bound $|\Pi^{\kappa}_{j}(x,y)| \le C|x-y|^{-4}$, uniform in the shell mass, for the current two-point function of the step-j lines. That bound is a pointwise statement about the line kernels: it needs, besides the decay of the kernel itself, a first difference at one end with one more power of decay, and a mixed second difference, one difference at each end of one line, with two more powers, on both sides of the shell boundary and uniformly in the shell mass. These are interior kernel estimates, which the estimates of [SW1] for bounded sources do not imply; they are proved by two independent arguments on the torus, the second of which also gives the restricted form used on the Neumann sub-domains, with both points at distance at least one block from the complement of the sub-domain.

(8h) is not an input. On a sub-domain with a Neumann face, the interior estimates of [SW1] are extended up to the face by reflecting the solution, and with it the gauge field, across the face. The reflected field has zero curvature on the plaquettes straddling the face, but the regularity condition of [K1, (3.5)] bounds every first difference of the field, and the odd reflection of the normal component, set to 0 on the bonds crossing the face, has a first difference of size $|A_m|/\eta$ there; for a constant normal component the reflected field is therefore not regular. What is used

is regularity after a gauge transformation on the face layer, which a smoothed gauge supplies, and the coercivity of the doubled operator, which holds for every field.

(8j), the light-side estimates on sub-domains, holds at the deep points near which every cube of Bałaban's decomposition that meets ∂U is a flat-face cube. Near the edges and corners of H' that condition fails, and the light-side estimates are not used there: the localisation lemma of [LT] reduces Bałaban's local inverse on any Neumann union of blocks, at a point of depth at least 4ρ, to the torus inverse for the data near the point, with an error exponentially small in ρ, and this supplies the forms that the consumers use, the payment of the far blocks included, (8k)(vi).

Every other step of our own on the chain of Theorem A is proved, or proved as an implication, and checked independently; the steps that the companion manuscripts [CT, H6, SW1, SW3] print below the grade [PROVED] are completed in [SUP], the determinant clause of (8k)(ii) with (DOM), and the removal of the tail of the background on the histories with proper propagator domains with (P9). Grouped by content, with the labels of Section 1.4, they are:

- *The shell mass inside Bałaban's estimates.* The transfer of Bałaban's inputs at block factor 5 to the model with the shell mass inside the covariance, (8e) = (8i); the Hölder bound for the external fields required by [Ba3, (1.33)], (F3); the constants of Bałaban's completions at block factor 5 and the uniformity of his unqualified bounds in the scale index, (5l)(vii); the cumulant bound that replaces the cumulant lemma of Benfatto et al., on cubic and on skew tori, (5l)(i); and the reading of the wall as constant on each cube in the light-side estimates, (8k)(iv).
- *The perturbative bound on sub-domains and its cut members.* The transfer of its torus form to the propagator domains, with its norm form and the amplitude bounds there and with the regrouping of the cut families of Bałaban's classes by cut datum, into products of full classes, (8k)(i); the bound on the cut members of the resummed photon two-leg kernel, the mismatch clause at King's backgrounds and the determinant clause on the torus and on the propagator domains, with the bound (DOM) for its first-order term, (8k)(ii).
- *The random walk and the edges of the source box in the first shell estimate.* The gradient bound for the simple random walk, (8k)(iii); and the far-block payment, with a radius free of the large-block integer and near the edges of H', (8k)(vi).
- *The restricted renormalisation theorem and the one-loop term in the κ-Cauchy estimate.* The degree counts for the classes with block vertices, in which King's credit for photon fields contracted outside a subgraph is replaced by a split count, (5l)(x); the restricted form of [K1, (3.38)] for the shell model, the interpolated members included, (10g)(i); and the one-loop z^2-term with the block fluctuation line, in both regimes and with the tadpole in g', (10g)(ii).
- *The Ward structure in the κ-Cauchy estimate.* The first-moment bound for the two-photon kernel of full lines that replaces the regrouped Ward remainder in the shell model, (10g)(vi); the photon-leg classes with an insertion, (10g)(iv); the members linear in the source at the light local vertex, (10g)(vii); and the field-dependent members that this vertex may bring into the closed class of (10d), (10g)(viii).
- *Scale and position sums in the κ-Cauchy estimate.* The Hölder remainder for lines used as legs with an open matter component, (10g)(iii); the lattice regime of the scale sum of the Hölder-leg remainders, (10g)(v); the bookkeeping of the crossing and winding loops in (10a); and the dead-shell chain (10c).
- *The own steps of the corrected torus theorem.* The absorption in the hybrid lower bound, (5l)(ii); the hybrid-flow form of [Ba3, Prop. 1], (5l)(iii); the two repair steps, the determinants included, (5l)(v), (vi); the transfer of Bałaban's Green's-function bounds to the tori of [CT], read at the printed page, (5l)(iv); the two statements of (3), (5l)(viii), and the four

steps of lower weight, (5l)(ix); and the Taylor step (4a), which both (K_0) and ($TIGHT_0$) use through (4).

The statements of the literature that the chain uses and that are asserted in print without a printed argument, or read off a printed construction, are all proved in the forms consumed. Four of them are proved as implications at block factor 5, in [AS] and [CA], and checked independently: Bałaban's same-scale estimates (item (1), the form for the operator with the scalar wall, (P1), included, except the six readings below, and on the skew tori with the change of domain (P7), for the contour families of the flows of [CT] and at the walls with Propositions 1.3 and 1.4 of Section 1.6), King's cube theorems that define King's state (item (13), with the long trees (P8) and Proposition 1.2), Bałaban's all-orders enumeration with one inserted vertex (item (10f)), and King's statement that the remaining bounds of his graph analysis follow in the same way [K1, pp. 675–676] (item (5k)). Of the others, two readings of Bałaban's papers, (B1) and (B4) of Section 9.2, the bounds for the vector-field propagators [Ba3, p. 426] on the torus ((10g)(ii)) and the vector half of the completion of [Ba2, §3C] on p. 591 there ((5l)(vii), hence (K_0) and (8)), are [PROVED] as implications and checked independently [SUP]. The rest are likewise [PROVED] as implications and checked independently [SUP], in the forms consumed: two statements of King, the source-free strict degree property of his renormalised graphs as rewritten by Bałaban [K1, p. 668], in the componentwise form of [Ba3, (3.5)] (items (4), (5g), (13)), and his treatment of the sources [K1, pp. 664–665, 677], as the small-field half of his upper bound at fixed sources (items (4), (5f), (5k), and the large-field clause of (1) at the walls); the photon legs of the per-step constants of the field-independent part of the perturbative bound of the shell model, as Bałaban's external fields in his norm ((12e), hence (8)); and four readings of Bałaban's papers, (B2), (B3), (B5), (B6) of Section 9.2: the pointwise bounds for the massive lattice Green's function and its differences [Ba3, p. 437] ((13)), the local coefficient of the classes with two scalar legs and one vector leg at zero shell mass [Ba3, (3.33)–(3.38)] ((8k)(ii), (10g)(iii), (iv), (8e), (K_0)) with the first moment of the vacuum-polarisation bubble at zero shell mass with the wall [Ba3, (3.26)–(3.28), p. 441] ((10)), and two readings of the construction of [Ba2], the regularity of the uncut background next to each propagator domain and the way a bound on the scalar block field near the shell is carried to the next step ((8k)(i), (13), (8e), (K_0)); with them the closure of his renormalised classes under his completion moves [Ba3, pp. 417–432], behind the regrouping of the cut families ((8k)(i), (13), (8e), (K_0)). Apart from the flagged steps of Section 9.6, which are carried at the grades of their completions, no input of Theorem A is unproved.

9.2. **Bałaban's asserted estimates.** Item (1) is the reading that every estimate of [Ba1, Ba2, Ba3, Ba4] asserted without a printed argument holds as asserted, at blocking factor 5, with per-step constants independent of the initial spacing. The asserted steps on the chain are, in the order of the papers: in [Ba1], the identities (2.39)–(2.42) ("after easy calculations", p. 612), the lower bound (2.33), the deferred statements (3.20)–(3.21) and (3.25), the perturbative formula (3.34)–(3.36) ("close inspection of the procedures", p. 618), the bound $O(\varepsilon^{\kappa})|T_1|$ on p. 615, which needs the order of the expansion to be at least 6, and the use of the cumulant lemma of Benfatto et al. (pp. 616, 623); the frame proof of (1.14); in [Ba2], the replacement of A'' by A' (pp. 561–562), the estimates of unnecessary terms through [Ba1, Props. 2.1–2.3] (p. 568; Proposition 2.2 of [Ba1] has no proof), the analysis of $W^{(j)}$ (p. 578), the transfer of [Ba1, Prop. 3.1] (p. 579) and Proposition 2.6 (p. 580, stated without proof), the inductive hypotheses (3.5), (3.7)–(3.8), the error arithmetic of (2.78), and Lemma 2.7 with (2.113), whose proof is omitted; in [Ba3], Proposition 1 with (1.33) and its consequences on pp. 421–422 ("rather easy consequences", "completely obvious"), and the Hölder clause on p. 412 ("similar considerations lead to the desired bounds"); in [Ba4], the scale-uniform analyticity neighbourhood of Lemma 2.4 (p. 586), the transfer of Corollary 2.3 to the mixed operators and the bounds (1.15), (1.16), (1.20) as its consequences (p. 588; the domain-difference clause (1.20) is not argued), the step from (3.10) to (1.14) (p. 589), the change of domain (5.5)

(p. 594), and (1.8) on general domains, whose proof is not printed and is completed in [CT, SW1]. Behind King's theorems stand the same estimates: [K1, Thm. 3.1] is proved there by a reference to [Ba1, Ba2, Ba3, Ba4] and to its Appendix, and [K2, (4.2)] by the sentence that the results of [Ba1, Ba2, Ba3, Ba4] apply to both models (used only for the lower bound at zero wall, item (5b)). All of these steps are proved at block factor 5, in the forms consumed, as implications from Bałaban's printed proofs in [AS], and checked independently, the renormalisation bound for the operator with the scalar wall in the form (P1) of Section 1.5 included, except the six readings (B1)–(B6) at the end of this subsection, [PROVED] as implications and checked independently [SUP]; the cumulant lemma of Benfatto et al., whose printed form covers the nearest-neighbour Markov field only, is replaced in [SW1] by a cumulant bound with constants free of the shell mass (item (5l)(i)). The repairs at block factor 5 of four printed steps are proved in [CT] (items (5l)(v), (vi)), and their re-runs for the shell model are (8e). Lemma 2.7 of [Ba2] with (2.113) holds at block factor 5 on the tori of the construction; read on King's cube with the Dirichlet weight, its printed form fails near the Dirichlet face and at the first scale, and the corrected form that King's cube limit uses is proved in [CA].

The step whose proof was completed last is Proposition 2.6 of [Ba2] for the interaction terms at a non-zero background. In the form proved in [AS] it reads, in the notation of [Ba2]: *in the inductive step $k \geq 1$ of the upper bound* [Ba2, §2] *at block factor 5, with $t = 5^k\varepsilon$, let $\tilde{B}$ be the background field on $B^k(\Lambda_0^{(k)})$ of* [Ba2, p. 579]*, and consider the interaction terms of that step localised in $B^k(\Lambda_7^{(k)})$, every vertex summation being restricted to $\Lambda_7^{(k)}$; their scalar lines are $G_k(\Omega_1, \tilde{B})$, $\Omega_1 := B^{k-1}(\Lambda_2^{(k-1)})$, the domain of the statement* [Ba2, Prop. 2.6, p. 580]*, their external scalar fields are $a_k G_k(\Omega_1, \tilde{B})f$ with $\|f\|_\infty \leq Cp\,\lambda(t)^{-1/4}$, and their other external fields have norms at most $Cp^a\lambda(t)^{-b}$. Replace $G_k(\Omega_1, \tilde{B})$ by $G_k(\Omega_2, B^{(k+1),\eta})$, $\Omega_2 := B^k(\Lambda_2^{(k)})$, in every scalar line and in every such external field. Then for every $\nu > 0$ there are $C_\nu < \infty$ and $t_0(\nu) > 0$ such that, for $t \leq t_0(\nu)$, the sum over all localisations of the absolute values of all terms containing at least one difference of the two propagators is at most $C_\nu t^\nu |\Lambda_7^{(k)}|$, with constants independent of the initial spacing and of the background $\tilde{B}$, the case $\tilde{B} \neq 0$ included.* In the proof the domain Ω_1 is written $B^k(\Lambda_2^{(k-1)\prime})$, as in [Ba2, (2.56)]; this is the same set, by the definitions of the prime and of B in [Ba1, (1.16)–(1.18)], since $\Lambda_2^{(k-1)}$ is a union of large blocks. The proof regroups the terms with at least one difference into contractions of products of full classes of Bałaban's graphs of the cut components, to which the renormalisation bound [Ba3, (1.33)] at block factor 5 applies at every background; no Ward identity for a fixed ordering is used. The replacement in the remaining part of the step, after [Ba2, (2.107)], is proved without the renormalisation bound. This is the form on the cubic tori and on the domains of the shell estimates; on the skew tori of [CT], which the torus comparison also uses, the decay of the difference lines with their depth is (P7) of Section 1.5, proved there by two independent arguments, and with it the same form holds on the skew tori.

Readings of Bałaban's papers. Six readings of Bałaban's papers that the chain uses, (B1)–(B6), some of them carrying more than one statement, are asserted there without a printed argument, or are read off his construction, and are not covered by [AS]. All are [PROVED] as implications and checked independently [SUP], in the forms stated:

(B1) the bounds [Ba3, (2.10)–(2.12)] for the propagators of the vector field: "Of course the same inequalities hold for vector field propagators" [Ba3, p. 426]: [PROVED] as an implication and checked independently [SUP], on the periodic torus fitted by blocks, for the sup bound, one first difference per endpoint, the Hölder bound (2.11) with one difference and the contour legs, uniformly in the spacing, the scales and the scale mass; used in (10g)(ii) through the sup and first-difference bounds of the photon fluctuation field, in both regimes, and in (12e) through (KA*), whose norm [Ba3, (1.32)] carries the Hölder quotient;

(B2) the pointwise bounds for the massive nearest-neighbour Green's function $C^\xi = (-\Delta^\xi + 1)^{-1}$ on $\xi\mathbb{Z}^3$ and for the kernels built from it, $|C^\xi(y - y')| \le O(1)e^{-|y-y'|/2}/|y - y'|$ "and the corresponding inequalities for derivatives" [Ba3, p. 437], with their first two differences and their differences between two scales: [PROVED] as implications and checked independently [SUP], the kernel bound, the first two differences and the differences between two scales with an unspecified rate $c_0 > 0$ in the exponential (Bałaban's printed rate $\frac{1}{2}$ is not claimed and not used); used in the boundary analysis of King's cube limit (13) and in the cut (P5); the reading stays listed for the block-spin inequality that one boundary bound of (13) takes from the same page;

(B3) Bałaban's treatment of the local term of the primitively divergent classes with two scalar legs and one vector leg at zero shell mass [Ba3, (3.33)–(3.38), pp. 443–444]: the coefficient of the local term vanishes for the graphs built from C^ξ, by (3.37), (3.38), the sign pairing of (2.18) and translation invariance, after a replacement of the propagators that leaves "a convergent expression": [PROVED] on the torus, as an implication and checked independently [SUP], for the window coefficient with a bounded potential and for the block term at any window; the coefficient vanishes for the graphs built from C^ξ and is bounded, in general non-zero, for the window lines; and the first-moment step of the same kind for the vacuum-polarisation bubble [Ba3, (3.26)–(3.28), p. 441] at zero shell mass with the wall: the first moment of the part built from C^ξ vanishes as printed, by the symmetry (3.27)–(3.28), and the remainder carrying the wall is bounded, uniformly in the spacing, the scale, the torus and the wall, with constants depending on the wall height n through $m^2 + n$, [PROVED] as an implication and checked independently [SUP], by two routes. The mixed-class step is used by the bounds for the mixed classes in the shell model, inside (8k)(ii) and in (10g)(iii), (iv), and through them by (8e) = (8i), (P3) and, through the repaired cube step of (1), by (K_0); the vacuum-polarisation step by the κ-Cauchy estimate (10);

(B4) the estimate of the integral over the vector fields in the completion of [Ba2, §3C], which [Ba2, p. 591] obtains "in a way similar to (3.22)", read componentwise: [PROVED] as an implication and checked independently [SUP], from (3.21) as printed and the scalar bound (3.39), on the cubic and the skew tori and for the model with the shell mass inside the covariance, whose vector action is unchanged, with the vector part of the constant a number independent of the spacing, the scale, the history and the torus; it gives the vector half of the constants (5l)(vii) and enters (K_0) and (8) through them;

(B5) the regularity, in the sense of [Ba1, Prop. 2.1] and [Ba4, (1.7)], of the gauge background on the blocks of scale k within distance one of each propagator domain Ω of that scale. Two statements: on Ω itself, [Ba2, Lemma 2.3] in the form that holds at block factor 5 (the printed lemma fails there for small couplings), with the regularity of the cut field; and off Ω, on the k-blocks within distance one of it, the regularity of the uncut background $A^{(k)}$ of [Ba2, (2.44), (2.54)], which equals the Ω-background on Ω (p. 567 there); the cut field $\theta_k A^{(k)}$ is not regular on the slice of the cut and is not used off Ω. Both are [PROVED] as implications and checked independently [SUP], the second from [Ba2, (2.55), (2.7)–(2.8)], the decay of the vector propagator and the rows of (1) on the small-field restrictions, which are imposed on Λ_0 and on its neighbourhood of thickness $r(\varepsilon)$ [Ba2, p. 558, (2.24)–(2.25), (2.95)–(2.98)]; used by (8k)(i) and, through it, by (13), and by (8e) = (8i) and (K_0) through the bound for the mixed classes;

(B6) the way a bound on the scalar block field near the shell, imposed at one step, is carried to the next step: it is inserted before the characteristic functions of that step are bounded by 1, as [Ba2, (2.55)] carries the restrictions given by the characteristic functions to the next step [Ba2, p. 570]: [PROVED] as an implication and checked independently [SUP] for the

inherited cap (an insertion lemma and an inherited-cap lemma); the gradient line of [Ba2, (2.55)] is carried at steps $k \geq 2$ on the domain $\Lambda_2^{(k-1)\prime}$ of [Ba2], by a transport theorem whose constant c_1 is split into its supremum, vector and scalar parts, of which the scalar part depends on the couplings as in [Ba2, (2.21)]; the supremum, $|A|$ and $|\partial A|$ lines hold on $\Lambda_0^{(k-1)\prime}$, and the printed domain Λ'_{-1} of the gradient line is not used; used by (8k)(i) and, through it, by (13), and by (8e) = (8i) and (K_0) through the bound for the mixed classes.

The regrouping of the cut families. Earlier versions listed, as a seventh reading, Bałaban's renormalised classes [Ba3, pp. 417–432] read as the classes closed under his completion moves, the moves listed in (P1)(b): the subgraphs are read as sets of lines; the move that forms the classes with two vector legs [Ba3, p. 431] acts at every divergent part with two vector legs, not only at the maximal ones; and in the picture of [Ba3, p. 417], in which the legs of a subgraph are moved to an anchor, the anchor is fixed by a rule intrinsic to the subgraph. What the regrouping of the cut families along the orbits of the moves uses is [PROVED] as an implication and checked independently [SUP]: the per-label full classes are closed under the completion moves, and every closed family obeys the bound; no reading of the printed classes is needed. It is used by the regrouping in (8k)(i) and in (13), through (8k)(i) by (13), by (8e) = (8i) and (K_0) through the bound for the mixed classes, and by the general-source extension through the ultraviolet construction on King's cube. Bałaban's sentence that the divergent vacuum graphs are cancelled by the vacuum-energy counterterms [Ba3, p. 445] holds at zero background only; at King's backgrounds the vacuum families with the mismatch vertex are bounded by (P2) of Section 1.5, whose proof does not use it, and those without it by the perturbative bound (8k)(ii), the one-loop determinant of the step excepted, which enters through the determinant clause of (8k)(ii); a bound of the part of these families that depends on the background is not printed in [Ba3], and none is taken from there.

9.3. **King's asserted steps.** *King's cube limit* (13), which defines King's state: [K2, Thm. 2.1] at $h = 0$. Its proof [K2, §3.1] transfers the torus proof of [K1] to Neumann–Dirichlet cubes with written modifications and states that "all the results of [K1] go through" (p. 332). The analysis at the sharp boundary is printed for one counterterm graph ("the other graphs in δm^2 can be treated similarly"), and for the photon it proceeds one ordering of the renormalised graphs at a time (pp. 331–332), which in that reading rests on the per-ordering cancellation discussed below; the replacement (3.54) is omitted (p. 330); the Hölder continuity behind the factor $5^{-\alpha k}$ of (3.13) is not stated; and the image-sum identity (3.9), which the transfer uses, is false as printed together with (3.3) and (3.8) for the weight printed there, and holds with a Dirichlet weight that suffices for the transfer. The bound at a fixed cube used in [K2, (3.41)] is this theorem at real coefficients. These steps are proved, in the forms consumed and on King's cubes C_J^K, as implications in [CA], checked independently, with the bound at long trees of Proposition 1.2 (item (13)): the reduction of the long trees of the localisations on the cube is the cut (P5), with (P8) of Section 1.5 for the long trees in the difference clause, and the ultraviolet construction on the cube uses the sub-domain form (8k)(i) of the perturbative bound at zero shell mass; there the photon boundary step uses the zeroth moment of the full photon kernel of the cube, with no identity for a fixed ordering.

King's renormalisation estimates (item (5k)). [K1, Thm. 3.1] is proved by a reference to [Ba1, Ba2, Ba3, Ba4] and to the Appendix of [K1]. [K1, Prop. 3.6] is stated on p. 662 and its proof (pp. 663–665) ends in an assertion; its one-flow instance is proved as an implication (5g), from the strict degree property of King's renormalised graphs as rewritten by Bałaban, [Ba3, (3.5)], [K1, (3.78)], which King asserts on p. 668 ("(3.38) relies solely on the cancellation of divergences") and which is [PROVED] as an implication, in the componentwise form of [Ba3, (3.5)], and checked independently [SUP]; a reading of the print finds [K1, (3.78)] false on certain vacuum graphs, whose contribution is a removed constant and is not used. [K1, Thm. 3.5(i)] is false as printed and is

replaced by [RS] (5h); [K1, Thm. 3.4], false as printed, is not used, the form needed at a fixed wall being proved as an implication in [H6]. The propagator bounds [K1, Props. 3.7–3.10, Lemma 4.5, (3.72), (3.88)] and the remainder bounds of [K1, Thm. 3.3] enter [CT, H6, RS] through (5k): [K1, Prop. 3.7] is a printed sketch ("follows immediately from Theorem 3.3 and the scaling"); in the proofs of [K1, Props. 3.8, 3.9] only the last line of (3.71) and the first line of (3.73) are printed, the derivative line of (3.73), the contour form (3.74) and the Hölder forms (3.75) being covered by "the other bounds follow in the same way" (p. 675); [K1, Lemma 4.5] rests on [Ba4, §5] and on the step (4.36)–(4.37), which is false as printed (the block average is treated there as a Fourier multiplier), while its conclusion (4.33) holds (Proposition A.4); and the bound [K1, (4.21)] is false as printed for $m > 0$ and holds in a corrected form that is the one used. These bounds are proved in the forms consumed in [AS], checked independently; an extension of [K1, Prop. 3.7] beyond its printed scope is false and is used nowhere.

The vacuum polarisation. King states on [K1, p. 666] that the divergences cancel for each ordering of the renormalised graphs, with a reference to [Ba3], whose identities (2.24)–(2.27) are for full propagators and whose organisation sums over the scales first and keeps a convergent remainder. In its per-ordering reading the statement is false, the violation being of order one in exact computations; it is used nowhere on the chain: King's photon boundary step for (13) [K2, pp. 331–332], which works one ordering at a time, is replaced in [CA] by the zeroth moment of the full photon kernel of the cube. Four places in the companion manuscripts state a vacuum-polarisation identity per ordering or per scale: the ordering-restricted corollary for King's estimate in [CT], the vacuum-polarisation step of the strict degree property in [H6], the Ward identities with one insertion in [SW3], and the vanishing zeroth moment of the combined photon two-leg kernel at each scale in [SW1]. In [CT], [H6] and [SW3] the statement consumed is Bałaban's resummed form, and [H6] says so; in [SW1] the form consumed is not stated there, which is the flagged step (F5)(c); it is the window identity of (8k)(ii) for full lines.

King's Appendix and sources. The expansion to order 13 with (3.54) "still sufficient" (p. 676) is replaced for the shell model by [SW1, Lemma appclass] (8b) and is proved in [AS] elsewhere (items (5k), (12e)). The photon legs of the per-step constants of the field-independent part of the perturbative bound of the shell model [SW1, Prop. TD(d)] are handled without the moment bounds of the Appendix [K1, (A.2)–(A.6)]: the photon slice lines and King's photon legs are Bałaban's external fields in the norm [Ba3, (1.32)], (KA*), [PROVED] as an implication and checked independently [SUP] (item (12e)). The inclusion of sources ("straightforward", p. 677), the componentwise replacement of the sources (p. 665) and the uniformity of King's constants over the source family (pp. 664–665) are asserted in print; they are used in [H6] (item (4)), in (5f), in Remark 2.9 and in the large-field clause of (1) at the walls (Section 1.6), and they are [PROVED] as implications and checked independently [SUP] in the forms consumed: the inclusion of sources as the small-field half of [K1, Thm. 3.1] at every fixed source of bounded size, in King's vertex form, for King's flow and for the centred flow, and for the hybrid flow its lower half; the componentwise replacement by a cube-side argument that covers the consumed instances; and the uniformity over the source family through a log-convexity of the walled partition function in a constant scalar source, so that the consumers need King's bound at four fixed sources only. The photon clause of [K1, Thm. 3.3] (p. 656, "of course with $A = 0$") is a specialisation of two lines on the torus that is not written in print; it is used in (5a) and in [K1, (A.5)–(A.6)] (item (12e)) and is proved in [AS] (item (5k)). *King's near-face analysis* [K2, §5.2, (5.41)–(5.49)] is a one-page sketch; its transplant is (8c).

9.4. **Reisz, Brydges, Fröhlich and Seiler.** Reisz's theorems are used in the corrected three-dimensional form (5e): the transfer to other dimensions, which [Re1, p. 86] states can easily be made, is written and checked independently, and the naturalness clause, false as printed, is replaced by a condition proved for the graphs consumed. The forest identity [Re2, Lemma 5.1], whose proof there is a pointer to Zimmermann's paper, is Zimmermann's forest formula, cited

by name (Section 9.7). The membership of the subtracted integrands in Reisz's class, asserted at [Re2, (5-1)] and on p. 97 there, is proved in Remark 9.1. The reversed correlation inequalities of [BFS, Thm. 6.2(3)–(4)], sketched in print, and King's [K2, (3.28)], stated without proof, are proved in Lemma 2.12(d).

Remark 9.1 (membership in Reisz's class). In the notation of [Re1, Defs. 2.2–2.5], $\mathcal{C}_m$ is the set of functions $V = a^{-m}F(xa)$ of the momenta x with F smooth, $\mathcal{C}_m^c \subset \mathcal{C}_m$ the subset on which $\lim_{a\to 0} V$ exists for every x, $\mathcal{C}^c$ the finite sums, and Reisz's class consists of the quotients V/C with $V = \sum_i P_i(\mu)V_i$, $V_i \in \mathcal{C}^c_{m_i}$, P_i polynomials in the masses, and $C = \prod_i (e_i(l_i;a) + \mu_i^2)$, $l_i \neq 0$ linear in the loop momenta, $e_i = a^{-2}\eta_i(l_i a) \in \mathcal{C}_2^c$ with η_i periodic, positive off $2\pi\mathbb{Z}^3$, and $\lim e_i = l_i^2$. *Claim:* for the graphs consumed in [CT] and the operators used there (the evaluation of external momenta at 0, the first-order Taylor operator with $P(q;a) = a^{-1}\sin(qa)$, and derivatives in the masses and in the external momenta), every forest term of the subtracted integrand, and each of its derivatives, lies in Reisz's class and is $2\pi/a$-periodic in every loop momentum, the loop momenta of each subgraph being chosen among those of the graph as in [Re2, (2-24)]. *Proof.* (i) $a^{-m}F(xa)\,a^{-n}G(xa) = a^{-(m+n)}(FG)(xa)$, and limits multiply. $\partial_j[a^{-m}F(xa)] = a^{-(m-1)}(\partial_j F)(xa)$; if $\lim_a a^{-m}F(xa)$ exists for every x and $m \geq 1$, the Taylor terms of F at 0 of order less than m vanish (the least non-vanishing one would make $a^{-m}F(xa)$ diverge), so $a^{-(m-1)}(\partial_j F)(xa)$ converges to the derivative of the term of order m; for $m \leq 0$ it tends to 0. Evaluation at $q = 0$ and integer linear substitutions of the momenta preserve $\mathcal{C}_m^c$; $a^{-1}\sin(qa) \in \mathcal{C}_1^c$. (ii) A derivative of a propagator factor is $\partial(e+\mu^2)^{-1} = -(\partial e)(e+\mu^2)^{-2}$ with $\partial e \in \mathcal{C}_1^c$ by (i), and $\partial_{\mu^2}(e+\mu^2)^{-1} = -(e+\mu^2)^{-2}$; so derivatives of products of propagators have numerators in $\mathcal{C}^c$ over products of the same factors. (iii) On a subgraph γ, the Taylor operators turn a factor $e(k^\gamma+p)+\mu^2$ into $e(k^\gamma)+\mu^2$, which has the required form with $k^\gamma \neq 0$ as a linear form, since γ is one-particle irreducible (a line of γ carrying no loop momentum of γ would be a bridge); the numerators stay in $\mathcal{C}^c$ by (i). (iv) The line momenta have integer coefficients in the loop momenta, and η_i and sin are periodic, so every factor is $2\pi/a$-periodic in each loop momentum; $e + \mu^2 \geq \mu^2 > 0$. (v) $V_1/C_1 + V_2/C_2 = (V_1C_2 + V_2C_1)/(C_1C_2)$, where C_1C_2 is again a product of admissible factors and C_1, C_2 expand into elements of $\mathcal{C}^c$ with coefficients polynomial in the masses, by (i); induction over the finitely many forests gives the claim. □

9.5. **Derived statements.** The items that the companion manuscripts [CT, H6, RS, SW1, SW3], in the versions cited, grade [DERIVED] are written arguments, complete at the level of a careful sketch with every step named. On the chain of Theorem A each of them is proved as an implication and checked independently, in that manuscript or in [SUP], or is not used; the grade given in Section 1.4 is the grade so reached. The statements graded [DERIVED] in this paper are not used by Theorem A: the two sub-steps of (3) that no consumer uses and the quantitative fixed-spacing bounds of Section 8.

9.6. **Flagged steps of the companion manuscripts.** The following steps of [CT, H6, SW1, SW3] are written there without a complete justification. None of them is known to be false in the form in which the chain consumes it, and for each the completion is stated with its grade; the items that contain them are graded "given" them (Section 1.3).

(F1) In [CT], the decay bound $|\Gamma^F(z,w)| \leq C(1+|z-w|)^{-1}e^{-cm_k|z-w|}$ for the coarse covariance $\Gamma^F = H^F B^F H^{F*}$ (the kernel B^F conjugated by the external-line kernels H^F; the bound is checked for the conjugated kernel, the one the chain consumes, not for B^F alone) is derived from the Paley–Wiener theorem, which gives a constant depending on m_k; the product form does not follow, as written, from the minimum of the polynomial and the exponential bound, and the analyticity of the inverse symbol in the strip rests on a lower bound for its real part that is stated without proof. The bound holds in the product form, uniformly in m_k: the kernel is written as the difference of the free propagator and its fine

part, and the product bound for the free lattice propagator is obtained by a contour shift and an integration by parts, with no use of the Paley–Wiener theorem or of the strip bound ([PROVED] as an implication, checked independently; item (5j)).

(F2) In the skew-torus and boundary-box re-run of Bałaban's steps behind (1) and (3) in [CT], which [SW1] also uses for its $(\mathrm{BAL})_5$: at the boundary boxes the background is not constant near the Neumann faces, while [Ba4, Lemma 2.2] assumes it constant there, and the open-ended clause of [Ba1, Prop. 2.1] ("for some simple sets") is used for them; the cumulant lemma of Benfatto et al. is applied directly there; and the hybrid kernels need their own domain-difference identity, since [Ba4, (3.9)–(3.10)] cover a change of pinning only. Completions: no step of the re-run needs [Ba4, Lemma 2.2] on a boundary box at a non-constant background, and the boundary clause is re-stated in the restricted form that is proved; the cumulant lemma is replaced by the cumulant bound of [SW1], whose lattice hypotheses hold on the skew torus ([PROVED]); the transfer of the rest of its proof to the skew torus is [PROVED] as an implication, and so is that proof itself (item (5l)(i)); the domain-difference identity for the hybrid kernels is [PROVED]. The sign misprint of [Ba2, (2.99)] does not matter, since $W^{(j)}$ is defined there by "=:" and only $C(\tilde{B}+\tilde{A})^{-1} = C(\tilde{B})^{-1} - W^{(j)}$ is used, never a term of $W^{(j)}$ (items (1), (3), (8)).

(F3) In the shell estimates, the Hölder clause of [Ba3, p. 412] is completed by a derivation in which the restriction radius is at least Bałaban's localisation radius ([PROVED] as an implication, checked independently, with Bałaban's smooth cutoffs taken in an admissible form whose second differences are bounded uniformly in the scale; items (1), (8), and (5l)(iii), whose proof uses it).

(F4) In [H6], the Montel–Vitali step from the strip bounds needs their uniformity in the scale. It holds: $|r_K(z)| \le r_K(\mathrm{Re}\, z) \le \exp[C_G(\mathrm{Re}\, z)^2]$ for the walled moment functions r_K, with C_G the supremum over K of half the variance of $F(G)$ under the free photon, finite by the convergence of the lattice variances; this bound is part of the strip-bound lemma of [H6], so the step is complete (item (4)).

(F5) In [SW1]: (a) the clause of the sup bound with block data in L^1 is false uniformly in the spacing; every consumer uses bounded or block-L^2 data, the form that the printed proof proves ([PROVED] as an implication, checked independently). (b) The pointwise bound $|\Pi_j^{\kappa}(x,y)| \le C|x-y|^{-4}$ for the photon two-leg kernel is attributed to the sup and first-difference clauses of the bounded-source estimate, which do not give pointwise kernel bounds; the kernel bounds it needs are of Kato and Combes–Thomas type, and they are (8g), [PROVED] as an implication. (c) The vanishing zeroth moment of the combined photon two-leg kernel is stated at each scale, and the text does not say that the resummed form of Section 9.3 is the one consumed. Completion: the vanishing is used only for the window class of a divergent subgraph with two photon legs, all its internal lines summed over the scales below the window, through an exact window identity whose cut members are bounded in (8k)(ii); no identity for a fixed ordering or scale assignment is used ([PROVED] as an implication, checked independently). (d) The sup form of the light-side bounds at distance $\ge M$ from the faces of a sub-domain rests on a localisation statement that claims King's [K1, (3.7)] at all points of a union of large blocks, which fails near re-entrant edges; its restricted form, at distance at least Bałaban's radius from the faces, holds; whether the sub-domains used satisfy the corresponding face condition need not be decided, since where it fails, near the edges and corners of H', the localisation lemma of (8j) supplies the forms that the consumers use, the payment of the far blocks (8k)(vi) included ([PROVED] as an implication, checked independently). (e) The use of the external field of the Appendix class in the proof of the finite-domain perturbative bound is asserted to match (item (8b)). (f) Citations: the uniform exponent $\frac{1}{4} - \beta$ is attributed to [Ba1, (3.58)], which prints

$\frac{1}{2} - \beta$, while Bałaban prints $\frac{1}{4}$ for one vertex [Ba2, (2.21)–(2.22)] and asserts the uniform exponent at [Ba3, p. 421] with the count written for a few vertices only; its uniform form over all vertex families is [PROVED] as an implication, checked independently, its input from [SW1] being the norm form of the perturbative bound ((8k)(ii)). [BFS, Thm. 2.3] is cited where [BFS, §3.3] is meant, with an extension of three lines to a site-dependent mass that is not printed ([PROVED]: the random-walk representation of the printed proof of the diamagnetic inequality holds for every non-negative site-diagonal mass). [K1, (A.3)–(A.6)] are photon identities, cited there for the matter background; they carry no load, the matter content at those places being [Ba2, Lemma 2.4] (item (8k)). The constants of [Ba4, Lemma 2.2, Prop. 2.3] depend on the block factor, which is harmless at the fixed block factor 5. The image sums on the torus do not improve the kernel bounds for the massless kernel in three dimensions, where the bound is obtained by lifting to the infinite lattice (item (8)).

(F6) In [SW3], in the face term of the tail theorem: the ballot bound for the outer pieces of a walk is used without its uniformity in the starting distance, which fails in general and holds for a start next to the half-line, the case used; the edges of a lifted coordinate walk above the axis are identified with its visits to U, which holds as two inequalities; and the moment bound $(Cq)^{2q}$ for the martingale part is attributed to Burkholder's inequality with constant $O(q)$, which gives only $(Cq)^{3q}$, while the stated bound holds by an exponential-supermartingale estimate for martingales with predictable increment bounds. Each completion is [PROVED] by one derivation with one independent check, and the exponent c in (10a), which (10) uses only through $\tau(\kappa) \to 0$, is unchanged (item (10a)).

The configuration produced by Theorem 7.10 must satisfy at once the class (6), the block alignment of [SW1, SW3] and the provisos of [H6] on the torus size and the wall. This is an argument of this paper, [PROVED] in Remark 5.6: the provisos of [H6] are the clauses of the class (6), the wall condition is linear in the height at a fixed torus, and the torus, the block alignment and the grid offset of the corners of H' are choices made in Theorem 7.12.

9.7. **Textbook results.** Cited by name, with their exact hypotheses checked against the use: the local limit law and the Green's function asymptotics of the simple random walk [LL], the theorems of Montel and Vitali, the two-constants theorem, Kato's inequality, the Combes–Thomas estimate, the Paley–Wiener theorem (whose use in [CT] is the flagged step (F1)), the Chung–Feller theorem, the discrete Nash inequality and the heat-kernel bounds of Carlen, Kusuoka and Stroock and of Davies, the Neumann eigenvalues of the discrete cube, and Zimmermann's forest formula. Burkholder's inequality is cited in [SW3] for a bound that it does not give in the form stated; this is the flagged step (F6). The multiple-reflection bound of Glimm and Jaffe behind [K2, (3.43)] is not consumed: at $h = 0$ that bound is trivial.

9.8. **The general-source extension.** Theorem A concerns curvature sources. Its extension to scalar sources with $\int h \neq 0$ goes through the scheme hypothesis ($\mathrm{W}^\sharp$) of Section 10, which is [PROVED] as an implication on King's cubes, its part ($\mathrm{W}^\sharp 1$) on the domain $D(h)$ defined there; it is not an input of Theorem A, and it has no hypothesis of its own beyond those of Theorem A; it uses the bound at long trees of Proposition 1.2, at zero scalar source through (13) and at a non-zero scalar source through (P4). In plain words, the rate (SRC) of Section 10, proved in companion manuscripts, says that the rate at which the graphs with block vertices change between two cutoffs, which (P8) gives when no scalar source is present, holds also when the graphs carry a vertex of a non-zero scalar source, at the localisations whose tree is short.

10. The general-source extension

For scalar sources with $\int h \neq 0$ King's printed Wick ordering has no finite continuum limit beyond the curvature sector (Appendix A.3), and the object is King's state in an admissible scheme (Section 3). For the scheme $\Theta^\sharp$ the hypothesis that carries the extension is the following form of $(\mathrm{W})_a$ on the cone.

Hypothesis 10.1 ($(\mathrm{W}^\sharp)$)**.** ($\mathrm{W}^\sharp 1$) For every King cube C_J^K (Definition 2.15), every finite family of real 2-forms $f_j \in C_c^\infty$, sampled at plaquette centres in the lattice's own frame, and of scalar sources h_j, all supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, and every real (z, w), the limit $\lim_k Z_{k,J}^{a^\sharp,K}(\sum_j z_j f_j, \sum_j w_j h_j)$ of the functionals of C_J^K exists in $(0, \infty)$, locally uniformly in complex (z, w).

($\mathrm{W}^\sharp 2$) For every real $0 \leq q \in C_c^\infty$ with $\sup q < s_*(e)$, $\sup_{J \geq J_q} \limsup_k Z^{a^\sharp}_{\Lambda_k, C_J \cap \Lambda_k}(0, q) < \infty$; here $s_*(e) > 0$ is a threshold fixed by the bulk bounds of the scheme, positive at every point of King's range outside the set E_W of Proposition 1.4, the graph of one scalar mass over the couplings (e, λ, μ_0).

With $(\mathrm{W}^\sharp)$ in place of $(\mathrm{W})_a$, and with $(\mathrm{K})_a$ and $(\mathrm{TIGHT})_a$ for scalar sources, the arguments of Sections 3–6 apply on the corresponding cone of sources, run on King's cubes through Lemma 2.16 as in Proposition 3.11; by Lemma 2.16, ($\mathrm{W}^\sharp 2$) is the same statement for King's cubes and for the boxes. None of these hypotheses enters Theorem A, and this section records only the status of $(\mathrm{W}^\sharp)$.

For a family as in ($\mathrm{W}^\sharp 1$) split each $h_j = h_j^+ - h_j^-$ as in the proof of Theorem 3.8(iii), with coefficients $(z, w^+, w^-) \in \mathbb{C}^{3N}$ of the curvature source $\sum_j z_j f_j$ and of the scalar source $\sum_j (w_j^+ h_j^+ + w_j^- h_j^-)$, let $k_1 := \sum_j [(\mathrm{Re}\, w_j^+)_+ h_j^+ + (\mathrm{Re}\, w_j^-)_+ h_j^-]$ be the positive part of the split scalar source at the real parts of the coefficients, and let $D(h)$ be the set of coefficients with $\sup 2k_1 < s_*(e)$: the positive part of the split source is below half the threshold of ($\mathrm{W}^\sharp 2$), and its negative part is arbitrary.

Status: ($\mathrm{W}^\sharp 2$) is [PROVED] *as an implication, and ($\mathrm{W}^\sharp 1$) on $D(h)$, that is, the existence of the limit of the functionals of C_J^K locally uniformly on $D(h)$, in $(0, \infty)$ at its real points, is* [PROVED] *as an implication on King's cubes with the bound at long trees of Proposition 1.2, through (13) and (P4); the status has no hypothesis beyond those of Theorem A, and none of its own.* ($\mathrm{W}^\sharp 1$) at the points outside $D(h)$, where the positive part of the split source exceeds half the threshold, is not proved: it would need the large-field half of the cube bound above the threshold. The inputs are statements that are proved as implications and checked independently (the bulk bounds of the scheme at orders 1 and λ, the two bulk bounds with their repaired display, the averaging theorem for both weights, the threshold proposition on the cone, the degree count with the merged counterterm vertices and the ordering-restricted rewriting on the torus, the cube pieces, the classes with curvature and scalar vertices on the cube, with the cut (P5) and (P8) for their long trees, the nesting of King's cubes, the ultraviolet construction on King's cube with a scalar source of either sign, which like (13) is a statement at zero shell mass, the identification step at a non-zero scalar source in the form (P4) of Section 1.5, and a cube-free convention bridge, which gives ($\mathrm{W}^\sharp 2$) in the displayed lim sup form with no theorem on any cube), together with the following:

(i) the all-orders bound for the lattice Feynman integrals of the counterterm family in Reisz's corrected form, with the restriction to integrands with no further subtraction, a vertex of order $(1, 1)$ that an earlier count omitted, and the uniformity of the constants on a compact mass interval, which is not printed: [PROVED] as an implication and checked independently;

(ii) for ($\mathrm{W}^\sharp 1$), the transfer of the restricted form of [K1, (3.39)] from the torus to King's cubes: its photon boundary step is [PROVED] as an implication and checked independently, through the zeroth moment of the full photon kernel of the cube and with no Ward identity for a

fixed ordering; its matter part, in the forms used, is (P6) of Section 1.5, whose difference form rests on the rate (SRC) below for its members with block vertices;

(iii) one lemma in the proof of the averaging theorem, for the tails of the difference of a pair of contour kernels: [PROVED] as an implication and checked independently; a further clause, with the scheme constant depending on the photon mass, is written but not checked independently and is not part of ($W^\sharp$);

(iv) the named literature: (1) and the cube theorems (13), [PROVED] as implications in [AS, CA], (13) with Proposition 1.2, with the readings (B1)–(B6) of Bałaban's papers and the statements of King that they use [PROVED] as implications in the forms consumed; Reisz's lemmas used in [Re1, Re2] beyond the corrected theorems, read at the statement only; and Zimmermann's forest formula (well established, cited by name).

Proposition 10.2 ((SRC): the rate of the split degree count at a non-zero scalar source; [SUP])**.** *On King's cube C_J^K with the flow of weight f_D, the free-boundary δm_ε^2 and the free Wick constant, let $\varepsilon = 5^{-K}$, $\delta_k := 5^{k-K}$, $0 \le k \le K - J_*$ and $p(\delta)$ be as in (P5), and let $h \neq 0$ be a real scalar source as in (P6), with any curvature source. The difference clause of the split degree count of (P8) holds at the source h for the classes with block vertices that contain a vertex of h, with or without curvature vertices, at the localisations whose tree has length at most $p(\delta_k)$ in units of the blocks of scale k: the change of the sum of such a class over these localisations between the cutoffs (K, k) and $(K+1, k+1)$ of a diagonal pair is at most $C[5^{-\gamma' k}\delta_k^{-\beta'} + \delta_k^{\sigma'}]|C_J|$ on King's small-field set, with $\gamma', \beta', \sigma' > 0$ and C independent of k and K.*

[PROVED] as an implication in companion manuscripts [SUP], by three proofs, each checked independently: one runs the argument of (P8) for short trees with the vertices of h among the source vertices, one splits off the source as a frozen field, and one bounds each localisation on one box of the interior zone with King's replacement of the sources [K1, p. 665]. It is stated for the range of scales of (P5), and it carries the part of (P8) for long trees that covers a scalar source and, for the mixed classes, the strict degree property named in (4), [PROVED] as an implication. At zero scalar source the clause is (P8). These localisations are interior in the sense of (P6). (SRC) enters the difference form (F-diff) of (P6) for its members with block vertices and the classes with curvature and scalar vertices at the same localisations, and through them the general-source cube bounds at the given and at the heavier mass, hence ($W^\sharp 1$); it does not enter Theorem A, whose cube chain is at zero scalar source, nor (F-one) of (P6), which uses no rate.

11. Grades of all inputs

The table lists every input of the theorem with its grade (Section 1.3), its source, and the hypotheses on which the grade rests; the item numbers are those of Section 1.4.

Input	Grade	Source	Rests on
Proved inputs and readings (Sections 1.5, 9.2, 9.3)			

Input	Grade	Source	Rests on
(DOM) the first-order term of the step determinants, composed over the earlier steps, on the propagator domains of Bałaban's upper bound and on the torus, uniformly in the shell mass and with no bound on $\|B\|$: (a) the per-site form at every block; (b) the current in L^1 at the blocks of the vertices; (c) the torus	PROVED as an implication, by two independent arguments (each checked independently); used by the determinant clause of (8k)(ii)	[SUP]; [Ba2, (2.86), (2.98), (2.104)–(2.107)]; [Ba3, p. 412]; [Ba4, (1.8)]	(1); for (b) also (8g), (8k)(iii), (F3), (8d), (P3)
(P1) the renormalisation bound for the operator with the scalar wall, (a) at every regular background for Bałaban's families, (b) at zero background for the node-closed classes; (P2) the vacuum families with the mismatch vertex at King's backgrounds, on the torus and on the propagator domains; (P3) Bałaban's enumeration block by block at higher orders; (P4) the identification step at a non-zero scalar source	PROVED as implications (checked independently), (P4) with Proposition 1.2 for its families with block vertices at long trees; (P2), (P3) through the cube step of (8e), carrying the readings (B3), (B5), (B6) on which (8e) rests, PROVED as implications; the difference form of (P2) on the histories with proper propagator domains through (P9)	[AS] for (P1); [SUP] for (P2)–(P4); [Ba3, Prop. 1, Prop. 2.2, §§2–3]	(1), (8d), (8e), (B3); (P2) also (P9); (P4) also Proposition 1.2
(P5) King's cut of the long lines on King's cubes; (P6) the restricted renormalisation bound on King's cubes for general sources	PROVED as implications (checked independently), (P6) in the forms (F-one), (F-diff), the latter for its members with block vertices with the rate (SRC); (P5) on the torus in [AS]; its use at the long trees in the difference clause of (13) is (P8)	[CA] for (P5); [SUP] for (P6); [K1, (3.38), p. 666]; [RS]	(B2); (P6) also (P5), (P8), and (SRC) for (F-diff)

Input	Grade	Source	Rests on
(P7) the change of domain of Bałaban's propagators on the skew tori: decay of the difference with the depth of its two points; its inputs, Bałaban's bounds for the Green's functions and the block operators and the scope statements of his construction, on the skew tori	PROVED as an implication on the skew tori by two independent arguments (each checked independently): the transfer of the proof on the cubic tori, which uses no symmetry of the cubic lattice, and a reduction to the aligned cubic torus that covers the skew torus; PROVED as an implication on the cubic tori and on the domains of the shell estimates; enters (1) on the skew tori, hence (K_0), only	[AS]; [CT]; [Ba2, Prop. 2.6]; [Ba4, §2]	(5l)(iv)
(P8) the long trees in the difference clause of the split degree count on King's cubes, at the faces and away from them, at each of two cutoffs	PROVED as an implication (checked independently), the long trees by three proofs, each checked independently, one of which covers also a scalar source; enters (13), hence the object and ($TIGHT_0$), and the general-source extension	[CA]; [K1, (3.41)]	(P5), (5l)(x)
(P9) the sub-domain form (8k)(i) of the perturbative bound at order 14, for the families of the removal of the tail of the background, with and without the mismatch vertex	PROVED as an implication by two arguments (each checked independently); enters the upper bound of (8) only, hence (RES_0) and ($TIGHT_0$)	[SUP]; [SW1]; [Ba2, (2.89)–(2.90)]	(8k)(i), (8e), (8d)

Input	Grade	Source	Rests on
(B1)–(B6) readings of Bałaban's papers: the vector-field propagators on the torus; the massive lattice Green's function and its differences (unspecified rate $c_0 > 0$); the local coefficient of the classes with two scalar legs and one vector leg (torus, window coefficient with a bounded potential, block term at any window) and the first moment of the vacuum-polarisation bubble at zero shell mass with the wall (constants depending on the wall height); the vector part of the completion of [Ba2, §3C]; the regularity of the gauge background on the propagator domain ([Ba2, Lemma 2.3] at block factor 5) and of the uncut background within distance one of it; the carrying of the scalar block-field bound to the next step (inherited cap; gradient line at $k \geq 2$ on $\Lambda_2^{(k-1)\prime}$); and the closure of his per-label classes under the completion moves behind the regrouping of the cut families	PROVED as implications (checked independently)	[SUP]; [Ba3, pp. 426, 417–432, 437, 441, 443–444]; [Ba2, pp. 558, 562, 567, 570, 591, (2.21), (2.55)]	(1), (8), (10)
(KA*) the photon slice lines and King's photon legs as Bałaban's external fields in the norm [Ba3, (1.32)], for the per-step constants of the field-independent part of the perturbative bound of the shell model	PROVED as an implication from the bounds for the vector-field propagators on the periodic torus, the reading (B1), itself PROVED as an implication (checked independently), componentwise and uniformly in the spacing, the scale and the scale mass, in that norm (checked independently)	[SUP]; [SW1, Prop. TD(d)]; [Ba3, (1.32), p. 426]	(B1)
The four statements formerly carried as hypotheses, and the step formerly open (Section 1.6)			

Input	Grade	Source	Rests on
Block vertices at long trees, on King's cubes, in the families with one inserted external field (Proposition 1.2)	PROVED as an implication (checked independently): (P8)'s argument with the three rows the inserted field requires; enters (13), hence the object, (TIGHT$_0$), and (P4)	[SUP]; [K1, (3.41)]	(P5), (P8)
The fluctuation estimates of [Ba1, Ba2, Ba3] and [K1, Thm. 3.1] for the in-block contours rooted at the block centres and for the composite contours of the hybrid flow (Proposition 1.3)	PROVED as an implication (checked independently), by a census of the occurrences of the contours and of the operators built from them in the printed proofs and, for the hybrid family, a direct proof of [Ba2, (2.65)–(2.66)]; enters (K$_0$) only	[SUP]; [Ba1, (2.1)–(2.3)]; [K1, (2.12)]; [Ba4, p. 573]	(1)
The positivity condition on the constant Δ_T of the convention bridge, on the three pairs of a lattice and a torus, with M_1 in King's range (Proposition 1.4)	PROVED as an implication (checked independently; three arguments) at every point of King's range outside the set E_W; at E_W Theorem A by right-continuity of King's state in the scalar mass (item (15)), PROVED as an implication (checked independently); enters (1) at a non-zero wall, hence (H3), (4), (K$_0$) and (TIGHT$_0$)	[SUP]; [K1, (3.32)–(3.33)]	(1), (5f)
The final-scale cap $\varepsilon_0 \le \varepsilon_{\mathrm{sys}}(M, R, \rho, \delta_0)$ on the coarsest scale, J-free and closed downward, under which the large-field clause of (1) at the wall holds on the skew torus for every J of the class (6)	PROVED (checked independently); a proviso of the torus theorem on its parameters, met by the class (6); not a hypothesis of Theorem A; enters (H3) on D_J and (SK), hence (K$_0$)	[SUP]; [Ba1, (1.14)]; [Ba2, (1.3)]; Remark 5.6	–

Input	Grade	Source	Rests on
The odd-block clause of the degree census of the shell model, inside the κ-Cauchy estimate (item (10))	PROVED as an implication (checked independently); enters (10) only, hence (RES_0) and (TIGHT_0)	[SUP]; [SW3]; Section 1.6	–
The object and the closure			
Geometry: $5R$ integral, descent to D_J, $\lvert Rx\rvert_\infty \le \frac{7}{5}\lvert x\rvert_\infty$, density of G_0 (Lemma 2.2)	PROVED	here; [K2, pp. 337–338]	–
Correlation inequalities (Lemma 2.12)	(a)–(c) PROVED-LIT; (d) PROVED here (sketched in [BFS]; [K2, (3.28)] stated without proof); the killing clauses PROVED here, with no limit	[BFS, Thms. 6.1, 6.2, pp. 275–281]; [K2, Prop. 3.1, (3.24)–(3.29)]; here	–
Current representation, Gaussian domination (Lemma 2.13)	PROVED	here	–
Killing as a limit of masses (Lemma 2.7); cutting lemma (Lemma 2.14); passive covariance (Lemma 2.10)	PROVED	here; [K2, Thm. 3.2]	–
King's cubes and the boxes: interlacing, monotonicity in J, iterated limits (Lemma 2.16, Corollary 2.17)	PROVED	here; [G]; [K2, (3.3)–(3.9), pp. 328–334]	–
King's cube limit with curvature sources on King's cubes C_J^K, own-frame 2-form reading (item (13))	PROVED as an implication, at zero shell mass, the long trees by (P5) and (P8), with Proposition 1.2; (B2), (B5), (B6) and the regrouping of the cut families PROVED as implications; printed grade SKETCH-LIT (proof transferred from [K1] by assertion; the per-ordering photon boundary step and (3.9) flagged)	[CA]; [K2, Thm. 2.1, §3.1, pp. 331–332, (3.9)]; [K1, Thm. 3.1, Prop. 3.6, p. 665]	(1), (5k), (10f), (8k), (P5), (P8), Proposition 1.2

Input	Grade	Source	Rests on
King's state on curvature sources, [K2, Thms. 2.2, 2.3(iii)] and the parts of [K2, (3.40)–(3.44)] used, at $h = 0$, and the continuity lemma (Theorem 3.8(i)–(iv), Proposition 3.11, Lemma 3.7), on King's cubes	PROVED given (13)	here; [K2, Thm. 3.2]	(13), Lemma 2.16
Closure from the generators to $O(3)$, reflection (Lemma 6.2, Theorem 6.3)	PROVED (at $g = 0$ on King's cubes, given (13))	here; [G]	(13), Lemma 2.16
Translations (Theorem 3.8(v) at $h = 0$; item (14))	PROVED given (13)	here	(13)
Right-continuity of King's state in the scalar mass at the points of E_W (item (15)): monotonicity in the bare mass coefficient ([BFS, Thm. 6.1], [K2, (3.25)]) and in the cube ([K2, Thm. 3.2]), real-analyticity of the limiting difference of the bare mass coefficients, the cube against the dead-shell torus at a fixed stopping scale with the endpoint step of [SW1]	PROVED as an implication (checked independently); used for Theorem A at E_W only, for the $E(3)$ clause as for the rotations	[SUP]; [BFS, Thm. 6.1]; [K2, (3.25), Thm. 3.2]; [SW1]	(13), (8b), (12a)
Admissible scheme $(\mathrm{W})_a$ (Hypothesis 3.4)	OPEN; scalar sources only, not a hypothesis of Theorem A	Remark 3.6	–
$(\mathrm{W}^\sharp)$ on the cone (Hypothesis 10.1), the general-source extension	$(\mathrm{W}^\sharp 2)$ PROVED as an implication; $(\mathrm{W}^\sharp 1)$ PROVED as an implication on King's cubes on the domain $D(h)$, with Proposition 1.2, and not proved outside $D(h)$; no hypothesis of its own; not a hypothesis of Theorem A	Section 10	(1), (13), (P4)–(P6), (P8), (SRC), Proposition 1.2
(SRC) the difference clause of the split degree count at a non-zero scalar source, at the localisations with short trees (Proposition 10.2)	PROVED as an implication, by three proofs, each checked independently; used by the general-source extension only	[SUP]; Section 10; [K1, p. 665]	(P5), (P8)
The torus comparison (K_0)			
Involution S (Lemma 4.3); closing at one spacing (Theorem 4.5); Corollary 4.6	PROVED as implications from (F_0), (V)	here	–

Input	Grade	Source	Rests on
(F_0), (V) and the existence on the axis lattice for walled curvature ratios	supplied by Theorem 4.9	[CT]	(1)–(5)
(1) $(BAL)_5$ with the large-field clause of [K1, (3.4)], the walled form (P1) included	PROVED as an implication (printed grade SKETCH-LIT), the readings (B1)–(B6) PROVED; on the skew tori the replacement of the propagators of [Ba2, Prop. 2.6] through (P7); the repaired cube step through the cube step of (8e) at zero shell mass, with (B3), (B5), (B6) and the regrouping of the cut families, PROVED; its block-factor-5 repairs PROVED, (5l)(v), (vi) included; for the contour families of the flows of [CT] with Proposition 1.3; the large-field clause at a non-zero wall PROVED as an implication with $c_* < 1$, with Proposition 1.4 outside E_W, for the rotated lattice also Proposition 1.3, on the skew torus under the final-scale cap, with King's p. 677 sentence in its PROVED form (5f)	[AS]; [SUP]; [Ba1, Ba2, Ba3, Ba4]; [CT]	(8e); (P7) on the skew tori; Propositions 1.3, 1.4
(2) ($H1^*$): one weakened element, two flows, five sectors	PROVED as an implication	[CT]; [Ba3, Prop. 1]; [K1, p. 665]	(2a), (2b), (5)
(2a) first-layer slice-kernel bound (cubic-Hermite extension)	PROVED	[CT]	–
(2b) restricted contraction rule; local-term chain under it	PROVED as an implication	[CT]; [SUP]	–
(3) (B_Δ): hybrid block-spin kernels	PROVED given (F2), (5l)(viii), (ix) included; two DERIVED sub-steps, not consumed	[CT]; [K1, Thm. 3.3]	–

Input	Grade	Source	Rests on
(H3), (SK): King's upper bound, skew instances	PROVED as implications from (1), (5k) (RD_J) and relative to (1) (D_J), with (5l)(i); [SKEW-U] with (P7); with Propositions 1.3, 1.4 (outside E_W), on D_J under the final-scale cap of the class (6); posed in the Cameron–Martin form of the effective action, reached from King's vertex form by a PROVED bridge (5i)	[CT]; [K1, Thm. 3.1]	(1), (5f), (5k), (P7), Propositions 1.3, 1.4
(H4): hybrid lower bound (item (5i))	PROVED as an implication from (1), (3), (CT), [SKEW-L], with its own steps (5l)(i)–(iii), the lower half of (5f) for the hybrid flow and the bridge of (5i), with Proposition 1.3	[CT]; compare [K2, Prop. 4.1]	(1), (3), (5f), Proposition 1.3
(H5): polynomial lower bound, at $h = 0$ and at the wall	PROVED as an implication ($h = 0$, RD_J, item (5b), under (1); D_J; wall); printed grade of (5b) SKETCH-LIT	[Ba1, (1.14)]; [AS]; [CT]	(1)
(CT): convention bridge	(a) PROVED; (b) PROVED as an implication in the consumed form; (c) not an input	[CT]; [K1, (3.33), (2.8), (3.21), (3.35)]; [K2, p. 330]	(5f); its rate (4c) through (4) only
(4) (H6W): existence of the walled value on the axis lattice (the limit input of Theorem 7.10)	PROVED as an implication, (4a)–(4d) included, with Proposition 1.4 outside E_W through the large-field clause of (1) at the wall (not used at E_W); King's strict degree property and his treatment of the sources, which it uses, PROVED as implications	[H6]; [SUP]; [RS]; [K1, Thm. 2.1, Prop. 3.6, pp. 664–665, 668, 677]; Remark 7.11	(1), (5), (5k), Proposition 1.4

Input	Grade	Source	Rests on
(5a), (5c), (5e) block-spin operators, counterterm conditions and Ward identities, Reisz in corrected form	(5a), (5c) PROVED-LIT, the three asserted steps of (5a) PROVED as implications under (1), (5k); for the transfer of [Ba4, §§2–4] see (5l)(iv); (5e) PROVED as an implication from the printed proofs, checked independently (forest identity = Zimmermann's formula, cited; membership PROVED, Remark 9.1)	[K1, Thm. 3.3, (3.32)–(3.33)]; [Ba4, Lemmas 2.1, 2.2, 2.4]; [Ba3, (2.18)–(2.28), Prop. 2.1]; [Re1, Re2]; [CT]	(1), (5k)
(5b) [Ba1, (1.14)]; (5f) source-shift uniformity of King's constant	(5b) PROVED as an implication under (1); (5f) PROVED as an implication under (1), (5k), King's p. 677 sentence included, in the form of the small-field half of [K1, Thm. 3.1] at fixed sources (King's and the centred flow; lower half for the hybrid flow); printed grade SKETCH-LIT	[Ba1, (1.14)]; [K1, Thm. 3.1]; [AS]; [SUP]	(1); (5f) also (5k)
(5d) local limit law	textbook, cited	[LL, Chap. 2]	–
(5g) [K1, Prop. 3.6], one-flow instance	PROVED as an implication (its printed proof ends in an assertion); its degree count PROVED: the strict degree property in the componentwise form of [Ba3, (3.5)] for King's flow, the unweakened form ($D \geq 0$, bounded zero-degree count) for the hybrid flow; the strict form $D' > 0$ not consumed	[CT, H6]; [SUP]; [Ba3, (3.5)]; [K1, pp. 662–665]	(4), (5k)
(5h) [K1, Thm. 3.5(i)]: FALSE as printed; restricted form	PROVED as an implication	[RS]	(4b)

Input	Grade	Source	Rests on
(5i), (5j) hybrid lower bound; free-propagator bound, Bałaban-side completions, coarse-covariance decay	PROVED as implications (the last flagged, (F1), with its completion)	[CT]; [SUP]	(1)
(5k) King's asserted steps: [K1, Thm. 3.1]; [K1, Props. 3.7–3.10, Lemma 4.5, (3.72), (3.88)] and the remainder bounds of [K1, Thm. 3.3]; its photon clause; the source readings	PROVED as implications in the forms consumed, the source readings included; printed grade SKETCH-LIT; (4.33) PROVED, Proposition A.4	[AS]; [K1, pp. 656, 664–677]; here	(1)
(5l) the own steps of [CT]: the cumulant bound on cubic and skew tori; the absorption in (H4); [Ba3, Prop. 1] for the hybrid flow; the transfer of [Ba4, §§2–4], read at the printed page; two repair steps, the determinants included; the constants of [Ba2, §3C] at block factor 5; two statements of (3); four steps of lower weight; the degree counts for block-vertex classes	PROVED as implications (checked independently); the vector half of (vii) with the reading (B4), PROVED	[CT]; [SUP]; [Ba2, Ba3, Ba4]	(1), (3)
(K_0): the torus comparison, curvature sector, on the class (6)	PROVED as an implication (closing implication and own steps (5l) checked independently), with (P7) through (1) on the skew tori, with Propositions 1.3, 1.4, outside E_W, under the final-scale cap of the class (6); its literature inputs at the printed grade SKETCH-LIT all PROVED as implications: (B4), King's two statements and, through the end-functional bound of (8e) in the cube step of (1), (B3), (B5), (B6)	Theorem 4.9	(1)–(5), (P7), Propositions 1.3, 1.4
$(K)_a$: all sources	PROVED as an implication in [CT] from (H1)–(H5) and ($C^\sharp$); ($C^\sharp$) OPEN; outside Theorem A	[CT]	–

Input	Grade	Source	Rests on
Walled values, free direction, orientation independence at $g = 0$ (Definition 5.1, Lemmas 5.2, 5.4)	PROVED given (K_0)	here	(K_0)
Theorem 6.1 at $g = 0$, Theorem 6.4, Corollary 6.5	PROVED as implications, on King's cubes	here	(K_0), ($TIGHT_0$), (13), Lemma 2.16
Necessity of an input beyond the inequalities (Remark 5.8)	PROVED	here	–
The closing hypothesis ($TIGHT_0$)			
Photon boundary switch (Lemma 7.1)	PROVED	here; [LL, Chap. 4]	–
(SH) ⇒ (TIGHT) and the dead-shell bound (7.1) (Theorem 7.6)	PROVED as an implication	here	(K_0)
Two-level sandwich (Lemma 7.7); (SH) ⇒ (RES) (Proposition 7.9)	PROVED	here; [K2, (3.25)]	–
(RES_0) ⇒ ($TIGHT_0$) (Theorem 7.10)	PROVED as an implication	here	(K_0), (10c)
(7) the heavy-shell theorem: (8)–(11) ⇒ (RES_0) at one configuration (Theorem 7.12)	PROVED as an implication	here (Section 7.4)	(8)–(11)
(8) $(SW1^\circ)_{int}$: two-sided bounds on the capped stopping-scale integral, uniformly in $\kappa \geq \kappa_0$, for the free-boundary model	PROVED as an implication given (F2), (F3), (F5) (checked independently in full), the determinant clause of (8k)(ii) with (DOM), its upper bound on the histories with proper propagator domains with (P9); the photon legs of its per-step constants by (KA^*), PROVED	[SW1]; [SUP]	(8a)–(8k), (1), (12), (P1)–(P3), (P9), (DOM), (KA^*)
(8a) scale-free sub-domain bound; the earlier form FALSE at small c_0	PROVED as an implication	[SW1]	–
(8f) counterterm offset $(CT)_{sh}$	PROVED as an implication	[SW1, Lemma M(d)]	Δ_*

Input	Grade	Source	Rests on
(8b) the short extensions written in [SW1]; (8c) the near-face sketch transplanted; (8e) the shell-model re-runs of the repaired steps	PROVED as implications (DERIVED in [SW1]; checked independently), (8e) with the mixed-class bound and (8k)(i), and with (B3), (B5), (B6) and the regrouping of the cut families, PROVED; the input [Ba2, Lemma 2.7] of (8c) PROVED under (1)	[SW1]; [SUP]; [K2, §5.2]	(1); (8e) also (P1), (8k)(i), (B3)
(8d) Bałaban's enumeration as a frame	PROVED as an implication (printed grade SKETCH-LIT)	[AS]; [Ba3, §§2–3]; [SW1]	(1)
(8g) (K-pt): pointwise first and mixed second difference bounds of the step-j line kernels	PROVED as an implication (two arguments on the torus, checked independently; on the sub-domains in the restricted form used there)	[SW1, Lemma ward(b), Prop. TD(b)]	(8k)(iii)
(8h) regularity of the reflected gauge field in the sense of [K1, (3.5)]: not an input	FALSE as printed; the form used (regular after a gauge; coercivity of the doubled operator for every field) PROVED (checked independently)	[SW1, Lemma refl(a),(b)]	–
(8i) $(\mathrm{BAL})_5^{\kappa}$: the clause of (1) for the shell model (= (8e))	PROVED as an implication, all clauses (checked independently), as (8e); its steps of the lower and the upper bound use (DOM) through (8k)(ii); on the histories of the upper bound with proper propagator domains its removal of the tail of the background with (P9)	[SW1]; [SUP]	(1), (P1), (8k)(i), (P9)

Input	Grade	Source	Rests on
(8j) (LOC-R): light-side estimates on sub-domains at deep points where (L_x) holds; near the edges and corners of H' the localisation lemma, (8k)(vi) included	PROVED as implications (checked independently)	[LOC, Prop. global]; [LT]; [SW1]	(8k)(iv)
(8k) the further steps of [SW1]: the perturbative bound on sub-domains; the cut members and the mismatch clause; the random-walk gradient bound; the wall reading in [LOC]; the far-block payment	PROVED as implications (checked independently); the determinant clause of (ii) PROVED on the torus and on the propagator domains, with (DOM); (i) for the M of Definition 2.1 and the corners of H' in (6), with (B5), (B6) and the regrouping of the cut families, PROVED, at order at most 13 (at order 14 it is (P9)); (ii) with (P1)–(P3), and (B3), PROVED	[SW1]; [LOC]; [SUP]	(5l)(i), (vii), (F3), (P1)–(P3), (DOM)
(9) ($\mathrm{SW2}^{\min}$): the disc bound, at $g = 0$	PROVED (matter factor 1)	here (Lemma 2.13; Theorem 7.12, Step 0)	–
(10) $(\mathrm{SW3}^{\circ})_{\mathrm{int}}$: κ-Cauchy property of the reduced small-field action	PROVED as an implication given (F6) (checked independently in full)	[SW3]	(10a), (10b), (10d)–(10g), (12); (8g), (8i)–(8k) through [SW1]
(10a) tail of the block-free determinant factor, torus form with the winding bracket	PROVED as an implication; its winding bookkeeping PROVED as an implication; the form without the winding bracket FALSE on the torus	[SW3]	–
(10b) heavy-kernel bounds: kernel, mass, far pairs, heavy far field, inserted line, differentiated light estimate, near pairs in the dist^{-2} form	PROVED as implications; the Yukawa heavy-pair clause and the full-covariance sup bound next to ∂U FALSE, not used	[SW3]	(8)
(10c) dead-shell chain for $U = T \setminus H'$	PROVED as an implication (checked independently)	here (Theorem 7.6)	–

Input	Grade	Source	Rests on
(10d) the κ-derivative bound for deep clusters closed in matter	PROVED as an implication (checked independently)	[SW3, Lemma Σ(iv′)(d)]	(8g), (8i), (10g)(i), (vi), (5l)(x), (10f)
(10e) the derived inputs of [SW3]; (10f) Bałaban's enumeration with one inserted vertex	(10e) PROVED as implications (DERIVED in [SW3]); (10f) PROVED as an implication (printed grade SKETCH-LIT)	[SW3]; [AS]; [Ba3, §§2–3]	(1)
(10g) the further steps of [SW3]: the restricted [K1, (3.38)] for the shell model; the one-loop z^2-term with the block fluctuation line; the Hölder remainder with an open matter component; the photon-leg classes with an insertion; the lattice regime of the Hölder-leg scale sum; the first-moment bound in place of the regrouped Ward remainder; the source-linear members at the light local vertex; field-dependent members there	PROVED as implications (checked independently); (ii) uses [Ba3, p. 426], the reading (B1), PROVED; (iii), (iv) use (B3), PROVED	[SW3]; [SUP]; [Ba3, pp. 426, 441]	(1), (8)
(11) (SWT): threshold clause	PROVED (empty on the capped object)	[SW1, SW3]	–
(12) literature inputs of the shell estimates	(12a) PROVED-LIT; (12b) not an input (Proposition 3.11); (12c), (12d), (12e) printed grade SKETCH-LIT, PROVED as implications through (1), (5k), (8b), (8c), and (13) for (12d) on King's cubes; King's p. 677 sentence, (KA*) in (12e) and (B1)–(B6) in (12c) PROVED as implications	[K1, K2, BFS, Ba1, Ba2, Ba3, Ba4]; [AS]	(1), (5k); (13) only for (12b) and for (12d) on King's cubes, not used by (8)–(11)
(RES$_0$) at one configuration	PROVED as an implication, through (8)–(11), the upper bound of (8) with (P9)	Theorem 7.12	(8)–(11)

Input	Grade	Source	Rests on
(TIGHT$_0$)	PROVED as an implication, through (K$_0$), (13) and (8), with the statements of the literature at their printed grade PROVED-LIT, outside E_W	Theorems 7.12, 7.10	(K$_0$), (8)–(12), (13)
Continuum form: existence of $\Psi^0(n,K)$, monotonicity, equivalence (Theorem 7.15); cube sandwich (Proposition 7.14)	PROVED given (K)$_a$ (the sandwich of (d) at $g \neq 0$: also (W)$_a$); not used	here	(K)$_a$
(E$_\infty$): existence of the dead-shell torus value	SKETCH-LIT; not used	[K2, §5]	–
Markov layer (Lemma 7.17); (HEL) $\Rightarrow$ (RES) (Proposition 7.19); general identity (Proposition 7.20)	PROVED; not used	here	–
Fixed-spacing statement, shape of a counterexample (Proposition 8.1)	PROVED	here	–
No common lattice hyperplanes (Proposition 8.2)	PROVED	here	–
(SH)$_a$, (RES)$_a$ at $g > 0$, (RES^0), (HEL), (TIGHT)$_a$ at $g > 0$	OPEN; not hypotheses of Theorem A	Hypotheses 7.5, 7.8, 7.16, 7.18, 5.5	–
Appendix			
Large-field term in [K1, Thm. 3.1] (Proposition A.1)	PROVED	here	–
King's torus conjecture, scalar reading (Proposition A.2)	PROVED at $e = 0$: FALSE as printed	here; [K2, (2.12), Thm. 2.4]	–
The Wick logarithm (Proposition A.3)	PROVED in [CT]; computation sketched here	here; [CT]	–
King's bound (4.33) (Proposition A.4)	PROVED	here; [K1, (4.5), p. 674]	–
Membership in Reisz's class (Remark 9.1)	PROVED	here; [Re1, Defs. 2.2–2.5]; [Re2, (2-24), (5-1)]	–

In summary. Every implication in the chain

$$(1)\text{–}(5) \;\Rightarrow\; (\mathrm{K}_0), \qquad (8)\text{–}(11) \;\Rightarrow\; (\mathrm{RES}_0) \xrightarrow{(\mathrm{K}_0),\ (10c)} (\mathrm{TIGHT}_0),$$

$$(\mathrm{K}_0) + (\mathrm{TIGHT}_0) \;\Rightarrow\; \text{invariance under } \mathcal{R}_0 \;\Rightarrow\; O(3) \xrightarrow{(14)} E(3)$$

is proved: Theorem A is proved as an implication from the inputs of the list of Section 1.4, at every point of King's range, at the points of E_W by right-continuity in the scalar mass. These inputs are proved here, or proved as implications in companion manuscripts and checked independently, the bound (DOM) of Section 1.5 for the step determinants on the propagator domains and the statements (P7)–(P9) behind item (1) on the skew tori, King's cube limit (13) and the upper bound

of (8) included, or they are statements of the literature cited at their printed grade [PROVED-LIT]; the flagged steps (F1)–(F6) of the companion manuscripts are carried at the grades of their completions (Section 9.6). The four statements that earlier versions carried as hypotheses, behind King's cube limit (13), the contour families of the flows of [CT] and the large-field clause of (1) at the walls, are proved (Section 1.6), and so is the odd-block clause of the degree census of the shell model, inside the κ-Cauchy estimate (10): no step of the chain is unproved. The theorem of this paper is the implication from these statements of the literature; they are assumed, not asserted. Section 9 states every input in plain words.

Appendix A. Errata to King's papers

We list the points in [K1, K2] that affect the use of King's theorems in this paper. None of them changes the statement of King's cube limit at $h = 0$ that we use, item (13), which is carried at its own grade (Section 9).

A.1. **The large-field term in Theorem 3.1 of** [K1]**.** Theorem 3.1 of [K1] compares the lattice generating functional on a torus T with the small-field integral $I^{(k)} := \int (dA_k)(d\phi_k)\chi_k \exp[-S^{(k),1}]$ of the effective action after k block-spin steps, $\eta := 5^k \varepsilon_K$. As printed on p. 655 the upper bound reads

$$Z^{\varepsilon_K}(T_{\varepsilon_K}, g, h) \le \int (dA_k)(d\phi_k)\chi_k \exp\big[- S^{(k),1} + C\eta^\sigma |T|\big] + \exp\big[- p(\eta)^2 + C|T|\big], \qquad (3.4)$$

with $p(\eta) = b_0(1 + \log \eta^{-1})^p$, followed by "Once again, C depends on g, h and $|T|$". *Convention P.* The exponent p of the small-field function is a fixed integer, $p = 6$ throughout; any fixed integer $p \ge 4$ would do. [K1, (3.41)] prints "for any σ, since $p > 1$", King's standing inequality on p; $p \ge 2$ is forced by [Ba1, (3.54)] (kept by [K1, p. 676]) and $p \ge 4$ by [Ba2, pp. 557, 594] ("$p > 2$"; "$2p \ge (d+1)r$", $r \ge 2$, $d = 3$). The consumers of the large-field clause use p only through $p(\delta_k)^2$ growing at least quadratically in k; [K1, (3.41)] uses $p > 1$. The lower bound (3.3) has the small-field integral with the exponent $\mp C\eta^\sigma |T|$ (in a lower bound the sign must be $-$). The second term of (3.4) is an additive large-field remainder; it is used in [K1] only at fixed volume ((3.10)–(3.13) there), and in the same way in King's torus comparison [K2, (4.7)], where it is harmless.

Proposition A.1. *Suppose (3.3)–(3.4) hold with C independent of $|T|$, and $I^{(k)} \ge e^{-C''|T|}$. Then*

$$-C\eta^\sigma \le |T|^{-1} \log\big(Z/I^{(k)}\big) \le \max\{C\eta^\sigma,\ C + C'' - p(\eta)^2/|T|\} + |T|^{-1}\log 2.$$

At fixed η the right side tends to $C + C''$ as $|T| \to \infty$, not to $O(\eta^\sigma)$.

Proof. $Z/I \le e^{C\eta^\sigma |T|} + e^{-p^2 + (C+C'')|T|}$ and $\log(x+y) \le \max(\log x, \log y) + \log 2$. □

Hence the printed bounds do not give a two-sided representation of the free energy per unit volume by a small-field effective integral at a fixed coarse scale, uniformly in the volume, and arguments that need such a representation cannot be based on them. In this paper the volume is fixed wherever these bounds are used (in (F)), and the large-field term is absorbed on the left by a polynomial lower bound, as in [CT, Prop. 7.1]. The constant of [K1, Thm. 2.1(ii)] likewise depends on $|T|$, so King's theorems give no bound on torus functionals uniform in the torus; where such uniformity would be needed (Lemma 4.2(d)) we use only a fixed torus.

A.2. **King's torus conjecture: Theorem 2.4 of** [K2] **as printed.** In [K1, K2] the curvature source is $F(g) = \sum_p \varepsilon^3 g(p) F(p)$ with one function g on all plaquettes [K2, (2.12)], and in Theorem 2.4 of [K2] the source is rotated as a scalar.

Proposition A.2 (the scalar reading)**.** *Let $e = 0$ and $F^{sc}_\Lambda(g) := \sum_p \varepsilon^3 g(c(p)) F(p)$ with a single real function g on all plaquettes. Let $R = R_{12}(\theta_0)$, $\ell = 50MJ$ and $g(x) = \cos(2\pi x_1/\ell)$. Then* $\lim_k \langle e^{F^{sc}(g)} \rangle_{\Lambda_k, D_J} \ne \lim_k \langle e^{F^{sc}(g)} \rangle_{R\Lambda_k, D_J}$.

Proof. At $e = 0$ the photon is the centred Gaussian with covariance $\delta_{\alpha\beta}(-\Delta_\varepsilon + \mu_0^2)^{-1}$ on the torus, so $\langle e^{F^{sc}(g)}\rangle = e^{\frac{1}{2}\mathrm{Var}\, F^{sc}(g)}$. On a lattice with frame (f_α), $\sum_{\alpha<\beta} F(f_\alpha, f_\beta) = \langle u, B\rangle$ for a continuum field strength $F = dA$, $B = \star F$, where $u := \sum_{\alpha<\beta} f_\alpha \times f_\beta$. For the plane wave g only the momenta $\pm k_0$, $k_0 = (2\pi/\ell)e_1$, contribute, and the lattice symbols converge, so $\lim_k \mathrm{Var}\, F^{sc}(g) = \frac{\ell^3}{2}\,|k_0|^2(|u|^2 - u_1^2)/(|k_0|^2 + \mu_0^2)$. For Λ_k, $u = (1, -1, 1)$ and $|u|^2 - u_1^2 = 2$; for $R\Lambda_k$, $u = R(1, -1, 1) = (\frac{7}{5}, -\frac{1}{5}, 1)$ and $|u|^2 - u_1^2 = \frac{26}{25}$. The limiting variances differ by the factor $\frac{25}{13}$. □

With 2-form sources sampled in each lattice's own frame (Definition 2.8) the limit on either lattice is the variance of $\int\langle G, F\rangle$, which does not depend on the frame, and the obstruction disappears. This reading is consistent with King's proof, which treats the source components separately (Remark 2.9); it is the form of Hypothesis 4.1 and of Theorem 4.9. The second error in the printed statement concerns the Wick constant and is the subject of the next subsection. Since the printed statement is false on both counts, we refer to it as King's torus conjecture; the statement that is true, as an implication from the inputs of Section 4.3, is Theorem 4.9. We note also that Theorem 2.4 is not, by itself, a statement about King's infinite-volume state: the passage from tori to the state [K2, (2.24)–(2.25)] is explicitly conjectural there ("we can expect that a cluster expansion would give the following result"), and neither of the two routes of this paper uses it (Remark 4.11).

A.3. **The Wick square and the constant** $\ln 5$**.**

Proposition A.3 (the logarithm)**.** *In formal perturbation theory, King's model on $\varepsilon\mathbb{Z}^3$ with the free Wick constant C_ε alone satisfies, at order $e^2\lambda^0$,*

$$\langle :\phi^2:_{\varepsilon\mathbb{Z}^3}(h)\rangle = \frac{e^2}{4\pi^2}\,\ln(1/\varepsilon)\int h + O(1).$$

The coefficient does not depend on m or μ_0. (*A complete proof, with the lattice regularisation carried through, is* [CT, Thm. 9.2, App. B]*; we indicate the computation.*)

Computation. In the normalisation of Definition 2.4 each real component of ϕ has propagator $(p^2 + m^2)^{-1}$ at $e = 0$. Write the one-loop correction to the inverse propagator as $\Sigma(p)$; the mass counterterm removes $\Sigma(0)$. The photon-exchange part is $-e^2\int \frac{d^3k}{(2\pi)^3}\frac{(2p-k)^2}{((p-k)^2+m^2)(k^2+\mu_0^2)}$; writing $(2p - k)^2 = 2p^2 + 2(p - k)^2 - k^2$ and using $\int \frac{d^3k}{(2\pi)^3}\frac{1}{k^2(p-k)^2} = \frac{1}{8|p|}$, one finds $\Sigma(p) - \Sigma(0) = -\frac{e^2}{4}|p| + O(m + \mu_0)$ for $1 \ll |p| \ll \varepsilon^{-1}$, the lattice regularisation changing only momentum-independent terms and terms of order εp^2. Inserting, $\langle|\phi(0)|^2\rangle - 2\mathsf{C}_\varepsilon = -2\int \frac{d^3p}{(2\pi)^3}\frac{\Sigma(p)-\Sigma(0)}{(p^2+m^2)^2} + O(e^4) = \frac{e^2}{2}\cdot\frac{1}{2\pi^2}\ln(1/\varepsilon) + O(1)$ at order e^2. □

Consequences.

(a) For h with $\int h \neq 0$ the printed generating functionals with scalar sources have no finite continuum limit beyond the curvature sector, at least in perturbation theory: for $h \geq 0$, $h \not\equiv 0$, Jensen's inequality gives $\langle e^{:\phi^2:(h)}\rangle \geq e^{\langle:\phi^2:(h)\rangle} \to \infty$. This affects [K1, Thm. 2.1] and [K2, Thms. 2.1–2.3, (3.43), (4.2)] for scalar sources with $\int h \neq 0$. In the cube, King subtracts the Dirichlet constant [K2, (2.13)] and, at distance > 1 from the boundary, the free one, which differ by a convergent amount only. The statement of [K1, p. 677] that the sources are proportional to $(5^k\varepsilon)^\alpha$, $\alpha > 0$, in the small-field region does not hold for the marginal graph that produces the logarithm. We allow the additional vacuum counterterm of Section 3; walled ratios and the curvature sector are unaffected, since the h-linear c-numbers cancel in them.

(b) King's proof of [K2, Thm. 2.4] compares the effective action generated from spacing ε by a first block-spin step with rotated blocks with the effective action generated from the rotated

lattice of spacing 5ε [K2, (4.3)–(4.9)]. In a scheme a the two generating functionals carry the multipliers $e^{-wa_\varepsilon \mathsf{S}(h)}$ and $e^{-wa_{5\varepsilon}\mathsf{S}(h)}$. If a removes the logarithm, then $a_\varepsilon - a_{5\varepsilon} \to \frac{e^2}{4\pi^2}\ln 5$, so the comparison [K2, (4.6)] must carry the c-number $\frac{e^2}{4\pi^2}\ln 5 \int h$, and as printed it cannot hold for $\int h \neq 0$. Since a walled ratio contains the wall source $-w$ with $\int w > 0$, this enters the proof of walled comparisons even though their statement is scheme-free. In the equal-spacing formulation of Section 4 and of [CT] the two lattices carry the same a_k, and the issue does not arise.

A.4. **The bound (4.33) of** [K1]**.** In the proof of [K1, Lemma 4.5] (p. 674) the positivity (4.33) is derived through (4.36)–(4.37), which treat the block average as a Fourier multiplier; that step is false as printed. The conclusion holds. We use King's setting at the unit scale of step k: Ω is a periodic box of the unit lattice $\mathbb{Z}^3$ that is a union of blocks of side 5, the background vanishes, $\Delta^{(k)}$ and $\Delta^{(k+n)}$ are the translation-invariant operators of the flow on Ω, Q is the block average, so that $Q^*Q = P$ is the orthogonal projection onto block-constant functions, and $-\Delta$ is the lattice Laplacian of Ω. King's coefficients satisfy $a_j \geq \frac{24}{25}a$, and $\Delta^{(j)}(p)^{-1} = a_j^{-1} + \sum_l |u(p+l)|^2/\Delta^\eta(p+l)$ [K1, (4.5)], where u is the symbol of the average over a unit block of the fine lattice of spacing η, $\sum_l |u(p+l)|^2 = 1$, the sum runs over $l \in 2\pi\mathbb{Z}^3$ with $p+l$ in the Brillouin zone of the fine lattice, and $\Delta^\eta(q) = \sum_\mu 4\eta^{-2}\sin^2(\eta q_\mu/2) + m_k^2$ with $m_k^2 \geq 0$.

Proposition A.4 (the bound (4.33))**.** *For $s \in [0,1]$, all k, n and $m_k^2 \geq 0$, the operator $X(s) := s\Delta^{(k)} + (1-s)\Delta^{(k+n)} + \frac{a}{25}Q^*Q$ on $\ell^2(\Omega)$ satisfies $X(s) \geq \gamma_0$, where $\gamma_0 := \min\left(4c_\Delta \sin^2\frac{\pi}{10}, \frac{a}{25}\right)$ and $c_\Delta := \left(\frac{25\cdot 3\pi^2}{24a} + \frac{\pi^2}{4}\right)^{-1}$.*

Proof. (i) For $|t| \leq \pi$, $4\sin^2(t/2) \geq 4t^2/\pi^2$, so $\Delta^\eta(q) \geq 4|q|^2/\pi^2$ on the fine Brillouin zone; and $|p_\mu + l_\mu| \geq |p_\mu|$ for $|p_\mu| \leq \pi$, $l_\mu \in 2\pi\mathbb{Z}$. With $\sum_l |u(p+l)|^2 = 1$ this gives $\Delta^{(j)}(p)^{-1} \leq a_j^{-1} + \frac{\pi^2}{4}|p|^{-2}$, and since $|p|^2 \leq 3\pi^2$ and $a_j \geq \frac{24}{25}a$, $\Delta^{(j)}(p) \geq c_\Delta|p|^2 \geq c_\Delta \sum_\mu 4\sin^2(p_\mu/2)$, the symbol of $-\Delta$. So $s\Delta^{(k)} + (1-s)\Delta^{(k+n)} \geq c_\Delta(-\Delta)$. (ii) On one block $B = \{0,\dots,4\}^3$ the Neumann Laplacian is a sum of three path-graph Laplacians, whose smallest non-zero eigenvalue is $\lambda_1 = 4\sin^2\frac{\pi}{10}$; hence $\sum_{x\in B}|\phi(x) - \phi_B|^2 \leq \lambda_1^{-1}\sum_{\{x,x'\}\subset B}|\phi(x) - \phi(x')|^2$, the sum over nearest-neighbour pairs, ϕ_B the block mean. Summing over the blocks, whose inner bonds are bonds of Ω, gives $\|(1-P)\phi\|^2 \leq \lambda_1^{-1}\langle\phi, -\Delta\phi\rangle$. (iii) Therefore $\langle\phi, X(s)\phi\rangle \geq c_\Delta\lambda_1\|(1-P)\phi\|^2 + \frac{a}{25}\|P\phi\|^2 \geq \gamma_0\|\phi\|^2$. For the photon the same holds componentwise. □

A.5. **Minor points.** (i) In [K2, (3.41)] the real part of a scalar source is split as $h_+ - h_-$ with $h_\pm \geq 0$ and no further regularity stated; the argument of Theorem 3.8(iii) uses smooth non-negative parts. (ii) The inequalities [K2, (3.24)–(3.29)] are stated for ratios of expectations of $e^{r^2(h)+\mathrm{i}A(g)}$; we use them in the form of [BFS, Thms. 6.1, 6.2], which also covers killed sites and the two-level walls. (iii) King's bound [K2, (3.43)] is asserted for the Neumann–Dirichlet cubes by "a standard argument (see [GJ1])" (not printed); it is not a bound on torus functionals, which lie above the cube functionals on the cone (Lemma 2.14). The torus moment constant used in Propositions 7.19 and 7.20 is therefore taken at a fixed torus and comes from the existence of the torus limit, not from (3.43). (iv) King's vertex list [K1, (3.23)] is printed "for $2 < m$"; the action [K1, (2.1)–(2.4)], from which [K1, (3.14)–(3.17)] build the list, gives it for every even $m \geq 2$, so the seagull vertex of order two belongs to the model, and with the list as printed one renormalisation statement for the photon two-point graphs fails. (v) In the proof of [K2, Thm. 3.2] (p. 334) the sentence that the remaining expression "is exactly $\langle\mathcal{O}(ig,h)\rangle_{N,D}^{C_J}$" names the state of the box $C_J \cap \Lambda_k$ of Definition 2.5(a), in which every site of the smaller cube is alive; between two of King's renormalisation-group cubes it would be one layer off, and the inequalities (3.32)–(3.33) between them follow instead from Lemma 2.16 (Corollary 2.17).

References

[AS] Bałaban's and King's asserted steps at block factor 5: (I) Bałaban's same-scale estimates, with Proposition 2.6 of [Ba2] at a non-zero background and the change of domain of his propagators behind it, on the cubic and on the skew tori, and the renormalisation bound for the operator with a scalar wall; (II) Bałaban's enumeration of renormalised graphs as a frame, with one inserted vertex, and King's cut of long lines on the torus; (III) King's propagator and remainder bounds in the forms consumed. Proofs as implications from the printed proofs, each checked independently, for King's in-block contours; for the contour families of the flows of [CT] the estimates are Proposition 1.3 of Section 1.6, proved in [SUP]. Manuscripts, in their versions of 19 October 2026. The statements consumed from this collection are located in it through the dependency index of the record, which names, for each item of Sections 1.4 and 1.5, the manuscript and the statement in it.

[Ba1] T. Bałaban, $(\mathrm{Higgs})_{2,3}$ quantum fields in a finite volume. I. A lower bound, Commun. Math. Phys. **85** (1982), 603–626. DOI 10.1007/BF01403506.

[Ba2] T. Bałaban, $(\mathrm{Higgs})_{2,3}$ quantum fields in a finite volume. II. An upper bound, Commun. Math. Phys. **86** (1982), 555–594. DOI 10.1007/BF01214890.

[Ba3] T. Bałaban, $(\mathrm{Higgs})_{2,3}$ quantum fields in a finite volume. III. Renormalization, Commun. Math. Phys. **88** (1983), 411–445. DOI 10.1007/BF01213217.

[Ba4] T. Bałaban, Regularity and decay of lattice Green's functions, Commun. Math. Phys. **89** (1983), 571–597. DOI 10.1007/BF01214744.

[BIJ] T. Bałaban, J. Imbrie, A. Jaffe, Effective action and cluster properties of the abelian Higgs model, Commun. Math. Phys. **114** (1988), 257–315. DOI 10.1007/BF01225037.

[BFS] D. Brydges, J. Fröhlich, E. Seiler, On the construction of quantized gauge fields. I. General results, Ann. Physics **121** (1979), 227–284. DOI 10.1016/0003-4916(79)90098-8.

[CA] King's cube limit at zero scalar source on the renormalisation-group cube: the Cauchy argument of [K2, §3.1] in ratio form, with the ultraviolet construction on the cube, the scalar boundary remainders, the line bounds of the flow with the Dirichlet weight, the transfer [K1, (3.90)–(3.98)] to the cube, the classes with curvature vertices, the cut of the long lines on the cube, and the long trees in the difference of two cutoffs, at the faces of the cube and away from them. A proof as an implication, each step checked independently, with the bound at long trees for the families with one inserted field whose graphs contain block vertices (Proposition 1.2, proved in [SUP]). Manuscripts, in their versions of 19 October 2026; the statements consumed are located in the collection through the dependency index of the record, as for [AS].

[CT] The corrected torus theorem for the three-dimensional $U(1)$ Higgs model: the curvature sector, as one chain. Manuscript, version 4.5, as of 19 October 2026; the statements consumed are located in it through the dependency index of the record, as for [AS].

[G] The two lattice conventions for King's Neumann–Dirichlet cube: an interlacing lemma, with its corollaries. Manuscript, checked independently; as of 17 October 2026.

[H6] Existence of the walled torus limit on the axis lattice for curvature sources. Manuscript, version 3.8.

[K1] C. King, The $U(1)$ Higgs model. I. The continuum limit, Commun. Math. Phys. **102** (1986), 649–677. DOI 10.1007/BF01221651.

[K2] C. King, The $U(1)$ Higgs model. II. The infinite volume limit, Commun. Math. Phys. **103** (1986), 323–349. DOI 10.1007/BF01206942.

[LL] G. F. Lawler, V. Limic, Random Walk: A Modern Introduction, Cambridge Studies in Advanced Mathematics 123, Cambridge University Press, Cambridge, 2010. DOI 10.1017/CBO9780511750854.

[LOC] Bałaban's local inverses with the mass profile $m^2+n\psi+\kappa\mathbf{1}_U$ at a flat face, light-side half. Manuscript, version 3.

[LT] Deep-point reduction of Bałaban's local inverses on Neumann unions of blocks to the torus. Manuscript, as of 17 October 2026.

[Re1] T. Reisz, A power counting theorem for Feynman integrals on the lattice, Commun. Math. Phys. **116** (1988), 81–126. DOI 10.1007/BF01239027.

[Re2] T. Reisz, Renormalization of Feynman integrals on the lattice, Commun. Math. Phys. **117** (1988), 79–108. DOI 10.1007/BF01228412.

[RS] The restricted form of King's renormalisation theorem: Theorems R-i–R-iii with the degree count of the merged counterterm vertices. Manuscript, version 2.7.

[SUP] Completions of [CT, H6, SW1, SW3]: the residues named in Section 1.4, among them the sub-domain form of the perturbative bound, at order fourteen included, its cut members and mismatch clause at King's backgrounds, the vacuum families with the mismatch vertex, Bałaban's enumeration block by block at higher orders and the steps of the κ-Cauchy estimate, and, for the general-source extension, the identification step at a non-zero scalar source, the restricted renormalisation bound on King's cubes and the rate (SRC) at a non-zero scalar source; the large-field clause of King's upper bound at the walls; and the non-polynomial

terms of King's effective action at the frozen wall. Proofs as implications, each checked independently; the identification step at a non-zero scalar source with the bound at long trees of Proposition 1.2, and the large-field clause at the walls with Propositions 1.4 and 1.3 of Section 1.6, on the skew torus under the final-scale cap; the proofs of Propositions 1.2, 1.3, 1.4, of the final-scale cap, of the right-continuity of King's state in the scalar mass, of the readings (B1)–(B6), of the regrouping of the cut families, of (KA*) and of King's two statements in their consumed forms. Manuscripts, in their versions of 19 October 2026; the statements consumed are located in the collection through the dependency index of the record, as for [AS].

[SW1] King I Theorem 3.1 with the shell mass inside the covariance: a direct proof of $(\mathrm{SW1}^\circ)_{\mathrm{int}}$, both lines. Manuscript, version 2.8.17. It cites the two proofs of (8g), states the reflection lemma in the form used (8h), and states (LOC-R) with its flat-face condition (8j); it names the estimates near the edges and corners of H' as a further input, whose consumed forms (8j) supplies through [LT], (8k)(vi) included.

[SW3] The κ-Cauchy property of King's reduced small-field action on the capped stopping-scale set: a proof of $(\mathrm{SW3}^\circ)_{\mathrm{int}}$ as an implication. Manuscript, version 2.5v; it contains the proof of (10d) and has been checked independently in full. It uses the conventions and numbered results of [SW1] in version 2.8.7, which names the inputs of its main theorem explicitly and states its first-difference clauses on sub-domains at points away from the complement of the sub-domain; the statements of [SW1] cited in [SW3] are read with the inputs (8g), (8i)–(8k) (Section 9.1).

[TE] One configuration of the heavy-shell estimate implies the closing hypothesis. Manuscript, version 3.